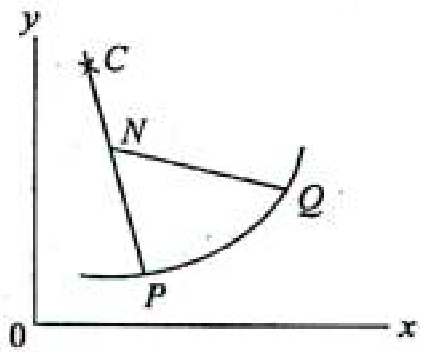
2. Curvature

Curvature measures the bend in a curve at a given point. The reciprocal of the radii of curvature at P gives curvature at P.

It C is the centre of curvature and CP is radius of curvature then the circle with centre at C and radius CP is known as circle of curvature at P. A chord drwan through P to the circle of curvature at P is known as chord of curvature at P.

(A) Formulas for Radius of curvature :

(i) If the intrinsic (s, ψ) equation of a plane curve is given, then radius of curvature is $\rho = \frac{ds}{d\psi}$



(ii) The radius of curvature for the curve $y = f(x)$ at any point (x, y) [cartesian co-ordinates] is given by :

$$\rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

on interchanging x and y we also have ρ as

$$\rho = \frac{\left[1+\left(\frac{dx}{dy}\right)^2\right]^{3/2}}{\frac{d^2x}{dy^2}}$$

(iii) For any curve $x = \phi(t)$, $y = \psi(t)$ the radius of curvature at any point 't' is given by :-

$$\rho = \frac{\left[x'^2 + y'^2\right]^{3/2}}{x'y'' - y'x''}$$

Where (') dash denotes differentiation with respect to 't'.

(B) Curvature at origin

(i) Expansion Method :

If the equation of the curve is of the form,

$$y = px + \frac{q}{\underline{|2}}x^2 + ...$$

then $\left(\frac{dy}{dx}\right)_{(0,0)} = p, \left(\frac{d^2y}{dx^2}\right)_{(0,0)} = q$

$$\therefore \quad (\rho)_{(0,0)} = \frac{(1+p^2)^{3/2}}{q}$$

(ii) Newton's Formulae

(i) When x – axis is tangent at the origin

$$(\rho)_{(0,0)} = \underset{\substack{x \to 0 \\ y \to 0}}{Lt} \frac{x^2}{2y}$$

(ii) When y – axis is the tangent at the origin

$$(\rho)_{(0,0)} = \underset{\substack{x \to 0 \\ y \to 0}}{Lt} \frac{y^2}{2x}$$

(C) The centre of curvature for any point (x, y) for the curve $f(x, y) = 0$ is

$$\left(x - \frac{y_1(1+y_1^2)}{y_2}, y + \frac{1+y_1^2}{y_2}\right) \text{ where } y_1 = \frac{dy}{dx};\ y_2 = \frac{d^2y}{dx^2}$$

(D) Length of chord of curvature parallel to x–axis $= \dfrac{2y_1(1+y_1^2)}{y_2}$ where $y_1 = \dfrac{dy}{dx}, y_2 = \dfrac{d^2y}{dx^2}$

Length of chord of curvature parallel to y axis $= \dfrac{2(1+y_1^2)}{y_2}$, where $y_1 = \dfrac{dy}{dx}, y_2 = \dfrac{d^2y}{dx^2}$

(E) Test for concavity, convexity and point of inflexion :

(i) A curve is convex or concave at P to the x – axis according as $y\dfrac{d^2y}{dx^2}$ is positive or negative at p.

(ii) A curve is convex or concave at P to the y – axis according as $x\dfrac{d^2x}{dy^2}$ is positive or negative at p.

(iii) If at any point $P(x, y)$ we have $\dfrac{d^2y}{dx^2} = 0$ and $\dfrac{d^3y}{dx^3} \neq 0$ then P is point of inflexion.

EXERCISE–2(A)

Find the radius of curvature at any point (s, ψ) on the following curves :

Ex.1. $s = 8a\sin^2\frac{1}{6}\psi$ **(intrinsic form of a cardioid)**

Sol. $\rho = \dfrac{ds}{d\psi} = 8a \cdot 2\sin\frac{1}{6}\psi \cdot \cos\frac{1}{6}\psi \cdot \frac{1}{6}$

$= \frac{8}{3}a\sin\frac{\psi}{6}\cos\frac{\psi}{6} = \frac{4}{3}a\sin\frac{\psi}{3}$ **Ans.**

Ex.2. $s = 4a\sin\psi$ **(cycloid)**

Sol. $\rho = \dfrac{ds}{d\psi} = 4a\cos\psi$ **Ans.**

Ex.3. $s = c\tan\psi$ **(catenary)**

Sol. $\rho = \dfrac{ds}{d\psi} = c\sec^2\psi$ **Ans.**

Find the radius of curvature at any point (x, y) on the following curves :

Ex.4. $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

Sol. $y^2 = -\dfrac{b^2}{a^2}x^2 + b^2$

$$\Rightarrow 2yy_1 = -\frac{2b^2}{a^2}x \Rightarrow yy_1 = -\frac{b^2}{a^2}x \Rightarrow y_1 = -\frac{b^2 x}{a^2 y}$$

$$\Rightarrow y_2 = -\frac{b^2}{a^2}\left(\frac{y - xy_1}{y^2}\right) = \frac{-b^2}{a^2y^2}\left(y + \frac{b^2x^2}{a^2y}\right) = -\frac{b^2(a^2y^2 + b^2x^2)}{a^4y^3}$$

$$= -\frac{b^2}{a^4y^3}\cdot a^2b^2 = \frac{-b^4}{a^2y^3}$$

$$\therefore \quad \rho = \frac{\left(1+y_1^2\right)^{3/2}}{y_2}$$

$$\Rightarrow \quad \rho = -\frac{\left(1+\frac{b^4x^2}{a^4y^2}\right)^{3/2}}{-\frac{b^4}{a^2y^3}} = \frac{a^2y^3}{b^4}\frac{\left(a^4y^2 + b^4x^2\right)^{3/2}}{a^6\cdot y^3}$$

$$\Rightarrow \quad \rho = \frac{(b^4x^2 + a^4y^2)^{3/2}}{a^4b^4} \quad \textbf{Ans.}$$

Ex.5. $x^2 = 4ay$

Sol. $\frac{dy}{dx} = y_1 = \frac{x}{2a};\ \frac{d^2y}{dx^2} = y_2 = \frac{1}{2a}$

$$\therefore \quad \rho = \frac{\left(1+y_1^2\right)^{3/2}}{y_2}$$

$$\Rightarrow \quad \rho = \frac{\left(1+\frac{x^2}{4a^2}\right)^{3/2}}{\frac{1}{2a}} = \frac{\left(1+\frac{y}{a}\right)^{3/2}}{\frac{1}{2a}} = 2a\cdot\frac{1}{a^{3/2}}(y+a)^{3/2}$$

$$\Rightarrow \quad \rho = \frac{2}{\sqrt{a}}(y+a)^{3/2} \quad \textbf{Ans.}$$

Ex.6. $xy = a^2$

Sol. $y_1 = \frac{-a^2}{x^2};\ y_2 = \frac{2a^2}{x^3}$

$$\therefore \quad \rho = \frac{\left(1+y_1^2\right)^{3/2}}{y_2}$$

$$\therefore \quad \rho = \frac{\left(1+\frac{a^4}{x^4}\right)^{3/2}}{\frac{2a^2}{x^3}} = \frac{x^3}{2a^2}\cdot\frac{1}{x^6}(x^4+a^4)^{3/2}$$

$$= \frac{1}{2a^2x^3}(x^4 + x^2y^2)^{3/2}$$

$$= \frac{1}{2a^2x^3}\cdot x^3(x^2+y^2)^{3/2} \Rightarrow \rho = \frac{1}{2a^2}(x^2+y^2)^{3/2}$$

Ans.

Ex.7. $y^2 = 4\ ax$

Sol. $y^2 = 4ax \Rightarrow y = 2\sqrt{ax}$

$$\Rightarrow \quad \frac{dy}{dx} = 2\sqrt{a}\cdot\frac{1}{2\sqrt{x}} = \sqrt{\frac{a}{x}} \Rightarrow \frac{d^2y}{dx^2} = -\frac{\sqrt{a}}{2x\sqrt{x}}$$

$$\therefore \quad \rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{\left(1+\frac{a}{x}\right)^{3/2}}{-\frac{\sqrt{a}}{2x\sqrt{x}}}$$

$$= -\frac{2x\sqrt{x}\left(\frac{x+a}{x}\right)^{3/2}}{\sqrt{a}} = -\frac{2(x+a)^{3/2}}{\sqrt{a}}$$

or, $\rho = \frac{2(x+a)^{3/2}}{\sqrt{a}}$ ($\because$ ρ is always positive) **Ans.**

Ex.8. Prove that the radius of curvature at any point θ, on the curve $x = a(3\cos\theta - \cos 3\theta)$, $y = a(3\sin\theta - \sin 3\theta)$ is $3a\sin\theta$.

Sol. Here, $\dot{x} = a(-3\sin\theta + 3\sin 3\theta)$; $\dot{y} = a(3\cos\theta - 3\cos 3\theta)$

$\ddot{x} = a(-3\cos\theta + 9\cos 3\theta)$; $\ddot{y} = a(-3\sin\theta + 9\sin 3\theta)$

$$\therefore \quad \rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\therefore \quad \rho = \frac{[9a^2(\sin^2\theta + \sin^2 3\theta - 2\sin\theta\sin 3\theta) + 9a^2(\cos^2\theta + \cos^2 3\theta - 2\cos\theta\cos 3\theta)]^{3/2}}{[a^2(9\sin^2\theta + 27\sin^2 3\theta - 36\sin\theta\sin 3\theta) - a^2(-9\cos^2\theta - 27\cos^2 3\theta + 36\cos\theta\cos 3\theta)]}$$

$$= \frac{27a^3[2 - 2\cos 2\theta]^{3/2}}{9a^2 + 27a^2 - 36a^2\cos 2\theta} = \frac{27a^3[2\cdot 2\sin^2\theta]^{3/2}}{36a^2\cdot 2\sin^2\theta}$$

$$= \frac{27\times 8a^3\sin^3\theta}{72a^2\sin^2\theta}$$

$\Rightarrow \quad \rho = 3a\sin\theta$ **Proved**

Ex.9. Prove that for the cycloid. $x = a(t + \sin t)$, $y = a(1 - \cos t)$, the radius of curvature at the point 't' is $4a\cos\frac{t}{2}$.

Sol. Here, $\dot{x} = a(1 + \cos t)$; $\dot{y} = a\sin t$

$\ddot{x} = -a\sin t$; $\ddot{y} = a\cos t$

$$\therefore \quad \rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\Rightarrow \quad \rho = \frac{\{a^2(1+\cos t)^2 + a^2\sin^2 t\}^{3/2}}{a^2\cos t(1+\cos t) + a^2\sin^2 t}$$

$$= \frac{(2a^2 + 2a^2\cos t)^{3/2}}{a^2 + a^2\cos t} = \frac{(2a^2\cdot 2\cos^2 t/2)^{3/2}}{a^2\cdot 2\cos^2 t/2} \quad \frac{8a^3\cos^3 t/2}{2a^2\cos^2 t/2}$$

$\Rightarrow \quad \rho = 4a\cos t/2$ **Proved**

Ex.10. For the curve $y = a\,e^{x/a}$, prove that

$\rho = a\sec^2\theta\,\mathrm{cosec}\,\theta$, where $\theta = \tan^{-1}(y/a)$

Sol. Here $y_1 = e^{x/a}$; $y_2 = \frac{1}{a}e^{x/a}$

$$\therefore \quad \rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

$$\Rightarrow \quad \rho = \frac{(1+e^{2x/a})^{3/2}}{\frac{1}{a}e^{x/a}} = \frac{a(1+e^{2x/a})^{3/2}}{e^{x/a}}$$

Put $\frac{y}{a} = \tan\theta \Rightarrow e^{x/a} = \tan\theta$

$$\therefore \quad \rho = \frac{a(1+\tan^2\theta)^{3/2}}{\tan\theta} = a\sec^3\theta\cdot\cos\theta\cdot\mathrm{cosec}\,\theta$$

$\Rightarrow \quad \rho = a\sec^2\theta\,\mathrm{cosec}\,\theta;\ \theta = \tan^{-1}\left(\frac{y}{a}\right)$ **Proved**

Ex.11. Show that in the rectangular hyperbola

$r^2\cos 2\theta = a^2, \rho = \frac{r^3}{a^2}$

Sol. $r^2\cos 2\theta = a^2$

$\Rightarrow \quad r^2(\cos^2\theta - \sin^2\theta) = a^2$ ....(1)

Let $x = r\cos\theta$, $y = r\sin\theta$

$\therefore$ (1) $\Rightarrow$ $x^2 - y^2 = a^2$

$\Rightarrow \quad y^2 = x^2 - a^2 \Rightarrow y = \sqrt{x^2 - a^2}$ ....(2)

$$\therefore \quad y_1 = \frac{x}{\sqrt{x^2 - a^2}};$$

$$\Rightarrow \quad y_2 = \frac{\sqrt{x^2 - a^2} - \frac{x^2}{\sqrt{x^2 - a^2}}}{(x^2 - a^2)} = \frac{-a^2}{(x^2 - a^2)^{3/2}}$$

$$\therefore \quad \rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\left(1+\frac{x^2}{x^2-a^2}\right)^{3/2}}{\frac{-a^2}{(x^2-a^2)^{3/2}}} = -\frac{(2x^2-a^2)^{3/2}}{a^2}$$

$$= \frac{-(x^2+x^2-a^2)^{3/2}}{a^2} = -\frac{(x^2+y^2)^{3/2}}{a^2} \qquad \text{(using 2)}$$

$$= -\frac{(r^2)^{3/2}}{a^2} = \frac{-r^3}{a^2} \quad (\because\ x^2+y^2=r^2)$$

$$\Rightarrow \quad \rho = \frac{r^3}{a^2} \quad (\because\ \rho \text{ is always positive}) \qquad \textbf{Proved}$$

EXERCISE–2(B)

Find the radius of curvature at the origin for the following curves :

Ex.1. $y = x^4 - 4x^3 - 18x^2$.

Sol. $y = x^4 - 4x^3 - 18x^2$

$$\therefore \quad p = \left(\frac{dy}{dx}\right)_{(0,0)} = (4x^3 - 12x^2 - 36x)_{x=0} = 0$$

$$q = \left(\frac{d^2y}{dx^2}\right)_{(0,0)} = (12x^2 - 24x - 36)_{x=0} = -36$$

$$\therefore \quad \rho = \frac{(1+p^2)^{3/2}}{q} = \frac{1}{-36} = \frac{1}{36} \quad (\because\ \pi \text{ is always positive})$$

Ex.2. $x^3 + y^3 = 3axy$ **Ans.**

Sol $x^3 + y^3 = 3axy$ .....(1)

Substitue $y = px + \frac{qx^2}{2!} + ...$ in equation (1), we get :

$$x^3 + \left(px + \frac{qx^2}{2!} + ...\right)^3 = 3ax\left(px + \frac{qx^2}{2!} + ...\right)$$

or, $$x^3 + (p^3x^3 + ...) = 3apx^2 + \frac{3aq}{2}x^3 + ...$$

or, $$-3apx^2 + \left(1 + p^3 - \frac{3aq}{2}\right)x^3 + ... = 0$$

equating to zero the coefficient of x^2 and x^3, we get

$$3ap = 0 \Rightarrow p = 0$$

and $$1 + p^3 - \frac{3aq}{2} = 0 \Rightarrow 1 - \frac{3aq}{2} = 0 \qquad (\because\ p = 0)$$

$$\Rightarrow \quad q = \frac{2}{3a}$$

$$\therefore \quad \rho = \frac{(1+p^2)^{3/2}}{q} = \frac{1}{\left(\frac{2}{3a}\right)}$$

or, $\rho = \frac{3a}{2}$ **Ans.**

Ex.3. $y^2 - 3xy - 4x^2 + x^3 + x^4y + y^5 = 0$

Sol. Substitute $y = px + q\frac{x^2}{2!} + ...$ in the given equation, we get

$$\therefore \quad \left(px + \frac{qx^2}{2!} + ...\right)^2 - 3x\left(px + \frac{q}{2!}x^2 + ...\right) - 4x^2 + x^3$$

$$+ x^4\left(px + \frac{q}{2!}x^2 + ...\right) + \left(px + \frac{qx^2}{2!} + ...\right)^5 = 0$$

or, $$x^2(p^2 - 3p - 4) + x^3\left(pq - \frac{3}{2}q + 1\right) + ... = 0$$

equating to zero the coefficient of x^2 and x^3 :

$$p^2 - 3p - 4 = 0 \Rightarrow (p+1)(p-4) = 0 \Rightarrow p = 4, -1$$

$$pq-\frac{3}{2}q+1=0$$

Now $p=-1$

$$\Rightarrow \quad -q-\frac{3}{2}q+1=0 \Rightarrow \frac{5}{2}q=1 \Rightarrow q=\frac{2}{5}$$

$$\therefore \quad \rho=\frac{(1+p^2)^{3/2}}{q}=\frac{2\sqrt{2}}{2/5}=5\sqrt{2} \qquad \textbf{Ans.}$$

Again,

$$p=4 \Rightarrow 4q-\frac{3}{2}q+1=0$$

$$\Rightarrow \quad \frac{5q}{2}=-1 \Rightarrow q=-\frac{2}{5}$$

$$\therefore \quad \rho=\frac{17\sqrt{17}}{-2/5}=\frac{85}{2}\sqrt{17} \qquad (\because \ \rho \text{ is always +ve}) \qquad \textbf{Ans.}$$

Ex.4 $(y-x)(y-2x)=x^3+y^3$

Sol. Given equation can be written as

$$2x^2+y^2-3xy-x^3-y^3=0 \qquad \text{....(1)}$$

substitute $y=px+\frac{q}{2!}x^2+$ .....in equation (1), we get

$$2x^2+\left(px+\frac{q}{2}x^2+.....\right)^2-3x\left(px+\frac{q}{2}x^2+...\right)$$

$$-x^3-\left(px+\frac{q}{2}x^2+...\right)^3=0$$

or, $\left(2+p^2-3p\right)x^2+\left(pq-\frac{3q}{2}-1-p^3\right)x^3+.....=0$

equating to zero the coeff of x^2 and x^3 :

$$p^2-3p+2=0 \Rightarrow (p-1)(p-2)=0 \Rightarrow p=1,2$$

and $pq-\frac{3q}{2}-1-p^3=0$

$$p=1 \rightarrow q-\frac{3}{2}q-1-1=0$$

$$\Rightarrow \quad -\frac{q}{2}=2 \Rightarrow q=-4$$

$$\therefore \quad \rho=\frac{2\sqrt{2}}{+4}=\frac{1}{\sqrt{2}} \qquad \left(\because \ \rho=\frac{(1+p^2)^{3/2}}{q}\right) \qquad \textbf{Ans.}$$

$$p=2 \rightarrow 2q-\frac{3q}{2}-1-8=0$$

$$\frac{q}{2}=9 \Rightarrow q=18$$

$$\therefore \quad \rho=\frac{5\sqrt{5}}{18} \qquad \textbf{Ans.}$$

Ex.5 $y^2=\frac{x^2(a+x)}{(a-x)}$

Sol. Given equation can be written as

$$y^2a-xy^2-x^2a-x^3=0 \qquad \text{....(1)}$$

substitute $y=px+\frac{qx^2}{2!}+.....$ in equation (1), we get

$$\left(px+\frac{qx^2}{2!}+...\right)^2a-x\left(px+\frac{qx^2}{2!}+...\right)^2-x^2a-x^3=0$$

or, $x^2(p^2a-a)+x^3(pqa-p^2-1)+...=0$

equating to zero the coeff of x^2 and x^3

$$p^2a-a=0 \Rightarrow p=\pm 1$$

and $pq\,a-p^2-1=0$

$$p=1 \rightarrow aq-1-1=0 \Rightarrow q=\frac{2}{a}$$

$$\therefore \quad \rho=\frac{(1+p^2)^{3/2}}{q}=\frac{2\sqrt{2}}{2/a}=a\sqrt{2} \qquad \textbf{Ans.}$$

$$p=-1 \rightarrow -aq-1-1=0 \Rightarrow q=\frac{-2}{a}$$

$$\therefore \quad \rho = \frac{2\sqrt{2}}{-2/a} = -a\sqrt{2} \qquad \textbf{Ans.}$$

Find the radius of curvature for the following curves at a poin[t] of the curve :

Ex.6. **Find the radius of curvature for the curve**

$$x = c\log\left\{s + \sqrt{s^2 + c^2}\right\},\ y = \sqrt{(s^2 + c^2)}$$

Sol. Now, $\frac{dx}{ds} = c\frac{1}{s + \sqrt{(c^2 + s^2)}}\left\{1 + \frac{s}{\sqrt{(c^2 + s^2)}}\right\} = \frac{c}{\sqrt{(c^2 + s^2)}}$

$$\Rightarrow \quad \frac{d^2x}{ds^2} = \frac{-cs}{(c^2 + s^2)^{3/2}}$$

Also, $\frac{dy}{ds} = \frac{s}{\sqrt{(c^2 + s^2)}}$

Now, $\rho = -\frac{\frac{dy}{ds}}{\left(\frac{d^2x}{ds^2}\right)}$

$$= -\frac{s/\sqrt{(c^2 + s^2)}}{\frac{-cs}{(c^2 + s^2)^{3/2}}} = \frac{s}{\sqrt{(c^2 + s^2)}} \times \frac{(c^2 + s^2)^{3/2}}{cs} = \frac{c^2 + s^2}{c}$$

$$\therefore \quad \rho = \frac{(c^2 + s^2)}{c} \qquad \textbf{Ans.}$$

Ex.7. $s\sqrt{8ay}$.

Sol. $s^2 = 8ay$

$$\Rightarrow \quad 2s = 8a\frac{dy}{ds} \Rightarrow \frac{dy}{ds} = \frac{s}{4a} \Rightarrow \sin\psi = \frac{s}{4a} \quad \left[\because \sin\psi = \frac{dy}{ds}\right]$$

$$\Rightarrow \quad s = 4a\sin\psi$$

$$\Rightarrow \quad \frac{ds}{d\psi} = 4a\cos\psi = 4a\sqrt{1 - \sin^2\psi} = 4a\sqrt{1 - \frac{s^2}{16a^2}}$$

$$\Rightarrow \quad \frac{ds}{d\psi} = \sqrt{16a^2 - s^2} = \sqrt{16a^2 - 8ay} = \sqrt{16a^2\left(1 - \frac{y}{2a}\right)}$$

$$\text{or,} \quad \frac{ds}{d\psi} = \rho = 4a\sqrt{\left(1 - \frac{y}{2a}\right)} \qquad \textbf{Ans.}$$

Ex.8 $s = ae^{x/a}$

Sol. Diff. w. r. t. 's' $\rightarrow 1 = a\frac{1}{a}e^{x/a}\frac{dx}{ds} = e^{x/a}\frac{dx}{ds}$

$$\text{or,} \quad 1 = \frac{s}{a}\cdot\cos\psi \quad \left[\because \frac{dx}{ds} = \cos\psi\right]$$

$$\text{or,} \quad s = a\sec\psi$$

$$\Rightarrow \quad \rho = \frac{ds}{d\psi} = a\sec\psi\tan\psi = s\sqrt{\sec^2\psi - 1} = s\sqrt{\left(\frac{s^2}{a^2} - 1\right)}$$

$$\Rightarrow \quad \rho = \frac{s}{a}\sqrt{s^2 - a^2} \qquad \textbf{Ans.}$$

Find the co-ordinates of centre of curvature at any point (x, y) on each of the following curves

Ex.9 $a^2 y = x^3$

Sol. Here $y = \frac{x^3}{a^2}$

$$\therefore \quad y' = \frac{3x^2}{a^2}; \quad y'' = \frac{6x}{a^2}$$

Now coordinates of centre of curvature is :

$$\alpha = x - \frac{y'(1 + y'^2)}{y''}; \quad \beta = y + \frac{(1 + y'^2)}{y''}$$

$$\therefore \quad \alpha = x - \frac{\frac{3x^2}{a^2}\left(1 + \frac{9x^4}{a^4}\right)}{6x/a^2} = x - \frac{x}{2}\left(1 + \frac{9x^4}{a^4}\right)$$

or, $\alpha = \frac{x}{2} - \frac{9x^5}{2a^4} = \frac{x}{2}\left(1 - \frac{9x^4}{a^4}\right)$

$$\beta = y + \frac{1 + 9x^4 / a^4}{6x / a^2} = \frac{x^3}{a^2} + \frac{a^2}{6x}\left(1 + \frac{9x^4}{a^4}\right)$$

$$= \frac{x^3}{a^2} + \frac{a^2}{6x} + \frac{3x^3}{2a^2} = \frac{5x^3}{2a^2} + \frac{a^2}{6x}$$

$\therefore$ centre : $\left[\frac{x}{2}\left(1 - \frac{9x^4}{a^4}\right), \frac{5x^3}{2a^2} + \frac{a^2}{6x}\right]$ **Ans.**

Porb.10 $x^2 = 4ay$

Sol. Here, $y = \frac{x^2}{4a}$

$\therefore$ $y' = \frac{dy}{dx} = \frac{x}{2a}$; $y'' = \frac{d^2y}{dx^2} = \frac{1}{2a}$

$\therefore$ centre of curvature is

$$\alpha = x - \frac{\frac{x}{2a}\left(1 + \frac{x^2}{4a^2}\right)}{\frac{1}{2a}} = x - x - \frac{x^3}{4a^2} = \frac{-x^3}{4a^2}$$

and $\beta = y + \frac{\left(1 + \frac{x^2}{4a^2}\right)}{\frac{1}{2a}} = y + 2a + \frac{x^2}{2a}$

$$= \frac{x^2}{4a} + 2a + \frac{x^2}{2a} \left(\because\ y = \frac{x^2}{4a}\right)$$

$$= 2a + \frac{3x^2}{4a}$$

$\therefore$ $\beta = 2a + \frac{3x^2}{4a}$

$\therefore$ coordinates of centre of curvature :

$$(\alpha, \beta) = \left(-\frac{x^3}{4a^2}, 2a + \frac{3x^2}{4a}\right) \quad \textbf{Ans.}$$

Ex.11 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Sol. $\frac{2x}{a^2} - \frac{2yy'}{b^2} = 0 \Rightarrow y' = \frac{b^2x}{a^2y}$

$$\therefore \quad y'' = \frac{b^2}{a^2}\left[\frac{y - xy'}{y^2}\right] = \frac{b^2}{a^2}\left(\frac{y - x\frac{b^2x}{a^2y}}{y^2}\right) = \frac{b^2(a^2y^2 - b^2x^2)}{a^4y^3}$$

$$= \frac{b^2(-a^2b^2)}{a^4y^3} \Rightarrow y'' = -\frac{b^4}{a^2y^3}$$

$$\therefore \quad \alpha = x - \frac{\frac{b^2x}{a^2y}\left(1 + \frac{b^4x^2}{a^4y^2}\right)}{-\frac{b^4}{a^2y^3}} = x + \frac{y^2x}{b^2}\frac{(a^4y^2 + b^4x^2)}{a^4y^2}$$

or $\alpha = x + \frac{x(a^4y^2 + b^4x^2)}{b^2a^4}$

or, $\alpha = x + x\left(\frac{y^2}{b^2} + \frac{b^2}{a^2}\cdot\frac{x^2}{a^2}\right)$

$$= x + x\left(\frac{x^2}{a^2} - 1 + \frac{b^2x^2}{a^4}\right) = x + \frac{x^3}{a^2} - x - \frac{b^2x^3}{a^4}$$

$\Rightarrow$ $\alpha = \frac{a^2 + b^2}{a^4}x^3$

and $\beta = y + \frac{1 + \frac{b^4x^2}{a^4y^2}}{-\frac{b^4}{a^2y^3}} = y - \frac{a^2y^3}{b^4}\left(\frac{a^4y^2 + b^4x^2}{a^4y^2}\right)$

$$\Rightarrow \quad \beta = y - \frac{y}{a^2 b^4}(a^4 y^2 + b^4 x^2) = y - y\left(\frac{a^2}{b^2}\cdot\frac{y^2}{b^2} + \frac{x^2}{a^2}\right)$$

$$\therefore \quad \beta = y - y\left(\frac{a^2 y^2}{b^4} + 1 + \frac{y^2}{b^2}\right) = -y^3\left(\frac{a^2+b^2}{b^4}\right)$$

$$\therefore \quad \text{centre} \rightarrow \left(\frac{a^2+b^2}{b^4}x^3, -\frac{(a^2+b^2)}{b^4}y^3\right) \qquad \textbf{Ans.}$$

Ex.12 $y = c\cos h\dfrac{x}{c}$

Sol. Here, $y' = c\cdot\dfrac{1}{c}\sinh\dfrac{x}{c} = \sinh\dfrac{x}{c}$

$$y'' = \frac{1}{c}\cosh\frac{x}{c}$$

$$\therefore \quad \alpha = x - \frac{\sinh\frac{x}{c}(1+\sinh^2 x/c)}{\frac{1}{c}\cosh x/c} = x - \frac{c\sinh\frac{x}{c}\cdot\cosh^2\frac{x}{c}}{\cosh\frac{x}{c}}$$

$$\alpha = x - c\sinh\frac{x}{c}\cosh\frac{x}{c}$$

$$= x - c\cdot\sqrt{\left(\cosh^2\frac{x}{c} - 1\right)}\cdot\cosh\frac{x}{c}$$

$$\left[\because \cosh^2 x/c - \sinh^2 x/c = 1\right]$$

$$= x - c\cdot\sqrt{\frac{y^2}{c^2} - 1}\cdot\frac{y}{c} = x - \frac{\sqrt{y^2-c^2}\cdot y}{c}$$

$$\Rightarrow \quad \alpha = x - \frac{y}{c}\sqrt{y^2 - c^2} \qquad \text{....(1)}$$

and $\beta = y + \dfrac{1+\sinh^2\frac{x}{c}}{\frac{1}{c}\cosh\frac{x}{c}} = y + c\dfrac{\cosh^2\frac{x}{c}}{\cosh\frac{x}{c}} = y + c\cosh\dfrac{x}{c}$

$$\therefore \quad \beta = y + y = 2y \qquad \text{....(2)}$$

$$\therefore \quad \text{centre}: \left(x - \frac{y}{c}\sqrt{y^2-c^2}, 2y\right) \qquad \textbf{Ans.}$$

Ex.13. Find the radii of curvature at the origin of the two branches of the curve given by the equations

$$y = t - t^3,\ x = 1 - t^2$$

Sol. Since, $y = t(1-t^2) = tx \qquad (\because\ x = 1 - t^2)$

or, $y^2 = t^2x^2 = (1-x)x^2 = x^2 - x^3$

or, $x^2 - y^2 - x^3 = 0 \qquad \text{....(1)}$

substitute $y = px + \dfrac{q}{2!}x^2 + \ldots$ in equation (1), we get

$$x^2 - \left(px + \frac{q}{2!}x^2 + \ldots\right)^2 - x^3 = 0$$

or, $(1-p^2)x^2 + (-pq-1)x^3 + \ldots = 0$

equating to zero the coeff of x^2 and x^3 :

$$1 - p^2 = 0 \Rightarrow p = \pm 1$$

and $-pq - 1 = 0$

Now, $p = 1 \Rightarrow q = -1$

$$\therefore \quad \rho = \frac{2\sqrt{2}}{-1} = -2\sqrt{2} \qquad \textbf{Ans.}$$

Also, $p = -1 \Rightarrow q = 1$

$$\therefore \quad \rho = 2\sqrt{2} \qquad \textbf{Ans.}$$

Find the centre of curvature at the point 't' on each of the following curves :

Ex.14. $x = a\cos t,\ y = b\sin t.$

Sol. $\dfrac{dx}{dt} = -a\sin t\ ; \quad \dfrac{dy}{dt} = b\cos t$

$$\frac{dy}{dx} = -\frac{b}{a}\cot t$$

$$\frac{d^2y}{dx^2}=\frac{b}{a}\operatorname{cosec}^2 t\cdot\frac{dt}{dx}=\frac{b}{a}\operatorname{cosec}^2 t\cdot\left(\frac{-1}{a\sin t}\right)=-\frac{b}{a^2}\operatorname{cosec}^3 t$$

$$\therefore\quad \alpha=x-\frac{-\frac{b}{a}\cot t\left(1+\frac{b^2}{a^2}\cot^2 t\right)}{-\frac{b}{a^2}\operatorname{cosec}^3 t}=$$

$$=\left(a\cos t-a\cos t\sin^2 t\left(1+\frac{b^2\cos^2 t}{a^2\sin^2 t}\right)\right)$$

$$=a\cos t-a\cos t\sin^2 t-\frac{b^2}{a}\cos^3 t=a\cos^3 t-\frac{b^2}{a}\cos^3 t$$

$$\therefore\quad \alpha=\frac{(a^2-b^2)}{a}\cos^3 t \qquad \text{....(1)}$$

and $$\beta=y+\frac{1+\frac{b^2}{a^2}\cot^2 t}{-\frac{b}{a^2}\operatorname{cosec}^3 t}=y-\frac{a^2}{b}\sin^3 t\left(1+\frac{b^2\cos^2 t}{a^2\sin^2 t}\right)$$

$$=b\sin t-\frac{a^2}{b}\sin^3 t-b\sin t\cos^2 t=b\sin^3 t-\frac{a^2}{b}\sin^3 t$$

$$\therefore\quad \beta=\frac{b^2-a^2}{b}\sin^3 t$$

$$\therefore\quad \text{centre}\rightarrow\left[\frac{(a^2-b^2)}{a}\cos^3 t,\frac{b^2-a^2}{b}\sin^3 t\right]$$ **Ans.**

Ex.15. $x=a\left(\cos t+\log\tan\frac{t}{2}\right), y=a\sin t$

Sol. $$\frac{dx}{dt}=a\left[-\sin t+\frac{1}{\tan t/2}\cdot\sec^2 t/2\cdot\frac{1}{2}\right];\ \frac{dy}{dt}=a\cos t$$

$$=a\left[-\sin t+\frac{1}{\sin t}\right]=a\frac{\cos^2 t}{\sin t}$$

$$\therefore\quad \frac{dy}{dx}=\frac{a\cos t}{a\cos^2 t/\sin t}=\tan t$$

$$\frac{d^2y}{dx^2}=\sec^2 t\cdot\frac{dt}{dx}=\sec^2 t\cdot\frac{\sin t}{a\cos^2 t}=\frac{\sin t}{a\cos^4 t}$$

$$\therefore\quad \alpha=x-\frac{\tan t(1+\tan^2 t)}{\frac{\sin t}{a\cos^4 t}}=a\left(\cos t+\log\tan\frac{t}{2}\right)$$

$$-a\cos^3 t\cdot\sec^2 t$$

$$=a\cos t+a\log\tan t/2-a\cos t$$

$$\therefore\quad \alpha=a\log\tan t/2 \qquad \text{....(1)}$$

$$\beta=y+\frac{1+\tan^2 t}{\frac{\sin t}{a\cos^4 t}}=a\sin t+a\frac{\cos^2 t}{\sin t}=a\operatorname{cosec} t$$

$$\therefore\quad \text{centre}\rightarrow\left(a\log\tan\frac{t}{2},a\operatorname{cosec} t\right)$$ **Ans.**

Ex.16. $x=at^2,\ y=2at$

Sol. $$\frac{dx}{dt}=2at;\ \frac{dy}{dt}=2a$$

$$\therefore\quad \frac{dy}{dx}=\frac{1}{t}$$

$$\Rightarrow\quad \frac{d^2y}{dx^2}=-\frac{1}{t^2}\cdot\frac{dt}{dx}=-\frac{1}{t^2}\cdot\frac{1}{2at}=-\frac{1}{2at^3}$$

$$\therefore\quad \alpha=x-\frac{\frac{1}{t}\left(1+\frac{1}{t^2}\right)}{-\frac{1}{2at^3}}=at^2+2at^2\left(1+\frac{1}{t^2}\right)=3at^2+2a$$

$$\beta=y+\frac{1+\frac{1}{t^2}}{-\frac{1}{2at^3}}=2at-2at^3-2at=-2at^3$$

$$\therefore\quad \text{centre}\rightarrow(2a+3at^2,-2at^3)$$ **Ans.**

Ex.17. Show that in the catenary $y = a\cosh\left(\frac{x}{a}\right)$, the chord of curvature parallel to the axis of x is of length $a\sinh\left(\frac{2x}{a}\right)$ and the chord of curvature parallel to the axis of y is double the ordinate.

Sol. $y = a\cosh\frac{x}{a}$

$\frac{dy}{dx} = \sinh\frac{x}{a}$; $\frac{d^2y}{dx^2} = \frac{1}{a}\cosh\frac{x}{a}$

Length of chord of curvature || to x – axis is

$$= \frac{2dy/dx\left[1+(dy/dx)^2\right]}{d^2y/dx^2}$$

$$= \frac{2\sinh x/a\left[1+\sinh^2 x/a\right]}{\frac{1}{a}\cosh\frac{x}{a}} = 2a\sinh\frac{x}{a}\cdot\frac{\cosh^2 x/a}{\cosh x/a}$$

$$= a\sinh\left(\frac{2x}{a}\right) \qquad \text{.....(1) Proved.}$$

Length of chord || to y – axis

$$= \frac{2\left[1+(dy/dx)^2\right]}{d^2y/dx^2}$$

$$= \frac{2\left(1+\sinh^2 x/a\right)}{\frac{1}{a}\cosh x/a} = \frac{2a\cosh^2 x/a}{\cosh x/a} = 2\cdot a\cosh x$$

$= 2\cdot y$

$= 2$ (ordinate) **Proved.**

12

Ex.18. In the curve $y = a\log\sec\left(\frac{x}{a}\right)$, prove that the chord of curvature parallel to the axis of y is of constant length.

Sol. $\frac{dy}{dx} = a\cdot\frac{1}{\sec x/a}\sec x/a\cdot\tan x/a\cdot\frac{1}{a} = \tan\frac{x}{a}$

$\frac{d^2y}{dx^2} = \frac{1}{a}\sec^2\frac{x}{a}$

$\therefore$ Length of chord || to y axis

$$= \frac{2\left[1+\tan^2 x/a\right]}{\frac{1}{a}\sec^2\frac{x}{a}}$$

$= 2a$

$=$ constant **Proved.**

EXERCISE–2(C)

Ex.1. Prove that, for the curve $y(a^2+x^2) = x^3$ there is a point of inflexion at the origin and also at the points for which $x = \pm a\sqrt{3}$.

Sol. Here $y = \frac{x^3}{a^2+x^2}$

$$\Rightarrow \quad y' = \frac{3x^2(a^2+x^2)-2x^4}{(a^2+x^2)^2} = \frac{x^4+3a^2x^2}{(a^2+x^2)^2}$$

$$\Rightarrow \quad y'' = \frac{(4x^3+6a^2x)(a^2+x^2)^{\cancel{2}} - 2\cdot 2x\cancel{(a^2+x^2)}(x^4+3a^2x^2)}{(a^2+x^2)^{\cancel{4}}}$$

$$= \frac{(4x^3a^2+\cancel{4x^5}+6a^4x+6a^2x^3)-\cancel{4x^5}-12a^2x^3}{(a^2+x^2)^3}$$

$$\text{or,} \quad y'' = \frac{6a^4x-2a^2x^3}{(a^2+x^2)^3} = \frac{2a^2(3a^2-x^2)x}{(a^2+x^2)^3}$$

$$\text{and} \quad y''' = \frac{2a^2(3a^2-x^2)}{(a^2+x^2)^3} - \frac{4x^2a^2}{(a^2+x^2)^3} - 12a^2x^2\frac{(3a^2-x^2)}{(a^2+x^2)^4}$$

Now, $y'' = 0 \Rightarrow x = 0, \pm a\sqrt{3}$

and $y''' \neq 0$ for $x = 0$, $\pm a\sqrt{3}$

$\Rightarrow$ Points of inflexion are at $x = 0,\ \pm a\sqrt{3}$

Also, for $x = 0$, $y = 0$

$\therefore$ point of inf. is at origin and also at the pts. for which

$x = \pm a\sqrt{3}$ **Proved.**

Ex.2. **Prove that the curve $x^3 + y^3 = a^3$ has points of inflexion at points where it crosses the coordinate axes.**

Sol. Curve crosses the x-axis at $y = 0$

$\Rightarrow$ $x^3 = a^3$ or $x = a$

Curve crosses the y axis at $x = 0$

$\Rightarrow$ $y^3 = a^3 \Rightarrow y = a$

Now, $y = (a^3 - x^3)^{1/3}$

$$y' = \frac{1}{3}\left(a^3 - x^3\right)^{-2/3}(-3x^2) = \frac{-x^2}{(a^3 - x^3)^{2/3}}$$

$$y'' = -\left[\frac{2x(a^3 - x^3)^{2/3} + x^2 \cdot \frac{2}{3}(a^3 - x^3)^{-1/3}(+3x^2)}{(a^3 - x^3)^{1/3}}\right]$$

$$= -\left[\frac{2x(a^3 - x^3)^{2/3} + 2x^4(a^3 - x^3)^{-1/3}}{(a^3 - x^3)^{1/3}}\right]$$

$$= -\left[\frac{2x(a^3 - x^3) + 2x^4}{(a^3 - x^3)^{2/3}}\right] = -\frac{2xa^3}{(a^3 - x^3)^{2/3}}$$

$$y''' = -2a^3\left[\frac{(a^3 - x^3)^{2/3} + x\frac{2}{3}(a^3 - x^3)^{-1/3}(+3x^2)}{(a^3 - x^3)^{1/3}}\right]$$

$$= -2a^3\left[\frac{(a^3 - x^3) + 2x^3}{(a^3 - x^3)^{2/3}}\right] = -\frac{2a^3(x^3 + a^3)}{(a^3 - x^3)^{2/3}}$$

Now $y'' = 0 \Rightarrow x = 0$ [y – axis]

and $y''' \neq 0$ for $x = 0$

$\therefore$ Point of inflexion are at points where curve crosses the coordinate axes. **Proved.**

Find the points of inflexion for each of the following curve :

Ex.3. $y = (x^2 + 4x + 5)\, e^{-x}$.

Sol. $y' = (2x + 4)e^{-x} - (x^2 + 4x + 5)e^{-x}$

$$y'' = 2e^{-x} - 2xe^{-x} - 2xe^{-x} + x^2e^{-x} - 4e^{-x} + 4xe^{-x} + 5e^{-x} - 4e^{-x}$$

$$= 3e^{-x} + x^2e^{-x} - 4e^{-x} = (x^2 - 1)e^{-x}$$

$$y''' = 2xe^{-x} - (-1 + x^2)e^{-x}$$

Now, $y'' = 0 \Rightarrow (-1 + x^2)e^{-x} = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$

Now, $y''' \neq 0$ at $x = \pm 1$

$\therefore$ pt of inflexion at $x = \pm 1$ **Ans.**

Ex.4. $y = 3x - 4\sin x + \sin 2x$.

Sol. $y' = 3 - 4\cos x + 2\cos 2x$

$y'' = +4\sin x - 4\sin 2x$

$y''' = 4\cos x - 8\cos 2x$

Now, for pt. of inflexion $\rightarrow$

$y'' = 0 \Rightarrow 4\sin x - 4\sin 2x = 0 \Rightarrow \sin x - 2\sin x\cos x = 0$

$\Rightarrow$ $\sin x = 0$ or $\cos x = \frac{1}{2}$ $\sin x(1 \mp 2\cos x) = 0$

$\sin x = 0$, $1 - 2\cos x = 0$

$\Rightarrow$ $x = 0$ or $x = \frac{\pi}{3}$

for both values, $y''' \neq 0$

$\therefore$ Point of inflexion is at $x = 0, \frac{\pi}{3}$ **Ans.**

Ex.5. $y = \dfrac{a^2x}{(a^2 + x^2)}$

Sol. $$y' = a^2\left[\frac{a^2 + x^2 - 2x^2}{\left(a^2 + x^2\right)^2}\right] = a^2\frac{\left(a^2 - x^2\right)}{\left(a^2 + x^2\right)^2}$$

$$y'' = a^2\left[\frac{(a^2+x^2)^2(-2x)-(a^2-x^2)2(a^2+x^2)\cdot 2x}{\left(a^2+x^2\right)^4}\right]$$

$$= a^2\left[\frac{(a^2+x^2)}{(a^2+x^2)^4}\left\{-2xa^2-2x^3-4a^2x+4x^3\right\}\right]$$

$$= a^2\frac{(a^2+x^2)}{(a^2+x^2)^4}(2x^3-6a^2x) = \frac{2xa^2(x^2-3a^2)}{(a^2+x^2)^3}$$

$$y''' = 2a^2\frac{(x^2-3a^2)}{(a^2+x^2)^3}+\frac{2xa^2(2x)}{(a^2+x^2)^3}+\frac{2xa^2(x^2-3a^2)\cdot(-3)(2x)}{(a^2+x^2)^4}$$

$$= \frac{2a^2(x^2-3a^2)+4x^2a^2}{(a^2+x^2)^3}-\frac{12x^2a^2(x^2-3a^2)}{(a^2+x^2)^4}$$

$$= \frac{6a^2(x^2-a^2)\left(a^2+x^2\right)-12x^2a^2(x^2-3a^2)}{(a^2+x^2)^4}$$

$$y''' = \frac{a^2(x^4-a^4)-12x^4a^2+36x^2a^4}{(a^2+x^2)^4}$$

Now, $y''=0 \Rightarrow x\left(x^2-3a^2\right)=0 \Rightarrow x=0, \pm\sqrt{3}a$ and $y'''\neq 0$ for $x=0, \pm\sqrt{3}a$

$\therefore$ Point of inflexion is at $x=0, \pm\sqrt{3}a$ **Ans.**

Ex.6. Show that every point in which the curve $y = c\sin\left(\frac{x}{a}\right)$ meets the axis of x is a point of inflexion.

Sol. $y = c\sin\frac{x}{a}$

$\Rightarrow y' = \frac{dy}{dx} = \frac{c}{a}\cos\frac{x}{a}$

$\Rightarrow y'' = \frac{d^2y}{dx^2} = -\frac{c}{a^2}\sin\frac{x}{a}$

$\Rightarrow \frac{d^3y}{dx^3} = -\frac{c}{a^3}\cos\frac{x}{a}$

Now, for point of inflexion–

$$\frac{d^2y}{dx^2}=0 \Rightarrow -\frac{c}{a^2}\sin\frac{x}{a}=0 \Rightarrow \frac{x}{a}=n\pi \Rightarrow x=an\pi$$

and $\left(\frac{d^3y}{dx^3}\right)_{x=an\pi} = -\frac{c}{a^3}(-1)^n \neq 0$

$\therefore$ for $x = an\pi$, $y = c\sin n\pi = 0$

which is the eq^n of x – axis. **Proved.**

Ex.7. In the curve $y(a^2+x^2) = 2x^2$, prove that for varying values of a, the locus of the points of inflexion is the straight line $y = 1/2$.

Sol. Here, $y = \frac{2x^2}{a^2+x^2}$ .....(1)

$$\Rightarrow y' = \frac{4x(a^2+x^2)-4x^3}{(a^2+x^2)^2} = \frac{4a^2x}{(a^2+x^2)^2}$$

$$\Rightarrow y'' = \frac{4a^2(a^2+x^2)^2-16a^2x^2(a^2+x^2)}{(a^2+x^2)^4} = \frac{4a^4-12a^4x^2}{(a^2+x^2)^3}$$

$$\Rightarrow y''' = \frac{-24a^2x(a^2+x^2)^3-6x(a^2+x^2)^2(4a^4-12a^2x^2)}{(a^2+x^2)^6}$$

$$= \frac{48(x^2-a^2)a^2x}{(a^2+x^2)^4}$$

Now, for point of inflexion

$4a^2(a^2-3x^2)=0$

$x^2=\frac{a^2}{3}$

$$y''=0 \Rightarrow \frac{4a^4-12a^2x^2}{(a^2+x^2)^3}=0 \Rightarrow 4a^4-12a^2x^2=0 \Rightarrow x^2=\frac{a^2}{3}$$

$\Rightarrow \quad x = \pm\dfrac{a}{\sqrt{3}}$

Also, $(y''')_{x=\pm\frac{a}{\sqrt{3}}} \neq 0$

$\therefore$ There is a point of infelxion at $x = \pm\dfrac{a}{\sqrt{3}}$

Also, for $x = \pm\dfrac{a}{\sqrt{3}}$; (1) gives :

$$y = \frac{\frac{2a^2}{3}}{a^2 + \frac{a^2}{3}} = \frac{2}{4} \Rightarrow y = \frac{1}{2}$$

which is a straight line. **Proved.**

Ex.8. **Find the points of inflexion of the curve**
$x = a\,(2\theta - \sin\theta)$, $y = a\,(2 - \cos\theta)$.

Sol. $x = a\,(2\,\theta - \sin\theta)$; $\quad y = a(2 - \cos\theta)$

$\Rightarrow \quad \dfrac{dx}{d\theta} = a(2-\cos\theta)$; $\quad \dfrac{dy}{d\theta} = a\sin\theta$

$y' = \therefore \quad \dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta} = \dfrac{\sin\theta}{2-\cos\theta}$

$y'' = \therefore \quad \dfrac{d^2y}{dx^2} = \left[\dfrac{(2-\cos\theta)\cos\theta - \sin\theta(\sin\theta)}{(2-\cos\theta)^2}\right] \times \dfrac{d\theta}{dx}$

$$= \frac{2\cos\theta - 1}{(2-\cos\theta)^2} \times \frac{1}{a(2-\cos\theta)} = \frac{2\cos\theta - 1}{a(2-\cos\theta)^3}$$

$$\frac{d^3y}{dx^3} = \frac{1}{a}\left[\frac{-2\sin\theta(2-\cos\theta)^3 - 3(2\cos\theta-1)(2-\cos\theta)^2(\sin\theta)}{(2-\cos\theta)^6}\right]\frac{d\theta}{dx}$$

$$= \frac{1}{a}\left[\frac{-4\sin\theta + 2\sin\theta\cos\theta - 6\sin\theta\cos\theta + 3\sin\theta}{(2-\cos\theta)^4}\right]\frac{1}{a(2-\cos\theta)}$$

$$= \frac{-[4\sin\theta\cos\theta - \sin\theta]}{a^2(2-\cos\theta)^5}$$

Now, $y'' = 0 \Rightarrow \cos\theta = \dfrac{1}{2} \Rightarrow \theta = \dfrac{\pi}{3}$

and $\quad y''' \neq 0$ for $\theta = \dfrac{\pi}{3}$

$\therefore$ Point of inflexion is at $\theta = \dfrac{\pi}{3}$ **Ans.**

Ex.9. **Show that the curve $y = e^x$ is at every point convex to the foot of the corresponding ordinate.**

Sol. Here $y = e^x$

$\Rightarrow \quad y' = e^x$ and $y'' = e^x$

$\therefore \quad y\dfrac{d^2y}{dx^2} = e^x \cdot e^x = e^{2x}$

= a positive quantity for all x.

$\Rightarrow$ curve is convex at every point to the foot of the corresponding ordinate. **Proved**

8. Partial Differential Equations

Exercise - 8(A)

Ex. 1 **Solve $z = (ax + y)^2 + b$**

Sol. : $p = \frac{\partial z}{\partial x} = (ax + y)\,2a$

and $q = \frac{\partial z}{\partial y} = (ax + y)$

than $px + qy = (ax + y)\,ax + y\,(ax + y)$

$= (ax + y)^2 = q^2$

Hence required differential equation is

$Px + qy = q^2$ Ans.

Ex. 2 **Solve $ax^2 + by^2 + z^2 = 1$**

Sol. : $p = \frac{\partial z}{\partial x} = \frac{-ax}{z}$ and $q = \frac{\partial z}{\partial y} = \frac{-yb}{z}$

$px + qy = \frac{-ax^2}{z} - \frac{y^2 b}{z} = \frac{-1}{z}(y^2 b + x^2 a)$

or $z\,(px + qy) = z^2 - 1$ Ans.

which is required differential equation.

Ex. 3 $Z = e^{mx} f(x + y)$

Sol. : $p = \frac{\partial z}{\partial x} = \frac{\partial}{\partial x}[\,e^{mx} f(x + y)] = me^{mx} f(x + y) + e^{mx} f'(x + y)$

$= mz + e^{mx} f'(x + y)$ ...(1)

and $q = \frac{\partial z}{\partial y} = \frac{\partial}{\partial y}[e^{mx} f(x + y)] = e^{mx} f'(x + y)$ ...(2)

using (2) in (1), we get

$p = mz + q,$ or $p - q = mz$ Ans.

Ex. 4. **Solve $\phi(x + y + z, x^2 + y^2 - z^2) = 0$**

Sol. : Let $x + y + z = u$ and $x^2 + y^2 - z^2 = v$.

Then the given equation reduce to $\phi(u, v) = 0$

Differentiating *it w.r.t.x* and *y* partially, we obtain

$$\frac{\partial \phi}{\partial u}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial x}\right) + \frac{\partial \phi}{\partial v}\left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z}\cdot\frac{\partial z}{\partial x}\right) = 0$$

$$\frac{\partial \phi}{\partial u}\left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial y}\right) + \frac{\partial \phi}{\partial v}\left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z}\cdot\frac{\partial z}{\partial y}\right) = 0$$

These give

$$\frac{\partial \phi}{\partial u}(1 + 1\,.\,p) + \frac{\partial \phi}{\partial v}(2x + (-2z)\,p) = 0$$

and $$\frac{\partial \phi}{\partial u}(1 + 1\,.\,q) + \frac{\partial \phi}{\partial v}(2x + (-2z)\,q) = 0$$

$\Rightarrow \frac{\partial \phi}{\partial u} + \frac{\partial \phi}{\partial v} = \frac{2\,(zp - x)}{p + 1}$ and $\frac{\partial \phi}{\partial u} + \frac{\partial \phi}{\partial v} = \frac{2\,(zq - y)}{q + 1}$

Eliminating f, we obtain

$\frac{2\,(zp - x)}{p + 1} = \frac{2\,(zq - y)}{q + 1}$ or $(y + z)\,p - (x + z)\,q = x -$ Ans.

Which is the required diff. equation.

Ex. 5. **Solve $yzp + zxq = xy$** ...(1)

Sol. : This is of the form $Pp + Qq = R$

Lagrange's auxiliary equation are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{ or } \frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$$

From first two fraction

$$\frac{dx}{yz} = \frac{dy}{zx} \text{ or } xdx = ydy$$

Integrating $x^2 - y^2 = C_1$

Also from first and third fraction

$$\frac{dx}{yz} = \frac{dz}{xy} \text{ or } xdx = zdz$$

integrating $x^2 - z^2 = C_2$

Hence the general integral is given by

$$f(x^2 - y^2, x^2 - z^2) = 0$$ Ans.

Ex. 6. **Solve** $\frac{y^2 z}{x} P + zxq = y^2$

(1) is the form of $Pp + Qq = R$

Lagrange's auxiliary equation are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{ or } \frac{dx}{y^2 z/x} = \frac{dy}{zx} = \frac{dz}{y^2}$$

From first two fraction $\frac{xdx}{y^2 z} = \frac{dy}{zx}$

or $x^2 dx = y^2 dy$

Integrating $x^3 - y^3 = C_1$

Also from first & third fraction $\frac{xdx}{y^2 z} = \frac{dz}{y^2}$

or $xdx = zdz$

integrating $x^2 - z^2 = C_2$

Hence, the general integral is given by

$$f(x^3 - y^3, x^2 - z^2) = 0$$

Ans.

Ex. 7. **Solve** $(y + z)\, p + (z + x)\, q = (x + y)$

Sol. : This is form of $Pp + Qq = R$

La grange's auxiliary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

or $$\frac{dx}{(y+z)} = \frac{dy}{(z+x)} = \frac{dz}{(x+y)} = \frac{dx - dy}{y - x} = \frac{dy - dz}{z - y} = \frac{dz - dx}{(x - z)}$$

$$= \frac{dx + dy + dz}{2(x + y + z)}$$

Taking (iv) & (vii) fraction

$$-\frac{dx - dy}{(x - y)} = \frac{dx + dy + dz}{2(x + y + z)}$$

Integrating

$$-2 \log (x - y) = \log (x + y + z) - \log C_1$$

or $$C_1 = (x + y + z)(x - y)^2$$

Taking (iv) and (v) fraction

$$-\left(\frac{dx - dy}{x - y}\right) = -\left(\frac{dy - dz}{y - z}\right)$$

Integrating $\log (x - y) = \log (y - z) + \log C_2$

or $$C_2 = \left(\frac{x - y}{y - z}\right)$$

Hence, the general integrating given by

$$f\left\{(x - y)^2 (x + y + z),\ \left(\frac{x - y}{y - z}\right)\right\} = 0$$

$$(x - y)^2 (x + y + z) = f\left(\frac{x - y}{y - z}\right)$$

Ans.

Ex. 8. **Solve** $z - xp - yq = a\sqrt{x^2 + y^2 + z^2}$

Sol. : The given equation can be written as

$$px + qy = z - a(x^2 + y^2 + z^2)^{1/2}$$

La grange's auxiliary equations are

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a(x^2 + y^2 + z^2)^{1/2}}$$

Also each fraction $= \frac{xdx + ydy + zdz}{x^2 + y^2 + z^2 - az(x^2 + y^2 + z^2)^{1/2}}$

from (i) and (ii)

$\Rightarrow$ $$\frac{dx}{x} = \frac{dy}{y}$$

Integrating $\log x = \log y + \log C_1$ or $x = yC_1$ or $C_1 = \frac{x}{y}$

from (iii) and (iv)

$$= \frac{dz}{z - a(x^2 + y^2 + z^2)^{1/2}}$$

$$= \frac{xdx + ydy + zdz}{x^2 + y^2 + z^2 - az(x^2 + y^2 + z^2)^{1/2}}$$

or $$\frac{dz}{z - at} = \frac{\cancel{t}dt}{t^{\cancel{2}} - az\cancel{t}}$$ [putting $x^2 + y^2 + z^2 = t^2$]

or $$\frac{dz}{z - at} = \frac{\cancel{t}dt}{t - az} = \frac{dz + dt}{z - at + t - za} = \frac{dz + dt}{(z + t)(1 - a)} = \frac{dx}{x}$$

$$= \frac{dz + dt}{z + t} = (1 - a)\frac{dx}{x}$$

Integrating $\log (z + t) = (1 - a) \log x + \log C_2$

$$z + t = C_2 x^{(1-a)} \text{ or } \frac{z + \sqrt{x^2 + y^2 + z^2}}{x^{(1-a)}} = C_2$$

Hence, the general solution is

$$f\left[\left(\frac{x}{y}\right), \frac{z+\sqrt{x^2+y^2+z^2}}{x^{(1-a)}}\right]=0$$

or $$x^{(1-a)}=\left(z+\sqrt{x^2+y^2+z^2}\right) f\left(\frac{x}{y}\right)$$ Ans.

Ex. 9 **Solve $p+3q=5z+\tan(y-3x)$**

Sol. : Lagrange's subsidiary equations are

$$\frac{dx}{1}=\frac{dy}{3}=\frac{dz}{5z+\tan(y-3x)} \quad ...(1)$$

From first two fractions, $y-3x=C_1$.

$$\frac{dx}{1}=\frac{dz}{5z+\tan(y-3x)}=\frac{dz}{5z+\tan C_1}$$

$\Rightarrow$ $x=\frac{1}{5}\log(5z+\tan C_1)-\frac{1}{5}\log C_2$

$\Rightarrow$ $5x=\log\{5z+\tan(y-3x)/C_2\}$

$\Rightarrow$ $e^{-5x}[5x+\tan(y-3x)]=C_2$.

Hence General solution is

$$f[(y-3x), e^{-5x}[5z+\tan(y-3x)]]=0$$ Ans.

Ex. 10 **Solve$(mz-ny)p+(nx-lz)q=ly-mx$**

Sol. : (1) is of the form $Pp+Qq=R$.

Auxiliary equations are

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}=i.e., \frac{dx}{mz-ny}=\frac{dy}{nx-lz}=\frac{dz}{ly-mx}.$$

Obviusly each fraction

$$=\frac{ldx+mdy+ndz}{0}=\frac{xdx+ydy+zdz}{0}$$

$\Rightarrow$ $ldx+mdy+ndz=0$ and $xdx+y\,dy+zdz=0$

$\Rightarrow lx+my+nz=C_1$ and $x^2+y^2+z^2=C_2$.

Hence, general solution is

$$f(lx+my+nz, x^2+y^2+z^2)=0.$$ Ans.

Ex. 11. **Solve $(x^2-yz)p+(y^2-zx)q=z^2-xy$**

Sol. : (1) is of the form $Pp+Qq=R$.

Auxiliary equations are

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R} \text{ or } \frac{dx}{x^2-yz}=\frac{dy}{y^2-zx}=\frac{dz}{z^2-xy} \quad ...(2)$$

Now each fraction

$$=\frac{dx-dy}{(x-y)(x+y+z)}=\frac{dy-dz}{(y-z)(x+y+z)}=\frac{dz-dx}{(z-x)(x+y+z)}$$
(i) (ii) (iii)

(i) and (ii) give $\frac{dx-dy}{x-y}=\frac{dy-dz}{y-z}\Rightarrow\frac{x-y}{y-z}=C_1$. ...(3)

Similarly (ii) and (iii) give $\frac{y-z}{z-x}=C_2$ or $\frac{z-x}{y-z}=C_2$ ...(4)

Hence general integral is

$$f\left\{\frac{(x-y)}{(y-z)}, \frac{(z-x)}{(y-z)}\right\}=0 \text{ or } \left(\frac{x-y}{y-z}\right)=f\left(\frac{z-x}{y-z}\right)$$ Ans.

Ex. 12. **Solve $p\cos(x+y)+q\sin(x+y)=z$**

Sol. : this is of the form $Pp+Qq=R$

Lagrange's auxiliary equations are

$$\frac{dx}{\cos(x+y)}=\frac{dy}{\sin(x+y)}=\frac{dz}{z}$$

obviously $=\frac{dx+dy}{\cos(x+y)+\sin(x+y)}=\frac{dx-dy}{\cos(x+y)-\sin(x+y)}$

Now (iv) and (v)

$\Rightarrow$ $\frac{\{\cos(x+y)-\sin(x+y)\}(dx+dy)}{\cos(x+y)+\sin(x+y)}=dx-dy$

or $d[\log[\cos(x+y)+\sin(x+y)]]=dx-dy$

Integrating, $\log\{\cos(x+y)+\sin(x+y)\}=x-y+\log C_1$

$$\cos(x+y)+\sin(x+y)=C_1e^{x-y}$$

or $\{\cos(x+y)+\sin(x+y)\}e^{y-x}=a$

Further $\frac{dx+dy}{\cos(x+y)+\sin(x+y)}=\frac{dz}{z}$

or $\frac{dt}{\cos t+\sin t}=\frac{dz}{z}$ putting $[x+y=t]$

or $\frac{dt}{\sqrt{2}\sin(t+\pi/4)}=\frac{dz}{z}$ or $\frac{1}{\sqrt{2}}\operatorname{cosec}\left(t+\frac{\pi}{4}\right)dt=\frac{dz}{z}$

Integrating, $\frac{1}{\sqrt{2}}\log\tan\left(\frac{x+y}{2}+\frac{\pi}{8}\right)=\log C_2z$

$$\tan\left(\frac{x+}{2}+\frac{\pi}{8}\right)=(C_2z)^{\sqrt{2}}$$

$$\tan\left(\frac{x+y}{2}+\frac{\pi}{8}\right) z^{-\sqrt{2}} = C_2^{\sqrt{2}}$$

or $$C_2^{-\sqrt{2}} = b = z^{\sqrt{2}} \cot\left(\frac{x+y}{2}+\frac{\pi}{8}\right)$$

Hence the general intigral is

$$f\left[\{\cos(x+y)+\sin(x+y)\,e^{y-x}\},\ \left\{z^{\sqrt{2}}\cot\left(\frac{x+y}{2}+\frac{\pi}{8}\right)\right\}\right]$$

or $$\cos(x+y)+\sin(x+y) = e^{x-y} f\left[z^{\sqrt{2}}\cot\left(\frac{x+y}{2}+\frac{\pi}{8}\right)\right]$$ Ans.

Ex. 13. **Solve $z(xp - yq) = (y^2 - x^2)$**

Sol. : Lagrange's auxiliary equation is

$$\frac{dx}{x} = -\frac{dy}{y} = \frac{zdz}{y^2-x^2} = \frac{dx-dy}{(x+y)}$$

from (i) and (ii) $\frac{dx}{x} = -\frac{dy}{y}$

integrating $\log x = -\log y + C_1$ or $xy = C_1$

from (iii) and (iv) fraction

$$\frac{(dx-dy)}{(x+y)} = \frac{zdz}{(y^2-x^2)} \text{ or } dx-dy = \frac{zdz}{(y-x)}$$

or $$(y-x)(dx-dy) = zdz$$

$$ydx - xdx - ydy + xdy = zdz$$

$$-xdx - ydy + (ydx + xdy) = zdz$$

$$-(xdx + ydy) + d(xy) = zdz \qquad [\because d(xy) = dC_1 = 0]$$

integrating

$$-x^2 - y^2 = z^2 - C_2$$

$$C_2 = z^2 + x^2 + y^2$$

Hence general integral is

$$f[xy, x^2+y^2+z^2] = 0$$ Ans.

Ex. 14. **Solve $(x^3 + 3xy^2)p + (y^3 + 3x^2y)q = 2(x^2+y^2)z$**

Sol. : The lagrange's auxiliary equation is

$$\frac{dx}{(x^3+3xy^2)} = \frac{dy}{(y^3+3x^2y)} = \frac{dz}{2z(x^2+y^2)}$$

$$\frac{dx/x}{(x^2+3y^2)} = \frac{dy/y}{(y^2+3x^2)} = \frac{dz/z}{2(x^2+y^2)}$$

$$= \frac{\frac{dx}{x}+\frac{dy}{y}}{4(x^2+y^2)}$$

from (iii) and (iv) fraction

$$\frac{dx/x + dy/y}{4(x^2+y^2)} = \frac{dz/z}{2(x^2+y^2)}$$

or $$\frac{dx}{x}+\frac{dy}{y} = 2\frac{dz}{z}$$

Integrating

$$\log(xy) = \log z^2 C_1 \text{ or } xy = z^2 C_1$$

or $$C_1 = \frac{xy}{z^2} \qquad \ldots(2)$$

$$\frac{dx}{(x^3+3xy^2)} = \frac{dy}{(y^3+3x^2y)} = \frac{dx+dy}{(x+y)^3} = \frac{dx-dy}{(x-y)^3}$$

then $$\frac{dx+dy}{(x+y)^3} = \frac{dx-dy}{(x-y)^3}$$

integrating $-\frac{1}{2}(x+y)^2 = -\frac{1}{2}(x-y)^2 - \frac{1}{2}C_2$

$$C_2 = \frac{1}{(x+y)^2} - \frac{1}{(x-y)^2}$$

Hence general integral is

$$f\left[\left\{\frac{1}{(x+y)^2}-\frac{1}{(x-y)^2}\right\}, \frac{xy}{z^2}\right] = 0$$

or $$\frac{1}{(x+y)^2}-\frac{1}{(x-y)^2} = f\left(\frac{xy}{z^2}\right)$$ Ans.

Ex. 15. **Solve $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = xyz$**

Sol. : lgrange's auxiliary equation is

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} = \frac{du}{xyz}$$

from (1) & (ii) fraction

$$\frac{dx}{x} = \frac{dy}{y}$$

integrating $\log x = \log C_1 y$ or $\frac{x}{y} = C_1$

from (ii) & (iii) fraction

$$\frac{dy}{y} = \frac{dz}{z}$$

Scanned with

integrating $\log y = \log z\, C_2$ or $\frac{y}{z} = C_2$

Also $\frac{yz\,dx + zx\,dy + yx\,dz}{3xyz} = \frac{du}{xyz}$

or $d(xyz) = 3du$, integrating $xyz - 3u = C_3$

Hence general integral is

$$f\left(\frac{x}{y}, \frac{y}{z}, xyz - 3u\right) = 0$$ Ans.

Ex. 16. **Solve $(x + 2z)\,p + (4zx - y)\,q = 2x^2 + y$**

Sol. : The auxiliary equation are

$$\frac{dx}{(x + 2z)} = \frac{dy}{(4zx - y)} = \frac{dz}{(2x^2 + y)}$$

or $$\frac{ydx}{xy + 2zy} = \frac{xdy}{4zx^2 - xy} = \frac{2z\,dz}{4x^2z + y \times 2z}$$

$\therefore$ $ydx + xdy - 2z\,dz = 0$

[Taking multipliers $y, x, -2z$]

integrating $yx - z^2 = C_1$ $[ydx + xdy = d(xy)]$

Also [Taking multiplier $(2x, -1, -1)$

$$\frac{2x\,dx}{2x^2 + 4zx} = \frac{dy}{4zx - y} = \frac{dz}{2x^2 + y}$$

So $2x\,dx - dy - dz = 0$

integrating $x^2 - y - z = C_2$

Hence, general integral is

$$f\{(yx - z^2), (x^2 - y - z)\} = 0$$ Ans.

Ex. 17. **$p \tan x + q \tan y = \tan z$**

Sol. : Lagrange's auxiliary equations is

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

(in (i) and (ii) *fraction*

$$\frac{dx}{\tan x} = \frac{dy}{\tan y}$$

integrating $\log \sin x = \log \sin y + \log C_1$

or $C_1 = \frac{\sin x}{\sin y}$

in (ii) and (iii) fraction $\frac{dy}{\tan y} = \frac{dz}{\tan z}$

integrating $\log \sin y = \log \sin z + \log C_2$ or $C_2 = \frac{\sin y}{\sin z}$

Hence general integral is

$$f\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0$$

or $$\frac{\sin x}{\sin y} = f\left(\frac{\sin y}{\sin z}\right)$$ Ans.

Ex. 18. **Solve $x(z^2 - y^2)\,p + y(x^2 - z^2)\,q = z(y^2 - x^2)$**

Sol. : Lagrange's auxiliary equations are

$$\frac{dx}{x(z^2 - y^2)} = \frac{dy}{y(x^2 - z^2)} = \frac{dz}{z(y^2 - x^2)}$$

or $$\frac{dx/x}{(z^2 - y^2)} = \frac{dy/y}{(x^2 - z^2)} = \frac{dz/z}{(y^2 - x^2)}$$

or $$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

integrating $\log x + \log y + \log z = \log C_1$

$C_1 = xyz$

fraction (i), (ii) and (iii) multipliers $[x, y, z]$

$$\frac{xdx}{x^2(z^2 - y^2)} = \frac{ydy}{y^2(x^2 - z^2)} = \frac{zdz}{z^2(y^2 - x^2)}$$

or $xdx + ydy = zdz = 0.$

Integrating

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{C_2}{2}$$

$$x^2 + y^2 + z^2 = C_2$$

Hence general integration is

$$f\{(xyz), x^2 + y^2 + z^2\} = 0$$

or $$(x^2 + y^2 + z^2) = f(xyz)$$ Ans.

Ex. 19. **Solve $(z^2 - 2yz - y^2)\,p + (xy + zx)\,q = xy - zx$**

Sol. : Lagrange auxiliary equations are

$$\frac{dx}{(z^2 - 2yz - y^2)} = \frac{dy}{(xy + zx)} = \frac{dz}{(xy - zx)}$$

taking (ii) and (iii) fraction

$$\frac{dy}{y + z} = \frac{dz}{(y - z)}$$ or $ydy - zdy = ydz + zdz$

or $\quad ydy - zdz - (ydz + zdy) = 0$

or $\quad \frac{1}{2}d(y^2) - \frac{1}{2}d(z^2) - d(yz) = 0$

integrating $\frac{y^2}{2} - \frac{z^2}{2} - yz = C_1$ or $y^2 - z^2 - 2yz = C_1$

Also $\quad \frac{xdx}{(z^2 - 2yz - y^2)} = \frac{dy}{(y+z)} = \frac{dz}{(y-z)} = \frac{y\,dy}{y^2+zy} = \frac{zdz}{yz - z^2}$

$xdx + ydy + zdz = 0$

integrating

$x^2 + y^2 + z^2 = C_2$

Hence general integral is

$f[(y^2 - z^2 - 2yz), (x^2 + y^2 + z^2)] = 0$

or $\quad (x^2 + y^2 + z^2 = f(y^2 - z^2 - 2yz)$ Ans.

Exercise - 8(B)

Ex. 1. **Solve $\sqrt{p} + \sqrt{q} = 1$**

Sol. : The complete integral is

$z = ax + by + c$ where $\sqrt{a} + \sqrt{b} = 1$

($\because p = a$ and $q = b$) *i.e.*, $b = (1 - \sqrt{a})^2$

Putting the value of b, in the complete integral taken from

$z = ax + (1 - \sqrt{a})^2 y + c$ Ans.

Ex. 2. **Solve $p^2 + q^2 = npq$**

Sol. : The complete integral is :

$z = ax + by + c$ where $a^2 + b^2 = nab$ (by putting $p = a$ and $q = b$ in given eqn.) or $b^2 - nab + a^2 = 0$

$\therefore b = \frac{na \pm \sqrt{n^2a^2 - 4a^2}}{2} = \left(\frac{n \pm \sqrt{n^2 - 4}}{2}\right)a$

Putting the value of b, the complete integral taken from

$z = ax + \frac{1}{2}\left\{na \pm a\left(\sqrt{n^2 - 4}\right)\right\} y + c$ Ans.

Ex. 3. **Solve $q = e^{-p/a}$**

Sol. : The complete integral is

$z = mx + ny + c$ where $n = e^{-m/a} = q$

Putting the value of n, the complete integral taken from

$z = mx + e^{-m/a} y + c$ Ans.

Ex. 4. **Solve $z^2(p^2 + q^2 + 1) = c$**

Sol. : putting $Z = z^2/_2$ so that

$\frac{\partial Z}{\partial x} = \frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial x} = zp$ and $\frac{\partial Z}{\partial y} = \frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial y} = zq$

the given equation becomes

$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 2z = c$

Let $z = f(X)$ where $X = x + ay$

So that $\frac{\partial z}{\partial x} = \frac{dz}{dX}$ and $\frac{\partial z}{\partial y} = \frac{adz}{dX}$

By these substitutions (2) reduces to

$\left(\frac{dz}{dX}\right)^2 (1 + a^2) + 2z = c$ or $\frac{\sqrt{1+a^2}}{\sqrt{c - 2z}} dz = dX$

Integrating $-\sqrt{1+a^2}\,.\sqrt{c - 2z} = X + b$

or $(1 + a^2)(c - z^2) = (x + ay + b)^2$ Ans.

$z^2p^2 + z^2q^2 + z^2 = 1$

$z^2(p^2+q^2) = 1 - z^2$

$p^2 + q^2 = \frac{1-z^2}{z^2}$

Let $\frac{\partial z}{\partial x} = \frac{dz}{dx}$

$\frac{\partial z}{\partial y} = a\frac{dz}{dx}$

$\left(\frac{dz}{dx}\right)^2 + a^2\left(\frac{dz}{dx}\right)^2 = \frac{1-z^2}{z^2}$

$\left(\frac{dz}{dx}\right)^2(1+a^2) = \frac{1-z^2}{z^2}$

$\sqrt{(1+a^2)}\frac{dz}{dx} = \frac{\sqrt{1-z^2}}{z}$

$\int \sqrt{1+a^2}\frac{z}{\sqrt{1-z^2}}dz = \int dx + c$

$-\sqrt{1+a^2}\sqrt{1-z^2} = x + c$

$(1+a^2)(1-z^2) = (x+c)^2$

$(1+a^2)(1-z^2) = (x+ay+c)^2$

A

Ex. 5. **Solve $p^3 + q^3 = 3pqz$**

Sol. : It is from of $f(z, p, q) = 0$

putting $\frac{dz}{dX}$ *for p* and $\frac{adz}{dX}$ *for q*, the equation *becomes*

$\left(\frac{dz}{dX}\right)^3 + a^3\left(\frac{dz}{dX}\right)^3 = 3a\left(\frac{dz}{dX}\right)^2 z$ or $\left(\frac{dz}{dX}\right) = \frac{3a}{1+a^3}z$

or $(1 + a^3)(dz/z) = 3adX$

Integrating, $(1 + a^3)\log z = 3aX + c$

putting value of $X = (x + ay)$

$(1 + a^3)\log z = 3a(x + ay) + c$ Ans.

Ex. 6. **Solve $p^2 = z^2(1 - pq)$**

Sol. : This is of the form $f(z, p, q) = 0$

Letting $z = f(x + ay) = f(X)$ and so putting $\frac{dz}{dX}$ for p and $a\left(\frac{dz}{dX}\right)$ for q, the equation becomes, where $X = x + ay$

$\left(\frac{dz}{dX}\right)^2 = z^2\left[1 - a\left(\frac{dz}{dX}\right)^2\right]$ or $\left(\frac{dz}{dX}\right)^2(1 + az^2) = z^2$

or $\frac{dz}{dX} = \frac{\pm z}{\sqrt{1 + az^2}}$ or $\pm dX = \frac{\sqrt{(1 + az^2)}}{z}dz$

$$\text{or } dX = \frac{1 + az^2}{z\sqrt{(1 + az^2)}}\,dz = \frac{1}{z\sqrt{1 + az^2}}\,dz + \frac{az\,dz}{\sqrt{(1 + az^2)}}$$

$$\frac{1}{z\sqrt{1 + az^2}}\,dz \text{ put } z = \frac{1}{t} \text{ then } \int \frac{-\frac{1}{t^2}\,dt}{\frac{1}{t}\sqrt{1 + \frac{a}{t^2}}} = -\int \frac{dt}{\sqrt{t^2 + a}}$$

$$= -\log\left\{t + \sqrt{t^2 + a}\right\} = \log\left\{\frac{1}{\sqrt{t^2 + a} + t} \times \frac{\sqrt{t^2 + a} - t}{\sqrt{t^2 + a} - t}\right\}$$

$$= \log\left\{\frac{\sqrt{t^2 + a} - t}{a}\right\} = \log\left\{\frac{\sqrt{1 + az^2} - 1}{az}\right\}$$

Integrating :

$$(X + c) = \log\left\{\frac{\sqrt{1 + az^2} - 1}{az}\right\} + \sqrt{1 + az^2}$$

$$= \log\left[\frac{\sqrt{1 + az^2} - 1}{z}\right] + \sqrt{1 + az^2} \qquad (\log a \text{ is merged in } c)$$

Now putting $(x + ay)$ for x, we get

$$(x + ay) + c = \log\left[\frac{\sqrt{1 + az^2} - 1}{z}\right] + \sqrt{1 + az^2}$$ **Ans.**

Ex. 7. **Solve $p\,(1 + q) = qz$**

Sol. : This is of the form $f(z, p, q) = 0$

Let $z = f(x + ay) = f(X)$ and so putting $\frac{dz}{dX}$ for p and $\frac{a\,dz}{dX}$ for q.

the equation become where $= x + ay$

$$\frac{dz}{dX} + a\left(\frac{dz}{dX}\right)^2 = az\,\frac{dz}{dX}$$

$$a\left(\frac{dz}{dX}\right)^2 = \frac{dz}{dX}(az - 1)$$

$$a\,\frac{dz}{dX} = (az - 1)$$

$$\frac{dz}{dX} = \frac{1}{a}(az - 1) \text{ or } \frac{a\,dz}{(az - 1)} = dX$$

integrating $\log(az - 1) = X + c$

Putting value of X

$\log(az - 1) = x + ay + c$ **Ans.**

Ex. 8. **Solve $\sqrt{p} + \sqrt{q} = x + y$**

Sol. : We have $\sqrt{p} - x = y - \sqrt{q} = a$ (say)

$\therefore p = (a + x)^2$ and $q = (y - a)^2$

Now $dz = p\,dx + q\,dy = (a + x)^2\,dx + (y - a)^2\,dy$

Integrating :

$$z = \frac{1}{3}(a + x)^3 + \frac{1}{3}(y - a)^3 + c$$

$3z = (a + x)^3 + (y - a)^3 + c$ **Ans.**

Ex. 9. **solve $q = 2y\,p^2$**

Sol. : We have $2p^2 = \frac{q}{y} = 2a^2$ (say)

$\therefore p = a$ and $q = 2a^2y$

Now $dz = p\,dx + q\,dy = a\,dx + 2a^2y\,dy$

integrating $z = ax + a^2y^2 + c$ **Ans.**

Ex. 10. **Solve $z = px + qy - 2\sqrt{pq}$**

Sol. : The Complete integral is :

$z = ax + by - 2\,(ab)^{1/2}$ --- (1) **Ans.**

Differentiating (1) partially w.r.t. a and b, we get

$$0 = x - (ab)^{-1/2} \times b \text{ or } 0 = y - \left(\frac{a}{b}\right)^{1/2}$$

Hence $x = \left(\frac{b}{a}\right)^{1/2}$ and $y = \left(\frac{a}{b}\right)^{1/2}$

Now eliminating a & b between these equation, we get :

Singular integral $= xy = \sqrt{\frac{b}{a}} \times \sqrt{\frac{a}{b}} = 1$ **Ans.**

Ex. 11. **Solve $z = px + qy + pq$**

Sol. : The complete integral is :

$z = ax + by + ab$ ... (1) **Ans.**

Differentiating (1), partially w.r.t. a and b, we get

$0 = x + b$ and $0 = y + a$

$x = -b$ ... (2)

$y = -a$ ... (1)

putting a, b value in equation (1) we get S. I.

$z = -xy - xy + xy$

S.I. is $z = -xy$ **Ans.**

Ans.

Ex. 15. **Solve $yp = 2xy + \log q$**

Sol. :

$$yp = 2xy + \log q$$

or, $$y(p - 2x) = \log q$$

or. $$p - 2x = \frac{1}{y}\log q \quad \text{[standard form } f(p, x) = F(q, y)]$$

Let $$p - 2x = \frac{1}{y}\log q = a$$

$$\Rightarrow \quad p = a + 2x \quad \text{...(1)}$$

and $$\log q = ay \Rightarrow q = e^{ay} \quad \text{...(2)}$$

Since, $$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

or, $$dz = p\,dx + q\,dy$$

$$\Rightarrow \quad dz = (a + 2x)\,dx + e^{ay}\,dy \quad \text{[by (1) and (2)]}$$

On integrating :

$$z = ax + x^2 + \frac{e^{ay}}{a} + b$$

or, $$az = a^2x + x^2a + e^{ay} + ab$$ Ans.

Scanned with

From above $P = (a + x)^{1/2}$ and $Q = (y + a)^{1/2}$

Now $dz = Pdx + Qdy = (a + x)^{1/2}\,dx + (y + a)^{1/2}\,dy$

Integrating $u + \frac{2}{3}b = \frac{2}{3}(a + x)^{3/2} + \frac{2}{3}(y + a)^{3/2}$

or $\quad \frac{2}{3}z^{3/2} + \frac{2}{3}b = \frac{2}{3}(a + x)^{3/2} + \frac{2}{3}(y + a)^{3/2}$

or $\quad z^{3/2} + b = (a + x)^{3/2} + (y + a)^{3/2}$ **Ans.**

Ex. 17. **Solve $x^2y^3p^2q = z^3$**

putting $u = \log z$, $\frac{\partial u}{\partial x} = \frac{1}{z}\frac{\partial z}{\partial x} = P$

$$\frac{\partial u}{\partial y} = \frac{1}{z}\frac{\partial z}{\partial y} = Q$$

$\therefore \quad x^2y^3\left(\frac{1}{z}\frac{\partial z}{\partial x}\right)^2\left(\frac{1}{z}\frac{\partial z}{\partial y}\right) = 1$

or $\quad x^2y^3\left(\frac{\partial u}{\partial x}\right)^2\left(\frac{\partial u}{\partial y}\right) = 1$

or $\quad x^2y^3P^2Q = 1$, where $P = \frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y} = Q$

$x^2P^2 = \frac{1}{y^3Q} = a$ (say) [Standard Form]

$\therefore \quad P = \sqrt{\frac{a}{x^2}}$, $Q = \frac{1}{ay^3}$

Now $du = Pdx + Qdy = \frac{\sqrt{a}}{x}dx + \frac{1}{ay^3}dy$

Integrating :

$$u = \sqrt{a}\log x - \frac{1}{2ay^2} + b$$

$\log z = \lambda \log x - \frac{1}{2\lambda^2y^2} + b$ **Ans.** $[\because \sqrt{a} = \lambda]$

Ex. 18. **Solve $(x^2 + y^2)(p^2 + q^2) = 1$** ... (1)

Sol. : Put $x = r\cos\theta$, $y = r\sin\theta$

i.e. $r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$ or $\theta = \tan^{-1}(y/x)$

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta}\cdot\frac{\partial \theta}{\partial x} = \frac{\partial z}{\partial r}\cdot\frac{x}{r} + \frac{\partial z}{\partial \theta}\left(\frac{-y}{x^2 + y^2}\right)$$

$$= \cos\theta\frac{\partial z}{\partial r} - \frac{\sin\theta}{r}\cdot\frac{\partial z}{\partial \theta}$$

and $q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial r}\cdot\frac{\partial r}{\partial y} + \frac{\partial z}{\partial \theta}\cdot\frac{\partial \theta}{\partial y} = \frac{\partial z}{\partial r}\cdot\frac{y}{r} + \frac{\partial z}{\partial \theta}\cdot\frac{x}{x^2 + y^2}$

$= \sin\theta\frac{\partial z}{\partial r} + \frac{\cos\theta}{r}\cdot\frac{\partial z}{\partial \theta}$

Substituting these value in (1) we get

$$r^2\left\{\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2\right\} = 1$$

or $\quad r^2\left(\frac{\partial z}{\partial r}\right)^2 = 1 - \left(\frac{\partial z}{\partial \theta}\right)^2$ or $r^2P^2 = 1 - Q^2$ ... (2)

$$\left[\text{Where } P = \frac{\partial z}{\partial r},\ Q = \frac{\partial z}{\partial \theta}\right]$$

from (2) let $r^2P^2 = 1 - Q^2 = a^2$ (Say)

$$P = \frac{a}{r} \text{ and } Q = \sqrt{1 - a^2}$$

Now $dz = Pdr + Qd\theta = \frac{a}{r}\,dr + \sqrt{1 - a^2}\,d\theta$

integrating :

$$+ b = a\log r + \theta\left(\sqrt{1 - a^2}\right)$$

or $z + b = a\log\sqrt{x^2 + y^2} + \sqrt{1 - a^2}\,\tan^{-1}\frac{y}{x}$ **Ans.**

Ex. 19. **Solve $z(p^2 - q^2) = x - y$**

Sol. : Same as Ex. No. 16

Ex. 20. **Solve $z = px + qy + p^2q^2$**

Since given p.d.e. is in clairaut's form

Sol. : the complete integral is

$Z = ax + by + a^2b^2$ ... (1) **Ans.**

Ex. 21. **Solve $z^2(p + q) = x^2 + y^2$** ... (1)

Let $u = \frac{z^3}{3}$

$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial x} = z^2p$ and $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial y} = z^2q$

Putting these value in equation (1)

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = x^2 + y^2$$

$P + Q = x^2 + y^2$ ... (2) $\quad \left[P = \frac{\partial u}{\partial x},\ Q = \frac{\partial u}{\partial y}\right]$

Letting $P - x^2 = y^2 - Q = a^2$ (Say)

$P = a^2 + x^2$ and $Q = y^2 - a^2$

Now $du = Pdx + Qdy$

$du = (a^2 + x^2)\,dx + (y^2 - a^2)\,dy$

Integrating :

$$u + b = a^2x + \frac{x^3}{3} + \frac{y^3}{3} - a^2y$$

or $z^3 + 3b = x^3 + y^3 + 3a^2x - 3a^2y$

Ex. 22. **Solve $pq = x^m y^n z^l$**

Sol. : The given equation can be written a

$$\left(\frac{dz}{dx}\right)\left(\frac{dz}{dy}\right) = x^m y^n z^l \text{ or } \left(\frac{1}{x^m}\frac{dz}{dx}\right)\left(\frac{1}{yn}\frac{dz}{dy}\right) = z^l$$

Now put $\frac{x^{m+1}}{m+1} = u$ and $\frac{y^{n+1}}{n+1} = v$

$$\therefore \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} = x^m\frac{\partial z}{\partial u}$$

or $\left(\frac{\partial z}{\partial u} = \frac{1}{x^m}\frac{\partial z}{\partial x}\right)$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y} = y^n\frac{\partial z}{\partial v} \text{ or } \frac{\partial z}{\partial v} = \frac{1}{y^n}\frac{\partial z}{\partial y}$$

By these substitutions (*i*) given

$$\frac{\partial z}{\partial u}\cdot\frac{\partial z}{\partial v} = z^l \text{ or } PQ = z^l$$

$$\left[\text{Where } P = \frac{\partial z}{\partial u} \text{ and } Q = \frac{\partial z}{\partial v}\right]$$

$PQ = z^l$ ... (2) (Standard

(2) is the *form* of $f(z, p, q)$. Hence putting $\frac{dz}{dX}$ *for*

P and $\left(a\frac{dz}{dX}\right)$ *for* Q, the equation (2) become

(where $X = u$

$$\frac{dz}{dX}\cdot\frac{a\,dz}{dX} = z^l \text{ or } \left(\frac{dz}{dX}\right)^2 = \frac{z^l}{a}$$

or $\frac{dz}{dX} = \left(\frac{z^l}{a}\right)^{1/2}$

or $\frac{a^{1/2}\,dz}{z^{l/2}} = dX$

Integrating $\frac{a^{1/2}\,z^{-(l/2)+1}}{\left(1-\frac{l}{2}\right)} = X + c$

or $\frac{z^{-(l/2)+1}}{1-\frac{l}{2}} = \frac{(u + av + c)}{a^{1/2}}$

$Z = \frac{1}{a^{1/2}}\,u + a^{1/2}\,v + b$ Ans.

where $Z = \left(\frac{z^{-l/2+1}}{1-\frac{l}{2}}\right)$ $u = \frac{x^{m+1}}{m+1}$ and $v = \frac{y^{n+1}}{n+1}$

Exercise - 8(C)

Ex. 1 **Solve $P = (qy + z)^2$**

Sol. : Here $p - (qy + z)^2 = 0$ and Let $f = p - (qy + z)^2$ ...(1)

By charpit method the auxiliary equation are

$$\frac{dp}{0 + 2p\,(qy + z)} = \frac{dq}{4q\,(qy + z)} = \frac{dy}{-2y(qy + z)} = \ldots\ldots$$

From (*i*) and (*iii*),

$$\frac{dp}{p} = -\frac{dy}{y}$$

integrating,

$$\log p = -\log y + \log a$$

or $py = a$ or $p = \frac{a}{y}$

putting this value of p in (1), we get

$$\frac{a}{y} = (qy + z)^2 \text{ or } \left\{\sqrt{\frac{a}{y}} - z\right\}\frac{1}{y} = q$$

Now $dz = pdx + qdy$

$$= \frac{a}{y} \times dx + \frac{1}{y}\left\{\sqrt{\frac{a}{y}} - z\right\} dy$$

$$= \frac{a dx}{y} - \frac{z dy}{y} + \frac{\sqrt{a}}{y^{3/2}}\,dy$$

or $\frac{y dz + z dy}{y} = \frac{a dx}{y} + \frac{\sqrt{a}}{y\sqrt{y}}\,dy$

or $\quad d(yz) = a\,dx + \sqrt{a}\, y^{-1/2}\, dy$

Integrating, we get

$$yz + b = ax + 2\sqrt{a}\, y^{1/2}$$ Ans.

Ex. 2. **Solve** $pxy + pq + qy = yz$

Sol. : Here $pxy + pq + qy - yz = 0$ *and let* $f = pxy + pq + qy - yz$

By charpit method the auxiliary equation are

$$\frac{dp}{py - py} = \frac{dq}{(px + y) - py} = \ldots\ldots$$

$\therefore \quad dp = 0$ or $p = a$

Putting this value of p in (1), we

$$axy + aq + qy - yz = 0 \text{ or } q = \frac{yz - axy}{(a + y)}$$

Also $dz = pdx + qdy = adx + \dfrac{yz - axy}{a + y} dy$

or $\quad (dz - adx) = y\dfrac{(z - ax)}{(a + y)} dy$

or $\quad \dfrac{dz - adx}{z - ax} = \dfrac{y}{(a + y)} dy = \left(1 - \dfrac{a}{a + y}\right) dy$

integrating,

$\log(z - ax) + b = y - a\log(a + y)$

or $\quad \log[(z - ax)(a + y)^a] = y - b$

or $\quad (z - ax)(a + y)^a = e^{y - b} = b' e^y$

or $\quad z = ax + b' e^y (a + y)^{-a}$ Ans.

Ex. 3 **Solve** $2xz - px^2 - 2q\,xy + pq = 0$

Sol. : Let $f = 2xz - px^2 - 2q\,xy + pq$

By charpit's method the auxiliary equation are

$$\frac{dp}{2z - 2qy - 2xp + 2xp} = \frac{dq}{-2qx + 2qx} = \ldots\ldots$$

$$\frac{dp}{2z - 2qy} = \frac{dq}{0} \quad \therefore dq = 0, \text{ which gives } q = a$$

Putting this value of q in (1), we get

$$2xz - px^2 - 2axy + ap = 0 \text{ or } p = \frac{2x(z - ay)}{x^2 - a}$$

Now $dz = pdx + qdy = \dfrac{2x(z - ay)}{x^2 - a} dx + a\,dy$

or $\quad \dfrac{dz - ady}{(z - ay)} = \dfrac{2x}{(x^2 - a)} dx$

Integrating,

$\log(z - ay) - \log b = \log(x^2 - a)$

$\dfrac{(z - ay)}{b} = (x^2 - a)$

$z = ay + b(x^2 - a)$ Ans.

Ex. 4. **Solve** $(p^2 + q^2)y = qz$ ...(1)

Sol. : Here $(p^2 + q^2)y - qz = 0$ and Let $f = (p^2 + q^2)y - qz$

Now by charpit's method, the auxiliary equations are

i.e. $\quad \dfrac{dp}{0 + p(-q)} = \dfrac{dq}{(p^2 + q^2) + q(-q)}$

$$= \frac{dz}{-p(2py) - q(2qy - z)} = \ldots\ldots$$

$\Rightarrow \quad \dfrac{dp}{-pq} = \dfrac{dq}{p^2}$ or $\dfrac{dp}{-q} = \dfrac{dq}{p}$

or $\quad pdp + qdq = 0 \Rightarrow p^2 + q^2 = a'$

Putting the value of $(p^2 + q^2)$ in (1), we get

$$ya' = qz \text{ or } q = \left(\frac{ya'}{z}\right)$$

Now $dz = pdx + qdy = \left(\sqrt{a' - q^2}\right) dx + \dfrac{ya'}{z} dy$

$$= \left(\sqrt{a' - \frac{y^2 a'^2}{z^2}}\right) dx + \frac{ya'}{z} dy$$

or $\quad zdz = \left(\sqrt{z^2 a' - y^2 a'^2}\right) dx + ya'\, dy$

or $\quad 2(a'z\,dz - a'^2 y\,dy).(a'z^2 - y^2 a'^2)^{-1/2} = 2a'\,dx$

integrating, we get $(a'z^2 - y^2 a'^2)^{1/2} + b' = a'x$

or $\quad \dfrac{(a'z^2 - y^2 a'^2)^{1/2}}{a'^{1/2}} + \dfrac{b'}{a'^{1/2}} = a'^{1/2} x$

or $\quad (z^2 - y^2 a')^{1/2} + \dfrac{b'}{a'^{1/2}} = a'^{1/2} x$

put $\quad a' = a^2$ and $\dfrac{b'}{a} = -b$

$z^2 = a^2 y^2 + (ax + b)^2$ Ans.

Scanned with

Ex. 5. **Solve** $z = p^2x + q^2y$

Sol. : Here $p^2x + q^2y - z = 0$ ...(1)

and Let $f = p^2x + q^2y - z$

$\therefore$ The charpit's auxiliary equation are

$$\frac{dp}{p^2 + p(-1)} = \frac{dq}{q^2 + q(-1)} = \frac{dz}{(-p \cdot 2px - q \cdot 2qy)} = \frac{dx}{-2px}$$

$$= \frac{dy}{-2qy} = \frac{dF}{0}$$

$$\Rightarrow \quad \frac{dp}{p^2 - p} = \frac{dx}{-2px} = \frac{dq}{q^2 - q} = \frac{dy}{-2qy}$$

$$\Rightarrow \quad \frac{p^2dx + 2px\,dp}{p^2x} = \frac{q^2dy + 2qy\,dq}{q^2y}$$

Integrating,

$$\log p^2x = \log q^2y + \log a \quad \text{or} \quad p^2x = q^2y \cdot a \qquad ...(2)$$

Now using (2), we obtain $aq^2y + q^2y = z$

$$q = \left\{\frac{z}{(1+a)y}\right\}^{1/2} \text{ and hence } p = \left\{\frac{az}{(1+a)x}\right\}^{1/2}$$

Putting this value in $dz = pdx + qdy$, we get

$$dz = \left\{\frac{az}{(1+a)x}\right\}^{1/2} dx + \left\{\frac{z}{(1+a)y}\right\}^{1/2} dy$$

or $$\sqrt{1+a}\,\frac{dz}{\sqrt{z}} = \sqrt{a}\,\frac{dx}{\sqrt{x}} + \frac{dy}{\sqrt{y}}$$

Integrating, $\sqrt{(1+a)z} = \sqrt{ax} + \sqrt{y} + b$ **Ans.** ...(1)

Ex. 6. **Solve** $p^2 + q^2 - 2px - 2qy + 1 = 0$

Sol. : Let $f = p^2 + q^2 - 2px - 2qy + 1$

By Charpit's method the auxiliary equation are

$$\frac{dp}{-2p + p(0)} = \frac{dq}{-2q + q(0)} = \ldots\ldots$$

or $$\frac{dp}{-p} = \frac{dq}{q} \Rightarrow p = qa$$

Putting this value of p in (1)

$$q^2a^2 + q^2 - 2qax - 2qy + 1 = 0$$

$$q^2(a^2+1) - 2q(ax + y) + 1 = 0$$

$$\therefore \quad q = \frac{2(ax+y) \pm 2\sqrt{(ax+y)^2 - (a^2+1)}}{2(a^2+1)}$$

$$q = \frac{(ax+y) \pm \sqrt{(ax+y)^2 - (a^2+1)}}{(a^2+1)}$$

Now, $dz = pdx + qdy = aq\,dx + qdy = q\,(adx + dy)$

Putting value of q

$$dz = \frac{(ax+y) \pm \sqrt{(ax+y)^2 - (a^2+1)}}{(a^2+1)}(adx + dy)$$

$$dz = \frac{u \pm \sqrt{u^2 - (a^2+1)}}{(a^2+1)} \cdot du \qquad \text{where } u = ax + y$$

or $$(a^2+1)\,dz = \left\{u \pm \sqrt{u^2 - (a^2+1)}\right\}du$$

Integrating,

$$(a^2+1)z + b = \frac{1}{2}u^2 \pm \left[\frac{1}{2}u\lambda - \frac{1}{2}(a^2+1)\log(u+\lambda)\right] \quad \textbf{Ans.}$$

where $\lambda = \sqrt{u^2 - a^2 - 1}$

Ex. 7. **Solve** $q = -xp + p^2$.

Sol. : Here $q + px - p^2 = 0$ and Let $f = q + px - p^2$

Now by charpit's method, the auxiliary equation are

$$\frac{dp}{(p - p \cdot 0)} = \frac{dq}{0} = \ldots\ldots \Rightarrow dq = 0 \text{ or } q = a$$

Putting their value of q, we get

$$a + px - p^2 = 0 \quad \text{or} \quad p^2 - px - a = 0$$

$$p = \frac{1}{2}\left\{x \pm \sqrt{x^2 + 4a}\right\}$$

Now $$dz = pdx + qdy = \frac{1}{2}\left\{x \pm \sqrt{x^2+4a}\right\}dx + ady$$

integrating

$$z = \frac{x^2}{4} \pm \frac{1}{2}\left[\frac{x}{2}\sqrt{x^2+4a} + \frac{4a}{2}\log\left\{x + \sqrt{x^2+4a}\right\}\right] + ay + b$$

$$\text{or } z = \frac{1}{4}\left[x^2 + \left\{x\left(\sqrt{x^2+4a}\right) + 4a\log\left(x + \sqrt{x^2+4a}\right)\right\}\right] + ay + b \quad \textbf{Ans.}$$

11. Equation of First Order and First Degree

Differential equation : A relation involving independent and dependent variables and the derivaties of the dependent variable is called a differential equation.

Order of a differential equation : The order of the highest derivative present in the differential equation is termed as its order.

Degree of a differential equation : The degree of the highest order derivative present in a differential equation, after it is cleared of radicals and fractions so far as the derivatives are concerned is known as its degree.

Differential equations of first order and first degree : General form of differential equations of first order and first degree is :

$$\frac{dy}{dx}+f(x,y)=0 \text{ or}$$

$Mdx + Ndy = 0$ where M and N are functions of x and y.

Method of solution of differential equations of Ist order and Ist degree :

(i) **Seperation of variables**

(ii) **Homogeneous equations**

A differential equation of the form $\frac{dy}{dx}=\frac{f_1(x,y)}{f_2(x,y)}$ .....(1)

where f_1 and f_2 are homogeneous functions of x and y of same degree is called homogeneous different equation.

Solution : Put $y = vx$ in (i) and solve by seperating the variables.

(iii) **Equations reducible to homogeneous form :** They can be classified as :

(a) $\frac{dy}{dx}=\frac{ax+by+c}{Ax+By+C}$ .....(2)

where a, b, c and A, B, C are constants.

Let $x = X + h$, $y = Y + k$ substitute in (2) make it homogeneous and solve it.

(b) $\dfrac{dy}{dx} = \dfrac{ax + by + c}{m(ax + by) + C}$ ...(3)

Put $ax + by = w$ in (3), seperate the variables and solve it.

(c) $\dfrac{dy}{dx} = \dfrac{ax + by + c}{-bx + By + C}$

Similarly seperate the variables and solve.

Note that $d(x\,y) = xdy + dyx$

(iv) **Linear equations**

(a) $\dfrac{dy}{dx} + Py = Q$

which is Linear in y where P *and* Q are functions of x only

Find I.F. $= e^{\int P dx}$

$\therefore$ Solution is :

$$y \times (\text{I.F.}) = \int Q(\text{I.F.})\,dx + C$$

(b) $\dfrac{dx}{dy} + P_1 x = Q_1$

which is linear in x where P_1 *and* Q_1 are functions of y only

Here I.F. $= e^{\int P_1 dy}$

$\therefore$ Solution is :

$$x \times (\text{I.F.}) = \int Q_1(\text{I.F.})\,dy + C_1$$

(v) **Equations reducible to linear form (Bernoulli's equation) :**

$$\frac{dy}{dx} + Py = Qy^n \quad ...(4)$$

where $n \neq 1$ and P and Q are functions of x only.

Divide whole equation (4) by $y^n \Rightarrow$

$$\frac{1}{y^n}\frac{dy}{dx} + P\frac{1}{y^{n-1}} = Q \quad ...(5)$$

Let $\dfrac{1}{y^{n-1}} = v$ in (5) which reduces it into linear form.

(vi) **Exact differential equation :**

The equation $M(x, y)\,dx$ ...(6)

is said to be exact iff. $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$

Method of Solution

(a) Find $U(x, y) = \int M\,dx$ (keeping y constant)

(b) Find $\dfrac{\partial U}{\partial y}$

(c) Find $V(y) = \int\left(N - \dfrac{\partial U}{\partial y}\right)dy$

$\therefore$ Solution is :

$U(x, y) + V(y) = C$

(vii) **Equations reducible to exact equations :** Differential equations which is not exact can be made exact by multiplying it with a suitable function of x and y known as integrating Factor (I. F.).

Methods for finding I. F. :

(a) By inspection

(b) If the differential equation (6) has the form

$f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0$

then it's I.F. $= \dfrac{1}{Mx - Ny}$ where $Mx - Ny \neq 0$.

(c) If $\dfrac{1}{N}\left(\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}\right) = f(x)$

Then I.F. $= e^{\int f(x)dx}$

(d) If $\dfrac{1}{M}\left(\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}\right) = \phi(y)$

Then I.F. $= e^{\int \phi(y)dy}$

(e) For a diff. eq. of the form

$$x^a y^b (my\,dx + nx\,dy) + x^c y^d (py\,dx + qx\,dy) = 0$$

where a, b, m, n, c, d, p, q are constants

I.F. $= x^h y^k$

where h and k are determined by the condition of exactness of a differential equation i.e. $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

(viii) By change of variables

(1) $\int \cos x\,dx = \sin x$

(2) $\int \sin x\,dx = -\cos x$

(3) $\int \sec^2 x\,dx = \tan x$

(4) $\int \operatorname{cosec}^2 x\,dx = -\cot x$

(5) $\int \sec x \tan x\,dx = \sec x$

(6) $\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x$

(7) $\int e^x\,dx = e^x$

(8) $\int \frac{1}{x}\,dx = \log x$

(9) $\int a^x\,dx = \frac{a^x}{\log a}$

(10) $\int x^n\,dx = \frac{x^{n+1}}{n+1};\ n \neq -1$

(11) $\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1} x$ and $\int \frac{-1}{\sqrt{1-x^2}}\,dx = \cos^{-1} x$

(12) $\int \frac{1}{1+x^2}\,dx = \tan^{-1} x$

(13) $\int \tan x\,dx = \log \sec x = -\log \cos x$

(14) $\int \cot x\,dx = \log \sin x$

(15) $\int \sec x\,dx = \log(\sec x + \tan x) = \log \tan\left(\frac{x}{2} + \frac{\pi}{4}\right)$

(16) $\int \operatorname{cosec} x\,dx = -\log(\operatorname{cosec} x + \cot x)$

$$= \log \tan\left(\frac{x}{2}\right)$$

(17) $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right)$

(18) $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log\left\{x + \sqrt{x^2 - a^2}\right\} = \cosh^{-1}\left(\frac{x}{a}\right)$

(19) $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log\left\{x + \sqrt{x^2 + a^2}\right\} = \sinh^{-1}\left(\frac{x}{a}\right)$

(20) $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right)$

(21) $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$

(22) $\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$

(23) $\int \sqrt{x^2 - a^2}\,dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2}\log\left\{x + \sqrt{x^2 - a^2}\right\}$

(24) $\int \sqrt{x^2 + a^2}\,dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2}\sinh^{-1}\left(\frac{x}{a}\right)$

(25) $\int u\,v\,dx = u\int v\,dx - \int\left\{\frac{du}{dx}\int v\,dx\right\}dx$

Ex-24

EXERCISE–1(A)

Ex.1 $(3 + 2\sin x + \cos x)\,dy = (1 + 2\sin y + \cos y)\,dx.$

Sol. Separating variables, we have

$$\frac{dy}{(1+2\sin y+\cos y)}=\frac{dx}{(3+2\sin x+\cos x)}$$

$$=\frac{dx}{\left(2+\cos^2\frac{x}{2}+\sin^2\frac{x}{2}\right)+\left(4\sin\frac{x}{2}\cos\frac{x}{2}\right)+\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)}$$

$$\Rightarrow\quad \frac{dy}{2\cos^2\frac{y}{2}+4\sin\frac{y}{2}\cos\frac{y}{2}}=\frac{dx}{2+2\cos^2\frac{x}{2}+4\sin\frac{x}{2}\cos\frac{x}{2}}$$

$$\Rightarrow\quad \frac{\sec^2\frac{y}{2}\,dy}{2\left(1+2\tan\frac{y}{2}\right)}=\frac{dx}{2\left[\cos^2\frac{x}{2}+\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2\right]}$$

$$\Rightarrow\quad \frac{\sec^2\frac{y}{2}\,dy}{2\left(1+2\tan\frac{y}{2}\right)}=\frac{\sec^2\frac{x}{2}\,dx}{2\left[1+\left(1+\tan\frac{x}{2}\right)^2\right]}$$

$$\Rightarrow\quad \frac{\sec^2\frac{y}{2}\,dy}{2\left(1+2\tan\frac{y}{2}\right)}=\frac{\sec^2\frac{x}{2}\,dx}{2\left[1+\left(1+\tan\frac{x}{2}\right)^2\right]} \qquad \text{....(1)}$$

Let $\tan\frac{y}{2}=u$ and $\tan\frac{x}{2}=v$

then $\frac{1}{2}\sec^2\frac{y}{2}dy=du$ and $\frac{1}{2}\sec^2\frac{x}{2}dx=dv$

put these value in equation (1), we get :

$$\frac{du}{(1+2u)}=\frac{dv}{\left[1+(1+v)^2\right]}$$

integrating, we get :

$$\frac{1}{2}\log(1+2u)=\tan^{-1}(1+v)+C_1$$

Put value of u and v, we get

$$\log\left(1+2\tan\frac{y}{2}\right)=2\tan^{-1}\left[\tan\frac{x}{2}+1\right]+C \quad \text{Ans.}$$

Ex.2 $\operatorname{cosec} x \log y\, dy + x^2y^2\, dx = 0.$

Sol. Separating the variables, we get :

$$\frac{\log y\,dy}{y^2}+\frac{x^2\,dx}{\operatorname{cosec} x}=0$$

$$\Rightarrow\quad \frac{1}{y^2}\log y\,dy+x^2\sin x\,dx=0$$

Integrating, we get

$$\Rightarrow\quad -\frac{1}{y}\log y+\int\frac{1}{y^2}dy-x^2\cos x+2\int x\cos x\,dx=C$$

$$\Rightarrow\quad -\frac{1}{y}\log y-\frac{1}{y}-x^2\cos x+2x\sin x+2\cos x=C$$

or, $\frac{1}{y}(\log y+1)+x^2\cos x-2x\sin x-2\cos x=C$ **Ans.**

Ex.3 $(1 + x)\,y\,dx + (1 - y)\,x\,dy = 0.$

Sol. Separating variables, we get :

$$\left(\frac{1+x}{x}\right)dx+\left(\frac{1-y}{y}\right)dy=0$$

$$\left(\frac{1}{x}+1\right)dx+\left(\frac{1}{y}-1\right)dy=0$$

Integrating

$$\log x + x + \log y - y = \log C$$

or, $\log\frac{xy}{C}=y-x$

or, $xy = C\,e^{y-x}$ **Ans.**

Scanned with

Ex.4 $y(2\log y + 1)\,dy = (\sin x + x\cos x)\,dx$.

Sol. we have

$(\sin x + x\cos x)\,dx = y\,(2\log y + 1)$

Integrating, we get –

$$-\cos x + x\sin x + \cos x = 2\left[\frac{y^2}{2}\log y - \frac{y^2}{4}\right] + \frac{1}{2}y^2 + c$$

or, $x\sin x = y^2\log y + c$

Ans.

Ex.5 $\dfrac{dy}{dx} = e^{x-y} + x^2e^{-y}$.

Sol. Separating variables, we have –

$e^y\,dy = (e^x + x^2)\,dx$

integrating, we get

$$e^y = e^x + \frac{x^3}{3} + c$$

Ans.

Ex.6 Solve $x\cos^2 y\,dy = y\cos^2 x\,dy$.

Sol. Separating variable, we get –

$x\sec^2 x\,dx = y\sec^2 y\,dy$

Integrating, we get

$$x\tan x - \int \tan x\,dx = y\tan y - \int \tan y\,dy + C$$

$\Rightarrow$ $x\tan x - \log\sec x = y\tan y - \log\sec y + C$ Ans.

Ex.7 Solve $\sqrt{(a+x)}\,dy + x\,dx = 0$.

Sol. We can write it as

$$dy = \frac{-x\,dx}{\sqrt{a+x}}$$

$$\Rightarrow\ dy = \frac{-(x+a)+a}{\sqrt{a+x}}\,dx$$

$$\Rightarrow\ dy = \left[-\sqrt{x+a} + \frac{a}{\sqrt{x+a}}\right]dx$$

Integrating, we get

$$y = -\frac{2}{3}(x+a)^{3/2} + 2a(x+a)^{1/2} + C$$

$$\Rightarrow\ y = -\frac{2}{3}(x+a)^{1/2}[x+a-3a] + C$$

$$\Rightarrow\ y + \frac{2}{3}(x-2a)\left(\sqrt{x+a}\right) = C \quad \text{Ans.}$$

Ex.8 $x\,dx + y\,dy = a\,(x\,dy - y\,dx)$.

Sol. It can be written as :

$$\frac{1}{2}d\left(x^2+y^2\right) = ax^2 d\left(\frac{y}{x}\right) \qquad ...(1)$$

changing to polar, put $x = r\cos\theta$, $y = r\sin\theta$

$$\Rightarrow\ \frac{y}{x} = \tan\theta,\ x^2+y^2 = r^2$$

Put these values in equation (1), we get :

$$\frac{1}{2}d\left(r^2\right) = a\,r^2\cos^2\theta\,d(\tan\theta)$$

$$\Rightarrow\ \frac{1}{2}\times 2r\,dr = ar^2\cos^2\theta\sec^2\theta\,d\theta \Rightarrow \frac{dr}{r} = a\,d\theta$$

Integrating, we get :

$$\int\frac{dr}{r} = a\int d\theta$$

or, $\log r = a\theta + C$

or $\dfrac{1}{2}\log\left(x^2+y^2\right) = a\tan^{-1}\dfrac{y}{x} + C$ $\qquad \left[\because r = \sqrt{x^2+y^2}\right]$

Ans.

Ex.9 $\dfrac{dy}{dx} = \dfrac{xy+y}{xy+x}$.

Sol. Separating variables, we get :

$$\frac{dy}{dx} = \frac{y(x+1)}{x(y+1)}$$

or $\left(\dfrac{y+1}{y}\right)dy = \left(\dfrac{x+1}{x}\right)dx$

or $\left(1+\dfrac{1}{y}\right)dy = \left(1+\dfrac{1}{x}\right)dx$

Integrating, we get

$$y + \log y = x + \log x + \log C$$

$$\Rightarrow \quad y - x = \log \frac{xC}{y} \quad \text{Ans.}$$

Ex.10 $(1-x^2)(1-y)\,dx = xy(1+y)\,dy$: given that $x = 1, y = 0$.

Sol. Separating variables, we get

$$\left(\frac{1-x^2}{x}\right)dx = \left(\frac{1+y}{1-y}\right)y\,dy$$

$$\Rightarrow \quad \left(\frac{1-x^2}{x}\right)dx = \left(\frac{y^2+y}{1-y}\right)dy$$

Integrating

$$\Rightarrow \quad \int\left(\frac{1}{x}-x\right)dx = \int\frac{y^2-y+2y-2+2}{(1-y)}$$

$$\Rightarrow \quad \int\left(\frac{1}{x}-x\right)dx = \int\left(-y-2+\frac{2}{1-y}\right)dy$$

$$\Rightarrow \quad \log x - \frac{x^2}{2} + C = \frac{-y^2}{2} - 2y - 2\log(1-y)$$

$\Rightarrow$ Putting the values $x = 1, y = 0$ (as given), we get

$$\Rightarrow \quad 0 - \frac{1}{2} + C = 0$$

$$\Rightarrow \quad C = \frac{1}{2}$$

$$\therefore \quad \log x - \frac{x^2}{2} + \frac{1}{2} = \frac{-y^2}{2} - 2y - 2\log(1-y)$$

or, $\quad 2\log[x(1-y)^2] = x^2 - y^2 - 4y - 1 \quad$ **Ans.**

Ex.11 $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$.

Sol. Separating the variables, we get :

$$\frac{\sec^2 x}{\tan x}dx = -\frac{\sec^2 y}{\tan y}dy$$

Put $\tan x = p$ and $\tan y = q$

$\Rightarrow \quad \sec^2 x\, dx = dp$ and $\sec^2 y\, dy = dq$

$$\Rightarrow \quad \frac{dp}{p} = -\frac{dq}{q}$$

Integrating, we get

$$\log p = -\log q + \log C$$

or, $\quad \log \tan x = -\log \tan y + \log C$

$$\log(\tan x \tan y) = \log C$$

$\tan x \cdot \tan y$ [illegible] Ans.

EX-25

Exercise - 11([illegible])

Ex.1 $(xy - x^2)dy = y^2 dx$.

Sol. It can be written as :

$$\frac{dy}{dx} = \frac{y^2}{xy - x^2} \qquad \text{...(1)}$$

This is homogeneous equation, so put $y = vx$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Put the value of y and $\frac{dy}{dx}$ in equation (1)

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v^2}{(v-1)}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v}{(v-1)}$$

$$\Rightarrow \quad \frac{(v-1)dv}{v} = \frac{dx}{x}$$

$$\Rightarrow \quad \frac{(v-1)dv}{v} = \frac{dx}{x}$$

$$\Rightarrow \quad dv - \frac{1}{v}dv = \frac{dx}{x}$$

On integration, we get :

$$v - \log v = \log x + \log C$$

$$\Rightarrow \quad v = \log (v \times x \times C)$$

Put value of v i.e. $\frac{y}{x}$

$$\Rightarrow \quad \frac{y}{x} = \log yC$$

$$\Rightarrow \quad yc = e^{\left(\frac{y}{x}\right)} \quad \textbf{Ans.}$$

Ex.2 $x\,dy = \left(y - x \tan \frac{y}{x}\right) dx.$

Sol. It can be written as :

$$\frac{dy}{dx} = \frac{y}{x} - \tan \frac{y}{x} \qquad ...(1)$$

This is homogeneous equation, so put $y = vx$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Put value of y and $\frac{dy}{dx}$ in equation (1)

$$\Rightarrow \quad v + x\frac{dv}{dx} = v - \tan v$$

$$\Rightarrow \quad x\frac{dv}{dx} = -\tan v$$

$$\Rightarrow \quad -\frac{dx}{x} = \frac{dv}{\tan v} \Rightarrow -\frac{dx}{x} = \cot v\,dv$$

On integration, we get :

$$-\log x + \log C = \log \sin v$$

put value of $v = \frac{y}{x}$, we get :

$$\log C = \log \sin\left(\frac{y}{x}\right) + \log x$$

$$\Rightarrow \quad x \sin\left(\frac{y}{x}\right) = C \quad \textbf{Ans.}$$

Ex.3 Solve $x + y\frac{dy}{dx} = 2y$.

Sol. Rewrite as $\frac{dy}{dx} = \frac{(2y - x)}{y}$ ...(1)

This is homogeneous equation so put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Put values of y and $\frac{dy}{dx}$ in equation (1)

$$\Rightarrow \quad v + x\frac{dv}{dx} = \left(\frac{2v-1}{v}\right)$$

$$\Rightarrow \quad x\frac{dv}{dx} = \left(\frac{2v - 1 - v^2}{v}\right)$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{(v-1)^2}{v}$$

$$\Rightarrow \quad \frac{dx}{x} = -\frac{v}{(v-1)^2}dv$$

$$\Rightarrow \quad \frac{dx}{x} = -\left[\frac{1}{(v-1)^2} + \frac{1}{(v-1)}\right]dv$$

On integration, we get :

$$\log x = -\left[\frac{-1}{(v-1)} + \log(v-1)\right] + C$$

$$\Rightarrow \quad \log x + \log(v-1) = \frac{1}{(v-1)} + C$$

$$\Rightarrow \quad \log\{x(v-1)\} = \frac{1}{(v-1)} + C$$

$$\Rightarrow \quad \log(y - x) = \frac{x}{(y-x)} + C \qquad \left(\because v = \frac{y}{x}\right) \quad \text{Ans.}$$

Ex.4 $xdy - ydx = \sqrt{(x^2 + y^2)}\,dx$ ...(1)

Sol. We have, $\frac{dy}{dx} = \frac{y + \sqrt{(x^2 + y^2)}}{x}$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + \frac{x\,dv}{dx}$

∴ (1) reduces to

$$v + x\frac{dv}{dx} = v + \sqrt{(1+v^2)}$$

or, $x\frac{dv}{dx} = \sqrt{(1+v^2)}$

or, $\frac{dv}{\sqrt{(1+v^2)}} = \frac{dx}{x}$

integrating, we get

$$\log\left\{v + \sqrt{(1+v^2)}\right\} = \log x + \log c$$

or, $v + \sqrt{(1+v^2)} = cx$

or, $\frac{y}{x} + \sqrt{\left(1+\frac{y^2}{x^2}\right)} = cx \quad \left(\because\ v = \frac{y}{x}\right)$

or, $y + \sqrt{(x^2+y^2)} = cx^2$ **Ans.**

Ex.5 $\left(x\cos\frac{y}{x} + y\sin\frac{y}{x}\right)y - \left(y\sin\frac{y}{x} - x\cos\frac{y}{x}\right)x\frac{dy}{dx} = 0$.

Sol. We can write it as

$$\frac{dy}{dx} = \frac{\left(x\cos\frac{y}{x} + y\sin\frac{y}{x}\right)y}{\left(y\sin\frac{y}{x} - x\cos\frac{y}{x}\right)x} \qquad \text{...(1)}$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

∴ (i) reduces to

$$v + x\frac{dv}{dx} = \frac{(\cos v + v\sin v)v}{(v\sin v - \cos v)}$$

or, $x\frac{dv}{dx} = v\left[\frac{\cos v + v\sin v - v\sin v + \cos v}{(v\sin v - \cos v)}\right] = \frac{2v\cos v}{(v\sin v - \cos v)}$

or, $\frac{(v\sin v - \cos v)}{v\cos v}dv = \frac{2dx}{x}$

or, $\left(\tan v - \frac{1}{v}\right)dv = \frac{2dx}{x}$

integrating, we get–

$$(\log\sec v - \log v) = 2\log x + \log c$$

or, $\frac{\sec v}{v} = cx^2$

or, $\frac{\sec(y/x)}{(y/x)} = cx^2 \quad (\because\ y = vx)$

or, $\sec\left(\frac{y}{x}\right) = cxy$ **Ans.**

Ex.6 $x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\frac{y}{x} - x\right)dx$.

Sol. Rewrite as $\frac{dy}{dx} = \frac{\left(y\sin\frac{y}{x} - x\right)}{x\sin\left(\frac{y}{x}\right)}$ ...(1)

This is homogeneous equation, so put $y = vx$

$\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Put the values of y and $\frac{dy}{dx}$ in equation (1), we get :

$$v + x\frac{dv}{dx} = \frac{(v\sin v - 1)}{\sin v}$$

$\Rightarrow x\frac{dv}{dx} = \frac{-1}{\sin v}$

$\Rightarrow -\sin v\,dv = \frac{dx}{x}$

On integration, we get :

$$\log x = \cos v + c$$

or, $\log x = \cos\left(\frac{y}{x}\right) + c \quad \left(\because\ v = \frac{y}{x}\right)$ **Ans.**

Ex.7 $x\dfrac{dy}{dx} = y + 2\sqrt{y^2 - x^2}$.

Sol. Rewrite as

$$\frac{dy}{dx} = \frac{y + 2\sqrt{y^2 - x^2}}{x} \quad \text{....(1)}$$

This is homogeneous equation so put $y = vx$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Put these values in equation (1), we get :

$$\Rightarrow \quad \left(v + x\frac{dv}{dx}\right) = \frac{vx + 2\sqrt{v^2x^2 - x^2}}{x}$$

$$\Rightarrow \quad v + x\frac{dv}{dx} = v + 2\sqrt{v^2 - 1} \Rightarrow x\frac{dv}{dx} = 2\sqrt{v^2 - 1}$$

$$\Rightarrow \quad \frac{1}{2}\frac{dv}{\sqrt{v^2 - 1}} = \frac{dx}{x}$$

On integration, we get :

$$\frac{1}{2}\log\left[v + \sqrt{v^2 - 1}\right] = \log x + \log C_1$$

$$\Rightarrow \quad \log\left[v + \sqrt{v^2 - 1}\right]^{1/2} = \log xC_1$$

$$\Rightarrow \quad \left[v + \sqrt{v^2 - 1}\right]^{1/2} = \log xC_1 \quad \text{....(2)}$$

Substituting $v = \dfrac{y}{x}$ in equation (2), we get :

$$\left[\frac{y}{x} + \sqrt{\frac{y^2}{x^2} - 1}\right]^{1/2} = xC_1$$

or, $$\left[\frac{y}{x} + \frac{1}{x}\sqrt{y^2 - x^2}\right] = x^2C_1^2$$

$$\Rightarrow \quad \left[y + \sqrt{y^2 - x^2}\right] = x^3C \qquad [\text{Put } C_1^2 = C] \quad \textbf{Ans.}$$

Ex.8 Solve $x^2y\,dx - (x^3 + y^3)\,dy = 0$.

Sol. Rewrite as $\dfrac{dy}{dx} = \dfrac{x^2y}{(x^3 + y^3)}$ ....(1)

This is homogeneous equation so put $y = vx$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Put value of y and $\dfrac{dy}{dx}$ in equation (1), we get :

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v}{1 + v^3}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v - v - v^4}{1 + v^3}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{-v^4}{1 + v^3}$$

$$\Rightarrow \quad -\frac{1 + v^3}{v^4}dv = \frac{dx}{x}$$

$$\Rightarrow \quad \frac{-1}{v^4}dv - \frac{v^3}{v^4}dv = \frac{dx}{x} \Rightarrow -\frac{1}{v^4}dv - \frac{1}{v}dv = \frac{dx}{x}$$

On integration, we get :

$$\frac{1}{3v^3} - \log v = \log x - \log C$$

$$\Rightarrow \quad \frac{1}{3v^3} = \log x - \log C + \log v$$

$$\Rightarrow \quad \frac{1}{3v^3} = \log\left(\frac{xv}{C}\right) \quad \text{....(2)}$$

Substituting $v = \dfrac{y}{x}$ in equation (2), we get :

$$\frac{1}{3}\frac{x^3}{y^3} = \log\left(\frac{x}{C}\frac{y}{x}\right)$$

$$\Rightarrow \quad \frac{1}{3}\frac{x^3}{y^3} = \log\left(\frac{y}{C}\right)$$

$$\Rightarrow \quad y = Ce^{\frac{1}{3}\left(\frac{x^3}{y^3}\right)} \quad \textbf{Ans.}$$

Ex.9 $\left(x^2+y^2\right)\frac{dy}{dx} = 2xy$.

Sol. Above equation is homogeneous so put $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Rewrite above differential equation

$$\frac{dy}{dx} = \frac{2xy}{\left(x^2+y^2\right)} \qquad \text{—(1)}$$

Put value of y and $\frac{dy}{dx}$ in equation (1), we get

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{2v}{1+v^2}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{2v - v - v^3}{1+v^2}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v - v^3}{1+v^2}$$

$$\Rightarrow \quad \frac{(1+v^2)}{v(1-v^2)}dv = \frac{dx}{x}$$

$$\Rightarrow \quad \frac{1+2v^2-v^2}{v(1-v^2)}dv = \frac{dx}{x}$$

$$\Rightarrow \quad \left[\frac{1}{v} + \frac{2v}{1-v^2}\right]dv = \frac{dx}{x}$$

On integration, we get :

$$\log v - \log(1 - v^2) = \log x - \log C$$

$$\Rightarrow \quad \log\frac{v}{(1-v^2)} = \log\frac{x}{C}$$

$$\Rightarrow \quad \frac{v}{1-v^2} = \frac{x}{C}$$

$$\Rightarrow \quad vC = x(1 - v^2)$$

Substituting $v = \frac{y}{x}$

$$\frac{yC}{x} = x\left(1 - \frac{y^2}{x^2}\right)$$

$$yC = (x^2 - y^2) \quad \textbf{Ans.}$$

Ex.10 $2x^2\frac{dy}{dx} = y(x+y)$.

Sol. Rewrite as

$$\frac{dy}{dx} = \frac{y(x+y)}{2x^2} \qquad \text{—(1)}$$

This is homogeneous differential equation to solve it put $y = vx$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Put these values in equation (1), we get

$$v + x\frac{dv}{dx} = \frac{vx(x+vx)}{2x^2}$$

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v}{2} + \frac{1}{2}v^2$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{1}{2}v^2 - \frac{v}{2} = \frac{v(v-1)}{2}$$

$$\Rightarrow \quad \frac{dx}{2x} = \frac{dv}{v(v-1)}$$

$$\Rightarrow \quad \frac{dx}{x} = 2\left[\frac{1}{(v-1)} - \frac{1}{v}\right]dv \qquad \text{(By partial fraction)}$$

On integration, we get

$$\log x + \log C = 2\,[\log (v-1) - \log v]$$

$$\Rightarrow \quad \log xC = \log\left[\frac{(v-1)}{v}\right]^2$$

$$\Rightarrow \quad xC = \left[\frac{v-1}{v}\right]^2$$

Substituting, the value of $v = \frac{y}{x}$ in it, we get

$$xC = \left[\frac{y-x}{y}\right]^2$$

$$\Rightarrow \quad (y-x)^2 = y^2\, x\, C \quad \text{Ans.}$$

Ex-26

Exercise – 11 (C)

Ex.1 $(2x+3y-5)\frac{dy}{dx} + (3x+2y-5) = 0.$

Sol. $\frac{dy}{dx} = -\frac{(3x+2y-5)}{(2x+3y-5)}$

Substituting $x = X + h$, $y = Y + k$, where h and k are to be determined, the given differential equation becomes :

$$\frac{dY}{dX} = -\frac{3(X+h)+2(Y+k)-5}{2(X+h)+3(Y+k)-5} = \frac{3X+2Y+(3h+2k-5)}{2X+3Y+(2h+3k-5)}$$

Now h and k choose such that

$$3h + 2k - 5 = 0$$

$$2h + 3k - 5 = 0$$

$$\Rightarrow \quad h = k = 1$$

Now $\frac{dY}{dX} = -\frac{3X+2Y}{2X+3Y}$

Put $Y = vX$, we get $\frac{dY}{dX} = v + X\frac{dv}{dX}$

$$\therefore \quad v + X\frac{dv}{dX} = -\frac{3+2v}{2+3v}$$

$$\Rightarrow \quad X\frac{dv}{dX} = +\frac{-3-2v-2v-3v^2}{2+3v}$$

$$\Rightarrow \quad X\frac{dv}{dX} = -\frac{3v^2+4v+3}{2+3v}$$

$$\Rightarrow \quad \frac{(2+3v)dv}{(3v^2+4v+3)} = -\frac{dX}{X}$$

Put $3v^2 + 4v + 3 = t$ then $(6v + 4)\,dv = dt$

or, $2(3v + 2)\,dv = dt$

$$\therefore \quad \frac{dt}{2t} = -\frac{dX}{X}$$

On integration, we get

$$\frac{1}{2}\log t + \log X = \log C$$

$$\Rightarrow \quad \frac{1}{2}\log(3v^2+4v+3) + \log X = \log C$$

$$\Rightarrow \quad \frac{1}{2}\log\left(\frac{3Y^2}{X^2}+\frac{4Y}{X}+3\right) + \log X = \log C$$

$$\Rightarrow \quad \frac{1}{2}\log(3Y^2+4YX+3X^2) - \log X + \log X = \log C$$

$$\Rightarrow \quad \frac{1}{2}\log(3Y^2+4YX+3X^2) = \log C$$

$$\Rightarrow \quad 3Y^2 + 4YX + 3X^2 = C^2$$

Put $Y = (y-1)$ and $X = (x-1)$, we get :

$$3(y-1)^2 + 4(x-1)(y-1) + 3(x-1)^2 = C^2$$

$$\Rightarrow \quad 3y^2 + 3 - 6y + 3x^2 + 3 - 6x + 4xy - 4x - 4y + 4 = C^2$$

$$\Rightarrow \quad 3y^2 + 3x^2 - 10x - 10y + 10 + 4xy = C^2$$

$$\Rightarrow \quad \frac{3}{2}(x^2+y^2) - 5x - 5y + 2xy = \frac{C^2}{2} \qquad \left[\frac{C^2}{2} = C_1\right]$$

or, $\frac{3}{2}(x^2+y^2) - 5x - 5y + 2xy = C_1$ Ans.

Ex.2 $\frac{dy}{dx}=\frac{2+y-x}{x-2y-3}$.

Sol. Let $x=X+h$, $y=Y+k$ then given equation reduces to :

$$\frac{dY}{dX}=\frac{2+Y+k-X-h}{X+h-2Y-2k-3}=\frac{Y-X+(k-h+2)}{X-2y+(-2k+h-3)}$$

Choose h and k such that $k-h+2=0$

and $-2k+h-3=0$

$\Rightarrow$ $k=-1$ and $h=1$

Now $\frac{dY}{dX}=\frac{Y-X}{X-2Y}$

Put $Y=vX$, we get

$$\frac{dY}{dX}=v+X\frac{dv}{dX}$$

$$\Rightarrow \quad v+X\frac{dv}{dX}=\frac{v-1}{1-2v}$$

$$\Rightarrow \quad X\frac{dv}{dX}=\frac{v-1-v+2v^2}{1-2v}$$

$$\Rightarrow \quad X\frac{dv}{dX}=\frac{2v^2-1}{1-2v} \Rightarrow \frac{dX}{X}=\left(\frac{1-2v}{2v^2-1}\right)dv$$

On integration, we get :

$$\int\frac{dX}{X}=\frac{1}{2}\int\frac{1}{v^2-\left(\frac{1}{\sqrt{2}}\right)^2}dv-\int\frac{2v}{2v^2-1}dv$$

$$\Rightarrow \quad \log X+C=\frac{1}{2\sqrt{2}}\log\left(\frac{v-\frac{1}{\sqrt{2}}}{v+\frac{1}{\sqrt{2}}}\right)-\frac{1}{2}\log\left(2v^2-1\right)$$

$$\left[\because \int\frac{1}{x^2-a^2}dx=\frac{1}{2a}\log\left(\frac{x-a}{x+a}\right)\right]$$

$$\Rightarrow \quad \log(x-1)+C=\frac{1}{2\sqrt{2}}\log\left(\frac{v-\frac{1}{\sqrt{2}}}{v+\frac{1}{\sqrt{2}}}\right)-\frac{1}{2}\log\left(2v^2-1\right)$$

$(\because X=x-1)$

where $v=\frac{y+1}{x-1}$ **Ans.**

Ex. 3 $\frac{dy}{dx}=\frac{x-y+3}{2x-2y+5}$.

Sol. Put $x-y=v$, we have

$$1-\frac{dy}{dx}=\frac{dv}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx}=1-\frac{dv}{dx}$$

Put these value in above differential equation, we get :

$$\left(1-\frac{dv}{dx}\right)=\frac{v+3}{2v+5}$$

$$\Rightarrow \quad 1-\left(\frac{v+3}{2v+5}\right)=\frac{dv}{dx}$$

$$\Rightarrow \quad \left(\frac{v+2}{2v+5}\right)=\frac{dv}{dx}$$

$$\Rightarrow \quad \left(\frac{2v+5}{v+2}\right)dv=dx$$

$$\Rightarrow \quad \left[2+\frac{1}{v+2}\right]dv=dx$$

On integration, we get :

$$2v+\log(v+2)=x+C$$

or, $\quad x-2y+\log(x-y+2)=C$ **Ans.**

Ex.4 Solve $(x+y)(dx-dy)=dx+dy$.

Sol. Rewrite $(x+y)\,dx-dx=dy+(x+y)\,dy$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(x+y-1)}{(x+y+1)} \qquad ...(1)$$

Substituting $x + y = V$

$$\Rightarrow \quad 1 + \frac{dy}{dx} = \frac{dV}{dx}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{dV}{dx} - 1$$

Put value of $(x + y)$ and $\frac{dy}{dx}$ in equation (1)

$$\Rightarrow \quad \frac{dV}{dx} - 1 = \frac{V-1}{V+1}$$

$$\Rightarrow \quad \frac{dV}{dx} = \frac{V-1}{V+1} + 1$$

$$\Rightarrow \quad \frac{dV}{dx} = \frac{2V}{V+1}$$

$$\Rightarrow \quad \left(\frac{V+1}{V}\right)dV = 2dx$$

Integrating, we get

$$V + \log V - \log C = 2x$$

Substituting $V = x + y$

$$x + y + \log(x+y) - \log C = 2x$$

$$\Rightarrow \quad \log(x+y) - \log C = x - y$$

$$\Rightarrow \quad x + y = Ce^{x-y} \quad \textbf{Ans.}$$

Ex.5 $\frac{dy}{dx} = \frac{y-x+1}{x+y+5}$. [Raj., 2002]

Sol. $\frac{dy}{dx} = \frac{y-x+1}{x+y+5}$

Putting $x = X + h$, $y = Y + k$, where h and k are to be determined. The given differential equation becomes,

$$\frac{dY}{dX} = \frac{Y - X + (k-h+1)}{X + Y + (h+k+5)}$$

Now h and k are chosen such that

$$k - h + 1 = 0$$

and $h + k + 5 = 0$

$$\Rightarrow \quad k = -3, h = -2$$

Now, $\frac{dY}{dX} = \frac{Y-X}{X+Y}$ ...(1)

Put $Y = vX$

then $\frac{dY}{dX} = v + X\frac{dv}{dX}$

$\therefore$ (1) reduces to

$$\Rightarrow \quad v + X\frac{dv}{dX} = \frac{v-1}{1+v}$$

$$\Rightarrow \quad X\frac{dv}{dX} = \frac{v-1}{1+v} - v$$

$$\Rightarrow \quad X\frac{dv}{dX} = -\frac{v^2+1}{v+1}$$

$$\Rightarrow \quad \left(\frac{v+1}{v^2+1}\right)dv = -\frac{dX}{X}$$

$$\Rightarrow \quad \left(\frac{1}{1+v^2} + \frac{v}{1+v^2}\right)dv = -\frac{dX}{X}$$

Integrating, we get;

$$\tan^{-1} v + \frac{1}{2}\log(1+v^2) = -\log X + C$$

Substituting $v = \frac{Y}{X}$, we get

$$\Rightarrow \quad \tan^{-1}\left(\frac{Y}{X}\right) + \frac{1}{2}\log\left(\frac{X^2+Y^2}{X^2}\right) = -\log X + C$$

$$\Rightarrow \quad \tan^{-1}\left(\frac{Y}{X}\right) + \log\left(\frac{\sqrt{X^2+Y^2}}{X}\right)X = C$$

$$\Rightarrow \quad \tan^{-1}\left(\frac{Y}{X}\right) + \log\sqrt{X^2+Y^2} = C$$

Since $Y = y + 3$ and $X = x + 2$

$$\Rightarrow \quad \tan^{-1}\left(\frac{y+3}{x+2}\right) + \log\sqrt{(x+2)^2+(y+3)^2} = C$$

$$\Rightarrow \quad \tan^{-1}\left(\frac{y+3}{x+2}\right) + \frac{1}{2}\log\left[(x+2)^2+(y+3)^2\right] = C \quad \textbf{Ans.}$$

Ex.6 $\dfrac{dy}{dx} = \dfrac{(1-2x-3y)}{(2x+3y-5)}$.

Sol. $\dfrac{dy}{dx} = \dfrac{(1-2x-3y)}{(2x+3y-5)}$ ...(1)

Substituting $2x + 3y = v$

$$2 + 3\frac{dy}{dx} = \frac{dv}{dx}$$

i.e. $\dfrac{dy}{dx} = \dfrac{1}{3}\left(\dfrac{dv}{dx} - 2\right)$

$\therefore$ (1) reduces to :

$$\frac{1}{3}\left(\frac{dv}{dx} - 2\right) = \frac{(1-v)}{(v-5)}$$

$$\Rightarrow \quad \frac{dv}{dx} - 2 = \frac{3(1-v)}{(v-5)}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{3(1-v)}{(v-5)} + 2$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{3-3v+2v-10}{(v-5)}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{7+v}{5-v}$$

$$\Rightarrow \quad -dx = \frac{v-5}{v+7}dv$$

$$\Rightarrow \quad -dx = \left(\frac{v+7-12}{v+7}\right)dv$$

$$\Rightarrow \quad -dx = \left(1 - \frac{12}{v+7}\right)dv$$

On integration, we get :

$$-x + C = v - 12\log(v+7)$$

Substituting $2x + 3y = v$, we get

$$-x + C = 2x + 3y - 12\log(2x+3y+7)$$

$$C + 4\log(2x+3y+7) = x + y \quad \text{Ans.}$$

Ex.7 $\dfrac{dy}{dx} = \dfrac{x+2y-3}{2x+y-3}$.

Sol. Putting $x = X + h$, $y = Y + k$ we get

$$\frac{dY}{dX} = \frac{(X+2Y)+(h+2k-3)}{(2X+Y)+(2h+k-3)} \qquad ...(1)$$

choose h and k such that

$h + 2k - 3 = 0$

$2h + k - 3 = 0$

$\Rightarrow \quad h = 1, k = 1$

$$\therefore \quad (1) \Rightarrow \quad \frac{dY}{dX} = \frac{X+2Y}{2X+Y} \qquad ...(2)$$

(homogeneous form)

Put $Y = vX \Rightarrow \dfrac{dY}{dX} = v + X\dfrac{dv}{dX}$

$$\therefore \quad (2) \Rightarrow \quad v + X\frac{dv}{dX} = \frac{1+2v}{2+v}$$

$$\Rightarrow \quad X\frac{dv}{dX} = \frac{1+2v-2v-v^2}{2+v} = \frac{1-v^2}{2+v}$$

$$\Rightarrow \quad \left(\frac{2+v}{1-v^2}\right)dv = \frac{dX}{X}$$

$$\Rightarrow \quad \left(\frac{2}{1-v^2}\right)dv + \frac{v}{(1-v^2)}dv = \frac{dX}{X}$$

integrating, we get

$$\log\left(\frac{1+v}{1-v}\right) - \frac{1}{2}\log(1-v^2) = \log X + c'$$

$$\Rightarrow \quad \left(\frac{1+v}{1-v}\right)\frac{1}{\sqrt{(1-v^2)}} = \sqrt{c}X\,; \qquad c' = \frac{1}{2}\log c$$

or, $\dfrac{1+v}{(1-v)^3} = cX^2$

or, $\dfrac{X+Y}{(X-Y)^3} = c \left(\because \; v = \dfrac{Y}{X}\right)$

Since, $X = x - h = x - 1$
and $Y = y - k = y - 1$, we get

$$\frac{x+y-2}{(x-y)^3} = c \qquad \text{Ans.}$$

Ex.8 $\dfrac{dy}{dx} = \dfrac{x+y+1}{2x+2y+3}$.

Sol. Putting $x + y = v$

$\Rightarrow \quad 1 + \dfrac{dy}{dx} = \dfrac{dv}{dx}$

$\therefore$ given eqn reduces to

$$\frac{dv}{dx} - 1 = \frac{v+1}{2v+3}$$

or, $\dfrac{dv}{dx} = \dfrac{v+1+2v+3}{2v+3} = \dfrac{3v+4}{2v+3}$

or, $\dfrac{2v+3}{3v+4}dv = dx$

$\Rightarrow \quad \dfrac{1}{3}\left(2 + \dfrac{1}{3v+4}\right)dv = dx$

integrating, we get

$$2v + \frac{1}{3}\log(3v+4) = 3x + c'$$

$\Rightarrow \quad \dfrac{1}{3}\log\left(\dfrac{3x+3y+4}{c}\right) = 3x - 2(x+y)\,; \; c' = \dfrac{1}{3}\log c$

$\Rightarrow \quad \log\left(\dfrac{3x+3y+4}{c}\right) = 3x - 6y = 3(x-2y)$

or, $3x + 3y + 4 = ce^{3(x-2y)}$ Ans.

Ex.9 $\dfrac{dy}{dx} = \dfrac{x+y+1}{x+y-1}$.

Sol. $\dfrac{dy}{dx} = \dfrac{x+y+1}{x+y-1}$ ...(1)

Put $x + y = V$

then $1 + \dfrac{dy}{dx} = \dfrac{dV}{dx}$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{dV}{dx} - 1$

$\therefore$ (1) reduces to

$\Rightarrow \quad \dfrac{dV}{dx} - 1 = \dfrac{V+1}{V-1}$

$\Rightarrow \quad \dfrac{dV}{dx} = \dfrac{V+1+V-1}{V-1}$

$\Rightarrow \quad \dfrac{dV}{dx} = \dfrac{2V}{V-1}$

$\Rightarrow \quad \left(\dfrac{V-1}{2V}\right)dV = dx$

$\Rightarrow \quad dV - \dfrac{1}{V}dV = 2dx$

On integration, we get,

$$V - \log V = 2x + C$$

put $V = x + y$

$\therefore \quad x + y - \log(x+y) = 2x + C \Rightarrow y - x - \log(x+y) = C$ Ans.

Ex.10 $\dfrac{dy}{dx} = \dfrac{3y+2x+4}{4x+6y+5}$.

Sol. Substituting $3y + 2x = v$

$$\frac{dv}{dx} = \frac{3dy}{dx} + 2$$

$\Rightarrow \quad \dfrac{dy}{dx} = \left(\dfrac{1}{3}\dfrac{dv}{dx} - \dfrac{2}{3}\right)$

$\Rightarrow \quad \left(\dfrac{1}{3}\dfrac{dv}{dx} - \dfrac{2}{3}\right) = \dfrac{v+4}{2v+5}$ (from given equation)

$\Rightarrow \quad \frac{dv}{dx}=\frac{3(v+4)}{(2v+5)}+2$

$\Rightarrow \quad \frac{dv}{dx}=\frac{3v+12+4v+10}{(2v+5)}$

$\Rightarrow \quad \frac{dv}{dx}=\frac{7v+22}{2v+5}$

$\Rightarrow \quad dx=\left(\frac{2v+5}{7v+22}\right)dv$

$\Rightarrow \quad dx=\frac{2}{7}\frac{(7v+22-22)}{(7v+22)}dv+\frac{5}{(7v+22)}dv$

$\Rightarrow \quad dx=\frac{2}{7}\left(1-\frac{22}{7v+22}\right)dv+\frac{5}{(7v+22)}dv$

$\Rightarrow \quad dx=\frac{2}{7}dv-\frac{9}{7(7v+22)}dv$

On integration, we get

$$x+C=\frac{2}{7}v-\frac{9}{49}\log(7v+22)$$

Put value of $v=2x+3y$, we get :

$$x+C=\frac{2}{7}(2x+3y)-\frac{9}{49}\log[14x+21y+22] \quad \text{Ans.}$$

Exercise - 11(D)

Ex-27

Solve :

Ex.1 $\frac{dy}{dx}=y\tan x-2\sin x$.

Sol. Rewrite as $\frac{dy}{dx}-y\tan x=-2\sin x$

This is linear in y and in form of $\frac{dy}{dx}+Py=Q$

$$\text{I.F.}=e^{\int -\tan x\,dx}=e^{\log\cos x}=\cos x$$

Hence, solution becomes

$$y\cos x=\int \cos x(-2\sin x)dx+C$$

$\Rightarrow \quad y\cos x=\cos^2 x+C$

$\Rightarrow \quad y=\cos x+C\sec x \quad$ Ans.

Ex.2 $\frac{dy}{dx}+y\tan x=\sec x$.

Sol. $\frac{dy}{dx}+(\tan x)y=\sec x$

This is linear in y :

$$\text{I.F.}=e^{\int \tan x\,dx}=e^{\log\sec x}=\sec x$$

Hence solution is :

$$y\sec x=\int \sec x\cdot\sec x\,dx+C=\int \sec^2 x\,dx+C$$

$\Rightarrow \quad y\sec x=\tan x+C$

or, $\quad y=\sin x+C\cos x \quad$ Ans.

Ex.3 Solve $\frac{dy}{dx}+\frac{y}{x}=x^2$ given $y=1$ when $x=1$.

Sol. $\frac{dy}{dx}+\left(\frac{1}{x}\right)y=x^2$

This is linear diff. equation

$\therefore \quad \text{I.F.}=e^{\int \frac{1}{x}dx}=e^{\log x}=x$

Hence, solution $yx=\int x\cdot x^2 dx+C$

$$yx=\frac{x^4}{4}+C$$

Put $y=1$, when $x=1$

$$1=\frac{1}{4}+C$$

Then $C=\frac{3}{4}$

$\therefore \quad yx = \dfrac{x^4}{4} + \dfrac{3}{4}$

or, $4yx = x^4 + 3$ **Ans.**

Ex.4 $\dfrac{dr}{d\theta} = \alpha\theta^n - \dfrac{r}{\theta}$.

Sol. We can write as $\dfrac{dr}{d\theta} + \left(\dfrac{1}{\theta}\right) r = \alpha\theta^n$

This is linear in r :

$$\text{I.F.} = e^{\int \frac{1}{\theta} d\theta} = e^{\log\theta} = \theta$$

Hence, solution

$$\Rightarrow \quad r\theta = \int \theta \times \alpha\theta^n d\theta + C$$

$$\Rightarrow \quad r\theta = \frac{\alpha\theta^{n+2}}{n+2} + C \quad \textbf{Ans.}$$

Ex.5 $\dfrac{dy}{dx} + \dfrac{3x^2}{1+x^3} y = \dfrac{\sin^2 x}{1+x^3}$.

Sol. This is linear in y, so that

$$\text{I.F.} = e^{\int \frac{3x^2}{1+x^3} dx} = e^{\log(1+x^3)} = (1+x^3)$$

Hence solution is $y(1+x^3) = \displaystyle\int \frac{\sin^2 x}{(1+x^3)} \times (1+x^3) dx + C$

$$y(1+x)^3 = \int \sin^2 x \, dx + C \qquad \left[\because \sin^2 x = \frac{1-\cos 2x}{2}\right]$$

$$\Rightarrow \quad y(1+x^3) = \int \left(\frac{1-\cos 2x}{2}\right) dx + C$$

$$\Rightarrow \quad y(1+x^3) = \frac{1}{2}x - \frac{1}{4}\sin 2x + C \quad \textbf{Ans.}$$

Ex.6 $x(x-1)\dfrac{dy}{dx} - (x-2)y = x^3(2x-1)$.

Sol. Rewrite as

$$\frac{dy}{dx} - \frac{(x-2)}{x(x-1)} y = \frac{x^2(2x-1)}{(x-1)}$$

This is linear in y

$$\text{I.F.} = e^{\int -\frac{x-2}{x(x-1)} dx} = e^{\int \left[-\frac{1}{(x-1)} + \frac{2}{x(x-1)}\right] dx}$$

$$\text{I.F.} = e^{\int -\frac{1}{(x-1)} + 2\left\{\frac{1}{x-1} - \frac{1}{x}\right\} dx} = e^{[-\log(x-1) + 2\log(x-1) - 2\log x]}$$

$$= e^{\log\frac{(x-1)}{x^2}} = \frac{x-1}{x^2}$$

Hence solution is

$$y \cdot \frac{(x-1)}{x^2} = \int \frac{x-1}{x^2} \times x^2 \frac{(2x-1)}{(x-1)} dx + C = \int (2x-1) dx + C$$

$$\Rightarrow \quad y \cdot \frac{(x-1)}{x^2} = x^2 - x + C$$

or, $(x-1)y = x^2(x^2 - x + C)$ **Ans.**

Ex.7 $(x^2+1)\dfrac{dy}{dx} + 2xy = 4x^2$.

Sol. It can be written as –

$$\frac{dy}{dx} + \frac{2x}{x^2+1} y = \frac{4x^2}{x^2+1}$$

which is a linear diff. equation.

$$\therefore \quad \text{I.F.} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1 + x^2$$

$\therefore$ solution is

$$y(1+x^2) = \int \frac{4x^2}{(x^2+1)} (1+x^2) dx + c$$

$$\Rightarrow \quad y(1+x^2) = \int 4x^2 dx + c$$

or, $y(1+x^2) = \dfrac{4}{3}x^3 + c$ **Ans.**

Ex.8 $\cos^2 x \dfrac{dy}{dx} + y = \tan x$.

Sol. It can be written as

$$\frac{dy}{dx} + y\sec^2 x = \sec^2 x \tan x \qquad \text{(linear in } x)$$

$\therefore$ I. F. $= e^{\int \sec^2 x dx} = e^{\tan x}$

$\therefore$ solution is

$$y(e^{\tan x}) = \int \sec^2 x \tan x \, e^{\tan x} dx + c$$

$$\Rightarrow \quad y(e^{\tan x}) = \int t e^t dt + c \,;\ \tan x = t \Rightarrow \sec^2 x \, dx = dt$$

$$\Rightarrow \quad y e^{\tan x} = e^t (t-1) + c$$

or, $\quad y e^{\tan x} = e^{\tan x}(\tan x - 1) + c$

or, $\quad y = \tan x - 1 + c e^{-\tan x}$ **Ans.**

Ex.9 $x \cos x \dfrac{dy}{dx} + y(x \sin x + \cos x) = 1$.

Sol. Rewrite as $\dfrac{dy}{dx} + \dfrac{(x \sin x + \cos x)}{x \cos x} y = \dfrac{1}{x \cos x}$

This is linear is y, so that

$$\text{I.F.} = e^{\int \left(\tan x + \frac{1}{x}\right) dx} = e^{\log \sec x + \log x}$$

$$= x \sec x$$

Hence solution is

$$y x \sec x = \int \frac{1}{x \cos x} \times x \sec x \, dx + C$$

or, $\quad y x \sec x = \int \sec^2 x \, dx + C$

$\Rightarrow \quad x y \sec x = \tan x + C$ **Ans.**

Ex.10 $x \log x \dfrac{dy}{dx} + y = 2 \log x$. [Raj., 2002]

Sol. Rewrite as $\dfrac{dy}{dx} + \left(\dfrac{1}{x \log x}\right) y = \dfrac{2}{x}$

This is linear in y :

let $\log x = t \Rightarrow \frac{1}{x} dx = dt$

$$\text{I.F.} = e^{\int \left(\frac{1}{x \log x}\right) dx} = e^{\log(\log x)} = \log x$$

$\Rightarrow$ I.F. $= e^{\int \frac{1}{t} dt}$

I.F. $= e^{\log t}$

I.F. $= e^{\log(\log x)}$

I.F. $= \log x$

Hence, solution

$$y \cdot \log x = \int \frac{2}{x} \cdot \log x \, dx + C$$

$$y \log x = (\log x)^2 + C \quad \textbf{Ans.}$$

Ex.11 $\sin y \dfrac{dy}{dx} = \cos y (1 - x \cos y)$.

[MREC (Auto), 2000; JNVU, 97]

Sol. Rewrite as $\dfrac{\sin y}{\cos^2 y} \dfrac{dy}{dx} - \dfrac{1}{\cos y} = -x$

$$\Rightarrow \quad \tan y \sec y \frac{dy}{dx} - \sec y = -x$$

Substituting $\sec y = v$

$\Rightarrow \quad \tan y \sec y \, dy = dv$

$\therefore \quad \dfrac{dv}{dx} - v = -x$

This is linear in v

$$(\text{I.F.}) = e^{\int -dx} = e^{-x}$$

Hence, solution is $v e^{-x} = -\int x e^{-x} dx + C$

$\Rightarrow \quad v e^{-x} = x e^{-x} + e^{-x} + C$

Replacing v by $\sec y$

$$\sec y \, e^{-x} = x e^{-x} + e^{-x} + C$$

or $\quad \sec y = x + 1 + C e^x$ **Ans.**

Ex.12 $\sin 2x \dfrac{dy}{dx} = y + \tan x$.

Scanned with

Sol. Rewrite as $\frac{dy}{dx}+\left(-\frac{1}{\sin 2x}\right)y=\frac{\tan x}{\sin 2x}$

or, $\frac{dy}{dx}+\left(-\frac{1}{\sin 2x}\right)y=\frac{1}{2}\sec^2 x$

This is linear in y and in form of $\frac{dy}{dx}+Py=Q$

$$\text{I.F.}=e^{\int -\frac{1}{\sin 2x}dx}=e^{\int -\operatorname{cosec} 2x\,dx}=e^{-\frac{1}{2}\log\tan x}$$

$$\left(\because \int \operatorname{cosec} x\,dx=\log\tan\frac{x}{2}\right)$$

$$=\sqrt{\cot x}$$

Hence, solution is

$$y\sqrt{\cot x}=\int \frac{1}{2}\sec^2 x\cdot\sqrt{\cot x}\,dx+C$$

$$y\sqrt{\cot x}=\int \frac{1}{2}\frac{\sec^2 x}{\sqrt{\tan x}}\,dx+C$$

Put $\tan x=t \Rightarrow \sec^2 x\,dx=dt$

$$\Rightarrow\quad y\sqrt{\cot x}=\int \frac{1}{2}\frac{dt}{\sqrt{t}}+C$$

$$\Rightarrow\quad y\sqrt{\cot x}=\frac{1}{2}\times 2\sqrt{t}+C \qquad \text{Replacing } t \text{ by } \tan x$$

$$\Rightarrow\quad y\sqrt{\cot x}=\sqrt{\tan x}+C$$

or, $y=\tan x+C\sqrt{\tan x}$ **Ans.**

Ex.13 $(1+x^2)\frac{dy}{dx}+2yx-4x^2=0$ **and obtain the cubic curve satisfying this equation and passing through the origin.**

Sol. Rewrite as $\frac{dy}{dx}+\frac{2x}{1+x^2}y=\frac{4x^2}{1+x^2}$

This is linear in y, so that

$$\text{I.F.}=e^{\int\frac{2x}{1+x^2}dx}=e^{\log(1+x^2)}$$

$$=(1+x^2)$$

Hence solution is

$$y(1+x^2)=\int\frac{4x^2}{(1+x^2)}\times(1+x^2)\,dx$$

$$\Rightarrow\quad y(1+x^2)=\frac{4x^3}{3}+C$$

As per question curve passing through origin (0, 0), so that $C=0$

$\therefore$ curve equation becomes

$$y(1+x^2)=\frac{4x^3}{3}$$

$$\Rightarrow\quad 4x^3=3(1+x^2)y \quad \textbf{Ans.}$$

Ex.14 Solve $(x+2y^3)\frac{dy}{dx}=y$. **[Raj., 2003]**

Sol. Rewrite as $\frac{dx}{dy}=\frac{(x+2y^3)}{y}$

or, $\frac{dx}{dy}-\left(\frac{1}{y}\right)x=2y^2$

This is linear in x

$$\text{I.F.}=e^{\int\left(-\frac{1}{y}\right)dy}=e^{-\log(y)}=\frac{1}{y}$$

Hence, solution

$$x\left(\frac{1}{y}\right)=\int(2y^2)\frac{1}{y}dy+C=2\int y\,dy+C$$

$$\Rightarrow\quad \frac{x}{y}=y^2+C$$

$$\Rightarrow\quad x=y^3+Cy \quad \textbf{Ans.}$$

Ex.15 $(y-x)\dfrac{dy}{dx}=a^2$.

Sol. Rewrite as $\dfrac{dx}{dy}=\dfrac{y-x}{a^2}$

or, $\dfrac{dx}{dy}+\dfrac{x}{a^2}=\dfrac{y}{a^2}$

This is linear in x, so that

$$\text{I.F.}=e^{\int \frac{1}{a^2}dy}=e^{\frac{y}{a^2}}$$

Hence solution is

$$xe^{\frac{y}{a^2}}=\int e^{\frac{y}{a^2}}\cdot\frac{y}{a^2}\,dy+C$$

$$\Rightarrow\quad xe^{\frac{y}{a^2}}=a^2\int e^t\cdot t\,dt+C \qquad \left[\text{Put } \frac{y}{a^2}=t\Rightarrow dy=a^2dt\right]$$

$$\Rightarrow\quad xe^{\frac{y}{a^2}}=a^2\left[te^t-e^t\right]+C$$

$$\Rightarrow\quad xe^{\frac{y}{a^2}}=a^2\left[\frac{y}{a^2}e^{\frac{y}{a^2}}-e^{\frac{y}{a^2}}\right]+C$$

$$\Rightarrow\quad x=a^2\left[\frac{y}{a^2}-1\right]+Ce^{-\frac{y}{a^2}}$$

or, $x=(y-a^2)+Ce^{-\frac{y}{a^2}}$ **Ans.**

Ex.16 $2(x-5y^3)\dfrac{dy}{dx}+y=0$. **[MREC (Auto), 2001]**

Sol. Rewrite as $\dfrac{dx}{dy}+\left(\dfrac{2}{y}\right)x=10y^2$

This is linear in x, so that

$$\text{I.F.}=e^{\int\left(\frac{2}{y}\right)dy}=e^{\log y^2}=y^2$$

$\therefore$ Solution is

$$xy^2=\int y^2\cdot 10y^2\,dy+C$$

$$xy^2=10\frac{y^5}{5}+C$$

$$xy^2=2y^5+C$$

$$x-2y^3=\frac{C}{y^2}\quad \textbf{Ans.}$$

Ex.17 $\dfrac{dy}{dx}+\dfrac{(1-2x)}{x^2}y=1$.

Sol. This is linear in y

$$\text{I.F.}=e^{\int\frac{(1-2x)}{x^2}dx}$$

$$=e^{-\frac{1}{x}-\log(x^2)}$$

$$=\frac{e^{-\frac{1}{x}}}{e^{(\log x^2)}}=\frac{e^{-\frac{1}{x}}}{x^2}$$

Hence, solution is

$$y\times\frac{e^{-\frac{1}{x}}}{x^2}=\int\frac{e^{-\frac{1}{x}}}{x^2}dx+C$$

or, $y\dfrac{e^{-\frac{1}{x}}}{x^2}=e^{-\frac{1}{x}}+C$

or, $\dfrac{y}{x^2}=1+Ce^{\frac{1}{x}}$ **Ans.**

Ex.18 $\dfrac{dy}{dx}+\dfrac{x}{1+x^2}y=\dfrac{1}{2x(1+x^2)}$.

Sol. This equation is linear in y being of the form $\frac{dy}{dx}+Py=Q$

$$\therefore \quad \text{I.F.}=e^{\int \frac{x}{1+x^2}dx}=e^{\frac{1}{2}\log(1+x^2)}=e^{\log\sqrt{1+x^2}}=\sqrt{1+x^2}$$

Hence, solution is

$$y\cdot(\text{I.F.})=\int Q(\text{I.F.})\,dx+C$$

$$\Rightarrow \quad y\sqrt{1+x^2}=\int \frac{1}{2x(1+x^2)}\times\sqrt{1+x^2}\,dx+C$$

$$\Rightarrow \quad y(1+x^2)^{1/2}=\int \frac{1}{2x\sqrt{1+x^2}}\,dx+C$$

Put $x=\tan\theta$, $dx=\sec^2\theta\,d\theta$, we get ; (In RHS)

$$y(1+x^2)^{1/2}=\int \frac{\sec^2\theta\,d\theta}{2\tan\theta\sec\theta}+C$$

$$y(1+x^2)^{1/2}=\frac{1}{2}\int \text{cosec}\,\theta\,d\theta+C$$

$$y(1+x^2)^{1/2}=\frac{1}{2}\log(\text{cosec}\,\theta-\cot\theta)+C$$

$$=\frac{1}{2}\log\left(\sqrt{1+\cot^2\theta}-\cot\theta\right)+C \quad (\because \text{cosec}^2\theta=1+\cot^2\theta)$$

$$=\frac{1}{2}\log\left(\sqrt{1+\frac{1}{x^2}}-\frac{1}{x}\right)+C \quad \left(\because x=\tan\theta\Rightarrow\cot\theta=\frac{1}{x}\right)$$

$$\text{or,}\quad y\sqrt{(1+x^2)}=\frac{1}{2}\log\left[\left(\frac{\sqrt{(1+x^2)}-1}{x}\right)\right]+C \quad \textbf{Ans.}$$

Ex.19 $(1+y+x^2y)\,dx+(x+x^3)\,dy=0$.

Sol. Rewriting it as $\frac{dy}{dx}=-\frac{1+y+x^2y}{(x+x^3)}$

$$\text{or}\quad \frac{dy}{dx}+\frac{1+x^2}{x(1+x^2)}y=-\frac{1}{x(1+x^2)}\Rightarrow\frac{dy}{dx}+\frac{1}{x}y=\frac{-1}{x(1+x^2)}$$

This is linear in y and in form of $\frac{dy}{dx}+Py=Q$

$$\therefore \quad \text{I.F.}=e^{\int\frac{1}{x}dx}=e^{\log x}=x$$

The solution becomes

$$x\cdot y=-\int\frac{1}{x(1+x^2)}\cdot x\,dx+C$$

$$\Rightarrow \quad xy=-\int\frac{1}{(1+x^2)}\,dx+C$$

$$\Rightarrow \quad yx=-\tan^{-1}x+C \quad \textbf{Ans.}$$

Ex.20 $(1+y^2)+(x-e^{-\tan^{-1}y})\frac{dy}{dx}=0$.

Sol. $\frac{dy}{dx}=-\frac{1+y^2}{x-e^{-\tan^{-1}y}}$

$$\Rightarrow \quad \frac{dx}{dy}=\frac{e^{-\tan^{-1}y}-x}{1+y^2}$$

$$\Rightarrow \quad \frac{dx}{dy}+\frac{x}{1+y^2}=\frac{e^{-\tan^{-1}y}}{1+y^2}$$

This is linear in x

$$\Rightarrow \quad \text{I.F.}=e^{\int\frac{1}{1+y^2}dy}=e^{\tan^{-1}y}$$

Hence solution is :

$$xe^{\tan^{-1}y}=\int\frac{e^{-\tan^{-1}y}\times e^{\tan^{-1}y}}{1+y^2}\,dy+C=\int\frac{1}{1+y^2}\,dy+C$$

$$\Rightarrow \quad xe^{\tan^{-1}y}=\tan^{-1}y+C \quad \textbf{Ans.}$$

Exercise - 11(E)

Ex-28

Solve :

Ex.1 $x\dfrac{dy}{dx}+y=xy^3$.

Sol. Rewrite as $\dfrac{1}{y^3}\dfrac{dy}{dx}+\dfrac{1}{xy^2}=1$

Put $\dfrac{1}{y^2}=V \Rightarrow \dfrac{-2}{y^3}\dfrac{dy}{dx}=\dfrac{dV}{dx}$

$\Rightarrow \quad \dfrac{1}{y^3}\dfrac{dy}{dx}=\dfrac{-1}{2}\dfrac{dV}{dx}$

$\Rightarrow \quad \dfrac{dV}{dx}-\dfrac{2V}{x}=-2$

This is linear in V, so that

$$\text{I.F.}=e^{-\int \frac{2}{x}dx}=e^{\log x^{-2}}=\frac{1}{x^2}$$

Hence solution is

$$\frac{1}{x^2}V=\int \frac{1}{x^2}(-2)\,dx+C$$

$$\frac{V}{x^2}=\frac{2}{x}+C$$

Replacing V by $\dfrac{1}{y^2}$

$$\frac{1}{x^2y^2}=\frac{2}{x}+C$$

or $(2+Cx)\,xy^2=1$ **Ans.**

Ex.2 $\dfrac{dy}{dx}+\dfrac{y}{x}=x^2y^6$.

Sol. Rewrite, it as

$$\frac{1}{y^6}\frac{dy}{dx}+\frac{1}{xy^5}=x^2 \qquad \text{....(1)}$$

Put $\dfrac{1}{y^5}=V \Rightarrow \dfrac{-5}{y^6}\dfrac{dy}{dx}=\dfrac{dV}{dx}$

$\therefore$ (1) reduces to

$$\frac{-1}{5}\frac{dV}{dx}+\frac{V}{x}=x^2$$

$\Rightarrow \quad \dfrac{dV}{dx}-\left(\dfrac{5}{x}\right)V=-5x^2$

This is linear in V

$$\text{I.F.}=e^{\int -\frac{5}{x}dx}=x^{-\log x^5}=\frac{1}{x^5}$$

Hence solution

$$\frac{1}{x^5}V=\int \frac{1}{x^5}(-5x^2)\,dx+C$$

$$\frac{V}{x^5}=-5\left(\frac{-1}{2x^2}\right)+C$$

$$\frac{V}{x^5}=\frac{5}{2x^2}+C$$

Replacing $V=\dfrac{1}{y^5}$

$$y^{-5}x^{-5}=\frac{5}{2x^2}+C \quad \textbf{Ans.}$$

Ex.3 $(1-x^2)\dfrac{dy}{dx}+xy=xy^2$.

Sol. Rewrite as $\dfrac{1}{y^2}\dfrac{dy}{dx}+\dfrac{x}{(1-x^2)}\dfrac{1}{y}=\dfrac{x}{(1-x^2)}$

Put $\dfrac{1}{y}=V$

$\Rightarrow \quad \dfrac{-1}{y^2}\dfrac{dy}{dx}=\dfrac{dV}{dx}$

$$-\frac{dV}{dx}+\frac{x}{1-x^2}V=\frac{x}{(1-x^2)}$$

$$\frac{dV}{dx}-\frac{x}{1-x^2}V=\frac{-x}{(1-x^2)}$$

This is linear in V

$$\text{I.F.} = e^{-\int \frac{x}{1-x^2}dx}$$

$$\text{I.F.} = e^{\int \frac{1}{2}\frac{1}{t}dt} \qquad [1-x^2 = t \qquad -2x\,dx = dt]$$

$$\text{I.F.} = e^{\log t^{\frac{1}{2}}} = t^{\frac{1}{2}}$$

Put $t = (1-x^2)$

$$\text{I.F.} = (1-x^2)^{\frac{1}{2}}$$

$$\therefore \quad (1-x^2)^{\frac{1}{2}}V = -\int (1-x^2)^{\frac{1}{2}} \frac{x}{(1-x^2)}dx + C$$

$$(1-x^2)^{\frac{1}{2}}V = -\int \frac{x\,dx}{(1-x^2)^{\frac{1}{2}}} + C$$

Put $1-x^2 = u$ then $-x\,dx = \frac{1}{2}du$

$$(1-x^2)^{\frac{1}{2}}V = \frac{1}{2}\int u^{-\frac{1}{2}}du + C$$

$$\Rightarrow \quad (1-x^2)^{\frac{1}{2}}V = u^{\frac{1}{2}} + C$$

Replace value of V and u, we get :

$$(1-x^2)^{\frac{1}{2}}\frac{1}{y} = (1-x^2)^{\frac{1}{2}} + C$$

$$\Rightarrow \quad \sqrt{(1-x^2)}(1-y) = Cy \quad \textbf{Ans.}$$

Ex.4 $\dfrac{dy}{dx} + \dfrac{y}{x} = y^2 \sin x$.

Sol. Rewrite as $\dfrac{1}{y^2}\dfrac{dy}{dx} + \dfrac{1}{xy} = \sin x$ ...(1)

Put $\dfrac{1}{y} = V \Rightarrow -\dfrac{1}{y^2}\dfrac{dy}{dx} = \dfrac{dV}{dx}$

$$\Rightarrow \quad \frac{1}{y^2}\frac{dy}{dx} = -\frac{dV}{dx}$$

$$(1) \Rightarrow \quad \frac{-dV}{dx} + \frac{V}{x} = \sin x$$

$$\Rightarrow \frac{dV}{dx} - \frac{V}{x} = -\sin x$$

This is linear in V, so that

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$$

Hence solution is

$$\frac{1}{x}V = \int -\frac{1}{x}\sin x\,dx + C$$

Replacing V by $\dfrac{1}{y}$, we get :

$$\frac{1}{xy} = C - \int x^{-1}\sin x\,dx \quad \textbf{Ans.}$$

Ex.5 $\dfrac{dy}{dx} - \dfrac{y}{x} = \dfrac{y^2}{x^2}$.

Sol. It can be written as

$$\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{x}\cdot\frac{1}{y} = \frac{1}{x^2} \qquad \text{...(1)}$$

put $\dfrac{1}{y} = v \Rightarrow -\dfrac{1}{y^2}\dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$\therefore \quad (1) \Rightarrow \quad -\frac{dv}{dx} - \frac{1}{x}v = \frac{1}{x^2}$$

$$\text{or,} \quad \frac{dv}{dx} + \frac{v}{x} = -\frac{1}{x^2} \qquad \text{(linear in } v)$$

$\therefore$ $\text{I. F.} = e^{\int \frac{1}{x}dx} = e^{\log x} = x$

$\therefore$ solution is

$$v(x) = \int -\frac{1}{x^2}(x)dx + c = -\int \frac{1}{x}dx + c$$

Scanned with

or, $vx = -\log x + c$ $\left(\because \ v = \frac{1}{y}\right)$

$\Rightarrow \frac{x}{y} = -\log x + c$ **Ans.**

Ex.6 $\frac{3dy}{dx} + \frac{2}{(x+1)}y = \frac{x^3}{y^2}$.

Sol. In can be written as

$$3y^2 \frac{dy}{dx} + \frac{2y^3}{(x+1)} = x^3 \qquad \text{...(1)}$$

put $y^3 = v \Rightarrow 3y^2 \frac{dy}{dx} = \frac{dv}{dx}$

$\therefore$ (1) $\Rightarrow \frac{dv}{dx} + \frac{2v}{x+1} = x^3$ (linear in v)

$\therefore$ I. F. $= e^{\int \frac{2}{(x+1)}dx} = e^{2\log(x+1)} = (x+1)^2$

$\therefore$ Solution is

$$v(x+1)^2 = \int x^3(x+1)^2 dx + c = \int (x^5 + 2x^4 + x^3)dx + c$$

or, $v(x+1)^2 = \frac{x^6}{6} + \frac{2}{5}x^5 + \frac{x^4}{4} + c$

or, $y^3(x+1)^2 = \frac{1}{6}x^6 + \frac{2}{5}x^5 + \frac{1}{4}x^4 + c$ **Ans.**

Ex.7 $\frac{dy}{dx} + \frac{1}{x}\tan y = \frac{1}{x^2}\frac{\sin^2 y}{\cos y}$.

Sol. In above equation, on multiply by $\cot y$ $\operatorname{cosec} y$, we get

$$\cot y \operatorname{cosec} y \frac{dy}{dx} + \frac{1}{x}\operatorname{cosec} y = \frac{1}{x^2} \qquad \text{...(1)}$$

Put $\operatorname{cosec} y = V$ then $-\operatorname{cosec} y \cot y \frac{dy}{dx} = \frac{dV}{dx}$

$\therefore$ (1) $\Rightarrow -\frac{dV}{dx} + \frac{V}{x} = \frac{1}{x^2}$

$\Rightarrow \frac{dV}{dx} - \frac{V}{x} = -\frac{1}{x^2}$

This is linear in V

$$\text{I.F.} = e^{\int -\frac{1}{x}dx} = e^{\log\left(\frac{1}{x}\right)} = \frac{1}{x}$$

Hence solution in

$$V \cdot \frac{1}{x} = \int -\frac{1}{x} \times \frac{1}{x^2} dx + C$$

$$\frac{V}{x} = \frac{1}{2x^2} + C$$

$\Rightarrow \frac{\operatorname{cosec} y}{x} = \frac{1}{2x^2} + C$

or, $\frac{1}{x \sin y} = \frac{1}{2x^2} + C$ Ans.

Ex.8 $\frac{dy}{dx} = \frac{1}{xy(x^2y^2+1)}$. [Raj., 2001]

Sol. Rewrite it as $\frac{dx}{dy} = x^3y^3 + xy$

or, $\frac{1}{x^3}\frac{dx}{dy} - \frac{y}{x^2} = y^3$

$$-\frac{1}{x^2} = V \Rightarrow \frac{1}{x^3}\frac{dx}{dy} = \frac{1}{2}\frac{dV}{dy}$$

$\therefore \frac{1}{2}\frac{dV}{dy} + yV = y^3 \Rightarrow \frac{dV}{dy} + 2yV = 2y^3$

This is linear in V, so that

$$\text{I.F.} = e^{\int 2y\,dy} = e^{y^2}$$

$$V \cdot e^{y^2} = \int e^{y^2} \cdot 2y^3 dy + C$$

Put $y^2 = t \Rightarrow 2y\,dy = dt$

$$V\cdot e^{y^2} = \int e^t t\,dt + C = e^t(t-1) + C$$

Replace value of V and t

$$\Rightarrow \quad \frac{-e^{y^2}}{x^2} = y^2 e^{y^2} - e^{y^2} + C$$

$$\Rightarrow \quad \frac{-1}{x^2} = (y^2-1) + Ce^{-y^2} \quad \textbf{Ans.}$$

Ex.9 $\cos x\frac{dy}{dx} = y(\sin x - y)$.

Sol. Rewrite it as $\cos x\frac{dy}{dx} - y\sin x = -y^2$

divide both sides by $-y^2\cos x$, we get :

$$-\frac{1}{y^2}\frac{dy}{dx} + \frac{\tan x}{y} = \sec x$$

Put $\frac{1}{y} = V$ then $-\frac{1}{y^2}\frac{dy}{dx} = \frac{dV}{dx}$

$$\Rightarrow \quad \frac{dV}{dx} + (\tan x)V = \sec x$$

This is linear in V, so that

$$\Rightarrow \quad \text{I.F.} = e^{\int \tan x\,dx} = e^{\log\sec x} = \sec x$$

$$\Rightarrow \quad \sec x\cdot V = \int \sec x\cdot\sec x\,dx + C = \int \sec^2 x\,dx + C$$

$$\Rightarrow \quad \frac{V}{\cos x} = \tan x + C$$

$$\Rightarrow \quad V = \sin x + C\cos x$$

$$\Rightarrow \quad \frac{1}{y} = (\sin x + C\cos x) \quad \textbf{Ans.}$$

Ex.10 $y(2xy + e^x)dx - e^x dy = 0$. **[Raj., 2000]**

Sol. We can write it as

$$2xy^2 + e^x y = e^x\frac{dy}{dx}$$

or, $\frac{dy}{dx} - y = 2y^2 x e^{-x}$

or, $\frac{1}{y^2}\frac{dy}{dx} - \frac{1}{y} = 2xe^{-x}$ ...(1)

Put $\frac{-1}{y} = V$

$$\Rightarrow \quad \frac{1}{y^2}\frac{dy}{dx} = \frac{dV}{dx}$$

∴ (1) reduces to

$$\frac{dV}{dx} + V = 2xe^{-x}$$

This is linear in V, so that

$$\text{I.F.} = e^{\int 1\,dx} = e^x$$

Hence solution is

$$Ve^x = \int e^x\cdot 2xe^{-x}dx - C$$

$$\Rightarrow \quad Ve^x = x^2 - C$$

Put value of $V = -\frac{1}{y}$

$$\therefore \quad -\frac{e^x}{y} = x^2 - C$$

$$e^x = (C - x^2)y \quad \textbf{Ans.}$$

Ex-29

Exercise 11(F)

Show that the following equations are exact, and solve them :

Ex.1 $\left(1+4xy+2y^2\right)dx+\left(1+4xy+2x^2\right)dy=0$.

Sol Here $M = 1+4xy+2y^2$; $N = 1+4xy+2x^2$

Now, $\dfrac{\partial M}{\partial y}=4x+4y$; $\dfrac{\partial N}{\partial x}=4y+4x$

$\Rightarrow \dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$, $\therefore$ given equation is exact

Now, $U=\int M\,dx=\int(1+4xy+2y^2)dx=x+2x^2y+2xy^2$

$\Rightarrow \dfrac{\partial U}{\partial y}=2x^2+4xy$

$\therefore V=\int\left(N-\dfrac{\partial U}{\partial y}\right)dy=\int(1)dy=y$

$\therefore$ Solution is $\rightarrow U+V=C$

$\Rightarrow x+2x^2y+2xy^2+y=C$. Ans.

Ex.2 $\left(x^4-2xy^2+y^4\right)dx-\left(2x^2y-4xy^3+\sin y\right)dy=0$.

Sol. Here, $M=x^4-2xy^2+y^4$; $N=-2x^2y+4xy^3-\sin y$

$\therefore \dfrac{\partial M}{\partial y}=-4xy+4y^3$; $\dfrac{\partial N}{\partial x}=-4xy+4y^3$

$\therefore \dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}\Rightarrow$ given equation is exact

Now, $U=\int M\,dx=\int\left(x^4-2xy^2+y^4\right)dx=\dfrac{x^5}{5}-x^2y^2+xy^4$

$\Rightarrow \dfrac{\partial U}{\partial y}=-2x^2y+4xy^3$

$\therefore V=\int\left(N-\dfrac{\partial U}{\partial y}\right)dy=\int(-\sin y)dy=\cos y$

$\therefore$ Solution is $\rightarrow U+V=C$

$\Rightarrow \dfrac{x^5}{5}-x^2y^2+xy^4+\cos y=C$ Ans.

Ex.3 $x\left(x^2+3y^2\right)dx+y\left(y^2+3x^2\right)dy=0$.

Sol. Here, $M=x(x^2+3y^2)$; $N=y(y^2+3x^2)$

$\therefore \dfrac{\partial M}{\partial y}=6xy$; $\dfrac{\partial N}{\partial x}=6xy$

$\therefore \dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}\Rightarrow$ given equation is exact

Now, $U=\int M\,dx=\int(x^3+3xy^2)dx=\dfrac{x^4}{4}+\dfrac{3x^2y^2}{2}$

$\Rightarrow \dfrac{\partial U}{\partial y}=3x^2y$

$\therefore V=\int\left(N-\dfrac{\partial U}{\partial y}\right)dy=\int y^3dy=\dfrac{y^4}{4}$

$\therefore$ Soln. is $\rightarrow U+V=C$

$\Rightarrow \dfrac{x^4}{4}+\dfrac{y^4}{4}+\dfrac{3x^2}{2}y^2=C$

or, $x^4+y^4+6x^2y^2=C'$ Ans.

Ex.4 $(2ax+by)\,y\,dx+(ax+2by)\,x\,dy=0$.

Sol. Here, $M=(2ax+by)\,y$; $N=(ax+2by)x$

$\therefore \dfrac{\partial M}{\partial y}=2ax+2by$; $\dfrac{\partial N}{\partial x}=2ax+2by$

$\Rightarrow \dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$

$\therefore$ given eqn is exact

Now $U=\int M dx$, (treating y as constant)

$$= \int (2ax + by)y\,dx$$

$$\Rightarrow \quad U = ayx^2 + by^2x \qquad \text{....(1)}$$

$$\therefore \quad \frac{\partial U}{\partial y} = ax^2 + 2byx$$

$$\therefore \quad V = \int \left(N - \frac{\partial U}{\partial y}\right) dy = \int (0)\,dy = c' \qquad \text{....(2)}$$

$\therefore$ Solution is

$U + V = c''$

$\Rightarrow \quad a\,x^2\,y + b\,x\,y^2 = c$ **Ans.**

Ex.5 $(x^2 - ay)\,dx = (ax - y^2)\,dy.$

Sol. It can be written as

$(x^2 - ay)\,dx + (y^2 - ax)\,dy = 0$

Here, $M = x^2 - ay$; $N = y^2 - ax$

$$\therefore \quad \frac{\partial M}{\partial y} = -a\,; \quad \frac{\partial N}{\partial x} = -a$$

$$\Rightarrow \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ given eq^n is exact

$$\therefore \quad U = \int M dx = \int (x^2 - ay)dx = \frac{x^3}{3} - axy \qquad \text{....(1)}$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = -ax$$

$$\therefore \quad V = \int \left(N - \frac{\partial U}{\partial y}\right) dy = \int y^2\,dy = \frac{y^3}{3} \qquad \text{....(2)}$$

$\therefore$ solution is

$U + V = C$

$$\text{or,} \quad \left(\frac{x^3}{3} - axy\right) + \frac{y^3}{3} = c \qquad \textbf{Ans.}$$

Ex.6 $\cos x(\cos x - \sin\alpha \sin y)\,dx + \cos y(\cos y - \sin\alpha \sin x)\,dy = 0.$

Sol. Here, $M = \cos x(\cos x - \sin\alpha \sin y)$; $N = \cos y(\cos y - \sin\alpha \sin x)$

$$\frac{\partial M}{\partial y} = -\sin\alpha \cos x \cos y\,; \quad \frac{\partial N}{\partial x} = -\sin\alpha \cos x \cos y$$

$$\therefore \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{given eqn. is exact}$$

$$\text{Now, } U = \int M\,dx = \int (\cos^2 x - \sin\alpha \sin y \cos x)\,dx$$

$$= \int \left(\frac{1 + \cos 2x}{2} - \sin\alpha \sin y \cos x\right) dx$$

$$\Rightarrow \quad U = \frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right) - \sin\alpha \sin y \sin x$$

$$\therefore \quad \frac{\partial U}{\partial y} = -\sin\alpha \cos y \sin x$$

$$\therefore \quad V = \int \left(N - \frac{\partial U}{\partial y}\right) dy = \int \cos^2 y\,dy = \frac{1}{2}\left(y + \frac{1}{2}\sin 2y\right)$$

$\therefore$ Solution is $\rightarrow U + V = C$

$\Rightarrow \quad 2(x + y) + \sin 2x + \sin 2y - \sin\alpha \sin x \sin y = C$ **Ans.**

Ex.7 $(e^y + 1)\cos x\,dx + e^y \sin x\,dy = 0.$

Sol. Here $M = (e^y + 1)\cos x$; $N = e^y \sin x$

$$\therefore \quad \frac{\partial M}{\partial y} = e^y \cos x\,; \quad \frac{\partial N}{\partial x} = e^y \cos x$$

$$\Rightarrow \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{given eqn. is exact.}$$

$$\text{Now, } U = \int M\,dx = \int (e^y + 1)\cos x\,dx = (e^y + 1)\int \cos x\,dx = (e^y + 1)\sin x$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = e^y \sin x$$

$$\therefore \quad V = \int \left(N - \frac{\partial U}{\partial y}\right) dy = \int (0)\,dy = C$$

$\therefore$ Solution is $\rightarrow U + V = C$

$\Rightarrow \quad (e^y + 1)\sin x = C$ **Ans.**

Ex.8 $(x^2+y^2)dx+2xy\,dy=0$.

Sol. Here $M=x^2+y^2$; $N=2xy$

$\Rightarrow \quad \frac{\partial M}{\partial y}=2y$; $\frac{\partial N}{\partial x}=2y$

$\therefore \quad \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x} \Rightarrow$ given eqn. is exact

Now $U=\int M\,dx$ (keeping y as constant)

$$=\int (x^2+y^2)\,dx=\frac{x^3}{3}+xy^2$$

$\Rightarrow \quad \frac{\partial U}{\partial y}=2xy$

$\therefore \quad V=\int\left(N-\frac{\partial u}{\partial y}\right)dx=\int (0)\,dy=C'$

$\therefore$ solution is $\rightarrow U+V=C$

$\Rightarrow \quad \frac{x^3}{3}+xy^2=C'$ **Ans.**

Ex.9 $\left(1+e^{\frac{x}{y}}\right)dx+e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)dy=0$.

Sol. Here $M=1+e^{\frac{x}{y}}$; $N=e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)$

$$\frac{\partial M}{\partial y}=-\frac{x}{y^2}e^{\frac{x}{y}}\ ;\ \frac{\partial N}{\partial x}=-\frac{1}{y}e^{\frac{x}{y}}+\left(1-\frac{x}{y}\right)\left(\frac{1}{y}\right)e^{\frac{x}{y}}=-\frac{x}{y^2}e^{\frac{x}{y}}$$

$\therefore \quad \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x} \Rightarrow$ given eqn. is exact

Now, $U=\int M\,dx=\int\left(1+e^{\frac{x}{y}}\right)dx=x+ye^{\frac{x}{y}}$

$\Rightarrow \quad \frac{\partial U}{\partial y}=e^{\frac{x}{y}}+ye^{\frac{x}{y}}\left(\frac{-x}{y^2}\right)=e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)$

$\therefore \quad V=\int\left(N-\frac{\partial U}{\partial y}\right)dy=\int (0)\,dy=C'$

$\therefore$ Solution is $\rightarrow U+V=C$

$\Rightarrow \quad x+ye^{\frac{x}{y}}=C$ **Ans.**

Ex.10 $\left[y\left\{1+\frac{1}{x}\right\}+\cos y\right]dx+[x+\log x-x\sin y]dy=0$

Sol. Here, $M=y\left(1+\frac{1}{x}\right)+\cos y$; $N=x+\log x-x\sin y$

$\therefore \quad \frac{\partial M}{\partial y}=1+\frac{1}{x}-\sin y$; $\frac{\partial N}{\partial x}=1+\frac{1}{x}-\sin y$

$\Rightarrow \quad \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

$\therefore$ given equation is exact.

Now, $U=\int M\,dx$

$$=\int\left[y\left(1+\frac{1}{x}\right)+\cos y\right]dx=y(x+\log x)+x\cos y$$

$\therefore \quad \frac{\partial U}{\partial y}=x+\log x-x\sin y$

$\Rightarrow \quad V=\int\left(N-\frac{\partial U}{\partial y}\right)dy=\int (0)\,dy=c_1$

$\therefore$ solution is

$U+V=c_2$

or, $y(x+\log x)+x\cos y=c$; $c=c_2-c_1$ **Ans.**

Ex-30

Exercise - 11

Find the integrating factor of each of the following differential equations and solved them :

[MREC (Auto), 2000]

Ex.1 $\left(x^3e^x-my^2\right)dx+mxy\,dy=0$.

Sol. Let $M = \left(x^3e^x - my^2\right)$ and $N = mxy$

$$\frac{\partial M}{\partial y} = -2my \text{ and } \frac{\partial N}{\partial x} = my$$

So, $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{mxy}(-2my - my)$

$= \frac{-3}{x}$ this is function of x alone

$\therefore$ I.F. $= e^{\int \frac{-3}{x}dx} = e^{-3\log x} = \frac{1}{x^3}$

Multiplying given equation by $\left(\frac{1}{x^3}\right)$, we get

$$\left(e^x - \frac{my^2}{x^3}\right)dx + \frac{my}{x^2}dy = 0$$

This is exact here $M' = e^x - \frac{my^2}{x^3}$ and $N' = \frac{my}{x^2}$

Integration M' w.r.t. x, keeping y as constant

$$U = \int M'dx = \int\left(e^x - \frac{my^2}{x^3}\right)dx$$

$\therefore \quad U = e^x + \frac{my^2}{2x^2}$

$\Rightarrow \quad \frac{\partial U}{\partial y} = \frac{my}{x^2}$

$\therefore \quad V = \int\left(N' - \frac{\partial U}{\partial y}\right)dy = \int (0)dy = C$

Hence solution is $\Rightarrow U + V = C$

$\Rightarrow \quad e^x + \frac{my^2}{2x^2} = C$ **Ans.**

Ex.2 $y^2 = (xy - x^2)\left(\frac{dy}{dx}\right)$.

Sol. Rewrite as $\frac{dy}{dx} = \frac{y^2}{(xy - x^2)}$

Let $y = vx$ them $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dv}{dx} = \frac{v^2x^2}{vx^2 - x^2}$

$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{v^2}{v-1}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{v^2 - v^2 + v}{v-1}$

$\Rightarrow \quad (v-1)\frac{dv}{v} = \frac{dx}{x}$

on integration :

$$v - \log v = \log x + \log C$$

$\Rightarrow \quad \left(\frac{y}{x}\right) = \log\left(\frac{y}{x}\times x\right) + \log C$

or, $\quad y = x\log(yC)$ **Ans.**

Ex.3 $\left(3x + 2y^2\right)y\,dx + 2x\left(2x + 3y^2\right)dy = 0$

Sol. Rewrite as $x(3y\,dx + 4x\,dy) + y^2(2y\,dx + 6x\,dy) = 0$

Let I.F. $= x^hy^k$

Multiply given differential equation by x^hy^k, we get :

$$\left(3x^{h+1}y^{k+1} + 2y^{k+3}x^h\right)dx + \left(4x^{h+2}y^k + 6x^{h+1}y^{k+2}\right)dy = 0$$

since above equation is exact, then $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Here, $M = 3x^{h+1}y^{k+1} + 2y^{k+3}x^h$

and $\quad N = 4x^{h+2}y^k + 6x^{h+1}y^{k+2}$

$$\frac{\partial M}{\partial y} = 3(k+1)x^{h+1}y^k + 2(k+3)y^{k+2}x^h$$

$$\frac{\partial N}{\partial x} = 4(h+2)x^{h+1}y^k + 6(h+1)x^h y^{k+2}$$

then $3(k+1)x^{h+1}y^k + 2(k+3)y^{k+2}x^h$

$$= 4(h+2)x^{h+1}y^k + 6(h+1)x^h y^{k+2}$$

$\Rightarrow \quad 3(k+1)x + 2(k+3)y^2 = 4(h+2)x + 6(h+1)y^2$

$\Rightarrow \quad 3(k+1) = 4(h+2)$ and $2(k+3) = 6(h+1)$

$\Rightarrow \quad 3k - 4h = 5$ and $k - 3h = 0$

$\therefore \quad h = 1$ and $k = 3$ then I.F. $= xy^3$

Multiply given differential equation by xy^3, we get :

$$(3x^2y^4 + 2xy^6)\,dx + (4x^3y^3 + 6x^2y^5)\,dy = 0$$

$$M = 3x^2y^4 + 2xy^6 \text{ and } N = 4x^3y^3 + 6x^2y^5$$

$\Rightarrow \quad \dfrac{\partial M}{\partial y} = 12x^2y^3 + 12xy^5$ and $\dfrac{\partial N}{\partial x} = 12x^2y^3 + 12xy^5$

$\therefore \quad \dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} \Rightarrow$ given equation is exact

New integrate M w.r.t. x, keeping y as constant

$$U = \int M\,dx = \int (3x^2y^4 + 2xy^6)\,dx$$

$\Rightarrow \quad U = x^3y^4 + x^2y^6$

$\therefore \quad \dfrac{\partial U}{\partial y} = 4x^3y^3 + 6x^2y^5$

$\therefore \quad V = \displaystyle\int \left(N - \frac{\partial U}{\partial y}\right) dy = \int (0)\,dy = C$

Hence solution is $\rightarrow U + V = C$

$\Rightarrow \quad x^3y^4 + x^2y^6 = C$

or, $\quad x^2y^4(x + y^2) = C$ Ans.

Ex.4 $y\left(a\,xy + e^x\right)dx - e^x\,dy = 0$.

Sol. By inspection we find $\dfrac{1}{y^2}$ is I.F.

Multiply given differential equation by $\dfrac{1}{y^2}$

$$\left(ax + \frac{e^x}{y}\right)dx - \frac{e^x}{y^2}\,dy = 0 \quad \text{...(1)}$$

Now equation (1) become exact.

$$\left(ax + \frac{e^x}{y}\right)dx - \frac{e^x}{y^2}\,dy = 0$$

Here, $M = ax + \dfrac{e^x}{y}$ and $N = -\dfrac{e^x}{y^2}$

Integrate M w.r.t. x, keeping y as constant

$$U = \int M\,dx = \int \left(ax + \frac{e^x}{y}\right)dx$$

$\therefore \quad U = \dfrac{ax^2}{2} + \dfrac{e^x}{y}$

$\Rightarrow \quad \dfrac{\partial U}{\partial y} = \dfrac{-e^x}{y^2}$

$\therefore \quad V = \displaystyle\int \left(N_1 - \frac{\partial U}{\partial y}\right) dy = \int (0)\,dy = C$

Hence solution is $\rightarrow U + V = C$

$\Rightarrow \quad \dfrac{ax^2}{2} + \dfrac{e^x}{y} = C$

or $\quad ax^2y + 2e^x - 2C\,y = 0$

or $\quad ax^2y + 2e^x - C'\,y = 0$ Ans.

Ex.5 $\left(3x^2y^4 + 2xy\right)dx + \left(2x^3y^3 - x^2\right)dy = 0$.

Sol. Here $M = 3x^2y^4 + 2xy$; $N = 2x^3y^3 - x^2$

$$\therefore \quad \frac{\partial N}{\partial x} = 6x^2y^3 - 2x \text{ and } \frac{\partial M}{\partial y} = 12x^2y^3 + 2x$$

So that $$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \frac{-1}{(3x^2y^4 + 2xy)}\left[(6x^2y^3 + 4x)\right]$$

$$= -\frac{2}{y}$$

So $$\text{I. F.} = e^{\int -\frac{2}{y}dy} = \frac{1}{y^2}$$

Multiply above equation by $\frac{1}{y^2}$, we get

$$\left(3x^2y^2 + 2\frac{x}{y}\right)dx + \left(2x^3y - \frac{x^2}{y^2}\right)dy = 0$$

Let $M_1 = 3x^2y^2 + \frac{2x}{y}$ and $N_1 = 2x^3y - \frac{x^2}{y^2}$

Integrating M_1 *w.r.t.* x, keeping y as constant

$$U = \int M_1 dx = \int\left(3x^2y^2 + \frac{2x}{y}\right)dx.$$

$$\Rightarrow \quad U = x^3y^2 + \frac{x^2}{y}$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = 2x^3y - \frac{x^2}{y^2}$$

$$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int(0)dy = C$$

Hence solution is $U + V = C$

$$\Rightarrow \quad x^3y^2 + \frac{x^2}{y} = C$$

or $\quad x^3y^3 + x^2 = Cy$ **Ans.**

Ex.6 $y(xy + 2x^2y^2)\,dx + x\,(xy - x^2y^2)\,dy = 0$

Sol. Given equation is of the form –

$$f_1(xy)y\,dx + f_2(xy)x\,dy = 0$$

$$\therefore \quad \text{I. F.} = \frac{1}{Mx - Ny};\ Mx - Ny \neq 0$$

Here, $M = (xy + 2x^2y^2)y$; $N = (xy - x^2y^2)x$

$$\therefore \quad \text{I. F.} = \frac{1}{Mx - Ny} = \frac{1}{3x^3y^3}$$

Multiplying the given eqn by $\frac{1}{3x^3y^3}$, we get

$$= \frac{1}{3}\left(\frac{1}{x^2y^2} + \frac{2}{xy}\right)y\,dx + \frac{1}{3}\left(\frac{1}{x^2y^2} - \frac{1}{xy}\right)x\,dy = 0$$

or, $$\left(\frac{1}{x^2y} + \frac{2}{x}\right)dx + \left(\frac{1}{xy^2} - \frac{1}{y}\right)dy = 0$$

which is exact

$$\therefore \quad U = \int M'\,dx$$

$$= \int\left(\frac{1}{x^2y} + \frac{2}{x}\right)dx = -\frac{1}{xy} + 2\log x$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = \frac{1}{xy^2}$$

$$\therefore \quad V = \int\left(N' - \frac{\partial U}{\partial y}\right)dy = \int -\frac{1}{y}dy = -\log y$$

$\therefore$ solution is
$U + V = c$

or, $$-\frac{1}{xy} + 2\log x - \log y = c$$

or, $$\log\left(\frac{x^2}{y}\right) = \frac{1}{xy} + c \quad \textbf{Ans.}$$

Ex.7 $(2ydx+3xdy)+2xy(3ydx+4xdy)=0$.

Sol. Let I. F. $= x^h y^k$

Multiplying the given equation by $x^h y^k$,we get

$$\left(2x^h y^{k+1}+6x^{h+1}y^{k+2}\right)dx+(3x^{h+1}y^k+8x^{h+2}y^{k+1})dy=0 \qquad \text{—(1)}$$

Since, the above equation is exact,

$\therefore \quad \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

$\Rightarrow \quad 2(k+1)x^h y^k+6(k+2)x^{h+1}y^{k+1}$

$$=3(h+1)x^h y^k+8(h+2)x^{h+1}y^{k+1}$$

on equating coefficeints of like powers, we get

$2\,(k+1)=3\,(h+1)$

and, $6\,(k+2)=8\,(h+2)$

$\Rightarrow \quad h=1,\ k=2$

$\therefore \quad$ I. F. $= xy^2$

$\therefore \quad$ (1) gives –

$$(2xy^3+6x^2y^4)dx+(3x^2y^2+8x^3y^3)dy=0\ .$$

which is exact

$\therefore \quad U=\int Mdx=\int(2xy^3+6x^2y^4)dx=x^2y^3+2x^3y^4$

$\Rightarrow \quad \frac{\partial U}{\partial y}=3x^2y^2+8x^3y^3$

$\therefore \quad V=\int\left(N-\frac{\partial U}{\partial y}\right)dy=\int(0)dy=c_1$

$\therefore \quad$ solution is

$U+V=c_2$

or, $\quad x^2y^3+2x^3y^4=c \qquad$ **Ans.**

Ex.8 $2xy^2dx=e^x(dy-y\,dx)$.

Sol. Rewrite as $(2xy^2+e^x y)dx-e^x dy=0$

By inspection y^{-2} is I.F., so

Multiply above equation by $\frac{1}{y^2}$, we get

$$\left(2x+\frac{e^x}{y}\right)dx-\frac{e^x}{y^2}dy=0$$

Let, $\quad M=2x+\frac{e^x}{y}$ and $N=\frac{-e^x}{y^2}$

$\frac{\partial M}{\partial y}=-\frac{e^x}{y^2}$ and $\frac{\partial N}{\partial x}=-\frac{e^x}{y^2}$

$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$, here differential equation is exact.

Integration M w.r.t. x, keeping y as constant

$U=\int M\,dx=\int\left(2x+\frac{e^x}{y}\right)dx$

$=x^2+\frac{e^x}{y}$

$\Rightarrow \quad \frac{\partial U}{\partial y}=-\frac{1}{y^2}e^x$

$\therefore \quad V=\int\left(N-\frac{\partial U}{\partial y}\right)dy=\int(0)dy=C$

Hence solution is $\rightarrow U+V=C$

$\Rightarrow \quad x^2+\frac{e^x}{y}=C$

or, $\quad e^x+(x^2-C)y=0 \qquad$ **Ans.**

Ex.9 $\left(x^4y^4+x^2y^2+xy\right)y\,dx+\left(x^4y^4-x^2y^2+xy\right)x\,dy=0$.

Sol. Given equation is in form of

$$f_1(xy)y\,dx+f_2(xy)x\,dy=0$$

Hence I.F. $=\frac{1}{Mx-Ny}$ where $Mx-Ny\neq0$

Now, $M=\left(x^4y^4+x^2y^2+xy\right)y$ and $N=\left(x^4y^4-x^2y^2+xy\right)x$

$$\therefore \quad Mx - Ny = x^5y^5 + x^3y^3 + x^2y^2 - x^5y^5 + x^3y^3 - x^2y^2$$

$$= 2x^3y^3$$

So that I.F. $= \dfrac{1}{2x^3y^3}$

Multiply above equation by $\dfrac{1}{2x^3y^3}$, we get

$$\frac{1}{2x^3y^3}[(x^4y^4 + x^2y^2 + xy)y\,dx]$$

$$+\frac{1}{2x^3y^3}\left[\left(x^4y^4 - x^2y^2 + xy\right)x\,dy\right] = 0$$

or, $\left[xy^2 + \dfrac{1}{x} + \dfrac{1}{x^2y}\right]dx + \left[x^2y - \dfrac{1}{y} + \dfrac{1}{xy^2}\right]dy = 0$

$$M_1 = xy^2 + \frac{1}{x} + \frac{1}{x^2y} \text{ and } N_1 = x^2y - \frac{1}{y} + \frac{1}{xy^2}$$

Integrating M_1 w.r.t x, keeping y as constant

$$U = \int M_1 dx = \int\left(xy^2 + \frac{1}{x} + \frac{1}{x^2y}\right)dx$$

$$\therefore \quad U = \frac{x^2y^2}{2} + \log x - \frac{1}{xy}$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = x^2y + \frac{1}{xy^2}$$

$$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int -\frac{1}{y}dy = -\log y$$

Hence solution is $U + V = C$

or, $\dfrac{x^2y^2}{2} + \log\left(\dfrac{x}{y}\right) - \dfrac{1}{xy} = C$ **Ans.**

Ex.10 $\left(y^2 + 2x^2y\right)dx + \left(2x^3 - xy\right)dy = 0$.

Sol. Let I.F. $= x^hy^k$

Multiply given differential equation by x^hy^k, we get

$$\left(x^hy^{k+2} + 2x^{h+2}y^{k+1}\right)dx + \left(2x^{h+3}y^k - x^{h+1}y^{k+1}\right)dy = 0$$

For exact equation $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$

Here, $M = x^hy^{k+2} + 2x^{h+2}y^{k+1}$

and $N = 2x^{h+3}y^k - x^{h+1}y^{k+1}$

$$\therefore \quad \frac{\partial M}{\partial y} = (k+2)x^hy^{k+1} + 2(k+1)x^{h+2}y^k$$

$$\frac{\partial N}{\partial x} = 2(h+3)x^{h+2}y^k - (h+1)x^hy^{k+1}$$

Now, $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$

$$\Rightarrow (k+2)x^hy^{k+1} + 2(k+1)x^{h+2}y^k = -(h+1)x^hy^{k+1} + 2(h+3)x^{h+2}y^k$$

Comparing the cofficient, we have

$$k + 2 = -h - 1 \text{ and } 2(k+1) = 2(h+3)$$

we get $\quad h = -\dfrac{5}{2}$ and $k = -\dfrac{1}{2}$

Put values of h and k in equation (1), we get

$$\left(x^{-\frac{5}{2}}y^{\frac{3}{2}} + 2x^{-\frac{1}{2}}y^{\frac{1}{2}}\right)dx + \left(2x^{\frac{1}{2}}y^{-\frac{1}{2}} - x^{-\frac{3}{2}}y^{\frac{1}{2}}\right)dy = 0 \qquad \text{...(2)}$$

or, $\quad M_1 = x^{-\frac{5}{2}}y^{\frac{3}{2}} + 2x^{-\frac{1}{2}}y^{\frac{1}{2}}$ and $N_1 = 2x^{\frac{1}{2}}y^{-\frac{1}{2}} - x^{-\frac{3}{2}}y^{\frac{1}{2}}$

Now, $\dfrac{\partial M_1}{\partial y} = \dfrac{3}{2}x^{-\frac{5}{2}}y^{\frac{1}{2}} + x^{-\frac{1}{2}}y^{-\frac{1}{2}}$ and $\dfrac{\partial N_1}{\partial x} = \dfrac{3}{2}x^{-\frac{5}{2}}y^{\frac{1}{2}} + x^{-\frac{1}{2}}y^{-\frac{1}{2}}$

$\Rightarrow \quad \dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x}$, hence equation (2) is exact

Integration M_1 w.r.t. x keeping y as constant

$$U=\int M_1\,dx=\int\left(x^{-\frac{5}{2}}y^{\frac{3}{2}}+2x^{-\frac{1}{2}}y^{\frac{1}{2}}\right)dx$$

$$U=-\frac{2}{3}x^{-\frac{3}{2}}y^{\frac{3}{2}}+4x^{\frac{1}{2}}y^{\frac{1}{2}}$$

$$\frac{\partial U}{\partial y}=-x^{-\frac{3}{2}}y^{\frac{1}{2}}+2x^{\frac{1}{2}}y^{-\frac{1}{2}}$$

$$\therefore\quad V=\int\left(N_1-\frac{\partial U}{\partial y}\right)dy=\int(0)\,dy=C'$$

Hence solution is $\rightarrow U+V=C$

$$\Rightarrow\quad 4x^{\frac{1}{2}}y^{\frac{1}{2}}-\frac{2}{3}x^{-\frac{3}{2}}y^{\frac{3}{2}}=C \quad \textbf{Ans.}$$

Ex.11 $\left(x^2+y^2+2x\right)dx+2y\,dy=0$.

Sol. Let $M=x^2+y^2+2x$ and $N=2y$

$$\frac{\partial M}{\partial y}=2y \text{ and } \frac{\partial N}{\partial x}=0$$

Here $\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)$

$$=\frac{1}{2y}(2y-0)=1, \text{ so that I.F.}=e^{\int 1\,dx}=e^x$$

Multiplying given diff. equation by I.F., we get

$$e^x(x^2+y^2+2x)dx+e^x 2y\,dy=0 \qquad \ldots(1)$$

Let $M_1=(x^2e^x+y^2e^x+2xe^x)$ and $N_1=e^x 2y$

$$\therefore\quad \frac{\partial M_1}{\partial y}=2ye^x \text{ and } \frac{\partial N_1}{\partial x}=2ye^x$$

$$\Rightarrow\quad \frac{\partial M_1}{\partial y}=\frac{\partial N_1}{\partial x}, \text{ hence equation (1) is exact}$$

so that,

Integrate M_1, w.r.t. x, keeping y as constant

$$U=\int M_1\,dx=\int(x^2e^x+y^2e^x+2xe^x)\,dx$$

$$=x^2e^x-\int 2x\cdot e^x dx+y^2e^x+\int 2xe^x dx$$

$$\therefore\quad U=(x^2+y^2)e^x$$

$$\Rightarrow\quad \frac{\partial U}{\partial y}=2ye^x$$

$$\therefore\quad V=\int\left(N_1-\frac{\partial U}{\partial y}\right)dy=\int(0)\,dy=C'$$

Hence solution is $\rightarrow U+V=C$

$$\Rightarrow\quad (x^2+y^2)e^x=C \quad \textbf{Ans.}$$

Ex.12 Solve $x^2y\,dx-(x^3+y^3)\,dy=0$.

Sol. Here $M=x^2y$ and $N=-(x^3+y^3)$

$$\Rightarrow\quad \frac{\partial N}{\partial x}=-3x^2, \frac{\partial M}{\partial y}=x^2$$

$$\therefore\quad \frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)$$

$$=\frac{1}{x^2y}(-x^2-3x^2)=\frac{-4}{y}$$

$$\text{I.F.}=e^{\int-\frac{4}{y}dy}=\frac{1}{y^4}$$

Multiply given equation by $\frac{1}{y^4}$ then, we get

$$\frac{x^2}{y^3}dx-\left(\frac{x^3+y^3}{y^4}\right)dy=0$$

$$\therefore\quad M_1=\frac{x^2}{y^3} \text{ and } N_1=-\left(\frac{x^3+y^3}{y^4}\right)$$

$$\frac{\partial M_1}{\partial y} = -\frac{3x^2}{y^4};\ \frac{\partial N_1}{\partial x} = -\frac{3x^2}{y^4}$$

$\therefore \quad \dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x} \Rightarrow$ eqn. is exact

Integration M_1 w.r.t. x, keeping y as constant

$$U = \int M_1 dx = \int \frac{x^2}{y^3} dx = \frac{x^3}{3y^3}$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = -\frac{x^3}{y^4}$$

$$\therefore \quad V = \int \left(N_1 - \frac{\partial U}{\partial y} \right) dy = \int -\frac{1}{y} dy$$

$\Rightarrow \quad V = -\log y$

Hence solution is $\rightarrow U + V = C$

$$\Rightarrow \quad \frac{x^3}{3y^3} - \log y = C$$

or, $\quad y = C' e^{\frac{x^3}{3y^3}}$ **Ans.**

Ex.13 $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$.

Sol. Let $M = xy^3 + y$ and $N = 2(x^2y^2 + x + y^4)$

$$\frac{\partial M}{\partial y} = 3xy^2 + 1 \text{ and } \frac{\partial N}{\partial x} = 4xy^2 + 2$$

Here $\dfrac{1}{M}\left(\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y} \right) = \dfrac{1}{(xy^3 + y)}\left[4xy^2 + 2 - 3xy^2 - 1\right] = \dfrac{1}{y}$

so that I.F. $= e^{\int \frac{1}{y} dy} = y$

Multiply given equation by y, we get

$$\left(xy^4 + y^2\right)dx + 2\left(x^2y^3 + xy + y^5\right)dy = 0 \qquad \ldots(1)$$

Let $M_1 = xy^4 + y^2$ and $N_1 = 2(x^2y^3 + xy + y^5)$

$$\Rightarrow \quad \frac{\partial M_1}{\partial y} = 4xy^3 + 2y \text{ and } \frac{\partial N_1}{\partial x} = 4xy^3 + 2y$$

$\dfrac{\partial M_1}{\partial y} = \dfrac{\partial N_1}{\partial x}$, hence equation (1) is exact

so that,

Integration M_1, w.r.t. x, keeping y as constant

$$U = \int M_1 dx = \int \left(xy^4 + y^2\right) dx$$

$$\therefore \quad U = \frac{x^2}{2} y^4 + xy^2$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = 2x^2y^3 + 2xy$$

$$\therefore \quad V = \int \left(N_1 - \frac{\partial U}{\partial y} \right) dy = \int 2y^5 dy = \frac{1}{3} y^6$$

Hence solution is $\rightarrow U + V = C$

$$\Rightarrow \quad \frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = C$$

or, $\quad 3x^2y^4 + 6xy^2 + 2y^6 = C^1$ **Ans.**

Ex.14 $(y - 2x^3)dx - x(1 - xy)dy = 0$.

Sol. Here $M = y - 2x^3$, $N = -x(1 - xy)$

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{1}{-x(1 - xy)}[1 + 1 - 2xy] = -\frac{2}{x}$$

$$\left(\because \frac{\partial M}{\partial y} = 1,\ \frac{\partial N}{\partial x} = -1 + 2xy \right)$$

$$\text{I.F.} = e^{\int -\frac{2}{x} dx} = e^{\log \frac{1}{x^2}} = \frac{1}{x^2}$$

Multiply given equation by $\dfrac{1}{x^2}$, we get

$$\left(\frac{y}{x^2} - 2x \right) dx - \frac{1}{x}(1 - xy) dy = 0$$

It becomes exact equation

Let, $M_1 = \frac{y}{x^2} - 2x$ and $N_1 = \frac{-1}{x}(1 - xy)$

Integrate M_1 w.r.t. x, keeping y as constant

$\therefore \quad U = \int M_1\,dx = \int\left(\frac{y}{x^2} - 2x\right)dx$

$U = -\frac{y}{x} - x^2$

$\Rightarrow \quad \frac{\partial U}{\partial y} = -\frac{1}{x}$

$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int y\,dy = \frac{y^2}{2}$

Hence solution is $\rightarrow U + V = C$

or, $-\frac{y}{x} - x^2 + \frac{y^2}{2} = C$

or, $y^2x = 2xC + 2x^3 + 2y$

or, $y^2x = xC' + 2x^3 + 2y$ **Ans.**

Ex.15 $(xy\sin xy + \cos xy)y\,dx + (xy\sin xy - \cos xy)\,x\,dy = 0$.

Sol. Given differential equation is in the form

$[f(xy)]y\,dx + [f(xy)]x\,dy = 0$

so that, if $Mx - Ny \neq 0$ then I.F. is $\frac{1}{Mx - Ny}$

Here $M = (xy\sin xy + \cos xy)y$ and $N = (xy\sin xy - \cos xy)x$

$$Mx - Ny = x^2y^2\sin xy + xy\cos xy - x^2y^2\sin xy + xy\cos xy$$
$$= 2xy\cos xy \neq 0$$

Hence, I.F. $= \frac{1}{2xy\cos xy}$

Multiply given diff. equation by I.F. we get

$$\frac{1}{2}\left[\tan xy + \frac{1}{xy}\right]y\,dx + \frac{1}{2}\left[\tan xy - \frac{1}{xy}\right]x\,dy = 0$$

$\Rightarrow \quad \tan xy\,[y\,dx + x\,dy] + \frac{1}{x}dx - \frac{1}{y}dy = 0$

Let $xy = z$ so that $x\,dy + y\,dx = dz$

$\Rightarrow \quad \tan z\,dz + \frac{dx}{x} - \frac{dy}{y} = 0$

On integration, we get

$\log\sec z + \log x - \log y = \log C$

$\Rightarrow \quad \log\frac{x\sec z}{y} = \log C$

or, $\frac{x}{y}\sec(xy) = C$

or, $Cy\cos(xy) = x$ **Ans.**

M 18d. Ex-31

Exercise - 11(H)

Solve :

Ex.1 $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$.

Sol. $y\frac{dy}{dx} - \frac{y^2}{2x} = \frac{x^2 + 1}{2x}$

Substituting $y^2 = v$

$2y\frac{dy}{dx} = \frac{dv}{dx}$

$\Rightarrow \quad y\frac{dy}{dx} = \frac{1}{2}\frac{dv}{dx}$

$\frac{1}{2}\frac{dv}{dx} - \frac{v}{2x} = \frac{x^2 + 1}{2x}$

$\frac{dv}{dx} - \frac{v}{x} = \frac{x^2 + 1}{x}$

This is linear in v so that

$$\text{I. F.} = e^{\int -\frac{1}{x}dx} = \frac{1}{x}$$

Hence solution is

$$v\frac{1}{x} = \int \frac{1}{x} \times \frac{x^2+1}{x}\,dx + C$$

$$\Rightarrow \quad \frac{v}{x} = \int \left(1+\frac{1}{x^2}\right)dx + C$$

$$\Rightarrow \quad \frac{v}{x} = x - \frac{1}{x} + C$$

$$\Rightarrow \quad \frac{y^2}{x} = \frac{x^2-1+Cx}{x}$$

$$\Rightarrow \quad y^2 = x^2 + Cx - 1$$

$$\Rightarrow \quad y^2 - x^2 + 1 = Cx \quad \text{Ans.}$$

Ex.2 $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$.

Sol. Rewrite as

$$e^y \frac{dy}{dx} + e^x e^y = e^{2x}$$

Put $e^y = v$ then $e^y \frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore \quad \frac{dv}{dx} + e^x v = e^{2x}$$

This is linear in v so that

$$\text{I. F.} = e^{\int e^x dx} = e^{e^x}$$

Hence soluation is

$$v \cdot e^{e^x} = \int e^{e^x} e^{2x}\,dx + C$$

Put $e^x = t \Rightarrow e^x\,dx = dt$

$$\therefore \quad e^y e^{e^x} = \int e^t\, t\,dt$$

$$\Rightarrow \quad e^y e^{e^x} = \left(t e^t - e^t\right) + C$$

$$\Rightarrow \quad e^y e^{e^x} = \left(e^x e^{e^x} - e^{e^x}\right) + C$$

$$\text{or,} \quad \left(e^y + 1\right) = e^x + Ce^{-e^x} \quad \text{Ans.}$$

Ex.3 $\frac{dy}{dx} + \frac{y}{x}\log y = \frac{y}{x^2}(\log y)^2$.

Sol. Rewrite as

$$\frac{1}{y(\log y)^2}\frac{dy}{dx} + \frac{1}{x \log y} = \frac{1}{x^2}$$

Put $\frac{1}{\log y} = v$ then $\frac{-1}{y(\log y)^2}\frac{dy}{dx} = \frac{dv}{dx}$

$$\therefore \quad -\frac{dv}{dx} + \frac{v}{x} = \frac{1}{x^2}$$

$$\text{or,} \quad \frac{dv}{dx} - \frac{v}{x} = -\frac{1}{x^2}$$

This is linear in v

$$\text{I. F.} = e^{\int -\frac{1}{x}dx} = \frac{1}{x}$$

$$\therefore \quad \frac{1}{x}v = \int \frac{-1}{x^2}\cdot\frac{1}{x}\,dx + C$$

$$\text{or,} \quad \frac{v}{x} = \frac{1}{2x^2} + C$$

$$\text{or,} \quad \frac{1}{x \log y} = \frac{1}{2x^2} + C \quad \text{Ans.}$$

Ex.4 $\frac{dy}{dx} = (4x+y+1)^2$

Sol. Put $4x + y + 1 = v$

$$\Rightarrow \quad 4 + \frac{dy}{dx} = \frac{dv}{dx}$$

$\therefore$ given equation reduces to

$$\frac{dv}{dx} - 4 = v^2$$

$$\Rightarrow \quad \frac{dv}{dx} = v^2 + 4$$

$$\Rightarrow \quad \frac{dv}{v^2+4} = dx$$

integrating, we get

$$\frac{1}{2}\tan^{-1}\left(\frac{v}{2}\right) = x + \frac{c}{2}$$

or, $v = 2\tan(2x + c)$

or, $(4x + y + 1) = 2\tan(2x + c)$ **Ans.**

Ex.5 **Solve** $\frac{dy}{dx} = \frac{1}{x+y+1}$.

Sol. Put $1 + x + y = v$

$$\Rightarrow \quad 1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \quad \frac{dv}{dx} = \frac{1}{v} + 1$$

Separating variables $\frac{vdv}{1+v} = dx$

Integration, we get

$$\int \frac{v}{1+v}\,dv = \int dx$$

or, $\int \frac{1+v-1}{1+v}\,dv = \int dx$

or, $\int dv - \int \frac{1}{1+v}\,dv = x + \log C$

or, $v - \log(1+v) = x + \log C$

or, $(x+y+1) - \log(2+x+y) = x + \log C$

or, $y = \log(2+x+y) + \log C'$

or, $y = \log\,(2+x+y)\,C'$ or $(2+x+y) = C\,e^{y}$ **Ans.**

Ex.6 $(x^3 + y^2 + 2)dx + 2y\,dy = 0$.

Sol. Rewrite as $2y\frac{dy}{dx} + y^2 = -(x^3 + 2)$

Let $y^2 = v$

$$2y\frac{dy}{dx} = \frac{dv}{dx}$$

Hence we get

$$\frac{dv}{dx} + v = -(x^3 + 2)$$

This is linear in v so that

I. F. $= e^{\int 1\,dx} = e^x$

Hence solution is

$$\Rightarrow \quad ve^x = \int -(x^3+2)e^x\,dx$$

$$\Rightarrow \quad ve^x = -2e^x - \int x^3 e^x\,dx$$

$$\Rightarrow \quad ve^x = -2e^x - \left[x^3e^x - 3x^2e^x + 6xe^x - 6e^x\right] + C$$

$$\Rightarrow \quad y^2e^x = -2e^x - x^3e^x + 3x^2e^x - 6xe^x + 6e^x + C$$

$$\Rightarrow \quad y^2 = 3x^2 - x^3 - 6x + 4 + Ce^{-x} \quad \textbf{Ans.}$$

Ex.7 **Solve** $\sec^2 y\frac{dy}{dx} + 2x\tan y = x^3$.

Sol. Rewrite as $\sec^2 y\frac{dy}{dx} + 2x\tan y = x^3$

Let $v = \tan y$

$$\Rightarrow \quad \frac{dv}{dx} = \sec^2 y\frac{dy}{dx}$$

$\therefore$ (1) reduces to

$$\frac{dv}{dx} + 2xv = x^3$$

This is linear in v

$\therefore$ I. F $= e^{\int 2x\,dx} = e^{x^2}$

Hence solution is

$$ve^{x^2} = \int e^{x^2} x^3 dx$$

Put $x^2 = t$

$\Rightarrow$ $2x\ dx = dt$

$$\Rightarrow \quad ve^{x^2} = \int te^t \frac{dt}{2}$$

$$\Rightarrow \quad (\tan y)e^{x^2} = \frac{1}{2}\left[te^t - e^t\right] + C$$

$$\Rightarrow \quad (\tan y)e^{x^2} = \frac{1}{2}\left[x^2 e^{x^2} - e^{x^2}\right] + C$$

$$\Rightarrow \quad \tan y = \frac{1}{2}\left(x^2 - 1\right) + Ce^{-x^2} \quad \textbf{Ans.}$$

Ex.8. $(x+y)^2 \dfrac{dy}{dx} = 9$. **[MREC (Auto), 2002]**

Sol. Put $x + y = v$

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \quad v^2\left(\frac{dv}{dx} - 1\right) = 9$$

$$\Rightarrow \quad v^2 \frac{dv}{dx} - v^2 = 9$$

$$\Rightarrow \quad \frac{v^2}{9+v^2} dv = dx \quad \text{or} \quad \left(\frac{v^2+9-9}{v^2+9}\right) dv = dx$$

$$\Rightarrow \quad dv - \frac{9}{v^2+9} dv = dx$$

On Integration, we get

$$\int dv - \int \frac{9}{v^2+9} dv = \int dx$$

$$\Rightarrow \quad v - \frac{9}{3}\tan^{-1}\frac{v}{3} = x + C$$

$$\Rightarrow \quad x + y - 3\tan^{-1}\frac{(x+y)}{3} = x + C$$

$$\Rightarrow \quad y = 3\tan^{-1}\frac{(x+y)}{3} + C \quad \text{Ans.}$$

Ex.9 $x^3 + xy^4 = x^5 - 2y^3 \dfrac{dy}{dx}$.

Sol. $2y^3 \dfrac{dy}{dx} + xy^4 = x^5 - x^3$

Put $y^4 = v$ then $4y^3 \dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$\therefore \quad \frac{dv}{dx} + 2xv = 2(x^5 - x^3)$$

This is linear in v so that

I. F. $= e^{\int 2x\,dx} = e^{x^2}$

Hence soluation

$$ve^{x^2} = \int e^{x^2} \cdot 2\left(x^5 - x^3\right) dx + C$$

$$\Rightarrow \quad ve^{x^2} = \int e^{x^2} 2x^3 (x^2 - 1) dx$$

put $x^2 = t$, $2x\ dx = dt$

$$\therefore \quad ve^{x^2} = \int e^t \cdot t(t-1)\, dt$$

$$\Rightarrow \quad ve^{x^2} = \int e^t t^2 dt - \int e^t \cdot t\, dt$$

$$\Rightarrow \quad ve^{x^2} = t^2 e^t - \int 2t e^t\, dt - \int te^t\, dt$$

$$\Rightarrow \quad ve^{x^2} = t^2 e^t - 3\int te^t\, dt$$

$$\Rightarrow \quad ve^{x^2} = t^2 e^t - 3e^t t + 3e^t + C$$

Put value $v = y^4$

$$\therefore \quad y^4 e^{x^2} = x^4 e^{x^2} - 3e^{x^2}x^2 + 3e^{x^2} + C$$

$$\text{or,} \quad y^4 = \left(x^4 - 3x^2 + 3\right) + Ce^{-x^2} \quad \textbf{Ans.}$$

Ex.10 Solve $\dfrac{dy}{dx} - \dfrac{\tan y}{1+x} = (1+x)e^x \sec y$.

Sol. Rewrite as

$$\cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x)e^x$$

Put $\sin y = v$ then $\cos y \dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$\therefore \quad \frac{dv}{dx} - \frac{v}{(1+x)} = (1+x)e^x$$

This is linear in v.

$$\text{I. F.} = e^{\int -\frac{1}{(1+x)}dx} = \frac{1}{(1+x)}$$

Hence solution is

$$v\frac{1}{(1+x)} = \int (1+x)e^x \cdot \frac{1}{(1+x)}dx + C$$

$$\Rightarrow \quad \frac{v}{1+x} = e^x + C$$

$$\Rightarrow \quad \sin y = (1+x)(e^x + C) \quad \textbf{Ans.}$$

MU 81.

Ex-32

Exercise 11

Solve :

Ex.1 $\dfrac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$.

Sol. Rewrite as $\dfrac{dy}{dx} = e^{-2y}\left(e^{3x} + x^2\right)$

$$\Rightarrow \quad e^{2y}dy = \left(e^{3x} + x^2\right)dx$$

On integration, we get

$$\frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + C \quad \textbf{Ans.}$$

Ex.2 $3e^x \tan y + (1-e^x)\sec^2 y \dfrac{dy}{dx} = 0$.

Sol. Rewrite as

$$-\frac{3e^x}{(1-e^x)}dx = \frac{\sec^2 y}{\tan y}dy \qquad \text{...(1)}$$

For Integrating equation (1)

Put $(1-e^x) = u$ and $\tan y = v$

Then $-e^x dx = du$; $\sec^2 y\, dy = dv$

$$\therefore \quad \int \frac{3du}{u} = \int \frac{dv}{v}$$

$$\Rightarrow \quad 3\log u = \log v - \log C$$

$$\text{or,} \quad \log\frac{v}{u^3} = \log C \quad \text{or,} \quad v = u^3 C$$

Put value of v and u, we get :

$$\tan y = (1-e^x)^3 C \quad \textbf{Ans.}$$

Ex.3 $x^2 y \dfrac{dy}{dx} = xy^2 - e^{-\frac{1}{x^3}}$.

Sol. Multiply above equation by $\dfrac{1}{x^2}$

$$y\frac{dy}{dx} - \frac{y^2}{x} = -\frac{e^{-\frac{1}{x^3}}}{x^2}$$

Put $y^2 = v \Rightarrow y\dfrac{dy}{dx} = \dfrac{1}{2}\dfrac{dv}{dx}$

$$\Rightarrow \quad \frac{1}{2}\frac{dv}{dx} - \frac{v}{x} = \frac{-e^{-1/x^3}}{x^2} \Rightarrow \frac{dv}{dx} - \frac{2}{x}v = \frac{-2e^{-1/x^3}}{x^2}$$

This is linear in v

$$\text{I. F.} = e^{-\int \frac{2}{x}dx} = \frac{1}{x^2}$$

Here solution is

$$v \cdot \frac{1}{x^2} = \int -\frac{2e^{-1/x^3}}{x^2} \cdot \frac{1}{x^2}\,dx$$

or, $\dfrac{v}{x^2} = \int -\dfrac{2e^{-1/x^3}}{x^4}\,dx$, put $\dfrac{-1}{x^3} = t$

then $\dfrac{+3}{x^4}dx = dt$

$\therefore$ $\dfrac{v}{x^2} = -\dfrac{2}{3}\int e^t\,dt$

or, $\dfrac{v}{x^2} = -\dfrac{2}{3}e^t + C$

or, $\dfrac{y^2}{x^2} = -\dfrac{2}{3}e^{-\frac{1}{x^3}} + C$

or, $3y^2 = -2x^2e^{-\frac{1}{x^3}} + Cx^2$ **Ans.**

Ex.4 $x\dfrac{dy}{dx} + y\log y = xye^x$.

Sol. It can be written as

$$\frac{1}{y}\frac{dy}{dx} + \frac{1}{x}\log y = e^x \qquad \text{...(1)}$$

Put $\log y = t \Rightarrow \dfrac{1}{y}\dfrac{dy}{dx} = \dfrac{dt}{dx}$

$\therefore$ (1) reduces to

$\dfrac{dt}{dx} + \dfrac{1}{x}t = e^x$ (linear in t)

$$\text{I. F.} = e^{\int \frac{1}{x}dx} = e^{\log x} = x$$

$\therefore$ Solution is

$$t(x) = \int xe^x\,dx + c$$

or, $x\log y = e^x(x-1) + c$ Ans.

Ex.5 $\dfrac{dy}{dx} + \dfrac{x}{1-x^2}y = x\sqrt{y}$.

Sol. Rewrite as

$$\frac{1}{\sqrt{y}}\frac{dy}{dx} + \frac{x}{1-x^2}\sqrt{y} = x$$

Put $\sqrt{y} = u$

Then $\dfrac{1}{2\sqrt{y}}\dfrac{dy}{dx} = \dfrac{du}{dx}$

$\therefore$ $\dfrac{du}{dx} + \dfrac{x}{2(1-x^2)}u = \dfrac{x}{2}$

This is linear in u so that

$$\text{I. F.} = e^{\int \frac{x}{2(1-x^2)}dx} = e^{\int -\frac{1}{4}\frac{dt}{t}} = e^{-\frac{1}{4}\log t} = (t)^{-\frac{1}{4}}$$

[Put $1-x^2 = t$ so $-2x\,dx = dt$]

$$\text{I. F.} = (1-x^2)^{-\frac{1}{4}}$$

Hence solution is

$$u\cdot(1-x^2)^{-\frac{1}{4}} = \int (1-x^2)^{-\frac{1}{4}} \times \frac{x}{2}dx + C$$

[Put $1 - x^2 = v$ so $-2x\,dx = dv$]

$$u\cdot(1-x^2)^{-\frac{1}{4}} = \int \frac{-1}{4}v^{-\frac{1}{4}}dv + C$$

$\Rightarrow$ $u\left(1-x^2\right)^{\frac{-1}{4}} = \dfrac{-1}{4} \times \dfrac{4}{3}v^{\frac{3}{4}} + C$

Put value of u and v we get

$$\sqrt{y}\left(1-x^2\right)^{-\frac{1}{4}} = -\frac{1}{3}\left(1-x^2\right)^{\frac{3}{4}} + C$$

or, $3\sqrt{y} + (1-x^2) = C_1(1-x^2)^{\frac{1}{4}}$ **Ans.**

Ex.6 $x\cos x\dfrac{dy}{dx} + y(x\sin x + \cos x) = 1.$

Sol. Rewrite as

$$\frac{dy}{dx} + y(\tan x + \frac{1}{x}) = \frac{\sec x}{x}$$

This is linear in y ; so that

$$\text{I. F.} = e^{\int(\tan x+\frac{1}{x})dx} = e^{(\log\sec x+\log x)} = x\sec x$$

Hence solution is

$$y\cdot x\sec x = \int x\sec x \times \frac{\sec x}{x}\,dx + C = \int \sec^2 dx + C$$

or, $\quad y\,x\sec x = \tan x + C$

or, $\quad xy = \sin x + C\cos x$ **Ans.**

Ex.7 $\dfrac{(x+y-a)\,dy}{(x+y-b)\,dx} = \dfrac{(x+y+a)}{(x+y+b)}.$

Sol. Rewrite as

$$\frac{dy}{dx} = \frac{(x+y+a)}{(x+y+b)}\,\frac{(x+y-b)}{(x+y-a)}$$

Put $x + y = v$ then $1+\dfrac{dy}{dx} = \dfrac{dv}{dx}$

$$\therefore \quad \frac{dv}{dx} - 1 = \frac{(v+a)}{(v+b)} \times \frac{(v-b)}{(v-a)}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{(v+a)}{(v+b)} \times \frac{(v-b)}{(v-a)} + 1$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{2v^2-2ab}{v^2-v(a-b)-ab}$$

$$\Rightarrow \quad \frac{v^2-v(a-b)-ab}{2(v^2-ab)}\,dv = dx$$

$$\Rightarrow \quad \frac{1}{2}dv - \frac{v(a-b)}{2\left(v^2-ab\right)}\,dv = dx$$

On Integration, we get

$$\frac{1}{2}\left[v - \frac{(a-b)}{2}\log(v^2-ab)\right] = x + C$$

$$x + y + \frac{(b-a)}{2}\log[(x+y)^2-ab] = 2x + 2C$$

Hence solution is

$$(b-a)\log[(x+y)^2-ab] = 2[x-y] + C' \quad \text{Ans.}$$

Ex.8 $(y\log x - 1)y\,dx = x\,dy.$

Sol. Rewrite as

$$\frac{dy}{dx} + \frac{y}{x} = y^2\frac{\log x}{x}$$

$$\Rightarrow \quad \frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = \frac{\log x}{x}$$

Put $\dfrac{1}{y} = t$ then $-\dfrac{1}{y^2}\dfrac{dy}{dx} = \dfrac{dt}{dx}$

$$\therefore \quad \frac{-dt}{dx} + \frac{t}{x} = \frac{(\log x)}{x} \Rightarrow \frac{dt}{dx} - \frac{1}{x}t = \frac{-\log x}{x}$$

This is linear in t so that

$$\text{I. F.} = e^{\int -\frac{1}{x}dx} = \frac{1}{x}$$

Hence solution is

$$t\cdot\frac{1}{x} = -\int \frac{1}{x} \times \frac{(\log x)}{x}\,dx$$

Put $x = e^v$ then $v = \log x \Rightarrow dv = \frac{dx}{x}$

$\therefore \quad t \cdot \frac{1}{x} = -\int v \cdot e^{-v} dv$

$\Rightarrow \quad t \cdot \frac{1}{x} = +ve^{-v} + e^{-v} + C$

$\Rightarrow \quad \frac{1}{xy} = \frac{1}{x}\log x + \frac{1}{x} - C$

$\Rightarrow \quad (C)xy + 1 = (\log x + 1)y$ **Ans.**

Ex.9 $(2x + y + 1)dx + (4x + 2y - 1)dy = 0$.

Sol. Rewrite as

$$\frac{dy}{dx} = \frac{-(2x + y + 1)}{(4x + 2y - 1)}$$

Put, $2x + y + 1 = t$

$$2 + \frac{dy}{dx} = \frac{dt}{dx} \text{ or } \frac{dy}{dx} = \frac{dt}{dx} - 2$$

$\Rightarrow \quad \left(\frac{dt}{dx} - 2\right) = -\frac{t}{(2t - 3)}$

$\therefore \quad \frac{dt}{dx} = \frac{3t - 6}{(2t - 3)}$

$\Rightarrow \quad \frac{2t - 3}{3(t - 2)} dt = dx$

$\Rightarrow \quad \frac{2t - 4 + 1}{3(t - 2)} dt = dx$

On integration, we get

$$\int \frac{2(t - 2)}{3(t - 2)} dt + \int \frac{1}{3(t - 2)} dt = \int dx + C$$

or, $\quad \frac{2}{3}t + \frac{1}{3}\log(t - 2) = x + C$

Put $t = 2x + y + 1$

$\Rightarrow \quad \frac{2}{3}(2x + y + 1) + \frac{1}{3}\log(2x + y - 1) = x + C$

$\Rightarrow \quad 4x + 2y + \log(2x + y - 1) = 3x + 3C - 2$

or, $\quad x + 2y + \log(2x + y - 1) = C' \; [C' = 3C - 2]$ Ans.

Ex.10 $(1 + xy)y\,dx + (1 - xy)x\,dy = 0$. [MREC (Auto), 2001]

Sol. Let $M = y + xy^2$ and $N = x - x^2y$

$$Mx - Ny = xy + x^2y^2 - xy + x^2y^2 = 2x^2y^2 \neq 0$$

Hence I. F. $= \frac{1}{Mx - Ny} = \frac{1}{2x^2y^2}$

Multiply given equation by $\frac{1}{2x^2y^2}$, we get

$$\left(\frac{1}{x^2y} + \frac{1}{x}\right)dx + \left(\frac{1}{xy^2} - \frac{1}{y}\right)dy = 0$$

This equation is exact

Let $M_1 = \frac{1}{x^2y} + \frac{1}{x}$ and $N_1 = \frac{1}{xy^2} - \frac{1}{y}$

Integrate M_1 *w.r.t.* x, keeping y as constant

$$U = \int M_1 dx = \int\left(\frac{1}{x^2y} + \frac{1}{x}\right)dx$$

$\Rightarrow \quad U = -\frac{1}{xy} + \log x$

Also, $\frac{\partial U}{\partial y} = \frac{1}{xy^2}$

$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int -\frac{1}{y}dy = -\log y$

Hence solution is $\rightarrow U + V = C$

$\Rightarrow \quad \log\left(\frac{x}{y}\right) - \frac{1}{xy} = C$ Ans.

Ex.11 $xy^3(y\,dx+2x\,dy)+(3y\,dx+5x\,dy)=0$.

Sol. Rewrite as

$$\left(xy^4+3y\right)dx+\left(2x^2y^3+5x\right)dy=0 \quad \text{....(1)}$$

Multiply equation (1) by $x^h y^k$ for become exact then

$$(x^{h+1}y^{k+4}+3x^h y^{k+1})dx+(2x^{h+2}y^{k+3}+5x^{h+1}y^k)dy=0 \quad \text{.....(2)}$$

$$\frac{\partial M}{\partial y}=(k+4)x^{h+1}y^{k+3}+(k+1)3x^h y^k$$

and $\dfrac{\partial N}{\partial x}=2(h+2)x^{h+1}y^{k+3}+5(h+1)x^h y^k$

For exact equation, condition is $\dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$

$\Rightarrow$ $(k+4)x^{h+1}y^{k+3}+(k+1)3x^h y^k$

$$=2(h+2)x^{h+1}y^{k+3}+5(h+1)x^h y^k$$

On Comparing cofficient, we get

$k+4=2(h+2)$ and $3(k+1)=5(h+1)$

$\Rightarrow$ $k-2h=0$ and $3k-5h=2$

then $h=2$ and $k=4$

Put value of h and k in equation (2), we get.

$$(x^3y^8+3x^2y^5)dx+(2x^4y^7+5x^3y^4)dy=0$$

It is exact equation

Let $M_1=x^3y^8+3x^2y^5$ and $N_1=2x^4y^7+5x^3y^4$

Integration M_1 w.r.t. x, keeping y is constant

$$U=\int M_1dx=\int(x^3y^8+3x^2y^5)dx$$

$\Rightarrow$ $U=\dfrac{1}{4}x^4y^8+x^3y^5$

Also, $\dfrac{\partial U}{\partial y}=2x^4y^7+5x^3y^4$

$\therefore$ $V=\displaystyle\int\left(N_1-\frac{\partial U}{\partial y}\right)dy=\int(0)\,dy=C'$

$\Rightarrow$ $V=C'$

Hence solution is $\rightarrow U+V=C$

$\Rightarrow$ $\dfrac{1}{4}x^4y^8+x^3y^5=C$

or $x^4y^8+4x^3y^5=C'$ Ans.

Ex.12 $\left(e^{2x}+y^3\right)dy=y^3dx$.

Sol. Rewrite as

$$\frac{dx}{dy}=\frac{e^{2x}}{y^3}+\frac{1}{y}$$

$\Rightarrow$ $\dfrac{dx}{dy}-\dfrac{1}{y}=\dfrac{e^{2x}}{y^3}$

$\Rightarrow$ $e^{-2x}\dfrac{dx}{dy}-\dfrac{e^{-2x}}{y}=\dfrac{1}{y^3}$

put $e^{-2x}=t$ so that $-2e^{-2x}\dfrac{dx}{dy}=\dfrac{dt}{dy}$

Hence equation becomes

$$-\frac{1}{2}\frac{dt}{dy}-\frac{t}{y}=\frac{1}{y^3}$$

$\Rightarrow$ $\dfrac{dt}{dy}+\dfrac{2t}{y}=\dfrac{-2}{y^3}$

This is linear in t so that

I. F. $=e^{\int\frac{2}{y}dy}=y^2$

Hence solution is,

$$t\,y^2=\int y^2\times\left(-\frac{2}{y^3}\right)dy+C$$

or, $\quad t y^2 = -2\log y + C$

or, $\quad y^2 e^{-2x} + 2\log y = C \quad$ Ans.

Ex.13 $dx + (x\tan y - \sec y)\,dy = 0$.

Sol. Rewrite as

$$\frac{dx}{dy} + \tan y \cdot x = \sec y$$

This is linear in x, so that

$$\text{I. F.} = e^{\int \tan y\,dy} = \sec y$$

Hence solution is

$$x \cdot \sec y = \int \sec^2 y\,dy + C$$

or, $\quad x\sec y = \tan y + C \quad$ Ans.

Ex.14 $(2x + 3y - 5)\,dy + (2x + 3y - 1)dx = 0$.

Sol. Rewrite as $\dfrac{dy}{dx} = -\dfrac{(2x+3y-1)}{(2x+3y-5)}$

Let $2x+3y = v$ then $2 + \dfrac{3dy}{dx} = \dfrac{dv}{dx}$

so that

or, $\quad \frac{1}{3}\left(\frac{dv}{dx} - 2\right) = \frac{-(v-1)}{(v-5)}$

or, $\quad \frac{dv}{dx} = -\frac{3v-3}{v-5} + 2$

$\Rightarrow \quad \frac{dv}{dx} = \frac{-3v+3+2v-10}{(v-5)} = \frac{-v-7}{v-5}$

$\Rightarrow \quad \frac{dv}{dx} = -\left(\frac{v+7}{v-5}\right)$

$\Rightarrow \quad \left(\frac{v-5}{v+7}\right)dv = -dx$

$\Rightarrow \quad \left(1 - \frac{12}{v+7}\right)dv = -dx$

On Integrating, we get

$$v - 12\log(v+7) = -x + C$$

or, $\quad 2x + 3y - 12\log(2x+3y+7) = -x + C$

or, $\quad 3x + 3y - 12\log(2x+3y+7) = C$

or, $\quad x + y - 4\log(2x+3y+7) = C' \quad$ Ans.

Ex.15 $x\dfrac{dy}{dx} - y = x\sqrt{x^2+y^2}$.

Sol. It can be written as

$$\frac{xdy - ydx}{\sqrt{x^2+y^2}} = xdx$$

or, $\quad \dfrac{xdy - ydx}{x\sqrt{1+\left(y^2/x^2\right)}} = xdx$

or, $\quad \dfrac{\left(\dfrac{xdy - ydx}{x^2}\right)}{\sqrt{1+(y/x)^2}} = dx \quad$ ...(1)

Put $\dfrac{y}{x} = v \Rightarrow \dfrac{xdy - ydx}{x^2} = dv$

$\therefore$ (1) gives $\dfrac{dv}{\sqrt{1+v^2}} = dx$

on integrating, we get –

$\sin h^{-1} v = x + c$

or, $\quad v = \sinh(x+c)$

or, $\quad \dfrac{y}{x} = \sin h(x+c)$

or, $\quad y = x \sin h\,(x + c) \quad$ Ans.

Ex.16 $(x^2 + 2xy)\dfrac{dy}{dx} + (2xy + y^2 + 3x^2) = 0$.

Sol. Rewrite as

$$(2xy+y^2+3x^2)dx+(x^2+2xy)dy=0$$

Let $M=2xy+y^2+3x^2$ and $N=x^2+2xy$

$$\frac{\partial M}{\partial y}=2x+2y \text{ and } \frac{\partial N}{\partial x}=2x+2y$$

$\therefore$ $\frac{\partial M}{\partial y}=\frac{\partial M}{\partial x}$ so that differential equation is exact

On integration M w. r. t x, keeping y as constant

$$U=\int M\,dx=\int(2xy+y^2+3x^2)dx$$

$\therefore$ $U=x^2y+y^2x+x^3$

$\Rightarrow$ $\frac{\partial U}{\partial y}=x^2+2xy$

$\therefore$ $V=\int\left(N-\frac{\partial U}{\partial y}\right)dy=\int(0)dy=C'$

Hence solution is $\rightarrow U+V=C$

$\Rightarrow$ $x^2y+y^2x+x^3=C$

or, $x(xy+y^2+x^2)=C$ **Ans.**

Ex.17 $\left(x^2+2xy-y^2\right)dx+\left(y^2+2xy-x^2\right)dy=0$.

Sol. Rewrite as

$$\frac{dy}{dx}=-\frac{(x^2+2xy-y^2)}{(y^2+2xy-x^2)}$$

Put $y=vx$ then $\frac{dy}{dx}=v+x\frac{dv}{dx}$

$\Rightarrow$ $v+x\frac{dv}{dx}=-\left(\frac{x^2+2x^2v-x^2v^2}{x^2v^2+2x^2v-x^2}\right)$

$\Rightarrow$ $v+x\frac{dv}{dx}=-\frac{(1+2v-v^2)}{(v^2+2v-1)}$

$\Rightarrow$ $x\frac{dv}{dx}=\frac{v^2-2v-1}{v^2+2v-1}-v$

$\Rightarrow$ $x\frac{dv}{dx}=\frac{v^2-2v-1-v^3-2v^2+v}{(v^2+2v-1)}$

$\Rightarrow$ $x\frac{dv}{dx}=\frac{-v^3-v^2-v-1}{(v^2+2v-1)}$

$\Rightarrow$ $x\frac{dv}{dx}=\frac{-(v+1)(v^2+1)}{(v^2+2v-1)}$

$\Rightarrow$ $\frac{-(v^2+2v-1)}{(v+1)(v^2+1)}dv=\frac{dx}{x}$

On Inetgration, we get

$$\int\frac{v^2+1-\left(2v^2+2v\right)}{(v+1)\left(v^2+1\right)}dv=\int\frac{dx}{x}$$

or, $\int\frac{1}{(v+1)}dv-\int\frac{2v}{(v^2+1)}dv=\int\frac{1}{x}dx$

$\Rightarrow$ $\log(v+1)-\log(v^2+1)=\log x+\log C$

$\Rightarrow$ $\log\frac{(v+1)}{(v^2+1)}=\log xC$

$\Rightarrow$ $(v+1)=(v^2+1)xC$

Put $v=\frac{y}{x}$

Hence solution is

$$(y+x)=(x^2+y^2)C \quad \textbf{Ans.}$$

Q.18 $y^2(y\,dx+2x\,dy)-x^2(2y\,dx+x\,dy)=0$.

Scanned with

Sol. It is of form

$$x^a y^b(my\,dx + nx\,dy) - x^c y^d(py\,dx + qx\,dy) = 0$$

so multiply given equation by $x^h y^k$

$$x^h y^{k+2}(y\,dx + 2x\,dy) - x^{h+2}y^k(2y\,dx + x\,dy) = 0$$

$$\left(x^h y^{k+3} - 2x^{h+2}y^{k+1}\right)dx + \left(2x^{h+1}y^{k+2} - x^{h+3}y^k\right)dy = 0$$

$$\frac{\partial M}{\partial y} = (k+3)x^h y^{k+2} - 2(k+1)x^{h+2}y^k$$

$$\frac{\partial N}{\partial x} = 2(h+1)x^h y^{k+2} - (h+3)x^{h+2}y^k$$

For exactness of equation (1) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$(k+3)x^h y^{k+2} - 2(k+1)x^{h+2}y^k = 2(h+1)x^h y^{k+2} - (h+3)x^{h+2}y^k$$

Comparing cofficient, we get

$k+3 = 2(h+1)$ and $2(k+1) = -(h+3)$

$k - 2h = -1$ and $-2k + h = -1 \Rightarrow h = 1,\ k = 1$

Put value of h and k in equation ...(1)

$$(xy^4 - 2x^3y^2)dx + (2x^2y^3 - x^4y)dy = 0 \qquad \text{...(2)}$$

Let $M_1 = xy^4 - 2x^3y^2$ and $N_1 = 2x^2y^3 - x^4y$

Now equation (2) is exact

Integrate M_1 w. r.t x, keeping y as constant

$$U = \int M_1 dx = \int (xy^4 - 2x^3y^2)dx$$

$$\Rightarrow \quad U = \frac{x^2y^4}{2} - \frac{x^4y^2}{2}$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = 2x^2y^3 - x^4y$$

$$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int (0)dy = C'$$

Hence solution is $\rightarrow U + V = C$

$$\Rightarrow \quad \frac{x^2y^4}{2} - \frac{x^4y^2}{2} = C$$

or, $\quad x^2y^2(y^2 - x^2) = 2C$

or, $\quad xy\sqrt{y^2 - x^2} = C'$ **Ans.** [Put $\sqrt{2C} = C'$]

Ex.19 $\left(2x^2y - 3y^2\right)dx + \left(2x^3 - 12xy + \log y\right)dy = 0.$

Sol. Here $M = 2x^2y - 3y^2$ and $N = 2x^3 - 12xy + \log y$

$$\frac{\partial M}{\partial y} = 2x^2 - 6y \text{ and } \frac{\partial N}{\partial x} = 6x^2 - 12y$$

$\therefore$ Given differention equation is not exact

$$\text{Now, } \frac{1}{M}\left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] = \frac{1}{2x^2y - 3y^2}\left[6x^2 - 12y - 2x^2 + 6y\right]$$

$$= \frac{1}{\left(2x^2y - 3y^2\right)}\left[4x^2 - 6y\right] = \frac{2}{y}$$

$$\text{I. F.} = e^{\int \frac{2}{y}dy} = y^2$$

Multiply given equation by y^2

$$\left(2x^2y^3 - 3y^4\right)dx + (2x^3y^2 - 12xy^3 + y^2\log y)dy = 0 \qquad \text{...(1)}$$

Now equation (1) is exact

Let $M_1 = 2x^2y^3 - 3y^4$ and $N_1 = 2x^3y^2 - 12xy^3 + y^2\log y$

Integrate M_1 w. r. t x, keeping y as constant

$$U = \int M_1 dx = \int\left(2x^2y^3 - 3y^4\right)dx$$

$$\therefore \quad U = \frac{2}{3}x^3y^3 - 3xy^4$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = 2x^3y^2 - 12xy^3$$

$$\therefore \quad V = \int\left(N_1 - \frac{\partial U}{\partial y}\right)dy = \int\left(y^2\log y\right)dy$$

$$= \left[\frac{y^3}{3}\log y - \int\frac{y^2}{3}dy\right] = \frac{y^3\log y}{3} - \frac{y^3}{9}$$

$$\therefore \quad V = \frac{1}{3}y^3\log y - \frac{1}{9}y^3$$

Hence solution is $U + V = C$

$$\Rightarrow \quad \frac{2}{3}x^3y^3 - 3xy^4 + \frac{y^3}{3}\log y - \frac{1}{9}y^3 = C$$

$$\Rightarrow \quad 6x^3y^3 - 27xy^4 + 3y^3\log y - y^3 = C \quad \textbf{Ans.}$$

Ex.20 $\left(x^2 + y^2 + x\right)dx - \left(2x^2 + 2y^2 - y\right)dy = 0.$

Sol. Rewrite as

$$\left(x^2 + y^2\right)\left(dx - 2dy\right) + xdx + ydy = 0$$

$$\Rightarrow \quad dx - 2dy + \frac{xdx + ydy}{\left(x^2 + y^2\right)} = 0$$

On Integration

$$x - 2y + \int\frac{xdx + ydy}{\left(x^2 + y^2\right)} = C$$

Put $x^2 + y^2 = t \Rightarrow 2xdx + 2ydy = dt$

$$\Rightarrow \quad xdx + ydy = \frac{1}{2}dt$$

$$\therefore \quad x - 2y + \int\frac{1}{2}\frac{dt}{t} = C$$

$$\Rightarrow \quad x - 2y + \frac{1}{2}\log t = C$$

$$\Rightarrow \quad x - 2y + \frac{1}{2}\log\left(x^2 + y^2\right) = C \quad \textbf{Ans.}$$

Ex.21 $\dfrac{dy}{dx} + y\cot x = 2\cos x.$

Sol. Given equation is linear in y

$\therefore$ I. F. $= e^{\int\cot x\,dx} = \sin x$

So that solution is

$$y\cdot\sin x = \int 2\cos x\cdot\sin x\,dx + C = \int\sin 2x\,dx + C$$

or, $\quad y\sin x = -\dfrac{\cos 2x}{2} + C$

or, $\quad 2y\sin x + \cos 2x + C = 0 \quad$ **Ans.**

Ex.22 $\dfrac{1}{2x}\dfrac{dy}{dx} + \dfrac{x + y}{x^2 + y^2} = 0.$

Sol. Rewrite as $2x\,(x + y)\,dx + (x^2 + y^2)\,dy = 0$

Here $M = 2x\,(x + y)$ and $N = (x^2 + y^2)$

$$\therefore \quad \frac{\partial M}{\partial y} = 2x \quad \text{and} \quad \frac{\partial N}{\partial x} = 2x$$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, Hence equation is exact.

Integrate M $w.r.t.$, x, keeping y as constant

$$U = \int M\,dx = \int \left(2x^2 + 2xy\right)dx = \frac{2x^3}{3} + x^2 y$$

$$\Rightarrow \quad \frac{\partial U}{\partial y} = x^2$$

$$V = \int \left(N - \frac{\partial U}{\partial y}\right)dy = \int y^2\,dy = \frac{y^3}{3}$$

Hence solution is $\rightarrow$ U + V = C $\Rightarrow \frac{2x^3}{3} + x^2 y + \frac{y^3}{3} = C'$

or, $y^3 + 3x^2 y + 2x^3 = C$ **Ans.**

13. Linear Differential Equations of Second Order

Ex-36

Exercise - 13(A)

Ex. 1. Solve $\dfrac{d^2y}{dx^2} + \dfrac{1}{x}\dfrac{dy}{dx} = \dfrac{12\log x}{x^2}$

Sol. : $\dfrac{x^2d^2y}{dx^2} + \dfrac{xdy}{dx} = 12\log x$

Putting $x = e^z$, the given equation reduce to

$[D(D-1) + D]y = 12Z$, where $D \equiv d/dz$

A.E. is $m^2 = 0 \Rightarrow m = 0, 0$

$\therefore$ C.F. $= C_1 + C_2 Z$

P.I. $= \dfrac{1}{D^2} 12Z = 12\dfrac{1}{D}\left(\dfrac{Z^2}{2}\right) = 12.\dfrac{Z^3}{6} = 2Z^3$

Hence complete solution is

$y = C_1 + C_2 Z + 2Z^3$

Or $y = C_1 + C_2 (\log x) + 2(\log x)^3$ **Ans.**

Ex. 2. Solve $x^2\dfrac{d^2y}{dx^2} - 2y = x^2 + \dfrac{1}{x}$

Sol. : Putting $x = e^z$, the given equation reduces to

$[D(D-1) - 2]\, y = e^{2z} + e^{-z}$, where $D \equiv d/dz$.

A.E. is $m^2 - m - 2 = 0 \Rightarrow m = 2, -1$.

$\therefore$ C.F. $= C_1e^{2z} + C_2e^{-z}$

P.I. $= \dfrac{1}{(D-2)(D+1)}e^{2z} + \dfrac{1}{(D-2)(D+1)}e^{-z}$

$$= \frac{1}{(D-2)(2+1)} e^{2z} + \frac{1}{(-1-2)(D+1)} e^{-z}$$

$$= \frac{1}{3} e^{2z} \cdot \frac{1}{(D+2-2)} (1) - \frac{1}{3} e^{-z} \frac{1}{(D-1+1)} (1)$$

$$= \frac{1}{3} e^{2z} \cdot \frac{1}{D} (1) - \frac{1}{3} e^{-z} \cdot \frac{1}{D} (1) = \frac{1}{3} z e^{2z} - \frac{1}{3} z e^{-z}.$$

Hence complete solution is

$$y = C_1 e^{2z} + C_2 e^{-z} + \frac{1}{3} z e^{2z} - \frac{1}{3} z e^{-z}$$

Or $\quad y = C_1 x^2 + \frac{C_2}{x} + \frac{1}{3} x^2 \log x - \frac{1}{3x} \log x.$

Or $\quad y = C_2 x^{-1} + C_1 x^2 + 1/3\, x^2 \log x - \frac{1}{3x} \log x$ **Ans.**

Ex. 3. Solve $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x.$

Sol. : Putting $x = e^z$, the given equation reduces to

$$[D(D-1) - D + 2]\, y = e^z \cdot z, \; D \equiv d/dz$$

or $\quad (D^2 - 2D + 2)\, y = z e^z;$

A.E. is $\quad m^2 - 2m + 2 = 0 \Rightarrow m = 1 \pm i.$

$\therefore$ C.F. $= e^z (C_1 \cos z + C_2 \sin z).$

$$\text{P.I.} = \frac{1}{D^2 - 2D + 2} z e^z = e^z \frac{1}{(D+1)^2 - 2(D+1) + 2} z$$

$$= e^z \frac{1}{D^2 + 1} z = e^z (1 + D^2)^{-1} z = e^z (1 - D^2 \ldots) z = z e^z.$$

Hence complete solution is

$$y = e^z (C_1 \cos z + C_2 \sin z) + z e^z$$

Or $\quad y = x[C_1 \cos(\log x) + C_2 \sin(\log x)] + x \log x.$ **Ans.**

Ex. 4. Solve $x^3 \frac{d^2y}{dx^2} + 3x^2 \frac{dy}{dx} + xy = \sin(\log x)$

Sol. : $x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{x} \sin(\log x)$

Putting $x = e^z$, the given equation reduce to

$[D(D-1) + 3D + 1]\, y = \frac{1}{e^z} \sin z$, where $D = \frac{d}{dz}$

Or $[D^2 + 2D + 1]\, y = e^{-z} \sin z$

A.E. is $\quad m^2 + 2m + 1 = 0 \Rightarrow m = -1, -1$

$\therefore \quad$ C.F. $= (C_1 + C_2 z) e^{-z}$

$$\text{P.I.} = \frac{1}{(D^2 + 2D + 1)} e^{-z} \sin z$$

$$= e^{-z} \frac{1}{(D-1)^2 + 2(D-1) + 1} \sin z$$

$$= e^{-z} \frac{1}{D^2 + 1 - 2D + 2D - 2 + 1} \sin z$$

$$= e^{-z} \frac{1}{D^2} \sin z = e^{-z} \frac{1}{D} (-\cos z) = e^{-z} (-\sin z)$$

Hence complete solution is

$$y = (C_1 + C_2 z) e^{-z} - e^{-z} \sin z$$

$y = (C_1 + C_2 \log x)\, x^{-1} - \frac{1}{x} \sin(\log x)$ **Ans.**

Ex. 5. Solve $\frac{x^2 d^2y}{dx^2} - \frac{2x\, dy}{dx} + 2y = x^{-1}$

Sol. : Putting $x = e^z$, the given equation reduce to

$[D(D-1) - 2D + 2]\, y = e^{-z}$, where $D = \frac{d}{dz}$

$[D^2 - 3D + 2]\, y = e^{-z}$

A.E. is $\quad m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$

$\therefore \quad$ C.F. $= C_1 e^z + C_2 e^{2z}$

$$\text{P.I.} = \frac{1}{(D^2 - 3D + 2)} e^{-z}$$

$$= \frac{e^{-z}\, y}{[(-1)^2 - 3(-1) + 2]} = \frac{e^{-z}}{6}$$

Hence complete solution is

$$y = C_1 e^z + C_2 e^{2z} + \frac{e^{-z}}{6}$$

$= C_1 x + C_2 x^2 + \frac{x^{-1}}{6}$ **Ans.**

Ex. 6. Solve $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x.$

Sol. : Putting $x = e^z$ the given equation reduces to

$[D(D-1) + 4D + 2]\, y = e^{e^z}$, where $D \equiv d/dz$,

Or $\quad (D^2 + 3D + 2)\, y = e^{e^z} \Rightarrow m = -2, -1.$

$\therefore \quad$ C.F. $= C_1 e^{-z} + C_2 e^{-2z}.$

$$\text{P.I.} = \frac{1}{(D+2)(D+1)} e^{e^z}.$$

Let $\frac{1}{(D+1)} e^{e^z} = v \Rightarrow \frac{dv}{dz} + v = e^{e^z}$,

$$\therefore \text{I.F.} = e^{\int 1\, dz} = e^z.$$

Hence solution of this linear equation is given by

$$v e^z = \int e^z e^{e^z} dz = e^{e^z} \text{ or } v = e^{-z}. e^{e^z}.$$

Therefore $\frac{1}{(D+2)(D+1)} e^{e^z} = \frac{1}{(D+2)} v = \frac{1}{(D+2)}\left(e^{-z}. e^{e^z}\right)$.

Again let $\frac{1}{(D+2)}\left(e^{-z} e^{e^z}\right) = u$ or $(D+2)u = e^{-z} e^{e^z}$.

Or $\frac{du}{dz} + 2u = e^{-z} e^{e^z}$, $\therefore$ I.F. $= e^{\int 2\, dz} = e^{2z}$.

Integrating, $u e^{2z} = \int e^{2z}. e^{-z} e^{e^z}. dz = \int e^z. e^{e^z}. dz = e^{e^z}$.

$$\Rightarrow \quad u = e^{-2z} e^{e^z}. \quad \therefore \frac{1}{(D+2)(D+1)} e^{e^z} = u = e^{-2z} e^{e^z}.$$

Hence required complete solution is

$$y = C_1 e^{-z} + C_2 e^{-2z} + e^{-2z} e^{e^z} = \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{1}{x^2} e^x.$$

Or $y = C_1 x^{-1} + C_2 x^{-2} + x^{-2} e^x$ **Ans.**

Ex. 7. Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + y = \frac{\log x \sin(\log x) + 1}{x}$

Sol. : Putting $x = e^z$ the equation reduces to

$$[D(D-1) - 3D + 1]\, y = \frac{z \sin z + 1}{e^z} \text{ where } D \equiv \frac{d}{dz}$$

Or $$(D^2 - 4D + 1)\, y = (z \sin z + 1)\, e^{-z}$$

A.E. is $m^2 - 4m + 1 = 0$; on solving $m = 2 \pm \sqrt{3}$

$$\therefore \quad \text{C.F.} = C_1 e^{(2+\sqrt{3})z} + C_2 e^{(2-\sqrt{3})z}$$

$$\text{P.I.} = \frac{1}{(D^2 - 4D + 1)} z \sin z\, e^{-z} + \frac{1}{D^2 - 4D + 1} e^{-z}$$

Now $\frac{1}{(D^2 - 4D + 1)} e^{-z} = \frac{e^{-z}}{1 + 4 + 1} = \frac{1}{6} e^{-z}$, and

$$\frac{1}{D^2 - 4D + 1} z \sin z e^{-z} = e^{-z} \frac{1}{(D-1)^2 - 4(D-1) + 1} z \sin z$$

$$= e^{-z} \frac{1}{D^2 - 6D + 6} z \sin z = \text{I.P. in } e^{-z} \frac{1}{D^2 - 6D + 6} z\, e^{iz}$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{1}{(D+i)^2 - 6(D+i) + 6} z$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{1}{D^2 + D(2i-6) + (5-6i)} z$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{1}{(5-6i)} \left[1 + \frac{D(2i-6)}{5-6i} \dots\right]^{-1} z$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{1}{(5-6i)} \left[1 - \frac{D(2i-6)}{5-6i} \dots\right] z$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{(5+6i)}{25 - 36i^2} \left[z - \frac{2i-6}{5-6i}\right]$$

$$= \text{Imaginary part in } e^{-z}. e^{iz} \frac{5+6i}{61} \left[z - \frac{(2i-6)(5+6i)}{25 - 36i^2}\right]$$

$$= \text{Imaginary part in } \frac{1}{61} e^{-z}. e^{iz} (5+6i) \left[z + \frac{(42+26i)}{61}\right]$$

$$= \text{I.P. in } \frac{1}{61} e^{-z} (\cos z + i \sin z)(5+6i) \left[z + \frac{42}{61} + \frac{26i}{61}\right]$$

$$= \frac{1}{61} e^{-z} \left[(6z \cos z + 5z \sin z) + \frac{130}{61} \cos z - \frac{156}{61} \sin z + \frac{252}{61} \cos z + \frac{210}{61} \sin z\right]$$

$$= \frac{1}{61} z. e^{-z} (6 \cos z + 5 \sin z) + \frac{2e^{-z}}{3721} (191 \cos z + 27 \sin z).$$

Hence complete solution is :

$$y = C_1 e^{(2+\sqrt{3})z} + C_2 e^{(2-\sqrt{3})z} + \frac{1}{6} e^{-z} + \frac{1}{61} z\, e^{-z} (6z \cos z + 5z \sin z) + \frac{2e^{-z}}{3721} (191 \cos z + 27 \sin z)$$

Or $$y = C_1 x^{2+\sqrt{3}} + C_2 x^{2-\sqrt{3}} + \frac{1}{6x} + \frac{1}{61} \frac{\log x}{x} [6 \cos(\log x) + 5 \sin(\log x)] + \frac{2}{3721x} [191 \cos(\log x) + 27 \sin(\log x)]$$

Or $y = x^2\left(C_1 x^{\sqrt{3}} + C_2 x^{-\sqrt{3}}\right)$

$$+ x^{-1}\left[\frac{1}{61}\log x\{5\ \text{six}\ (\log x) + 6\cos(\log x)\}\right.$$

$$\left. + \frac{1}{3721}\{54\sin(\log x) + 282\cos(\log x)\} + \frac{1}{6}\right] \textbf{ Ans.}$$

Ex. 8. Solve $(x^2D^2 - xD + 4)\,y = \cos(\log x) + x\sin(\log x)$.

Sol. : Putting $x = e^z$, the equation reduces to

$$[D'(D'-1) - D' + 4]\,y = \cos z + e^z \sin z$$

where $D' \equiv d/dz$.

Or $$(D'^2 - 2D' + 4)\,y = \cos z + e^z\sin z.$$

A.E. is $m^2 - 2m + 4 = 0 \Rightarrow m = 1 \pm \sqrt{3}\,i$.

$\therefore$ C.F. $= e^z(C_1\cos\sqrt{3}\,z + C_2\sin\sqrt{3}\,z)$.

$$\text{P.I.} = \frac{1}{D'^2 - 2D' + 4}\cos z + \frac{1}{D'^2 - 2D' + 4}e^z\sin z$$

$$= \frac{1}{-1 - 2D' + 4}\cos z + \frac{1}{(D'+1)^2 - 2(D'+1) + 4}\sin z$$

$$= \frac{3 + 2D'}{9 - 4D'^2}\cos z + e^z\frac{1}{D'^2 + 3}\sin z.$$

$$= \frac{3\cos z - 2\sin z}{9 + 4} + e^z\frac{\sin z}{-1 + 3}$$

$$= \frac{1}{13}(3\cos z - 2\sin z) + \frac{1}{2}e^2\sin z.$$

Hence complete solution is

$$y = e^z(C_1\cos\sqrt{3}\,z + C_2\sin\sqrt{3}\,z) + \frac{1}{13}(3\cos z - 2\sin z) + \frac{1}{2}e^z\sin z$$

Or $y = x[C_1\cos(\sqrt{3}\log x) + C_2\sin(\sqrt{3}\log x)]$

$$+ \frac{1}{13}[3\cos(\log x) - 2\sin(\log x)] + \frac{z}{2}\sin(\log x) \textbf{ Ans.}$$

Ex. 9. Solve $(x^2D^2 - 3xD + 5)\,y = x^2\sin(\log x)$

Sol. : Putting $x = e^z$, the equation reduce to

$$[D'(D'-1) - 3D' + 5]\,y = e^{2z}\sin(z) \quad \text{where } D' = \frac{d}{dz}$$

Or $(D'^2 - 4D' + 5)\,y = e^{2z}\sin(z)$

A.E. is $m^2 - 4m + 5 = 0 \Rightarrow m = 2 \pm i$

$\therefore$ C.F. $= e^{2z}(C_1\cos z + C_2\sin z)$

$$\text{P.I.} = \frac{1}{(D'^2 - 4D' + 5)}e^{2z}\sin z$$

$$= e^{2z}\frac{1}{\{(D'+2)^2 - 4(D'+2) + 5\}}\sin z$$

$$= e^{2z}\frac{1}{(D'^2 + 1)}\sin z = e^{2z}\left(-\frac{z}{2}\cos z\right)$$

$$\left[\because \frac{1}{D^2 + a^2}\sin ax = -\frac{x}{2a}\cos x\right]$$

$$= -\frac{z}{2}e^{2z}\cos z$$

Hence complete solution is

$$y = e^{2z}(C_1\cos z + C_2\sin z) - e^{2z}\cdot\frac{z}{2}\cos z,\ z = \log x \qquad \textbf{Ans.}$$

Ex. 10. Solve $\dfrac{d^2y}{dx^2} + \dfrac{1}{x}\dfrac{dy}{dx} = \dfrac{12\log x}{x^2}$

Sol. : Same as Ex. No. 1.

Ex. 11. Solve $(x + a)^2\dfrac{d^2y}{dx^2} - 4(x + a)\dfrac{dy}{dx} + 6y = x$.

Sol. : Putting $x + a = e^z$, the equation reduces to

$$[D(D-1) - 4D + 6]\,y = e^z - a.$$

Or $(D^2 - 5D + 6)\,y = e^z - a$, where $D \equiv d/dz$.

A.E. is $m^2 - 5m + 6 = 0 \Rightarrow m = 2 \pm 3i$

$\therefore$ C.F. $= C_1e^{2z} + C_2e^{3z}$

$$\text{P.I.} = \frac{1}{D^2 - 5D + 6}e^z - \frac{1}{D^2 - 5D + 6}(a)$$

$$= \frac{e^z}{1 - 5 + 6} - \frac{1}{6}(1 + 5D\ldots)\,a = \frac{1}{2}e^z - \frac{1}{6}a.$$

Hence required complete solution is

$$y = C_1e^{2z} + C_2e^{3z} + \frac{1}{2}e^z - \frac{1}{6}a$$

Or $y = C_1(x + a)^2 + C_2(x + a)^3 + \frac{1}{2}(x + a) - \frac{1}{6}a$

Or $y = C_1(x + a)^2 + C_2(x + a)^3 + \frac{1}{6}(3x + 2a) \qquad$ **Ans.**

Ex. 12. Solve $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 4\cos\log(1+x)$.

Sol. : Putting $1 + x = e^z$, the given equation reduces to

$$[D(D-1) + D + 1]\,y = 4\cos z, \text{ where } D \equiv d/dz$$

Or $$(D^2 + 1)\,y = 4\cos z.$$

A.E. is $m^2 + 1 = 0 \Rightarrow m = \pm i$.

$\therefore$ C.F. $= C_1\cos z + C_2 \sin z$.

$$\text{P.I.} = \frac{1}{D^2+1}4\cos z = \text{Real part in } \frac{1}{D^2+1}4e^{iz}$$

$$= \text{Real part in } 4e^{iz}\frac{1}{(D+i)^2+1}(1)$$

$$= \text{Real part in } 4e^{iz}\frac{1}{D^2+2Di}(1)$$

$$= \text{Real part in } 4e^{iz}\frac{1}{2Di}\left[1 - \frac{D}{2i}\ldots\right](1)$$

$$= \text{Real part in } -2\,i\,e^{iz}z$$

$$= \text{Real part in } -2\,i(\cos z + i\sin z)z = 2z\sin z$$

Hence complete solution is

$$y = C_1\cos z + C_2\sin z + 2z\sin z$$

$$y = C_1\cos\log(1+x) + C_2\sin\log(1+x) + 2\log(1+x)\sin\log(1+x)$$

Or $$y = C_1\cos\{\log(1+x)\} + C_2\sin\{\log(1+x)\} + 2\log(1+x)\sin\{\log(1+x)\}$$ **Ans.**

Ex. 13. Solve $(1+2x)^2\frac{d^2y}{dx^2} - 6(1+2x)\frac{dy}{dx} + 16y = 8(1+2x)^2$

Sol. : Putting $1 + 2x = e^z$

and hence $(1+2x)\frac{d}{dx} = 2D$, $(1+2x)^2\frac{d^2}{dx^2} = 2^2 D(D-1)$

the above the equation reduces to

$$[4D(D-1) - 2.6\,D + 16]y = 8e^{2z},$$

Or $[D^2 - 4D + 4]\,y = 2e^{2z}$, where $D \equiv d/dz$.

A.E. is $m^2 - 4m + 4 = 0$, $\Rightarrow m = 2, 2$.

$$\therefore \text{C.F.} = (C_1 + C_2 z)\,e^{2z}$$

$$\text{Now P. I.} = \frac{1}{(D-2)^2}2e^{2z} = 2e^{2z}\frac{1}{(D+2-2)^2}(1)$$

$$= 2e^{2z}\frac{1}{D^2}(1) = 2e^{2z}\left(\frac{z^2}{2}\right) = z^2e^{2z}.$$

Hence required complete solution is

$$y = (C_1 + C_2 z)\,e^{2z} + z^2e^{2z},\ z = \log(1+2x)$$ **Ans.**

Ex. 14. Solve $(5+2x)^2\frac{d^2y}{dx^2} - 6(5+2x)\frac{dy}{dx} + 8y = 0$ ... (1)

Sol. : Let $5 + 2x = v \Rightarrow 2dx = dv$

$$\Rightarrow \frac{dy}{dx} = 2\frac{dy}{dv} \qquad \Rightarrow \frac{d^2y}{dx^2} = 2\frac{d^2y}{dv^2}\cdot\frac{dv}{dx} = \frac{4d^2y}{dv^2}$$

Hence eq. (1) can be written as :

$$4v^2\frac{d^2y}{dv^2} - 12v\frac{dy}{dv} + 8y = 0$$

$$\Rightarrow v^2\frac{d^2y}{dv^2} - 3v\frac{dy}{dv} + 2y = 0 \qquad \text{... (2)}$$

which is a linear homogeneous differential equation

$\therefore$ let $v = e^z$ $\Rightarrow z = \log v \Rightarrow v\frac{d}{dv} \equiv \frac{d}{dz} \equiv D$

and $v^2\frac{d}{dv^2} \equiv D(D-1)$

Hence eq. (2) can be written as :

$$[D(D-1) - 3D + 2]\,y = 0$$

$$\Rightarrow (D^2 - 4D + 2)\,y = 0 \qquad \text{... (3)}$$

A.E. of (3) is $m^2 - 4m + 2 = 0 \Rightarrow m = 2 \pm \sqrt{2}$

$\therefore$ C.F. $= C_1e^{2z}\cosh\left(\sqrt{2}\,z + C_2\right)$

& P.I $= 0$

$\therefore$ Solution for eq. (3) is $y = C_1e^{2z}\cosh(\sqrt{2z} + C_2)$

$\Rightarrow y = C_1v^2\cosh[\sqrt{2}(\log v) + C_2]$ $[\because v = e^z]$

$\Rightarrow y = C_1(5+2x)^2\cosh[\sqrt{2}\{\log(5+2x)\} + C_2]$ **Ans.**

$[\because v = 5 + 2x]$

Ex-37

Exercise - 13(B)

Ex. 1. Solve $(1+x^2)\dfrac{d^2y}{dx^2} + 3x\dfrac{dy}{dx} + y = 0$

Sol. : $P_0 = 1 + x^2,\ P_1 = 3x,\ P_2 = 1.$

The equation is exact if $P_2 - P_1' + P_0'' = 0$.

i.e. $1 - 3 + 2 = 0,$ which is true

Hence the equation is exact whose first integral is

$$P_0\frac{dy}{dx} + (P_1 - P_0')\,y = C_1,$$

i.e. $(1+x^2)\dfrac{dy}{dx} + (3x - 2x)\,y = C_1$ or $\dfrac{dy}{dx} + \dfrac{x}{1+x^2}\,y = \dfrac{C_1}{1+x^2}$

Which is a linear differential equation of first order

$$\text{I.F.} = exp.\left[\int\frac{x}{1+x^2}\,dx\right] = exp\left\{\frac{1}{2}\log(1+x^2)\right\} = \sqrt{(1+x^2)},$$

$\therefore$ General solution is

$$y\sqrt{(1+x^2)} = \int\frac{C_1}{(1+x^2)}\cdot\sqrt{(1+x^2)}\,dx + C_2$$

$$= C_1\log\left[x + \sqrt{(1+x^2)}\right] + C_2$$ **Ans.**

Ex. 2. Solve $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + y = 2x$

Sol. : Here, $P_0 = 1 - x^2,\ P_1 = -x,\ P_2 = 1$

Now, $P_2 - P_1' + P_0'' = 1 + 1 - 2 = 0$

Hence the differential equation is exact.

$\therefore$ The first integral is :

$$P_0\frac{dy}{dx} + (P_1 - P_0')\,y = 2\int x\,dx + C_1$$

$$\Rightarrow\ (1-x^2)\frac{dy}{dx} + (-x + 2x)\,y = x^2 + C_1$$

Or, $$\frac{dy}{dx} + \frac{x}{1-x^2}\,y = \frac{x^2}{1-x^2} + \frac{C_1}{1-x^2} \quad \text{.... (1)}$$

which is a linear differential equation of first order

$$\therefore \text{I.F.} = e^{\int\frac{x}{1-x^2}dx} = e^{-\frac{1}{2}\log(1-x^2)} = e^{\log(1-x^2)^{-1/2}} = \frac{1}{\sqrt{1-x^2}}$$

$\therefore$ Solution of (1) is :

$$y\frac{1}{\sqrt{1-x^2}} = \int\frac{x^2}{(1-x^2)^{3/2}}\,dx + C_1\int\frac{dx}{(1-x^2)^{3/2}} + C_2$$

In R.H.S. put $x = \sin\theta \Rightarrow dx = \cos\theta\, d\theta$

$$\frac{y}{\sqrt{1-x^2}} = \int\frac{\sin^2\theta.\cos\theta}{\cos^3\theta}\,d\theta + C_1\int\frac{\cos\theta\, d\theta}{\cos^3\theta} + C_2$$

$$= \int\tan^2\theta\, d\theta + C_1\int\sec^2\theta\, d\theta + C_2$$

$$= \int(\sec^2\theta - 1)\,d\theta + C_1\tan\theta + C_2$$

$$= \tan\theta - \theta + C_1\tan\theta + C_2$$

$$= (1 + C_1)\tan\theta - \theta + C_2$$

$$= C'_1\tan\theta - \theta + C_2$$

$$= C'_1\frac{x}{\sqrt{1-x^2}} - \sin^{-1}x + C_2$$

$$\Rightarrow\ y = C'_1 x + \sqrt{(1-x^2)}\,(C_2 - \sin^{-1}x)$$ **Ans.**

Ex. 3. Solve $\dfrac{d^2y}{dx^2} + 2e^x\dfrac{dy}{dx} + 2e^x y = x^2$

Sol. : $P_0 = 1,\ P_1 = 2e^x,\ P_2 = 2e^x$

The equation is exact if $P_2 - P_1' + P_0'' = 0$

i.e. $2e^x - 2e^x + 0 = 0$ which is true

Hence the equation is exact where first integral is

$$\Rightarrow\ P_0\frac{dy}{dx} + (P_1 - P_0')\,y = \int x^2 dx + C_1$$

Or $$\frac{dy}{dx} + (2e^x - 0)\,y = \frac{x^3}{3} + C_1$$

Or $$\frac{dy}{dx} + 2e^x y = \frac{x^3}{3} + C_1$$

which is a linear differential equation of first order

$$\text{I.F.} = exp.\left[\int 2e^x dx\right] = exp\,\{2e^x\} = e^{2e^x}$$

$\therefore$ General solution

$$ye^{2e^x} = \int\left(\frac{x^3}{3} + C_1\right)e^{2e^x}\,dx + C_2$$

$$ye^{2e^x} = \frac{1}{3}\int x^3 e^{2e^x}\,dx + C_1\int e^{2e^x}\,dx + C_2$$ **Ans.**

Scanned with

Ex. 4. **Solve** $(2x^2 + 3x)\dfrac{d^2y}{dx^2} + (6x + 3)\dfrac{dy}{dx} + 2y = (x + 1)e^x$

Sol. : $P_0 = 2x^2 + 3x$, $P_1 = 6x + 3$, $P_2 = 2$.

The equation is exact if

$$P_2 - P'_1 + P''_0 = 0, \text{ i.e. } 2 - 6 + 4 = 0, \text{ which is true.}$$

Hence the equation is exact whose first integral is

$$\frac{P_0 dy}{dx} + (P_1 - P'_0) = \int (x + 1) e^x \, dx + C_1$$

Or $$(2x^2 + 3x)\frac{dy}{dx} + [(6x + 3) - (4x + 3)]\, y = xe^x + C_1$$

Or $$\frac{dy}{dx} + \frac{2}{(3 + 2x)} y = \frac{e^x}{(3 + 2x)} + \frac{C_1}{x(3 + 2x)}$$

$$\text{I.F.} = exp\left[\int \frac{2}{(3 + 2x)} dx\right] = exp\,[\log(3 + 2x)] = 3 + 2x.$$

Hence the G. solution is

$$y(3 + 2x) = \int \frac{e^x}{(3 + 2x)}(3 + 2x)\, dx$$

$$+ C_1 \int \frac{1}{x(3 + 2x)}(3 + 2x)\, dx + C_2$$

Or $y(3 + 2x) = e^x + C_1 \log x + C_2$ **Ans.**

Ex. 5. **Solve** $x\dfrac{d^2y}{dx^2} + (1 - x)\dfrac{dy}{dx} - y = e^x$.

Sol. : $P_0 = x$, $P_1 = (1 - x)$, $P_2 = -1$.

$$\therefore \quad P_2 - P_1' + P_0'' = -1 + 1 + 0 = 0.$$

Hence the equation is exact whose first integral is

$$P_0 \frac{dy}{dx} + (P_1 - P_0')y = \int e^x \, dx + C_1$$

Or $$x\frac{dy}{dx} + [(1 - x) - 1]y = e^x + C_1$$

Or $$\frac{dy}{dx} - y = \frac{1}{x}e^x + \frac{C_1}{x}, \quad \therefore \text{ I.F.} = e^{\int -dx} = e^{-x}.$$

Hence the solution is

$$y.\, e^{-x} = \int \frac{1}{x} e^x \cdot e^{-x}\, dx + \int \frac{C_1}{x} e^{-x}\, dx + C_2$$

Or $$y = e^x \log x + C_1 e^x \int \frac{e^{-x}}{x} dx + C_2 e^x$$ **Ans.**

Ex. 6. **Solve** $(x^2 - x)\dfrac{d^2y}{dx^2} + 2(2x + 1)\dfrac{dy}{dx} + 2y = 0$

Sol. : $P_0 = (x^2 - x)$, $P_1 = 2(2x + 1)$, $P_2 = 2$

The equation is exact if $P_2 - P + P_0'' = 0$

$$2 - 4 + 2 = 0 \text{ which is true.}$$

Hence the equation is exact where first integral is

$$P_0 \frac{dy}{dx} + (P_1 - P'_0)y = C_1$$

i.e. $$(x^2 - x)\frac{d^2y}{dx^2} + [2(2x + 1) - 2x + 1]\, y = C_1$$

Or $$\frac{d^2y}{dx^2} + \frac{[2x + 3]}{(x^2 - x)} = \frac{C_1}{x^2 - x}$$

which is a linear differential equation of first order.

$$\text{I.F.} = exp.\left[\int \frac{2x + 3}{(x^2 - x)} dx\right] = exp\left[\int \frac{(2x + 3)}{x(x - 1)} dx\right]$$

$$= exp\left[\int \frac{5}{(x - 1)} dx - \int \frac{3}{x} dx\right] \text{ [by partial integration]}$$

$$\text{I.F.} = e^{\log (x-1)^5/x^3} = \frac{(x-1)^5}{x^3}$$

$\therefore$ General solution is

$$y\frac{(x-1)^5}{x^3} = \int \frac{C_1}{(x^2 - x)} \times \frac{(x-1)^5}{x^3} dx + C_2$$

$$= C_1 \int \frac{(x-1)^4}{x^4} dx + C_2$$

$$= C_1 \int \frac{x^4 - 4x^3 + 6x^2 - 4x + 1}{x^4} + C_2$$

Scanned with

$$= C_1 \int \left(1 - \frac{4}{x} + \frac{6}{x^2} - \frac{4}{x^3} + \frac{1}{x^4}\right) + C_2$$

$$y\,\frac{(x-1)^3}{x^3} = C_1\left[x - 4\log x - \frac{6}{x} + \frac{2}{x^2} - \frac{1}{3x^3}\right] + C_2$$

$$y\,(x-1)^3 = C_1\left[x^4 - 4x^3\log x - 6x^2 + 2x - \frac{1}{3}\right] + C_2x^3 \qquad \textbf{Ans.}$$

Ex. 7. Solve $x^4 \frac{d^2y}{dx^2} + x^2(x-1)\frac{dy}{dx} + xy = x^3 - 4.$

Sol. : Let the given equation which at present is not exact, becomes exact by multiplying it by x^m, i.e., the equation becomes

$$x^{m+4}\frac{d^2y}{dx^2} + x^{m+2}(x-1)\frac{dy}{dx} + x^{m+1}y = x^m(x^3-4). \qquad \dots (1)$$

Now by applying the condition of exactness in equation (1), we have $P_2 - P_1' + P_0'' = 0$

or $x^{m+1} - [(m+3)x^{m+2} - (m+2)x^{m+1}] + (m+4)(m+3)x^{m+2} = 0$

or $(m+3)x^{m+1} + (m+3)(m+4-1)x^{m+2} = 0$

or $(m+3)x^{m+1} + (m+3)^2x^{m+2} = 0$

or $(m+3)[x^{m+1} + x^{m+2}(m+3)] = 0$ or $m = -3.$

$\Rightarrow$ Putting $m = -3$, equation (1) reduces to

$$x\frac{d^2y}{dx^2} + \left(1 - \frac{1}{x}\right)\frac{dy}{dx} + \frac{1}{x^2} = 1 - \frac{4}{x^3} \qquad \text{(exact equation)}$$

Its first integral is

$$P_0\frac{dy}{dx} + (P_1 - P_0')y = \int\left(1 - \frac{4}{x^3}\right)dx + C_1$$

or $x\frac{dy}{dx} + \left[\left(1 - \frac{1}{x}\right) - 1\right]y = x + \frac{2}{x^2} + C_1$

or $\frac{dy}{dx} - \frac{1}{x^2}y = 1 + \frac{2}{x^3} + \frac{C_1}{x}$ (a linear eq.)

$$\therefore \text{ I.F. } \exp\left[\int -\frac{1}{x^2}dx\right] = e^{1/x}.$$

Hence the required G.S. is given by

$$y\,e^{1/x} = \int\left(1 + \frac{2}{x^3} + \frac{C_1}{x}\right)e^{1/x}dx + C_2$$

or $y\,e^{1/x} = \int\left(1 + \frac{C_1}{x}\right).e^{1/x}dx + 2\int\frac{1}{x^3}e^{1/x}dx + C_2$

$$= \int\left(1 + \frac{C_1}{x}\right).e^{1/x}dx - 2\int t\,e^t\,dt + C_2$$

[putting $(1/x) = t$ in second integral]
$-\frac{1}{x^2}dx = dt$

$$= \int\left(1 + \frac{C_1}{x}\right).e^{1/x}dx - 2(t\,e^t - e^t) + C_2$$

$$= \int\left(1 + \frac{C_1}{x}\right).e^{1/x}dx - 2\left(\frac{1}{x} - 1\right)e^{1/x} + C_2 \qquad \textbf{Ans.}$$

Ex. 8. Solve $2x^2\frac{d^2y}{dx^2} + 15x\frac{dy}{dx} - 7y = 3x^2$

Sol. : Here, $P_0 = 2x^2$, $P_1 = 15x$, $P_2 = -7$

Now, $P_2 - P_1' + P_0'' = -7 - 15 + 4 \neq 0$

$\therefore$ Given equation is not exact

Now, multiplying both sides of the equation by x^m, we have

$$2x^{m+2}\frac{d^2y}{dx^2} + 15x^{m+1}\frac{dy}{dx} - 7x^my = 3x^{m+2} \qquad \dots (1)$$

If the above equation is exact, then

$P_2 - P_1' + P_0'' = 0$

i.e. $-7x^m - 15(m+1)x^m + 2(m+2)(m+1)x^m = 0$

$\Rightarrow$ $x^m[-7 - 15m - 15 + 2m^2 + 6m + 4] = 0$

$\Rightarrow$ $x^m[2m^2 - 9m - 18] = 0$

now, $x^m \neq 0$

$\Rightarrow$ $2m^2 - 9m - 18 = 0 \Rightarrow (2m+3)(m-6) = 0$

$\Rightarrow$ $m = -\frac{3}{2}, 6$

for both values of m, eq. n (1) is exact.

We choose m = 6.

Hence the integrating factor is x^6.

Now, multiplying original equation by x^6, we have

$$2x^8\frac{d^2y}{dx^2} + 15x^7\frac{dy}{dx} - 7x^6y = 3x^8.$$

which is now exact. Its first integral is—

$$2x^8\frac{dy}{dx}+(15x^7-16x^7)\,y=3\int x^8\,dx+C_1$$

or $$2x^8\frac{dy}{dx}-x^7y=\frac{x^9}{3}+C_1$$

or $$\frac{dy}{dx}-\frac{y}{2x}=\frac{x}{6}+\frac{C_1}{2}x^{-8} \quad \text{.... (2) (linear difference equation)}$$

$$\text{I.F.}=e^{\int-\frac{1}{2x}dx}=\frac{1}{\sqrt{x}}$$

$\therefore$ Solution of Equation (2) is—

$$y.\frac{1}{\sqrt{x}}=\int\left(\frac{x}{6}+\frac{C_1}{2}x^{-8}\right)\frac{1}{\sqrt{x}}\,dx+C_2=\int\left(\frac{\sqrt{x}}{6}+\frac{C_1}{2}x^{-17/2}\right)dx+C_2$$

or $$=\frac{x^{3/2}}{9}-\frac{C_1}{15}x^{\frac{-15}{2}}+C_2 \Rightarrow y=\frac{x^2}{9}+C_1'x^{-7}+C_2\sqrt{x} \qquad \textbf{Ans.}$$

Ex. 9. Solve $\dfrac{d^2y}{dx^2}+2\dfrac{dy}{dx}\tan x+3y=\tan^2 x\ (\sec x)$

Sol. : Multiplying the given equation which is not exact by $\cos x$, it becomes

$$\cos x\frac{d^2y}{dx^2}+2\frac{dy}{dx}\sin x+3\,y\cos x=\tan^2 x \qquad \text{... (1)}$$

$$P_0=\cos x,\ P_1=2\sin x,\ P_2=3\cos x$$

$\therefore$ $P_2-P_1'+P_0''=3\cos x-2\cos x-\cos x=0.$

Hence equation (1) is exact whose first integral is

$$P_0\frac{dy}{dx}+(P_1-P_0')\,y=\int\tan^2 x\,dx+C_1$$

or $$\cos x\frac{dy}{dx}+(2\sin x+\sin x)\,y=\int(\sec^2 x-1)\,dx+C_1$$

or $$\cos x\frac{dy}{dx}+3\sin x.\,y=\tan x-x+C_1$$

or $$\frac{dy}{dx}+3\frac{\sin x}{\cos x}\,y=\tan x\sec x-x\sec x+C_1\sec x, \text{(linear)}$$

$$\text{I.F.}=\exp\left(\int\frac{3\sin x}{\cos x}\,dx\right)=e^{(-3\log\cos x)}=\sec^3 x$$

Hence the G. solution is,

$$y\sec^3 x=\int\sec^4 x\tan x\,dx$$
$$-\int x\sec^4 x\,dx+C_1\int\sec^4 x\,dx+C_2 \qquad \text{... (2)}$$

sec³x sec x tan x dx

Now $$\int\sec^4 x\tan x\,dx=\frac{\sec^4 x}{4} \qquad \text{... (3)}$$

let sec x = t
sec x tan x dx = dt

$$\int x\sec^4 x\,dx=\int x\,(1+\tan^2 x)\sec^2 x\,dx$$

$$=\int x\sec^2 x\,dx+\int x\,(\sec^2 x\tan^2 x)\,dx$$

$$=x\tan x-\int\tan x\,dx+x\frac{\tan^3 x}{3}-\int\frac{\tan^3 x}{3}\,dx \quad \text{(by parts)}$$

$$=x\left(\tan x+\frac{1}{3}\tan^3 x\right)-\int\tan x\,dx-\frac{1}{3}\int\tan x\,(\sec^2 x-1)\,dx$$

$$=x\left(\tan x+\frac{1}{3}\tan^3 x\right)-\frac{2}{3}\int\tan x\,dx-\frac{1}{3}\int\tan x\sec^2 x\,dx$$

$$=x\left(\tan x+\frac{1}{3}\tan^3 x\right)-\frac{2}{3}\log\sec x-\frac{1}{3}\cdot\frac{1}{2}\tan^2 x. \qquad \text{... (4)}$$

Also $$\int\sec^4 x\,dx=\tan x+\frac{1}{3}\tan^3 x \qquad \text{... (5)}$$

From (2), (3), (4) and (5), the solution is

$$y\sec^3 x=\frac{1}{4}\sec^4 x-x\left(\tan x+\frac{1}{3}\tan^3 x\right)$$
$$+\frac{2}{3}\log\sec x+\frac{1}{6}\tan^2 x+C_1\left(\tan x+\frac{1}{3}\tan^3 x\right)+C_2$$

($\because$ Put $(C_2+1/4=C_3)$ & $\sec^4 x=(1+\tan^2 x)^2$)

$$y\sec^3 x=\frac{2}{3}\tan^2 x+\frac{1}{4}\tan^4 x+\frac{2}{3}\log\sec x$$
$$+(C_1-x)\left(\tan x+\frac{1}{3}\tan^3 x\right)+C_3 \ \textbf{Ans.}$$

Ex. 10. Solve $\dfrac{d^2y}{dx^2}-2y\ \text{cosec}^2 x=0$

Sol. : Multiplying the above equation by $\cot x$, we get

$$\cot x\frac{d^2y}{dx^2}+0.\frac{dy}{dx}-2\cot x\,\text{cosec}^2 x.\,y=0 \qquad \text{... (1)}$$

Scanned with

$$P_0 = \cot x,\ P_1 = 0,\ P_2 = -2\cot x\,\mathrm{cosec}^2 x.$$

$$\therefore\ P_2 - P_1' + P_0'' = -2\cot x\,\mathrm{cosec}^2 x - 0 + 2\cot x\,\mathrm{cosec}^2 x = 0$$

Hence equation (1) is exact whose first integral is

$$P_0\frac{dy}{dx} + (P_1 - P_0')\,y = C_1$$

or $$\cot x\frac{dy}{dx} + (0 + \mathrm{cosec}^2 x)\,y = C_1$$

or $$\frac{dy}{dx} + \frac{\mathrm{cosec}^2 x}{\cot x}\,y = C_1 \tan x \text{ (a linear eq.)}$$

$$\text{I.F.} = \exp\left(\int\frac{\mathrm{cosec}^2 x}{\cot x}\,dx\right) = \exp(-\log\cot x) = \tan x.$$

Hence the solution is $y.\tan x = \int C_1\tan^2 x\,dx + C$

or $y\tan x = C_1\int(\sec^2 x - 1)\,dx + C_2 = C_1(\tan x - x) + C_2$. **Ans.**

Ex-38

EXERCISE 13(C)

Ex. 1. **Solve** $\dfrac{d^2y}{dx^2} - x^2\dfrac{dy}{dx} + xy = x.$

Sol. : Here $P = -x^2$, $Q = x$. By inspection $P + Qx = 0$.

Hence $y = x$ is a part of C.F. Put $y = vx$

$$\frac{dy}{dx} = x\frac{dv}{dx} + v,\ \frac{d^2y}{dx^2} = \frac{x\,d^2v}{dx^2} + \frac{2\,dv}{dx}$$

Putting these values in the given equation, it reduces to

$$\left(x\frac{d^2v}{dx^2} + 2\frac{dv}{dx}\right) - x^2\left(x\frac{dv}{dx} + v\right) + x^2v = x$$

or $$x\frac{d^2v}{dx^2} + (2 - x^3)\frac{dv}{dx} = x$$

or $$\frac{d^2v}{dx^2} + \left(\frac{2}{x} - x^2\right)\frac{dv}{dx} = 1 \quad \ldots(1)$$

Now putting $\dfrac{dv}{dx} = p,\ \dfrac{d^2v}{dx^2} = \dfrac{dp}{dx}$, (1) gives

$$\frac{dp}{dx} + \left(\frac{2}{x} - x^2\right)p = 1, \quad \text{(linear in } p) \quad \ldots(2)$$

$$\text{I.F.} = \exp\left\{\int\left(\frac{2}{x} - x^2\right)dx\right\} = \exp\left(2\log x - \frac{x^3}{3}\right) = x^2e^{-x^3/3}$$

Multiplying (2) by I.F. and then integrating,

$$p\,.\,x^2e^{-x^3/3} = \int x^2e^{-x^3/3}\,dx + C_1 = -e^{-x^3/3} + C_1$$

$$\Rightarrow \frac{dv}{dx} = p = -\frac{1}{x^2} + \frac{C_1e^{(x^3/3)}}{x^2} \Rightarrow v = \frac{1}{x} + C_1\int x^{-2}e^{(x^3/3)}\,dx + C_2$$

Hence $y = vx = 1 + C_1\,x\int x^{-2}e^{(x^3/3)}\,dx + C_2x.$ **Ans.**

This is the required complete solution.

Ex. 2. **Solve** $(1 - x^2)\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - y = x(1 - x^2)^{3/2}.$

Sol. : Writing the equation in the standard form, we have

$$\frac{d^2y}{dx^2} + \frac{x}{1 - x^2}\frac{dy}{dx} - \frac{1}{1 - x^2}\,y = x(1 - x^2)^{1/2} \quad \ldots(1)$$

$$\Rightarrow \quad P = \frac{x}{1 - x^2},\ Q = -\frac{1}{1 - x^2}.$$

$$\Rightarrow \quad P + Qx = 0. \ \Rightarrow y = x \text{ is a part of C.F.}$$

Let $y = vx \Rightarrow \dfrac{dy}{dx} = x\dfrac{dv}{dx} + v,\ \dfrac{d^2y}{dx^2} = x\dfrac{d^2v}{dx^2} + 2\dfrac{dv}{dx}.$

Putting these values in the equation (1), we have

$$\left(x\frac{d^2v}{dx^2} + 2\frac{dv}{dx}\right) + \frac{x}{1 - x^2}\left(x\frac{dv}{dx} + v\right) - \frac{1}{1 - x^2}\,vx = x(1 - x^2)^{1/2}$$

or $$\frac{d^2v}{dx^2} + \left(\frac{2}{x} + \frac{x}{1 - x^2}\right)\frac{dv}{dx} = (1 - x^2)^{1/2}.$$

Putting $\dfrac{dv}{dx} = p,\ \dfrac{d^2v}{dx^2} = \dfrac{dp}{dx}$ in this, we get

$$\frac{dp}{dx} + \left(\frac{2}{x} + \frac{x}{1 - x^2}\right)p = (1 - x^2)^{1/2} \quad \text{(linear d.e.)}$$

$$\text{I.F.} = \exp\left\{\int\left[\frac{2}{x} + \frac{x}{1 - x^2}\right]dx\right\}$$

$$= \exp\left\{2\log x - \frac{1}{2}\log(1 - x^2)\right\} = \frac{x^2}{\sqrt{(1 - x^2)}}.$$

$$\therefore p \cdot \frac{x^2}{\sqrt{(1-x^2)}} = \int \sqrt{(1-x^2)} \cdot \frac{x^2}{\sqrt{(1-x^2)}} dx + C_1 = \frac{1}{3}x^3 + C_1$$

or $$p = \frac{dv}{dx} = \frac{1}{3}x\sqrt{(1-x^2)} + \frac{C_1}{x^2}\sqrt{(1-x^2)}.$$

$$\therefore v = \frac{1}{3}\int x\sqrt{(1-x^2)}\,dx + C_1\int \frac{1}{x^2}\sqrt{(1-x^2)}\,dx + C_2$$

$$= -\frac{1}{3}\cdot\frac{1}{2}\frac{(1-x^2)^{3/2}}{\frac{3}{2}} + C_1\sqrt{(1-x^2)}\left(-\frac{1}{x}\right)$$

$$- C_1\int \frac{1}{2}(1-x^2)^{-1/2}(-2x)\left(-\frac{1}{x}\right)dx + C_2$$

$$= -\frac{1}{9}(1-x^2)^{3/2} - C_1\frac{\sqrt{(1-x^2)}}{x} - C_1\int \frac{1}{\sqrt{(1-x^2)}}dx + C_2$$

$$= -\frac{1}{9}(1-2x^2)^{3/2} - C_1\frac{\sqrt{(1-x^2)}}{x} - C_1\sin^{-1}x + C_2.$$

$$\therefore y = vx = -\frac{1}{9}x(1-x^2)^{3/2} - C_1\left[\sqrt{(1-x^2)} + x\sin^{-1}x\right] + C_2x.$$ **Ans.**

Ex. 3. Solve $x\dfrac{d^2y}{dx^2} - (2x-1)\dfrac{dy}{dx} + (x-1)y = 0.$

Sol. : Dividing the equation by x^2 and comparing it with the standard form, we get

$$\frac{d^2y}{dx^2} - \left(2 - \frac{1}{x}\right)\frac{dy}{dx} + \left(1 - \frac{1}{x}\right)y = 0.$$

and $$P = -\left(2 - \frac{1}{x}\right),\ Q = 1 - \frac{1}{x}$$

By inspection, we see that $1 + P + Q = 0$.

Hence $y = e^x$ is a part of C.F.

Now put $y = ve^x$. On differentiation, we obtain

$$\frac{dy}{dx} = ve^x + e^x\frac{dv}{dx} \text{ and } \frac{d^2y}{dx^2} = e^x\frac{d^2v}{dx^2} + 2e^x\frac{dv}{dx} + ve^x.$$

Putting these values in the given equation, we get

$$e^x\left(\frac{d^2v}{dx^2} + 2\frac{dv}{dx} + v\right) - \left(2 - \frac{1}{x}\right)e^x\left(v + \frac{dv}{dx}\right)$$
$$+ \left(1 - \frac{1}{x}\right)ve^x = 0.$$

$$\Rightarrow \frac{d^2y}{dx^2} + \left[2 - \left(2 - \frac{1}{x}\right)\right]\frac{dv}{dx} = 0$$

or $$\Rightarrow \frac{d^2y}{dx^2} + \frac{1}{x}\frac{dv}{dx} = 0$$

Now put $\dfrac{dv}{dx} = p,\ \therefore \dfrac{d^2y}{dx^2} = \dfrac{dp}{dx}$

Above equation gives

$$\frac{dp}{dx} + \frac{p}{x} = 0 \text{ or } \frac{dp}{p} = \frac{-dx}{x}$$

Integrating it, we get

$$\log p = -\log x + \log C_1 \text{ or } p = \frac{C_1}{x}.$$

$$\Rightarrow \quad p = \frac{dv}{dx} = \frac{C_1}{x} \text{ or } dv = \frac{C_1}{x}dx.$$

Integrating, we get $v = C_1\log x + C_2$.

$$\therefore \quad y = ve^x = (C_1\log x + C_2)e^x.$$ **Ans.**

Which is the required general solution of the given equation.

Ex. 4. Solve $\dfrac{d^2y}{dx^2} + (1 - \cot x)\dfrac{dy}{dx} - y\cot x = \sin^2 x.$

Sol. : $P = 1 - \cot x,\ Q = -\cot x.$

By inspection, $1 - P + Q = 0 \Rightarrow y = e^{-x}$ is a part of C.F.

Put $y = ve^{-x} \Rightarrow \dfrac{dy}{dx} = e^{-x}\dfrac{dv}{dx} - ve^{-x},$

$$\frac{d^2y}{dx^2} = e^{-x}\frac{d^2v}{dx^2} - 2e^{-x}\frac{dv}{dx} + ve^{-x}.$$

Putting these values in the given equation, we have

$$\left(\frac{d^2v}{dx^2} - 2\frac{dv}{dx} + v\right)e^{-x} + (1 - \cot x)\left(\frac{dv}{dx} - v\right)e^{-x}$$
$$- \cot x \cdot ve^{-x} = \sin^2 x$$

or $$\frac{d^2v}{dx^2} - (1 + \cot x)\frac{dv}{dx} = e^x\sin^2 x$$

Putting $\dfrac{dv}{dx} = p,\ \dfrac{d^2v}{dx^2} = \dfrac{dp}{dx}$ in this, we get

$$\frac{dp}{dx} - (1 + \cot x)p = e^x\sin^2 x.$$

$$\text{I.F.} = \exp\left[\int -(1+\cot x)\,dx\right]$$

$$= \exp(-x - \log \sin x) = \frac{e^{-x}}{\sin x}$$

$$\therefore \quad p \cdot \frac{1}{\sin x} e^{-x} = \int e^x \sin^2 x \cdot \frac{1}{\sin x} e^{-x}\,dx + C_1$$

$$= \int \sin x\,dx + C_1 = -\cos x + C_1$$

or $\quad p = \dfrac{dv}{dx} = -\sin x \cos x e^x + C_1 e^x \sin x.$

$$= -\frac{1}{2} e^x \sin 2x + C_1 e^x \sin x.$$

$$\therefore \quad v = -\frac{1}{2}\int e^x \sin 2x\,dx + C_1 \int e^x \sin x\,dx + C_2$$

$$= -\frac{1}{2} \cdot \frac{1}{5} e^x (\sin 2x - 2\cos 2x) + \frac{C_1}{2} e^x (\sin x - \cos x) + C_2$$

or $\quad y = v e^{-x} = -\dfrac{1}{10}(\sin 2x - 2\cos 2x)$

$$+ \frac{C_1}{2}(\sin x - \cos x) + C_2 e^{-x}$$

$$y = v e^{-x} = \frac{-1}{10}(\sin 2x - 2\cos 2x) + \frac{C_1}{2}(\sin x - \cos x) + C_2 e^{-x}$$

$$y = C_2 e^{-x} + \frac{C_1}{2}(\sin x - \cos x) - \frac{1}{10}(\sin 2x - 2\cos 2x) \quad \textbf{Ans.}$$

Ex. 5. $\quad x\dfrac{d^2y}{dx^2} - (2x+1)\dfrac{dy}{dx} + (x+1)y = (x^2+x-1)e^{2x}$

Sol. : The differential equation is

$$\frac{d^2y}{dx^2} - \left(2 + \frac{1}{x}\right)\frac{dy}{dx} + \left(1 + \frac{1}{x}\right)y = \left(x + 1 - \frac{1}{x}\right)e^{2x} \quad \ldots(1)$$

Here, $\quad P = -\left(2 + \dfrac{1}{x}\right),\ Q = 1 + \dfrac{1}{x}$

As, $1 + P + Q = 0$

$\Rightarrow \quad y = e^x$ is an integral of C.F.

Suppose $\quad y = ve^x$ be the complete solution, we have

$$\frac{dy}{dx} = e^x v + e^x \frac{dv}{dx}$$

and $$\frac{d^2y}{dx^2} = 2\frac{dv}{dx}e^x + \frac{d^2v}{dx^2}e^x + ve^x$$

Substituting the value of y, $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ in equation (1)

$$ve^x + 2e^x\frac{dv}{dx} + e^x\frac{d^2v}{dx^2} - \left(2 + \frac{1}{x}\right)\left(v + \frac{dv}{dx}\right)e^x$$

$$+ \left(1 + \frac{1}{x}\right)ve^x = \left(x + 1 - \frac{1}{x}\right)e^{2x}$$

or $e^x\left[\dfrac{d^2v}{dx^2} + \left(2 - 2 - \dfrac{1}{x}\right)\dfrac{dv}{dx} + \left(1 - 2 - \dfrac{1}{x} + 1 + \dfrac{1}{x}\right)v\right]$

$$= \left(x + 1 - \frac{1}{x}\right)e^{2x}$$

or $\quad \dfrac{d^2v}{dx^2} - \dfrac{1}{x}\dfrac{dv}{dx} = \left(x + 1 - \dfrac{1}{x}\right)e^x \quad \ldots(2)$

Put $\dfrac{dv}{dx} = z \Rightarrow \dfrac{d^2v}{dx^2} = \dfrac{dz}{dx}$

$\therefore$ (2) reduces to

$\dfrac{dz}{dx} - \dfrac{1}{x}z = \left(x + 1 - \dfrac{1}{x}\right)e^x$ (Linear diff. eq.)

$$\text{I.F.} = e^{-\int 1/x\,dx} = \frac{1}{x}$$

$\therefore$ Sol. is—

$$Z \cdot \frac{1}{x} = \int\left(x + 1 - \frac{1}{x}\right)e^x \cdot \frac{1}{x}dx + C_1 = \int\left(e^x + \frac{e^x}{x} - \frac{e^x}{x^2}\right)dx + C_1$$

$$= e^x + \frac{e^x}{x} + \int\frac{e^x}{x^2}dx - \int\frac{e^x}{x^2}dx + C_1$$

or $\quad \dfrac{Z}{x} = e^x + \dfrac{e^x}{x} + C_1$

or $\quad Z = x e^x + e^x + C_1 x$

or $\quad \dfrac{dv}{dx} = x e^x + e^x + C_1 x \quad \left(\because Z = \dfrac{dv}{dx}\right)$

or $\quad dv = (x e^x + e^x + C_1 x)\,dx$

on integrating

$$v = e^x(x-1) + e^x + C_1\frac{x^2}{2} + C_2 = x e^x + C_1' x^2 + C_2$$

$\therefore$ Complete solution is

$y = v e^x$

$\Rightarrow \quad y = (x e^x + C'_1 x^2 + C_2) e^x \quad$ **Ans.**

Scanned with

Ex. 6. **Solve** $x\,(x\cos x - 2\sin x)\dfrac{d^2y}{dx^2} + (x^2+2)\sin x\,\dfrac{dy}{dx} - 2\,(x\sin x + \cos x)\,y = 0.$

Sol. : Writing the equation in the standard form, we get

$$\frac{d^2y}{dx^2} + \frac{(x^2+2)\sin x}{x\,(x\cos x - 2\sin x)}\frac{dy}{dx} - \frac{2\,(x\sin x + \cos x)}{x\,(x\cos x - 2\sin x)}\,y = 0 \quad ...(1)$$

By inspection, $2 + 2P \,.\, x + Qx^2 = 0$.

Hence $y = x^2$ is a part of C.P.

Putting $y = v\,x^2$,

$$\frac{dy}{dx} = x^2\frac{dv}{dx} + 2v\,x,\quad \frac{d^2y}{dx^2} = x^2\frac{d^2v}{dx^2} + 4x\frac{dv}{dx} + 2v.$$

Putting these values in equation (1), we obtain

$$\left(x^2\frac{d^2v}{dx^2} + 4x\frac{dv}{dx} + 2v\right) + \frac{(x^2+2)\sin x}{x\,(x\cos x - 2\sin x)}\left(x^2\frac{dv}{dx} + 2v\,x\right) - \frac{2\,(x\sin x + \cos x)}{x\,(x\cos x - 2\sin x)}\,y = 0$$

or
$$\frac{d^2v}{dx^2} + \left[\frac{4}{x} + \frac{(x^2+2)\sin x}{x\,(x\cos x - 2\sin x)}\right]\frac{dv}{dx} = 0.$$

Put
$$\frac{dv}{dx} = p,\quad \frac{d^2v}{dx^2} = \frac{dp}{dx}.$$

$$\therefore \frac{dp}{dx} + \left[\frac{4}{x} + \frac{(x^2+2)\sin x}{x\,(x\cos x - 2\sin x)}\right]p = 0.$$

$$\therefore \frac{dp}{p} + \left[\frac{4}{x} + \frac{(x^2+2)\sin x + \cancel{2x\cos x} - \cancel{2x\cos x}}{x\,(x\cos x - 2\sin x)}\right]dx = 0.$$

or
$$\frac{dp}{p} + \left[\frac{4}{x} - \frac{-x^2\sin x - 2\sin x - 2x\cos x + 2x\cos x}{x\,(x\cos x - 2\sin x)}\right]dx = 0$$

Integrating, we get

$$\log p + 4\log x - \log x\,(x\cos x - 2\sin x) = \log C_1$$

$$\therefore p = \frac{dv}{dx} = C_1\frac{x^2\cos x - 2x\sin x}{x^4} \text{ or } dv = C_1\frac{d}{dx}\left(\frac{\sin x}{x^2}\right).$$

$$v = C_1\left(\frac{\sin x}{x^2}\right) + C_2. \quad \therefore y = C_1\sin x + C_2x^2.$$ **Ans.**

Ex. 7. **Solve** $(x+1)\dfrac{d^2y}{dx^2} - 2\,(x+3)\dfrac{dy}{dx} + (x+5)\,y = e^x.$

Sol. Writing the equation in the standard form, we have

$$\frac{d^2y}{dx^2} - 2\frac{(x+3)}{x+1}\frac{dy}{dx} + \frac{(x+5)}{x+1}\,y = \frac{e^x}{x+1} \quad ...(1)$$

$$\Rightarrow \quad P = -2\frac{(x+3)}{x+1},\; Q = \frac{x+5}{x+1}$$

By inspection $1 + P + Q = 0$. Hence $y = e^x$ is a part of C.F.

Put $y = v\,e^x$.

$$\frac{dy}{dx} = e^x\frac{dv}{dx} + v\,e^x,\quad \frac{d^2y}{dx^2} = e^x\frac{d^2v}{dx^2} + 2e^x\frac{dv}{dx} + v\,e^x.$$

Putting these values in the equation (1), it reduces to

$$e^x\left(\frac{d^2v}{dx^2} + 2\frac{dv}{dx} + v\right) - 2\frac{(x+3)}{x+1}e^x\left(\frac{dv}{dx} + v\right) + \frac{(x+5)}{x+1}v\,e^x = \frac{e^x}{x+1}$$

or the reduced equation is

$$\frac{d^2v}{dx^2} + \left\{2 - 2\frac{(x+3)}{x+1}\right\}\frac{dv}{dx} = \frac{1}{x+1}$$

or
$$\frac{d^2v}{dx^2} - \frac{4}{x+1}\frac{dv}{dx} = \frac{1}{x+1} \quad ...(2)$$

Putting $\dfrac{dv}{dx} = p,\ \dfrac{d^2v}{dx^2} = \dfrac{dp}{dx}$ (2) $\Rightarrow \dfrac{dp}{dx} - \dfrac{4}{x+1}p = \dfrac{1}{x+1}$,

which is a linear equation.

$$\text{I.F.} = \exp\left(\int -\frac{4}{x+1}\,dx\right) = \exp\{-4\log(x+1)\} = \frac{1}{(x+1)^4}$$

$$p\cdot\frac{1}{(x+1)^4} = \int\frac{1}{(x+1)}\cdot\frac{1}{(x+1)^4}\,dx + C_1$$

$$= -\frac{1}{4(x+1)^4} + C_1$$

or $p = -\dfrac{1}{4} + C_1\,(x+1)^4$ or $dv = \left[-\dfrac{1}{4} + C_1\,(x+1)^4\right]dx.$

Integrating, we get

$$v = -\frac{1}{4}x + \frac{C_1}{5}(x+1)^5 + C_2.$$

$$\therefore \quad y = v\,e^x = \left[-\frac{1}{4}x + \frac{C_1}{5}(x+1)^5 + C_2\right]e^x.$$

or $y = C'_1\,(x+1)^5e^x + C_2\,e^x - \dfrac{1}{4}x\,e^x.$ **Ans.**

Ex. 8. **Solve $\sin^2 x \dfrac{d^2y}{dx^2} = 2y$ given that $y = \cot x$ is a solution.**

Sol. Put $y = v \cot x$, $\dfrac{dy}{dx} = \cot x \dfrac{dv}{dx} - v \operatorname{cosec}^2 x$.

$$\frac{d^2y}{dx^2} = \cot x \frac{d^2v}{dx^2} - 2 \operatorname{cosec}^2 x \frac{dv}{dx} + 2v \operatorname{cosec}^2 x \cot x.$$

Putting these values in the given equation, we have

$$\sin^2 x \left[\cot x \frac{d^2v}{dx^2} - 2 \operatorname{cosec}^2 x \frac{dv}{dx} + 2v \operatorname{cosec}^2 x \cot x\right] - 2v \cot x = 0$$

or $\dfrac{d^2v}{dx^2} - \dfrac{2}{\sin x \cos x} \dfrac{dv}{dx} = 0.$

Put $\dfrac{dv}{dx} = p$, $\dfrac{d^2v}{dx^2} = \dfrac{dp}{dx}$.

$\therefore$ $\dfrac{dp}{dx} - \dfrac{2}{\sin x \cos x} p = 0$ or $\dfrac{dp}{p} = 4 \operatorname{cosec} 2x \, dx.$

Integrating, we get

$$\log p = 4 \cdot \frac{1}{2} \log \tan x + \log C_1$$

or $p = C_1 \tan^2 x$ or $\dfrac{dv}{dx} = C_1 \tan^2 x.$

$\therefore$ $v = \int C_1 \tan^2 x \, dx = C_1 \int (\sec^2 x - 1) \, dx$

$= C_1 \tan x - C_1 x + C_2$

or $y = v \cot x = C_1 - C_1 x \cot x + C_2 \cot x.$ **Ans.**

Ex. 9. **$(1 - x^2) \dfrac{d^2y}{dx^2} - x \dfrac{dy}{dx} - a^2 y = 0$, given that $y = e^{a \sin^{-1} x}$ is a solution.**

Sol. : Given differential eq. can be put in standard form

$$\frac{d^2y}{dx^2} - \frac{x}{1 - x^2} \frac{dy}{dx} - \frac{a^2}{1 - x^2} y = 0 \qquad \ldots(1)$$

It is given that $y = e^{a \sin^{-1} x}$ is a solution.

Suppose $y = v e^{a \sin^{-1} x}$ be the complete solution of (1), the

$$\frac{dy}{dx} = \frac{av \, e^{a \sin^{-1} x}}{\sqrt{(1 - x^2)}} + e^{a \sin^{-1} x} \frac{dv}{dx}$$

& $$\frac{d^2y}{dx^2} = \frac{2a \, e^{a \sin^{-1} x}}{\sqrt{1 - x^2}} \cdot \frac{dv}{dx} + e^{a \sin^{-1} x} \frac{d^2v}{dx^2} + av \, e^{a \sin^{-1} x} \left[\frac{a}{(1 - x^2)} + \frac{x}{(1 - x^2)^{3/2}}\right]$$

Putting these values of y, $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ in eq. (1), we get

$$\frac{2a \, e^{a \sin^{-1} x}}{\sqrt{1 - x^2}} \frac{dv}{dx} + e^{a \sin^{-1} x} \frac{d^2v}{dx^2} + a e^{a \sin^{-1} x} \left[\frac{a}{1 - x^2} + \frac{x}{(1 - x^2)^{3/2}}\right] v$$
$$- \frac{x}{1 - x^2} \left[\frac{av}{\sqrt{1 - x^2}} + \frac{dv}{dx}\right] e^{a \sin^{-1} x} - \frac{a^2}{(1 - x^2)} v \, e^{a \sin^{-1} x} = 0$$

or $$\frac{d^2v}{dx^2} + \left(\frac{2a}{\sqrt{1 - x^2}} - \frac{x}{1 - x^2}\right) \frac{dv}{dx} + \left[\frac{a^2}{1 - x^2} + \frac{ax}{(1 - x^2)^{3/2}} - \frac{ax}{(1 - x^2)^{3/2}} - \frac{a^2}{1 - x^2}\right] v = 0$$

or $$\frac{d^2v}{dx^2} + \left(\frac{2a}{\sqrt{1 - x^2}} - \frac{x}{1 - x^2}\right) \frac{dv}{dx} = 0 \qquad \ldots(2)$$

Put $\dfrac{dv}{dx} = z \Rightarrow \dfrac{d^2v}{dx^2} = \dfrac{dz}{dx}$

$\therefore$ (2) reduces to

$$\frac{dz}{dx} + \left(\frac{2a}{\sqrt{1 - x^2}} - \frac{x}{1 - x^2}\right) z = 0$$

or $\dfrac{dz}{z} + \left(\dfrac{2a}{\sqrt{1 - x^2}} - \dfrac{x}{1 - x^2}\right) dx = 0$

or $\log z + 2a \sin^{-1} x + \dfrac{1}{2} \log (1 - x^2) = \log C_1$

or $z \sqrt{1 - x^2} = C_1 e^{-2a \sin^{-1} x}$

$\Rightarrow$ $\dfrac{dv}{dx} = \dfrac{C_1 e^{-2a \sin^{-1} x}}{\sqrt{1 - x^2}} dx$

$\Rightarrow dv = c_1 \dfrac{e^{-2a \sin^{-1} x}}{\sqrt{1 - x^2}} dx$

On integrating

$$v = C_1 \int \frac{e^{-2a \sin^{-1} x}}{\sqrt{1 - x^2}} dx + C_2$$

Put $\qquad e^{-2a\sin^{-1}x} = t$

$$\frac{-2a\, e^{-2a\sin^{-1}x}}{\sqrt{1-x^2}}\,dx = dt$$

$\therefore \quad v = C_1 \int \frac{dt}{-2a} + C_2 = -\frac{C_1}{2a}\,t + C_2$

$\therefore \quad v = -\frac{C_1}{2a}\,e^{-2a\sin^{-1}x} + C_2$

$\therefore$ Complete solution is

$y = v\, e^{a\sin^{-1}x}$

or $\quad y = -\frac{C_1}{2a}\,e^{-a\sin^{-1}x} + C_2\, e^{a\sin^{-1}x}$

or $\quad y = C'_1\, e^{-a\sin^{-1}x} + C_2\, e^{a\sin^{-1}x}$ **Ans.**

Ex. 10. Solve $x^2 \frac{d^2y}{dx^2} + x\frac{dy}{dx} - 9y = 0$

Sol. : Let $y = v\,x^3$,

$\Rightarrow \frac{dy}{dx} = x^3\frac{dv}{dx} + 3v\,x^2,\ \frac{d^2y}{dx^2} = x^3\frac{d^2v}{dx^2} + 6x^2\frac{dv}{dx} + 6vx.$

Putting these values in the given equation, we have

$$x^2\left(x^3\frac{d^2v}{dx^2} + 6x^2\frac{dv}{dx} + 6v\,x\right) + x\left(x^3\frac{dv}{dx} + 3x^2\,v\right) - 9v\,x^3 = 0$$

or $\quad x^5\frac{d^2v}{dx^2} + (6x^4 + x^4)\frac{dv}{dx} = 0$ or $\frac{d^2v}{dx^2} + \frac{7}{x}\frac{dv}{dx} = 0$

Put $\frac{dv}{dx} = p,\ \frac{d^2v}{dx^2} = \frac{dp}{dx}.$

$\therefore \quad \frac{dp}{dx} + \frac{7}{x}p = 0$ or $\frac{dp}{p} = -\frac{7}{x}\,dx.$

Integrating, we have

$\log p = -7\log x + \log C_1\ ;\ \therefore p = C_1 x^{-7}$

or $\quad \frac{dv}{dx} = C_1 x^{-7}\ ;\ \therefore v = -\frac{C_1}{6}x^{-6} + C_2.$

$\therefore \quad y = vx^3 = -\frac{C_1}{6}x^{-3} + C_2 x^3 = Ax^{-3} + C_2 x^3.$ **Ans.**

Ex-39

Exercise - 13(D)

Ex. 1. Solve $\frac{d^2y}{dx^2} - 2\tan x\,\frac{dy}{dx} + 5y = 0$

Sol. : Here, $P = -2\tan x,\ Q = 5,\ X = 0$

Now, $I = Q - \frac{1}{2}\frac{dp}{dx} - \frac{1}{4}p^2 = 5 - \frac{1}{2}(-2\sec^2 x) - \frac{1}{4}(4\tan^2 x)$

$\therefore \quad I = 6$

Now, $\quad u = e^{-\frac{1}{2}\int p\,dx} = e^{-\frac{1}{2}\int -2\tan x\,dx} = e^{\log \sec x} = \sec x$

$\therefore \quad u = \sec x$

If $y = uv = v\sec x$ be the complete solution, v satisfies—

$$\frac{d^2v}{dx^2} + Iv = \frac{X}{u}$$

i.e. $\quad \frac{d^2v}{dx^2} + 6v = 0$

$\therefore \quad$ Auxiliary equation is :

$m^2 + 6 = 0 \Rightarrow \quad m = \pm i\sqrt{6}$

$\therefore \quad v = C_1\cos\left(\sqrt{6}\,x + C_2\right)$

$\therefore \quad$ Complete solution is :

$y = v\sec x$

or $\quad y = C_1 \sec x\cos\left(\sqrt{6}\,x + C_2\right)$ **Ans.**

Ex. 2. Solve $\frac{d^2y}{dx^2} - 2bx\frac{dy}{dx} + b^2x^2y = 0$

Sol. : Here $P = -2bx,\ Q = b^2x^2$ and $X = 0$

Now if we choose

$$u = \exp\left(-\frac{1}{2}\int p\,dx\right) = \exp\left(-\frac{1}{2}\int(-2bx)\,dx\right) = \exp\left(\frac{bx^2}{2}\right)$$

then on substituting vu for y in the original equation the first derivative $\frac{dv}{dx}$ disappears and hence the transformed equation gives,

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{p^2}{4} - \frac{1}{2}\frac{dp}{dx}\right] = X\exp\left[\frac{1}{2}\int p\,dx\right]$$

or $\quad \frac{d^2v}{dx^2} + v\left[b^2x^2 - \frac{4b^2x^2}{4} - \frac{1}{2}(-2b)\right] = 0$

or $\dfrac{d^2v}{dx^2} + v\,[b] = 0$ or $[D^2 + b]\,v = 0$

A.E. is $m^2 + b = 0$, giving $m = \pm i\sqrt{b}$

$\Rightarrow$ Its C.F. is $\left(C_1 \cos\sqrt{b}x + C_2 \sin\sqrt{b}x\right)$

Also P.I. $= \dfrac{1}{(D^2+b)} \cdot 0 = 0$

$\Rightarrow$ $v = C_1 \cos\sqrt{b}x + C_2 \sin\sqrt{b}\,x$

Hence general solution of the given equation is

$$y = vu = \left\{C_1 \cos\sqrt{b}x + C_2 \sin\sqrt{b}x\right\} \cdot e^{\frac{1}{2}bx^2}$$

or $$y = C_1\, e^{\frac{1}{2}bx^2} \cos\left(\sqrt{b}x + C_2\right)$$ **Ans.**

Ex. 3. Solve $\dfrac{d^2y}{dx^2} + \dfrac{1}{x^{1/3}}\dfrac{dy}{dx} + \left[\dfrac{1}{4x^{2/3}} - \dfrac{1}{6x^{4/3}} - \dfrac{6}{x^2}\right]y = 0.$

Sol. : Here $P = \dfrac{1}{x^{1/3}}$, $Q = \dfrac{1}{4x^{2/3}} - \dfrac{1}{6x^{4/3}} - \dfrac{6}{x^2}$ and $X = 0$.

Now if we take

$$u = exp\left(-\frac{1}{2}\int p dx\right) = exp\left(-\frac{1}{2}\int x^{-1/3}\, dx\right) = exp\left(-\frac{3}{4}x^{2/3}\right).$$

then on substituting $v\,u$ for y in the original equation, the first derivative $\dfrac{dv}{dx}$ disappears and hence the transformed equation gives

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = X\, exp\left(\frac{1}{2}\int p\, dx\right) = 0 \; (\because X = 0)$$

or $$\frac{d^2v}{dx^2} + v\left[\left\{\frac{1}{4x^{2/3}} - \frac{1}{6x^{4/3}} - \frac{6}{x^2}\right\} - \frac{1}{4x^{2/3}} - \frac{1}{2}\cdot\left(-\frac{1}{3x^{4/3}}\right)\right] = 0.$$

or $\dfrac{d^2v}{dx^2} - \dfrac{6}{x^2}v = 0$ or $x^2\dfrac{d^2v}{dx^2} - 6v = 0$

This is a homogenous equation; hence in order to solve it, we put $x = e^z$.

$\therefore$ $D(D-1)\,v - 6v = 0$ where $D \equiv d/dz$

or $(D+2)(D-3)\,v = 0.$ ... (2)

A.E. is $(m+2)(m-3) = 0$, giving $m = -2, 3.$

$$\therefore v = C_1 e^{-2z} + C_2 e^{3z} = C_1 \cdot \frac{1}{x^2} + C_2 \cdot x^3 \qquad (\text{as } x = e^z)$$

Therefore the complete solution is

$$y = v\,u = \left[\frac{C_1}{x^2} + C_2\, x^3\right] \cdot exp\left(-\frac{3}{4}x^{2/3}\right)$$

or $$y = e^{-3/4\, x^{2/3}}\left(C_1 x^{-2} + C_2\, x^3\right)$$ **Ans.**

Ex. 4. Solve $\dfrac{d^2y}{dx^2} - 2\tan x\dfrac{dy}{dx} + 3y = 2\sec x$

Sol. : Here $p = -2\tan x$, $Q = 3$, $X = 2\sec x$

Now if we take

$$u = \exp\left(-\frac{1}{2}\int p dx\right) = \exp\left(-\frac{1}{2}\int -2\tan x\, dx\right)$$

$= \exp[\log\sec x] = \sec x.$

then on substituting vu for y in the original equation, we get

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = \frac{X}{u}$$

or $$\frac{d^2v}{dx^2} + v\,[3 - \tan^2 x + \sec^2 x] = 2$$

or $\dfrac{d^2v}{dx^2} + 4v = 2$ or $D^2 + 4v = 2$ where $D = \dfrac{d}{dx}$

A.E. is $m^2 + 4 = 0$, giving $m = \pm 2i$

C.F. $= C_1 \cos 2x + C_2 \sin 2x$

$$\text{P.I.} = \frac{1}{D^2+4}2 = \frac{2}{4\left(1+\frac{D^2}{4}\right)} = \frac{1}{2}\left(1+\frac{D^2}{4}\right)^{-1}$$

$$= \frac{1}{2}\left[1 - \frac{D^2}{4}\right] = \frac{1}{2}$$

$$v = \text{C.F.} + \text{P.I.} = C_1\,[2\cos^2 x - 1] + 2\,C_2 \sin x \cos x + \frac{1}{2}$$

$$y = uv = \left\{C_1\,(2\cos^2 x - 1) + 2\,C_2 \sin x \cos x + \frac{1}{2}\right\}\sec x$$

$$y = 2C_1\left(\text{Cos}x - \frac{1}{2}\sec x\right) + 2C_2 \sin x + \frac{1}{2}\sec x$$

$$y = C'_1\left(\cos x - \frac{1}{2}\sec x\right) + C'_2 \sin x + \frac{1}{2}\sec x$$ **Ans.**

Ex. 5. Solve $\dfrac{d^2y}{dx^2} - \dfrac{2}{x}\dfrac{dy}{dx} + \left(1 + \dfrac{2}{x^2}\right) y = xe^x$

Sol. : Here $p = -2/x$, $Q = \left[1 + \dfrac{2}{x^2}\right]$ and $X = xe^x$

Now if we take

$$u = exp\left(-\frac{1}{2}\int p dx\right) = exp\left[-\frac{1}{2}\int(-2/x)\,dx\right] = e^{\log x} = x$$

then on substituting vu for y in the original equation, it reduce to

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = \frac{xe^x}{x}$$

or $$\frac{d^2v}{dx^2} + v\left[1 + \frac{2}{x^2} - \frac{1}{x^2} - \frac{1}{x^2}\right] = e^x$$

or $$\frac{d^2v}{dx^2} + v = e^x \quad \text{or } (D^2+1)v = 0 \text{ where } D = \frac{d}{dx}$$

A.E. is $m^2 + 1 = 0$, giving $m = \pm i$

$$\text{C.F.} = C_1 \cos x + C_2 \sin x$$

$$\text{P.I.} = \frac{1}{(D^2+1)}e^x = \frac{e^x}{2}$$

$$v = \text{C.F.} + \text{P.I.} = C_1 \cos^2 x + C_2 \sin x + \frac{e^x}{2}$$

Therefore the complete selection is given by

$$y = vu = C_1\, x \cos x + C_2\, x \sin x + \frac{xe^x}{2}$$ **Ans.**

Ex. 6. Solve $\dfrac{d^2y}{dx^2} - 4x\dfrac{dy}{dx} + (4x^2 - 3)\,y = e^{x^2}$.

Sol. : Here $p = -4x$, $Q = (4x^2 - 3)$ and $X = e^{x^2}$.

Now if we take

$$u = exp\left[-\frac{1}{2}\int p dx\right] = exp\left[-\frac{1}{2}\int(-4x\,dx)\right] = e^{x^2},$$

then on substituting vu for y in the original equation, it reduce to normal form as

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = X\, exp\left(\frac{1}{2}\int p dx\right)$$

$$= e^{x^2}.\,exp\left\{\frac{1}{2}\int(-4x dx)\right\} = exp\,(x^2 - x^2) = 1$$

or $$\frac{d^2v}{dx^2} + v\left[(4x^2 - 3) - \frac{16x^2}{4} - \frac{1}{2}(-4)\right] = 1$$

or $$\frac{d^2v}{dx^2} + v\,[-1] = 1$$

or $$(D^2 - 1)\,v = 1 \text{ where } D \equiv \frac{d}{dx} \qquad \ldots (1)$$

A.E. is $m^2 - 1 = 0$, giving $m = \pm 1$, C.F. $= C_1 e^x + C_2 e^{-x}$

$$\text{P.I.} = \frac{1}{D^2 - 1}(1) = -\left[(1 - D^2)^{-1}\right] 1 = -1.$$

$$\therefore \quad v = C_1 e^x + C_2 e^{-x} - 1.$$

Therefore the complete solution is

$$y = vu = (C_1 e^x + C_2 e^{-x} - 1)\, e^{x^2}.$$ **Ans.**

Ex. 7. Solve $\left(\dfrac{d^2y}{dx^2} + y\right)\cot x + 2\left(\dfrac{dy}{dx} + \tan x\right) = \sec x$

Sol. : The equation can be written as

$$\frac{d^2y}{dx^2} + 2\tan x\frac{dy}{dx} + (2\tan^2 x + 1)\,y = \sec x \tan x$$

Here $P = 2\tan x$, $Q = 2\tan^2 x + 1$, $X = \sec x \tan x$

Now if we take $u = exp\left\{-\dfrac{1}{2}\int p dx\right\}$

$$= exp\left\{-\frac{1}{2}\int 2\tan x\, dx\right\} = e^{-\log \sec x}$$

$$= \frac{1}{\sec x} = \cos x$$

then on substituting vu for y in the original equation, it gives

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = \frac{X}{u}$$

or $$\frac{d^2v}{dx^2} + v\,[2\tan^2 x + 1 - \tan^2 x - \sec^2 x] = \sec^2 x \tan x$$

or $$\frac{d^2v}{dx^2} = \sec^2 x \tan x \quad \text{where} \quad D \equiv \frac{d}{dx}$$

A.E. is $m^2 = 0$, giving $m = 0, 0$

$$\text{C.F.} = (C_1 + C_2 x)$$

$$\text{P.I.} = \frac{1}{D^2} \times \sec^2 x \tan x = \frac{1}{2}\tan x$$

$$v = \text{C.F.} + \text{P.I.} = \left(C_1 + C_2 x + \frac{1}{2}\tan x\right)$$

$$y = uv = \frac{1}{2}\sin x + C_1 \cos x + C_2 x \cos x \qquad \textbf{Ans.}$$

Ex. 8. Solve $x^2 \dfrac{d^2y}{dx^2} + (x - 4x^2)\dfrac{dy}{dx} + (1 - 2x + 4x^2)\, y = 0.$

Sol. : Here $P = \dfrac{x - 4x^2}{x^2}$, $Q = \dfrac{1 - 2x + 4x^2}{x^2}$ and $X = 0$.

Now if we take

$$u = \exp\left[-\frac{1}{2}\int p\,dx\right] = \exp\left[-\frac{1}{2}\int \frac{x - 4x^2}{x^2}\,dx\right] = x^{-1/2} \cdot e^{2x},$$

then on substituting vu for y in the original equation, it changes to

$$\frac{d^2v}{dx^2} + v\left[\frac{1 - 2x + 4x^2}{x^2} - \frac{1}{4} \cdot \left(\frac{1 - 4x}{x}\right)^2 + \frac{1}{2} \cdot \frac{1}{x^2}\right] = 0$$

or $\quad D^2 v + (5/4x^2)\, v = 0$ or $x^2 D^2 v + \dfrac{5}{4} v = 0.$

Putting $x = e^z$, we get new equation as

$$D'(D' - 1)\, v + \frac{5}{4} v = 0, \text{ or } (4D'^2 - 4D' + 5)\, v = 0,\ D' \equiv d/dz$$

A.E. is $4m^2 - 4m + 5 = 0$, giving $m = \dfrac{1}{2} \pm i$.

C.F. $= C_1 e^{z/2} \cos(z + C_2) = C_1 \sqrt{x} \cos(\log x + C_2)$.

$\therefore \quad v = C_1 \sqrt{x} \cos(\log x + C_2)$.

Therefore the complete solution is

$$y = vu = C_1 \sqrt{x} \cos(\log x + C_2) \cdot \frac{1}{\sqrt{x}} e^{2x} \qquad \textbf{Ans.}$$

Ex. 9. Solve $x^2 \dfrac{d^2y}{dx^2} - 2x(3x - 2)\dfrac{dy}{dx} + 3x(3x - 4)\, y = e^{3x}.$

Sol. : The equation can be written as

$$\frac{d^2y}{dx^2} - \frac{2(3x - 2)}{x}\frac{dy}{dx} + \frac{3}{x}(3x - 4)\, y = \frac{e^{3x}}{x^2}$$

Here $P = \dfrac{-2(3x - 2)}{x}$, $Q = \dfrac{3}{x}(3x - 4)$ and $X = \dfrac{e^{3x}}{x^2}$

Now if we take

$$u = \exp\left[-\frac{1}{2}\int p\,dx\right] = \exp\left[-\frac{1}{2}\int\left(\frac{4}{x} - 6\right)dx\right] = \frac{e^{3x}}{x^2}$$

then on substituting vu for y in the original equation, it changes to

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = X \exp\left(\frac{1}{2}\int p\,dx\right)$$

or $\quad \dfrac{d^2v}{dx^2} + v\left[\dfrac{3}{x}(3x-4) - \dfrac{4}{x^2}\dfrac{(3x - 2)^2}{4} - \dfrac{1}{2} \cdot \left(-\dfrac{4}{x^2}\right)\right]$

$$= \frac{e^{3x}}{x^2} \cdot \exp\left[\frac{1}{2}\int\left\{-2\left(3 - \frac{2}{x}\right)dx\right\}\right]$$

or $\quad \dfrac{d^2v}{dx^2} - \dfrac{2}{x^2} v = 1$ or $(x^2 D^2 v - 2v) = x^2$.

$\therefore \quad v = C_1 x^2 + \dfrac{C_2}{x} + \dfrac{x^2}{3} \log x.$

Therefore the complete solution is

$$y = vu = \left\{C_1 x^2 + \frac{C_2}{x} + \frac{x^2}{3}\log x\right\} \cdot \frac{e^{3x}}{x^2} \qquad \textbf{Ans.}$$

Ex. 10. Solve $\dfrac{d^2y}{dx^2} + 2x\dfrac{dy}{dx} + (x^2 + 5)\, y = e^{-x^2/2}.$

Sol. : Here $P = 2x$, $Q = x^2 + 5$ and $X = xe^{-x^2/2}$.

Now if we take

$$u = \exp\left[-\frac{1}{2}\int p\,dx\right] = \exp\left[-\frac{1}{2}\int 2x\,dx\right] = e^{-x^2/2},$$

then on substituting vu for y in the original equation, the first derivative dv/dx disappears and hence the transformed equation gives

$$\frac{d^2v}{dx^2} + v\left[Q - \frac{P^2}{4} - \frac{1}{2}\frac{dP}{dx}\right] = X \exp\left(\frac{1}{2}\int p\,dx\right)$$

or $\quad \dfrac{d^2v}{dx^2} + v\left[x^2 + 5 - \dfrac{4x^2}{4} - \dfrac{1}{2} \cdot 2\right] = x\, e^{-x^2/2} \cdot e^{x^2/2} = x$

or $\quad \dfrac{d^2v}{dx^2} + 4v = x$ or $(D^2 + 4)\, v = x$ where $D \equiv \dfrac{d}{dx}$

A.E. is $m^2 + 4 = 0$, giving $m = \pm 2i$.

$\therefore \quad$ C.F. $= C_1 \cos 2x + C_2 \sin 2x$. or $C_1 \cos(2x + C_2)$

$$\text{P.I.} = \frac{1}{D^2 + 4} x = \frac{1}{4}\left(1 - \frac{D^2}{4}\right)x = \frac{x}{4}$$

$\therefore \quad v = \text{C.F.} + \text{P.I.}$

Scanned with

Therefore the general solution of the given equation is

$$y = vw = \left\{ C_1 \cos 2x + C_2 \sin 2x + \frac{x}{4} \right\} \cdot e^{-x^2/2}$$

$$y = e^{-x^2/2} \left[C_1 \cos (2x + C_2) + \frac{x}{4} \right] \quad \textbf{Ans.}$$

EX-40

Exercise - 13(E)

Ex. 1. Solve $\frac{d^2y}{dx^2} + \cot x \frac{dy}{dx} + 4y \operatorname{cosec}^2 x = 0.$

Or $\sin^2 x y'' + \sin x \cos x y' + 4y = 0.$

Sol. : Comparing this equation with the equation

$$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = X, \text{ we get}$$

$$P = \cot x,\ Q = 4 \operatorname{cosec}^2 x \text{ and } X = 0.$$

Now changing the independent variable x to z, the given equation is transformed into

$$\frac{dy^2}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1 = 0, \quad \text{... (1)}$$

where $P_1 = \dfrac{\frac{d^2z}{dx^2} + P\frac{dz}{dx}}{(dz/dx)^2} = \dfrac{\frac{d^2z}{dx^2} + \cot x \frac{dz}{dx}}{(dz/dx)^2}$

$$Q_1 = \frac{Q}{(dz/dx)^2} = \frac{4\operatorname{cosec}^2 x}{(dz/dx)^2} \text{ and } X_1 = \frac{0}{(dz/dx)^2} = 0.$$

Apply here the second method, i.e. when Q is constant, equal to 1 say,

$$\therefore \quad \frac{4\operatorname{cosec}^2 x}{(dz/dx)^2} = 1 \text{ or } \frac{dz}{dx} = 2 \operatorname{cosec} x$$

$$dz = 2 \operatorname{cosec} x\, dx \Rightarrow z = 2 \log \tan \frac{x}{2}$$

or

$$P_1 = \frac{\frac{d^2z}{dx^2} + \cot x \frac{dz}{dx}}{(dz/dx)^2} = \frac{-2 \operatorname{cosec} x \cot x + 2 \operatorname{cosec} x \cot x}{(dz/dx)^2} = 0$$

Thus we see that by this choice of z, P_1 also vanishes.

Now putting the values of P_1, Q_1 in (1), we get

$$\frac{d^2y}{dz^2} + y = 0 \text{ or } (D^2 + 1) y = 0, \text{ where } D = \frac{d}{dz}$$

$$y = C_1 \cos z + C_2 \sin z \text{ or } y = C_1 \cos (z + C_2)$$

$$= C_1 \cos \left(2 \log \tan \frac{x}{2}\right) + C_2 \sin \left(2 \log \tan \frac{x}{2}\right)$$

$$y = C_1 \cos \left(2 \log \tan \frac{x}{2} + C_2\right) \quad \textbf{Ans.}$$

This is the required general solution

Ex. 2. Solve $x^6 \frac{d^2y}{dx^2} + 3x^5 \frac{dy}{dx} + a^2 y = \frac{1}{x^2}$

Sol. : The given equation can be written as

$$\frac{d^2y}{dx^2} + \frac{3}{x}\frac{dy}{dx} + \frac{a^2}{x^6} y = \frac{1}{x^8}$$

Here $P = \frac{3}{x}$, $Q = \frac{a^2}{x^6}$ and $X = \frac{1}{x^8}$

Changing the independent variable x to z, the given equation is transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1 \quad \text{... (1)}$$

where P_1, Q_1, X_1 are find as Ex. No. 1.

Now choose z in such a way that $Q_1 = a^2$.

$$\therefore \quad a^2 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{a^2}{x^6 \left(\frac{dz}{dx}\right)^2} \text{ or } \frac{dz}{dx} = \frac{1}{x^3}, \Rightarrow z = \frac{1}{2x^2}$$

With this substitution, the equation (1) is changed into

$$\frac{d^2y}{dz^2} + \frac{\frac{d^2z}{dx^2} + \frac{3}{x}\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} \cdot \frac{dy}{dz} + a^2 y = \frac{x^{-8}}{\left(\frac{dz}{dx}\right)^2}$$

or

$$\frac{d^2y}{dz^2} + \frac{-\frac{3}{x^4} + \frac{3}{x} \cdot \frac{1}{x^3}}{\left(\frac{dz}{dx}\right)^2} \cdot \frac{dy}{dz} + a^2 y = \frac{x^{-8}}{\frac{1}{x^6}}$$

or

$$\frac{d^2y}{dz^2} + a^2 y = \frac{1}{x^2} = -2z \qquad \left(\because z = -\frac{1}{2x^2}\right)$$

or $\qquad (D^2 + a^2)\, y = -2z$ where $D \equiv d/dz$. $\qquad$ ... (1)

A.E. is $m^2 + a^2 = 0$, $m = 0 \pm ai$.

$$\text{C.F.} = C_1 \cos az + C_2 \sin az = C_1 \cos (az - C_2)$$

$$= C_1 \cos \left(C^2 - \frac{a}{2x^2}\right)$$

$$\text{P.I.} = \frac{1}{D^2 + a^2}(-2z) = \frac{1}{a^2\left(1 + \frac{D^2}{a^2}\right)}(-2z)$$

$$= \frac{1}{a^2}\left\{\left[\left(1 - \frac{D^2}{a^2}\right)\right]\right\}(-2z) = -\frac{2z}{a^2} = \frac{1}{a^2 x^2}$$

Hence the complete solution is given by

$$y = \text{C.F.} + \text{P.I.} = C_1 \cos\left(C^2 - \frac{a}{2x^2}\right) + \frac{1}{a^2 x^2}$$ **Ans.**

Ex. 3. Solve $\dfrac{x d^2 y}{dx^2} - \dfrac{dy}{dx} - 4x^3 y = 8x^3 \sin x^2$.

Sol. : The given equation can be written as

$$\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx} - 4x^2 y = 8x^2 \sin x^2.$$

Here $P = -\dfrac{1}{x}$, $Q = -4x^2$, $X = 8x^2 \sin x^2$.

Changing the independent variable x to z, the given equation is transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1 \qquad \text{... (1)}$$

where P_1, Q_1, X_1 are find as Ex. No. 1

Now choose z in such a way that $Q_1 = -1$.

$$\therefore \quad -1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = -\frac{4x^2}{\left(\frac{dz}{dx}\right)^2}, \text{ or } \frac{dz}{dx} = 2x \text{ or } z = x^2.$$

With this substitution, the equation (1) is changed into

$$\frac{d^2y}{dz^2} + \frac{2 - \frac{1}{x} \cdot 2x}{\left(\frac{dz}{dx}\right)^2}\frac{dy}{dz} + (-1)\, y = \frac{8x^2 \sin x^2}{4x^2} = 2 \sin z$$

or $\quad \dfrac{d^2y}{dz^2} - y = 2 \sin z$ or $(D^2 - 1)\, y = 2 \sin z$, where $D \equiv \dfrac{d}{dz}$

A.E. is $m^2 - 1 = 0$, $m = \pm 1 \Rightarrow$ C.F. $= C_1 e^z + C_2 e^{-z}$

Now $\text{P.I.} = \dfrac{1}{D^2 - 1} 2 \sin z = \dfrac{1}{(-1) - 1} 2 . \sin z = -\sin x^2$

Hence the complete solution is given by

$$y = \text{C.F.} + \text{P.I.} = C_1 e^z + C_2 e^{-z} - \sin x^2$$

$$= C_1 e^{x^2} + C_2 e^{-x^2} - \sin x^2.$$ **Ans.**

Ex. 4. Solve $\dfrac{d^2y}{dx^2} + (\tan x - 3 \cos x)\dfrac{dy}{dx} + 2y \cos^2 x = \cos^4 x$.

Sol. : $P = (\tan x - 3 \cos x)$, $Q = 2 \cos^2 x$ and $X = \cos^4 x$.

In the transformed equation, choose z in such a way that

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = 2 \text{ or } \frac{2 \cos^2 x}{\left(\frac{dz}{dx}\right)^2} = 2$$

$$\Rightarrow \quad \frac{dz}{dx} = \cos x \text{ and } z = \sin x, \ \frac{d^2z}{dx^2} = -\sin x$$

The reduced equation is

$$\frac{d^2y}{dz^2} + \frac{-\sin x + (\tan x - 3 \cos x)\cos x}{\cos^2 x}\frac{dy}{dz} + 2y = \frac{\cos^4 x}{\cos^2 x}$$

or $\quad (D^2 - 3D + 2)\, y = \cos^2 x = 1 - z^2$, where $D \equiv \dfrac{d}{dz}$

A.E. is $m^2 - 3m + 2 = 0$, giving $m = 2$, $m = 1$

$$\text{C.F.} = C_1 e^z + C_2 e^{2z} = C_1 e^{\sin x} + C_2 e^{2 \sin x}$$

$$\text{P.I.} = \frac{1}{(D-2)(D-1)}(1 - z^2) = \frac{1}{2\left(1 - \frac{D}{2}\right)(1 - D)}(1 - z^2)$$

$$= \frac{1}{2}\left\{\left(1 - \frac{D}{2}\right)^{-1} . (1 - D)^{-1}\right\}(1 - z^2)$$

$$= \frac{1}{2}\left[\left(1 + \frac{D}{2} + \frac{D^2}{4} + \dots\right)(1 + D + D^2 + \dots)\right](1 - z^2)$$

$$= \frac{1}{2}\left(1 + \frac{3D}{2} + \frac{D^2}{2} + \frac{9D^2}{4} + \dots\right)(1 - z^2)$$

$$= \frac{1}{2}(1 - z^2) - \frac{3z}{2} - \frac{7}{4}$$

$$= -\frac{z^2}{2} - \frac{3z}{2} - \frac{5}{4} = -\frac{\sin^2 x}{2} - \frac{3 \sin x}{2} - \frac{5}{4}$$

Hence the G solution is

$$y = C_1 e^{\sin x} + C_2 e^{2\sin x} - \frac{\sin^2 x}{2} - \frac{3\sin x}{2} - \frac{5}{4}$$ **Ans.**

Ex. 5. **Solve** $\dfrac{d^2y}{dx^2} + \left(1 - \dfrac{1}{x}\right)\dfrac{dy}{dx} + 4x^2 y\, e^{-2x} = 4(x^2 + x^3)\, e^{-3x}$.

Sol. : Here $P = \left(1 - \dfrac{1}{x}\right)$, $Q = 4x^2 e^{-2x}$, $X = 4x^2 (1 + x)\, e^{-3x}$.

Changing the independent variable x to z, the given equation transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1 \qquad \dots (1)$$

where P_1, Q_1, R_1 are find as Ex. No. 1

Now choose z in such a way that $Q = 4$

$$\therefore \quad 4 = \frac{4x^2 e^{-2x}}{\left(\frac{dz}{dx}\right)^2} \text{ or } \frac{dz}{dx} = x e^{-x},$$

so that $z = -e^{-x}(1+x)$ and $\dfrac{d^2z}{dx^2} = e^{-x}(1+x)$

With this substitution, the equation (1) is changed into

$$\frac{d^2y}{dz^2} + \frac{\frac{d^2z}{dx^2} + \left(1 - \frac{1}{x}\right)\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} \cdot \frac{dy}{dz} + 4y = \frac{X}{\left(\frac{dz}{dx}\right)^2} = \frac{4x^2(1+x)\,e^{-3x}}{x^2 \cdot e^{-2x}}$$

$$\text{or } \frac{d^2y}{dz^2} + \frac{e^{-x}(1-x) + \frac{(x-1)}{x} \cdot x e^{-x}}{\left(\frac{dz}{dx}\right)^2} \cdot \frac{dy}{dz} + 4y = 4e^{-x}(1+x) = -4z$$

or $D^2y + 4y = -4z$ or $(D^2 + 4)\,y = -4z$, where $D \equiv d/dz$.

A.E. is $m^2 + 4 = 0$, giving $m = \pm 2i$.

$$\begin{aligned} \text{C.F.} &= C_1 \cos 2z + C_2 \sin 2z, \text{ or C.F.} = C_1 \cos(2z + C_2) \\ &= C_1 \cos\{-2e^{-x}(1+x)\} + C_2 \sin\{-2e^{-x}(1+x)\} \\ &= C_1 \cos\{2e^{-x}(1+x)\} - C_2 \sin\{2e^{-x}(1+x)\} \end{aligned}$$

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 4}(-4z) = -\frac{1}{1 + \frac{1}{4}D^2}\, z = -\left(1 - \frac{D^2}{4}\right) z \\ &= -z = e^{-x}(1+x). \end{aligned}$$

Hence the complete solution is given by

$$\begin{aligned} y &= \text{C.F.} + \text{P.I.} \\ &= C_1 \cos\{2e^{-x}(1+x)\} - C_2 \sin\{2e^{-x}(1+x)\} + e^{-x}(1+x) \end{aligned}$$

Or $y = C_1 \cos[C_2 - 2e^{-x}(1+x)] + (1+x)\,e^{-x}$ **Ans.**

Ex. 6. **Solve** $\dfrac{d^2y}{dx^2} - \dfrac{1}{x}\dfrac{dy}{dx} + 4x^2 y = x^4$

Sol. : Comparing this equation with equation

$$\frac{d^2y}{dx^2} + \frac{P\,dy}{dx} + Qy = X, \text{ we get}$$

$$P = -\frac{1}{x}, \; Q = 4x^2, \; X = x^4$$

Now changing the independent variable x to z, the given equation is transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1 \qquad \dots (1)$$

$$\textit{where } P_1 = \frac{\frac{d^2z}{dx^2} + P\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} = \frac{\frac{d^2z}{dx^2} - \frac{1}{x}\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2},$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{4x^2}{\left(\frac{dz}{dx}\right)^2}, \text{ and}$$

$$X_1 = \frac{X}{\left(\frac{dz}{dx}\right)^2} = \frac{x^4}{\left(\frac{dz}{dx}\right)^2}$$

Apply here the second method i.e. when Q is constant, equal to 1

$$\therefore \quad \frac{4x^2}{\left(\frac{dz}{dx}\right)^2} = 1 \text{ or } 4x^2 = \left(\frac{dz}{dx}\right)^2$$

$$\text{or} \quad \frac{dz}{dx} = 2x \text{ or } dz = 2x\,dx$$

$$\Rightarrow \quad z = x^2$$

$$\text{Now } P_1 = \frac{\frac{d^2z}{dx^2} - \frac{1}{x}\frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} = \frac{2 - \frac{1}{x} \times 2x}{(2x^2)} = 0$$

Thus we see that by this choice of Z, P, also vanishes.

Now putting the values of P_1, Q_1 in (1), we get

$$= \frac{d^2y}{dz^2} + y = \frac{z}{4} \text{ or } (D^2 + 1)\, y = \frac{z}{4}, \text{ where } D \equiv \frac{d}{dz}$$

A.E. is $m^2 + 1 = 0$, $m = \pm 1$

$$\begin{aligned} \text{C.F.} &= C_1 \cos z + C_2 \sin z \\ &= C_1 \cos x^2 + C_2 \sin x^2 \\ &= C_1 \cos (x^2 + C_2) \end{aligned}$$

$$\text{P.I.} = \frac{1}{(D^2 - 1)} \left(\frac{z}{4}\right) = \frac{z}{4} = \frac{x^2}{4}$$

Hence the complete solution is given by

$$y = \text{C.F.} + \text{P.I.} = C_1 \cos (x^2 + C_2) + \frac{x^4}{4}$$ **Ans.**

Ex. 7. ***Solve*** $x \dfrac{d^2x}{dy^2} + (4x^2 - 1) \dfrac{dy}{dx} + 4x^3 y = 2x^3$

Sol. : $P = \dfrac{4x^2 - 1}{x}$, $Q = 4x^2$, $X = 2x^2$.

In the transformed equation, choose z in such a way that

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = 1 \text{ or } \frac{4x^2}{\left(\frac{dz}{dx}\right)^2} = 1$$

$$\Rightarrow \quad \frac{dz}{dx} = 2x \Rightarrow z = x^2 \text{ and } \frac{d^2z}{dx^2} = 2$$

The reduced equation is

$$\frac{d^2y}{dz^2} + \frac{2 + \left(\frac{4x^2 - 1}{x}\right) \cdot 2x}{4x^2} \cdot \frac{dy}{dz} + y = \frac{2x^2}{(2x)^2}$$

or $$\frac{d^2y}{dz^2} + 2\frac{dy}{dz} + y = \frac{1}{2}$$

or $$(D^2 + 2D + 1)\, y = \frac{1}{2} \text{ where } D \equiv \frac{d}{dz}$$

A.E. is $(m + 1)^2 = 0$, giving $m = -1$ two times.

$\Rightarrow$ C.F. $= (C_1 + C_2 z)\, e^{-z}$. Also P.I. $= \dfrac{1}{(1 + D)^2} \left(\dfrac{1}{2}\right) = \dfrac{1}{2}$,

Hence the G. solution is given by

$$y = \text{C.F.} + \text{P.I.} = (C_1 + C_2 z)\, e^{-z} + \frac{1}{2} = (C_1 + C_2 x^2)\, e^{-x^2} + \frac{1}{2}.$$ **Ans.**

Ex. 8. **Solve** $\dfrac{d^2y}{dx^2} - \cot x \dfrac{dy}{dx} - \sin^2 x \,.\, y = \cos x - \cos^3 x$.

Sol. : Here $P = -\cot x$, $Q = -\sin^2 x$ and $X = \cos x - \cos^3 x$.

Changing the independent variable x to z, the given equation is transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1, \qquad \text{... (1)}$$

Where P_1, Q_1, X_1 are find as Ex. No. 1

Now choose z in such a way that $Q_1 = -1$

$$\therefore -1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} \text{ or } 1 = \frac{\sin^2 x}{\left(\frac{dz}{dx}\right)^2} \text{ or } \frac{dz}{dx} = \sin x.$$

With this substitution, the equation (1) is changed to

$$\frac{d^2y}{dz^2} + \frac{\cos x - \cot x \,.\, \sin x}{\sin^2 x} \frac{dy}{dz} - y = \frac{\cos x - \cos^3 x}{\sin^2 x}$$

or $$\frac{d^2y}{dz^2} - y = \frac{-(z - z^3)}{1 - z^2} = -z$$

or $$(D^2 - 1)\, y = -z, \qquad \text{where } D \equiv \frac{d}{dz}$$

A.E. is $m^2 - 1 = 0$, $m = \pm 1$.

$$\text{C.F.} = C_1 e^z + C_2 e^{-z} = C_1 e^{-\cos x} + C_2 e^{\cos x}.$$

$$\text{Also P.I.} = \frac{1}{D^2 - 1}(-z) = \frac{1}{1 - D^2} z = \{(1 - D^2)^{-1}\}\, z$$

$$= \{(1 + D^2 + D^4 + \ldots)\} \,.\, z = z = -\cos x.$$

Hence the complete solution is given by

$$y = \text{C.F.} + \text{P.I.} = C_1 e^{-\cos x} + C_2 e^{\cos x} - \cos x.$$ **Ans.**

Ex. 9. **Solve** $\dfrac{d^2y}{dx^2} + (3 \sin x - \cot x) \dfrac{dy}{dx} + 2 \sin^2 x \,.\, y = e^{-\cos x} \sin^2 x$.

Sol. : Here $P = (3 \sin x - \cot x)$, $Q = 2 \sin^2 x$ and $X = e^{-\cos x} \sin^2 x$. Chaning the independent variable x to z, the given equation is transformed into

$$\frac{d^2y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = X_1. \qquad \text{... (1)}$$

Where $P_1 = \dfrac{\dfrac{d^2z}{dx^2} + P\dfrac{dz}{dx}}{\left(\dfrac{dz}{dx}\right)^2} = \dfrac{\dfrac{d^2z}{dx^2} + (3\sin x - \cot x)\dfrac{dz}{dx}}{\left(\dfrac{dz}{dx}\right)^2}$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{2\sin^2 x}{\left(\frac{dz}{dx}\right)^2}, \text{ and } X_1 = \frac{e^{-\cos x}\sin^2 x}{\left(\frac{dz}{dx}\right)^2}$$

Now choose z in such a way that $Q_1 = 2$ (say).

$$\therefore \frac{2\sin^2 x}{\left(\frac{dz}{dx}\right)^2} = 2 \text{ or } \frac{dz}{dx} = \sin x, \text{ or } z = -\cos x.$$

With this substitution, the equation (1) is changed to

$$\frac{d^2y}{dz^2} + \frac{\cos x + (3\sin x - \cot x)\sin x}{\sin^2 x}\frac{dy}{dz} + 2y = \frac{e^z \cdot \sin^2 x}{\sin^2 x}$$

or $\dfrac{d^2y}{dz^2} + 3\dfrac{dy}{dz} + 2y = e^z$

or $(D^2 + 3D + 2)\,y = e^z$, where $D \equiv \dfrac{d}{dz}$

A.E. is given by $m^2 + 3m + 2 = 0 \Rightarrow m = -2, -1.$

Hence C.F. $= (C_1 e^{-z} + C_2 e^{-2z})$

Also P.I. $= \dfrac{1}{D^2 + 3D + 2}e^z = \dfrac{1}{1 + 3 + 2}e^z = \dfrac{1}{6}e^z.$

Hence the complete solution is given by

$$y = \text{C.F.} + \text{P.I.} = C_1 e^{-z} + C_2 e^{-2z} + \frac{e^z}{6}$$

$$= C_1 e^{\cos x} + C_2 e^{2\cos x} + \frac{e^{-\cos x}}{6} \quad \textbf{Ans.}$$

Ex-41

Exercise 13(F)

Solve by the method of variation of parameters

Ex. 1. Solve $\dfrac{d^2y}{dx^2} + y = \operatorname{cosec} x.$

Sol. : C.F. $= A\cos x + B\sin x$. Now assume A and B as the functions of x in such a way that the given a equation is satisfied by $y = A\cos x + B\sin x$.

$$\therefore \frac{dy}{dx} = -A\sin x + B\cos x + \cos x\frac{dA}{dx} + \sin x\frac{dB}{dx}$$

$$= -A\sin x + B\cos x, \quad \ldots (1)$$

provided $\cos x\dfrac{dA}{dx} + \sin x\dfrac{dB}{dx} = 0.$ ... (2)

Also $\dfrac{d^2y}{dx^2} = -A\cos x - B\sin x - \sin x\dfrac{dA}{dx} + \cos x\dfrac{dB}{dx}$

$$= -\sin x\frac{dA}{dx} + \cos x\frac{dB}{dx} - y \quad \ldots (3)$$

Now putting the values of $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ in the original equation, we get

$$= -\sin x\frac{dA}{dx} + \cos x\frac{dB}{dx} = \operatorname{cosec} x. \quad \ldots (4)$$

Solving (2) and (4), we again get

$$\frac{dA}{dx} = -1 \text{ and } \frac{dB}{dx} = \cot x.$$

Integrating these, $A = -x + C_1$ and $B = \log\sin x + C_2$.

Hence the complete solution is

$$y = (C_1 - x)\cos x + (C_2 + \log\sin x)\sin x$$

$= C_1\cos x + C_2\sin x - x\cos x + \sin x.\log\sin x.$ **Ans.**

Ex. 2. Solve $\dfrac{d^2y}{dx^2} + n^2y = \sec nx.$

Sol. : The complementary function is

$$y = A\cos nx + B\sin nx$$

Now assume A and B as the function of x in such a way that the given equation is satisfied by

$$y = A\cos nx + B\sin nx$$

$$\therefore \frac{dy}{dx} = -An\sin nx + Bn\cos nx + \cos nx \times \frac{dA}{dx} + \sin nx\frac{dB}{dx} \quad \ldots (1)$$

$$= -An\sin nx + Bn\cos nx,$$

assuming $\cos nx\dfrac{dA}{dx} + \sin nx\dfrac{dB}{dx} = 0.$ ... (2)

Also $\dfrac{d^2y}{dx^2} = -An^2\cos nx - Bn^2\sin nx - \dfrac{dA}{dx}n\sin nx$

$$+ \frac{dB}{dx}n\cos nx \quad \ldots (3)$$

Scanned with

Now putting the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the original equation, we get

$$\frac{-dA}{dx} \cdot n \sin nx + \frac{dB}{dx} n\cos nx = \sec nx. \quad \text{... (4)}$$

Solving (2) and (4), we again get

$$\frac{n dA}{dx} = -\tan nx \text{ and } \frac{n dB}{dx} = 1$$

Integrating $A = -\frac{1}{n}\int \tan nx \, dx + C_1 = \frac{1}{n^2} \log \cos nx + C_1$

and $B = \frac{1}{n}\int dx + C_2 = \frac{x}{n} + C_2$

Hence the complete solution of the given equation is

$$y = \left[\frac{1}{n^2} \log \cos nx + C_1\right] \cos nx + \left[\frac{x}{n} + C_2\right] \sin nx$$

$$y = C_1 \cos nx + C_2 \sin nx + \frac{x}{n} \sin nx + \frac{\cos nx}{n^2} \log \cos nx \quad \textbf{Ans.}$$

Ex. 3. **Solve** $\frac{d^2y}{dx^2} - y = \frac{2}{1 + e^x}$

Sol. : C.F. $= Ae^x + Be^{-x}$. Now assume A and B as the functions of x in such a way that the given equation is satisfied by $y = Ae^x + Be^{-x}$

$$\therefore \quad \frac{dy}{dx} = Ae^x - Be^{-x} + e^x \frac{dA}{dx} + e^{-x} \frac{dB}{dx}$$

$$= Ae^x - Be^{-x} \quad \text{... (1)}$$

resuming $e^x \frac{dA}{dx} + e^{-x} \frac{dB}{dx} = 0$ ... (2)

$$\text{Also } \frac{d^2y}{dx^2} = Ae^x + Be^{-x} + e^x \frac{dA}{dx} - e^{-x} \frac{dB}{dx} \quad \text{... (3)}$$

Now putting these values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the original equation, we get

$$= e^x \frac{dA}{dx} - e^{-x} \frac{dB}{dx} = \frac{2}{1 + e^x} \quad \text{... (4)}$$

Solving (2) and (4), we get

$$e^x \frac{dA}{dx} = \frac{1}{e^x + 1} \text{ or } \frac{dA}{dx} = \frac{e^{-x}}{e^x + 1}$$

and $\quad e^{-x} \frac{dB}{dx} = -\frac{1}{1 + e^x}$ or $\frac{dB}{dx} = -\frac{e^x}{1 + e^x}$

Integrating these,

$$A = \int \frac{e^{-x}}{e^x + 1} dx + C_1 = \int \frac{1}{z^2(1 + z)} dz + C_1$$

[putting $e^x = z$]

$$= \int \left\{\frac{1}{z^2} - \frac{1}{z} + \frac{1}{1 + z}\right\} dz + C_1$$

$$= -\frac{1}{z} - \log z + \log(1 + z) + C_1 = -\frac{1}{z} + \log \frac{1 + z}{z} + C_1$$

$$= -e^{-x} + \log \frac{1 + e^x}{e^x} + C_1$$

and $B = -\int \frac{e^x}{1 + e^x} dx + C_2 = -\log(1 + e^x) + C_2.$

Hence the solution is

$$y = \left\{-e^{-x} + \log \frac{1 + e^x}{e^x} + C_1\right\} e^x + [-\log(1 + e^x) + C_2]\, e^{-x} \quad \textbf{Ans.}$$

Ex. 4. **Solve** $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$

Sol. : First of all we shall find the C.F., i.e., the solution of

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0$$

This is a homogeneous equation, hence in order to solve it we get $x = e^z$ in (1). Then it reduces to

$$\therefore \quad D(D - 1)y + Dy - y = 0 \text{ where } D = d/dz.$$

$$(D^2 - 1)y = 0$$

$$\therefore \quad y = C_1 e^z + C_2 e^{-z} = C_1 x + \frac{C_2}{x}$$

Now assuming A and B as the functions of x in such a way that given equation is satisfied by

$$y = Ax + B/x.$$

$$\therefore \quad \frac{dy}{dx} = A - \frac{B}{x^2} + x\frac{dA}{dx} + \frac{1}{x}\frac{dB}{dx} = A - \frac{B}{x^2}.$$

assuming $$x\frac{dA}{dx}+\frac{1}{x}\frac{dB}{dx}=0 \quad \text{... (1)}$$

Also $$\frac{d^2y}{dx^2}=\frac{dA}{dx}+\frac{2B}{x^3}-\frac{1}{x^2}\frac{dB}{dx}$$

Putting the values in the original equation, we get

$$x^2\left[\frac{dA}{dx}+\frac{2B}{x^3}-\frac{1}{x^2}\frac{dB}{dx}\right]+x\left[A-\frac{B}{x^2}\right]-\left[Ax+\frac{B}{x}\right]=x^2e^x$$

or $$x^2\frac{dA}{dx}-\frac{dB}{dx}=x^2e^x \text{ or } x\frac{dA}{dx}-\frac{1}{x}\frac{dB}{dx}=xe^x. \quad \text{... (2)}$$

Solving (1), (2) $\frac{dA}{dx}=\frac{e^x}{2}$ and $\frac{dB}{dx}=-\frac{x^2}{2}e^x$.

Integrating these, we get $A=\frac{1}{2}e^x+C_1$

and $B=-\frac{1}{2}\int(x^2e^x)\,dx+C_2=-\frac{e^x}{2}(x^2-2x+2)+C_2$.

Hence the complete solution is given by

$$y=\left\{\frac{1}{2}e^x+C_1\right\}.x+\left\{-\frac{e^x}{2}(x^2-2x+2)+C_2\right\}.\frac{1}{x}$$

$$=C_1x+e^x-\frac{e^x}{x}+\frac{C_2}{x}$$ **Ans.**

Ex. 5. Solve $x^3\frac{d^2y}{dx^2}-2x(1+x)\frac{dy}{dx}+2(1+x)y=x^3$

Sol. : The equation in standard form is

$$\frac{d^2y}{dx^2}-\frac{2}{x}(1+x)\frac{dy}{dx}+\frac{2(1+x)}{x^2}y=x.$$

Here P + Qx vanishes. Hence $y=x$ is a part of its C.F. Now putting $y=vu$ in the eaution

$$\frac{d^2y}{dx^2}-\frac{2}{x}(1+x)\frac{dy}{dx}+\frac{2(1+x)}{x^2}y=0,$$

we get the new equation as

$$\frac{d^2v}{dx^2}+\left[P+\frac{2}{u}\frac{du}{dx}\right]\frac{dv}{dx}=0$$

or $$\frac{d^2v}{dx^2}+\left(\frac{-2-2x}{x}+\frac{2}{x}\cdot 1\right)\frac{dv}{dx}=0 \qquad [\because u=x]$$

or $(D^2-2D)v=0$ or $D(D-2)v=0$

$\therefore\ v=C_1+C_2e^{2x},\ y=vu=vx=C_1x+C_2xe^{2x}.$

Hence the C.F. of the given equation can be put in the form

$$y=Ax+B\,.\,xe^{2x},\ A, B \text{ being constants.}$$

No assume A and B as the functions of x in such a way that the given equation is satisfied by $y=Bx+A.\,xe^x$.

$$\therefore\quad \frac{dy}{dx}=2xAe^{2x}+Ae^{2x}+B+xe^{2x}\frac{dA}{dx}+x\frac{dB}{dx}$$

$$=2A\,xe^{2x}+Ae^{2x}+B, \quad \text{... (1)}$$

assuming $$xe^{2x}\frac{dA}{dx}+x\frac{dB}{dx}=0. \quad \text{... (2)}$$

Now $$\frac{d^2y}{dx^2}=4A\,xe^{2x}+4A\,e^{2x}+\frac{dB}{dx}+2xe^{2x}\frac{dA}{dx}+e^{2x}\frac{dA}{dx} \quad \text{... (3)}$$

Substituting these values in the given equations, we get

$$\frac{dB}{dx}+e^{2x}\frac{dA}{dx}(2x+1)=x. \quad \text{... (4)}$$

Solving (2) and (4),

$$\frac{dA}{dx}=\frac{1}{2}e^{-2x} \text{ and } \frac{dB}{dx}=-\frac{1}{2}$$

Integrating, $A=-\frac{1}{4}e^{-2x}+C_1$ and $B=-(x/2)+C_2$.

Hence the complete solution is

$$y=\left(C_1-\frac{1}{4}e^{-2x}\right)xe^{2x}+\left(C_2-\frac{1}{2}x\right)x$$

$$=C_1xe^{2x}+C_2x-\frac{x^2}{2}-\frac{x}{4}$$ **Ans.**

Ex. 6. Solve $\frac{d^2y}{dx^2}+(1-\cot x)\frac{dy}{dx}-y\cot x=\sin^2 x$.

Sol. : First of all we shall find the C.F., *i.e.*, the solution of

$$\frac{d^2y}{dx^2}+(1-\cot x)\frac{dy}{dx}-y\cot x=0. \quad \text{... (1)}$$

Here $P=1-\cot x,\ Q=-\cot x$

$\therefore\ 1-P+Q=1-1+\cot x-\cot x=0$, showing that $y=e^{-x}$ is a solution of (1).

Now let $y=ve^{-x}$.

By this substitution, the equation reduces to

$$\frac{d^2v}{dx^2}+\frac{dv}{dx}\left(1-\cot x-\frac{2}{e^{-x}}\cdot e^{-x}\right)=0 \text{ or } \frac{d^2v}{dx^2}=(1+\cot x)\frac{dv}{dx}$$

Scanned with

or $\quad \frac{dt}{dx} = (1 + \cot x).\, t$ or $\frac{dt}{t} = (1 + \cot x)\, dx$ if $\frac{dv}{dx} = t$

$\therefore \quad \log (tC_1) = x + \log \sin x \quad$ or $C_1 \frac{dv}{dx} = e^x \sin x$

$$\therefore \quad C_1 v = \int e^x \sin x \, dx = -\frac{e^x}{2} (\cos x - \sin x) + C_2.$$

and
$$y = \left\{ -\frac{e^x}{2C_1} (\cos x - \sin x) + \frac{C_2}{C_1} \right\} . e^{-x}$$
$$= A (\cos x - \sin x) + Be^{-x}.$$

Hence the complementary function of the given equation is $y = A (\cos x - \sin x) + Be^{-x}$. Now assume A and B as the functions of x in such a way that the given equation is satisfied by

$$y = A (\cos x - \sin x) + Be^{-x}$$

$$\therefore \quad \frac{dy}{dx} = A (-\sin x - \cos x) - Be^{-x} + (\cos x - \sin x) \frac{dA}{dx} + e^{-x} \frac{dB}{dx}$$

$$= -A (\sin x + \cos x) - Be^{-x}, \qquad \text{... (1)}$$

assuming $\quad (\cos x - \sin x) \frac{dA}{dx} + e^{-x} \frac{dB}{dx} = 0. \qquad$ ... (2)

Also $\frac{d^2y}{dx^2} = A (-\cos x + \sin x) + Be^{-x}$

$$- (\sin x + \cos x) \frac{dA}{dx} - e^{-x} \frac{dB}{dx} \qquad \text{... (3)}$$

putting the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the original equation, we get

$$- (\sin x + \cos x) \frac{dA}{dx} - e^{-x} \frac{dB}{dx} = \sin^2 x \qquad \text{... (4)}$$

Solving (2) and (4), $\frac{dA}{dx} = \frac{-1}{2} \sin x$

and $\quad \frac{dB}{dx} = e^x \frac{\sin 2x}{4} - e^x \frac{(1 - \cos 2x)}{4}$

Integrating these, we get

$$A = -\frac{1}{2} \int \sin x \, dx + C_1 = \frac{\cos x}{2} + C_1$$

and $B = \frac{e^x}{-20} (2 \cos 2x - \sin 2x) \frac{-e^x}{4} + \frac{e^x}{20} (\cos 2x + 2 \sin 2x) + C_2$

Hence the complete solution is

$$y = Au + Bv$$
$$= \left[\frac{\cos x}{2} + C_1 \right] . (\cos x - \sin x) + \left[\frac{e^x}{-20} (2 \cos 2x - \sin 2x) - \frac{e^x}{4} + \frac{e^x}{20} (\cos 2x + 2 \sin 2x) \right] e^{-x}$$
$$= C_1 (\cos x - \sin x) + C_2 e^{-x} - \frac{1}{10} [\sin 2x - 2 \cos 2x] \qquad \textbf{Ans.}$$

Ex. 7. Solve $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x$.

Sol. : For finding C.F. of the given equation first we solve the equation :

$$\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} = 0$$

A.E. $\to m^2 - 2m = 0 \quad \Rightarrow m = 0, 2$

$\therefore \quad$ C.F. is given by—

$$y = C_1 + C_2 e^{2x}$$

Let the complete solution of the given equation be :

$y = A + Be^{2x}$ ... (1); where A and B are functions of x

$$\Rightarrow \quad \frac{dy}{dx} = \frac{dA}{dx} + \frac{dB}{dx} e^{2x} + 2Be^{2x} \qquad \text{... (2)}$$

Now, we choose A and B such that :

$$\frac{dA}{dx} + \frac{dB}{dx} e^{2x} = 0 \qquad \text{... (3)}$$

$\therefore \quad$ (2) gives :

$$\frac{dy}{dx} = 2Be^{2x} \qquad \text{... (4)}$$

also, $\quad \frac{d^2y}{dx^2} = 2e^{2x} \frac{dB}{dx} + 4Be^{2x} \qquad$ ... (5)

using (4) & (5) in the given differential equation, we get :

$$2\left(e^{2x} \frac{dB}{dx} + 2Be^{2x} \right) - 4Be^{2x} = e^x \sin x$$

or $\quad \frac{dB}{dx} = \frac{e^{-x} \sin x}{2} \qquad$ ... (6)

or $\quad dB = \frac{e^{-x} \sin x}{2} dx$

or $\quad B = \frac{e^{-x}}{4}(-\sin x - \cos x) + C_1$ ... (7)

using (6) in (3), we get

$$\frac{dA}{dx} + \frac{e^x \sin x}{2} = 0 \Rightarrow A = -\frac{e^x}{4}(\sin x - \cos x) + C_2$$

$$\therefore \ (1) \Rightarrow y = -\frac{e^x}{4}\sin x + \frac{e^x}{4}\cos x + C_2 - \frac{e^x}{4}\sin x - \frac{e^x}{4}\cos x + C_1 e^{2x}$$

or $\quad y = C_1 e^{2x} - \frac{1}{2}e^x \sin x + C_2$ **Ans.**

Ex. 8. Solve $\frac{d^2y}{dx^2} + y = \sec x$.

Sol. : C.F. = $A\cos x + B\sin x$. Now assume A and B as the function of x in such a way that the given equation is satisfied by $y = A\cos x + B\sin x$.

$$\therefore \quad \frac{dy}{dx} = -A\sin x + B\cos x + \cos x\frac{dA}{dx} + \sin x\frac{dB}{dx}$$

$$= -A\sin x + B\cos x \quad \text{... (1)}$$

assume $\cos x\frac{dA}{dx} + \sin x\frac{dB}{dx} = 0$ ... (2)

Also $\quad \frac{d^2y}{dx^2} = -A\cos x - B\sin x - \sin x\frac{dA}{dx} + \cos x\frac{dB}{dx}$

$$= -y - \sin x\frac{dA}{dx} + \cos x\frac{dB}{dx} \quad \text{... (3)}$$

Now putting the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the original equation, we get

$$-\sin x\frac{dA}{dx} + \cos x\frac{dB}{dx} = \sec x \quad \text{... (4)}$$

Solving (2) and (4), we again get

$$\frac{dA}{dx} = -\tan x, \quad \frac{dB}{dx} = 1$$

Integrating then, $B = x + C_2$, $A = \log\cos x + C_1$

Hence the complete solution is

$y = (\log\cos x + C_1)\cos x + (x + C_2)\sin x$

$y = C_1\cos x + C_2\sin x + x\sin x + \cos x\log\cos x$ **Ans.**

□□□

3. Partial Differentiation

1. Euler's theorem on homogeneous functions :

If $f(x, y)$ be a homogeneous function in x and y of degree n, then

$$x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf$$

2. If 'u' is a composite function of t; defined by the relations $u = f(x, y)$, $x = \phi(t)$, $y = \psi(t)$, then

$$\frac{du}{dt}=\frac{\partial u}{\partial x}\cdot\frac{dx}{dt}+\frac{\partial u}{\partial y}\cdot\frac{dy}{dt}$$

3. If $f(x, y) = 0$

then $\frac{dy}{dx}=-\frac{f_x}{f_y}$ where $f_x=\frac{\partial f}{\partial x}, f_y=\frac{\partial f}{\partial y}$

EXERCISE–3(A)

Ex.1. If $z=\frac{x^2+y^2}{x+y}$, show that $\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)$.

Sol. $z=\frac{x^2+y^2}{x+y}$

$$\Rightarrow \frac{\partial z}{\partial x}=\frac{(x+y)(2x)-1(x^2+y^2)}{(x+y)^2}=\frac{x^2+2xy-y^2}{(x+y)^2} \quad .....(1)$$

and $$\frac{\partial z}{\partial y}=\frac{(x+y)(2y)-1(x^2+y^2)}{(x+y)^2}=\frac{y^2+2xy-x^2}{(x+y)^2} \quad .....(2)$$

$$\text{Now}\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=\left[\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}\right]^2$$

$$=\left[\frac{2(x^2-y^2)}{(x+y)^2}\right]^2=\left[\frac{2(x-y)}{(x+y)}\right]^2$$

$$\Rightarrow\quad\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=4\left(\frac{x-y}{x+y}\right)^2 \qquad \text{....(3)}$$

$$\text{Also, } 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)=4\left[1-\frac{x^2+2xy-y^2}{(x+y)^2}-\frac{y^2+2xy-x^2}{(x+y)^2}\right]$$

$$=4\left[1-\frac{4xy}{(x+y)^2}\right]=4\left[\frac{(x-y)^2}{(x+y)^2}\right]$$

$$\Rightarrow\quad 4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)=4\left(\frac{x-y}{x+y}\right)^2 \qquad \text{....(4)}$$

(3) and (4)

$$\Rightarrow\quad\left(\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right)^2=4\left(1-\frac{\partial z}{\partial x}-\frac{\partial z}{\partial y}\right) \qquad \textbf{Proved}$$

Ex.2 **Show that** $\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y\partial x}$ **where u is equal to**

(i) $\log\left(\frac{(x^2+y^2)}{xy}\right)$

(ii) $\log(y\sin x+x\sin y)$

(iii) $\log(x^2+y^2)$.

Sol. (i) $u=\log\left\{\frac{x^2+y^2}{xy}\right\}$ (1)

Diff (1) w.r.t (y) (P)

$$\frac{\partial u}{\partial y}=\frac{xy}{(x^2+y^2)}\cdot\left[\frac{2y^2x-x(x^2+y^2)}{x^2y^2}\right]$$

$$=\frac{xy^2-x^3}{xy(x^2+y^2)}=\frac{y^2-x^2}{y(x^2+y^2)}$$

$$\frac{\partial^2 u}{\partial x\partial y}=\frac{-2x(x^2+y^2)-(y^2-x^2)(2x)}{y(x^2+y^2)^2}=-\frac{\cancel{2}4xy^2}{\cancel{y}(x^2+y^2)^2}$$

$$=\frac{-\cancel{2}4xy}{(x^2+y^2)^2} \quad -(A) \qquad \text{....(1)}$$

$$\text{Now, } \frac{\partial u}{\partial x}=\frac{xy}{(x^2+y^2)}\cdot\left\{\frac{2x^2y-y(x^2+y^2)}{x^2y^2}\right\}$$

$$=\frac{x^2y-y^3}{xy(x^2+y^2)}=\frac{x^2-y^2}{x(x^2+y^2)}$$

$$\Rightarrow\quad\frac{\partial^2 u}{\partial y\partial x}=\frac{1}{x}\left[\frac{-2y(x^2+y^2)-(x^2-y^2)\cdot 2y}{(x^2+y^2)^2}\right]$$

$$=-\frac{\cancel{2}4x^{\cancel{2}}y}{\cancel{x}(x^2+y^2)^2}=\frac{-\cancel{2}4xy}{(x^2+y^2)^2} \quad -(B) \qquad \text{....(2)}$$

(A) = (B)

$$\text{(1) and (2)} \Rightarrow \frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y\partial x} \qquad \textbf{Proved}$$

(ii) $\frac{\partial u}{\partial x}=\frac{y\cos x+\sin y}{y\sin x+x\sin y}$

$$\frac{\partial^2 u}{\partial y\partial x}=\frac{(y\sin x+x\sin y)(\cos x+\cos y)-(y\cos x+\sin y)(\sin x+x\cos y)}{(y\sin x+x\sin y)^2} \qquad \text{....(1)}$$

$$=\frac{y\sin x\cos y+x\sin y\cos x+x\sin y\cos y-xy\cos x\cos y-\sin x\sin y-x\sin y\cos y}{(y\sin x+x\sin y)^2}$$

$$\therefore\quad\frac{\partial^2 u}{\partial y\partial x}=\frac{(y\sin x\cos y+x\sin y\cos x-xy\cos x\cos y-\sin x\sin y)}{(y\sin x+x\sin y)^2}$$

Scanned with

....(2)

and $\dfrac{\partial u}{\partial y} = \dfrac{\sin x + x\cos y}{y\sin x + x\sin y}$

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{(y\sin x + x\sin y)(\cos x + \cos y) - (\sin x + x\cos y)(y\cos x + \sin x)}{(y\sin x + x\sin y)^2}$$

....(3)

(1) and (3) $\Rightarrow \dfrac{\partial^2 u}{\partial x\partial y} = \dfrac{\partial^2 u}{\partial y\partial x}$ **Proved**

(iii) $u = \log(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2+y^2} \Rightarrow \frac{\partial^2 u}{\partial y\partial x} = -\frac{4xy}{(x^2+y^2)^2} \quad ....(1)$$

$$\frac{\partial u}{\partial y} = \frac{2y}{(x^2+y^2)} \Rightarrow \frac{\partial^2 u}{\partial x\partial y} = -\frac{4xy}{(x^2+y^2)^2} \quad ....(2)$$

(1) and (2) $\Rightarrow \dfrac{\partial^2 u}{\partial x\partial y} = \dfrac{\partial^2 u}{\partial y\partial x}$ **Proved**

Ex.3. If $u = x^2\tan^{-1}\left(\dfrac{y}{x}\right) - y^2\tan^{-1}\left(\dfrac{x}{y}\right)$, **prove that**

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{x^2-y^2}{x^2+y^2}.$$

Sol. $\dfrac{\partial u}{\partial y} = x^2\dfrac{1}{1+\left(\dfrac{y}{x}\right)^2}\cdot\dfrac{1}{x} - 2y\tan^{-1}\left(\dfrac{x}{y}\right) - y^2\cdot\dfrac{1}{1+\dfrac{x^2}{y^2}}\left(\dfrac{-x}{y^2}\right)$

$$= \frac{x^3}{x^2+y^2} + \frac{xy^2}{x^2+y^2} - 2y\tan^{-1}\left(\frac{x}{y}\right)$$

$\therefore \dfrac{\partial u}{\partial y} = x - 2y\tan^{-1}\left(\dfrac{x}{y}\right)$

$$\therefore \frac{\partial^2 u}{\partial x\partial y} = 1 - 2y\cdot\frac{1}{1+\dfrac{x^2}{y^2}}\cdot\frac{1}{y} = 1 - \frac{2y^2}{x^2+y^2}$$

$\dfrac{\partial^2 u}{\partial x\partial y} = \dfrac{x^2-y^2}{x^2+y^2}$ **Hence proved.**

Ex.4. If $\theta(x, t) = At^{-\frac{1}{2}}e^{-\frac{x^2}{4at}}$, **prove that** $\dfrac{\partial\theta}{\partial t} = a\dfrac{\partial^2\theta}{\partial x^2}$.

Sol. Given $\theta = At^{-\frac{1}{2}}e^{-\frac{x^2}{4at}}$

$$\Rightarrow \frac{\partial\theta}{\partial x} = At^{-\frac{1}{2}}e^{-\frac{x^2}{4at}}\left(\frac{-x}{2at}\right) = -\frac{A}{2a}t^{-\frac{3}{2}}xe^{-\frac{x^2}{4at}}$$

$$\Rightarrow \frac{\partial^2\theta}{\partial x^2} = -\frac{A}{2a}t^{-\frac{3}{2}}\left[e^{-\frac{x^2}{4at}} - \frac{x^2}{2at}e^{-\frac{x^2}{4at}}\right]$$

$$= \frac{A}{4a^2}t^{-\frac{5}{2}}x^2e^{-\frac{x^2}{4at}} - \frac{1}{2a}At^{-\frac{3}{2}}e^{-\frac{x^2}{4at}} \quad ....(1)$$

and $\dfrac{\partial\theta}{\partial t} = A\left(\dfrac{-1}{2}\right)t^{\frac{-3}{2}}e^{\frac{-x^2}{4at}} + At^{\frac{-1}{2}}e^{\frac{-x^2}{4at}}\left(\dfrac{x^2}{4at^2}\right)$

$$= \frac{A}{4a}t^{-\frac{5}{2}}x^2e^{-\frac{x^2}{4at}} - \frac{1}{2}At^{-\frac{3}{2}}e^{-\frac{x^2}{4at}}$$

or, $\dfrac{1}{a}\dfrac{\partial\theta}{\partial t} = \dfrac{A}{4a^2}t^{-\frac{5}{2}}x^2e^{-\frac{x^2}{4at}} - \dfrac{1}{2a}At^{-\frac{3}{2}}e^{-\frac{x^2}{4at}}$ ....(2)

(1) and (2) $\Rightarrow \dfrac{1}{a}\dfrac{\partial\theta}{\partial t} = \dfrac{\partial^2\theta}{\partial x^2}$

$$\Rightarrow \frac{\partial\theta}{\partial t} = a\frac{\partial^2\theta}{\partial x^2} \quad \textbf{Proved.}$$

Ex.5. If $u = e^{xyz}$, **show that** $\dfrac{\partial^3 u}{\partial x\partial y\partial z} = \left(1 + 3xyz + x^2y^2z^2\right)e^{xyz}$.

Sol. $u = e^{xyz}$

$$\Rightarrow \frac{\partial u}{\partial z} = xye^{xyz}$$

$$\Rightarrow \frac{\partial^2 u}{\partial y \partial z} = \frac{\partial}{\partial y}\left(xye^{xyz}\right) = xe^{xyz} + (xy)(xz)e^{xyz} = (x + x^2yz)e^{xyz}$$

$$\Rightarrow \frac{\partial^3 u}{\partial x \partial y \partial z} = \frac{\partial}{\partial x}\left[(x + x^2yz)e^{xyz}\right] = (1 + 2xyz)e^{xyz} + (x + x^2yz)yze^{xyz}$$

$$= (1 + 2xyz + xyz + x^2y^2z^2)e^{xyz}$$

$$\Rightarrow \frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}. \quad \textbf{Proved}$$

Ex.6. If $\theta(r,t) = t^n \cdot e^{\left(\frac{-r^2}{4t}\right)}$ find what value of n will make

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \cdot \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}.$$

Sol. $\frac{\partial \theta}{\partial t} = nt^{n-1}e^{\frac{-r^2}{4t}} + t^n e^{\frac{-r^2}{4t}}\left(\frac{r^2}{4t^2}\right)$

$$= \left(nt^{n-1} + \frac{r^2}{4}t^{n-2}\right)e^{\frac{-r^2}{4t}} \qquad \text{....(1)}$$

and $\frac{\partial \theta}{\partial r} = t^n\left(\frac{-r}{2t}\right)e^{-\frac{r^2}{4t}} = -\frac{r}{2}t^{n-1}e^{-\frac{r^2}{4t}}$

$$\Rightarrow r^2\frac{\partial \theta}{\partial r} = -\frac{t^{n-1}}{2}r^3 e^{-\frac{r^2}{4t}}$$

$$\therefore \frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = -\frac{t^{n-1}}{2}\left[3r^2 e^{-\frac{r^2}{4t}} - \frac{r^4}{2t}e^{-\frac{r^2}{4t}}\right]$$

$$= -\frac{t^{n-1}}{2}\left(3r^2 - \frac{r^4}{2t}\right)e^{-\frac{r^2}{4t}}$$

$$\therefore \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = -\frac{1}{2}t^{n-1}\left(3 - \frac{r^2}{2t}\right)e^{-\frac{r^2}{4t}}$$

$$= \left(-\frac{3}{2}t^{n-1} + \frac{r^2}{4}t^{n-2}\right)e^{-\frac{r^2}{4t}} \qquad \text{....(2)}$$

since, (1) = (2)

$$\Rightarrow n = -\frac{3}{2} \qquad \textbf{Ans.}$$

Ex.7. **Verify Euler's theorem for each of the following functions :**

(i) $f(x,y) = \sin^{-1}\left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}\right)$

(ii) $f(x,y) = \frac{(x+y)}{\sqrt{x} + \sqrt{y}}$

Sol. (i) Here, $f(x,y) = \sin^{-1}\left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}\right) = z \text{ (say)}$ ....(1)

$$\Rightarrow \sin z = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{1 - \sqrt{\left(\frac{y}{x}\right)}}{1 + \sqrt{\left(\frac{y}{x}\right)}} = \phi\left(\frac{y}{x}\right)$$

Let, $\sin z = u = \phi\left(\frac{y}{x}\right)$

then u is a homogeneous function of x and y of degree zero so by Euler's theorem :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0.u = 0$$

or, $x\frac{\partial}{\partial x}(\sin z) + y\frac{\partial}{\partial y}(\sin z) = 0$

or, $x\cos z\frac{\partial z}{\partial x} + y\cos z\frac{\partial z}{\partial y} = 0$

or, $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0$

or, $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=0$ ...(1)

$[\because f(x,y) = \ldots]$

Now, $f=\sin^{-1}\left(\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}\right)$

$$\Rightarrow \frac{\partial f}{\partial x}=\frac{1}{\sqrt{1-\left[\frac{\left(\sqrt{x}-\sqrt{y}\right)}{\left(\sqrt{x}+\sqrt{y}\right)}\right]^2}}$$

$$\left[\frac{\left(\sqrt{x}+\sqrt{y}\right)\frac{1}{2\sqrt{x}}+\left(\sqrt{x}-\sqrt{y}\right)\frac{-1}{2\sqrt{x}}}{\left(\sqrt{x}+\sqrt{y}\right)^2}\right]=\frac{\left(\frac{\sqrt{y}}{\sqrt{x}}\right)}{2\left(\sqrt{xy}\right)}\times\frac{1}{\left(\sqrt{x}+\sqrt{y}\right)^2}$$

$$\Rightarrow x\frac{\partial f}{\partial x}=\frac{\sqrt{xy}}{2\sqrt{\left(\sqrt{xy}\right)}}\cdot\frac{1}{\left(\sqrt{x}+\sqrt{y}\right)^2}=\frac{(xy)^{\frac{1}{4}}}{2\left(\sqrt{x}+\sqrt{y}\right)^2} \quad \text{...(2)}$$

$$\text{and } \frac{\partial f}{\partial y}=\frac{1}{\sqrt{1-\left(\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}\right)^2}}$$

$$\left[\frac{\left(\sqrt{x}+\sqrt{y}\right)\left(-\frac{1}{2\sqrt{y}}\right)-\left(\sqrt{x}-\sqrt{y}\right)\frac{1}{2\sqrt{y}}}{\left(\sqrt{x}+\sqrt{y}\right)^2}\right]=\frac{-\left(\sqrt{\frac{x}{y}}\right)}{2\sqrt{\left(\sqrt{xy}\right)}}\cdot\frac{1}{\left(\sqrt{x}+\sqrt{y}\right)^2}$$

$$\therefore y\frac{\partial f}{\partial y}=\frac{-(xy)^{\frac{1}{4}}}{2\left(\sqrt{x}+\sqrt{y}\right)^2} \quad \text{...(3)}$$

$(2)+(3)\Rightarrow x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=0$ ...(4)

This verifies (1). **Proved**

(ii) $f=\frac{x+y}{\sqrt{x}+\sqrt{y}}=x^{\frac{1}{2}}\left(\frac{1+\frac{y}{x}}{1+\sqrt{\frac{y}{x}}}\right)=x^{\frac{1}{2}}\phi\left(\frac{y}{x}\right)$

$\therefore$ By Euler's theorem $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=\frac{1}{2}f$ ...(1)

$$\frac{\partial f}{\partial x}=\frac{\left(\sqrt{x}+\sqrt{y}\right)-(x+y)\frac{1}{2\sqrt{x}}}{\left(\sqrt{x}+\sqrt{y}\right)^2}$$

$$x\frac{\partial f}{\partial x}=\frac{2\left(x\sqrt{x}+x\sqrt{y}\right)-\left(x\sqrt{x}+y\sqrt{x}\right)}{2\left(\sqrt{x}+\sqrt{y}\right)^2}=\frac{x\sqrt{x}+2x\sqrt{y}-y\sqrt{x}}{2\left(\sqrt{x}+\sqrt{y}\right)^2} \quad \text{...(2)}$$

Similarly,

$$y\frac{\partial f}{\partial y}=\frac{y\sqrt{y}+2y\sqrt{x}-x\sqrt{y}}{2\left(\sqrt{x}+\sqrt{y}\right)^2} \quad \text{...(3)}$$

$$(2)+(3)\Rightarrow x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=\frac{x\sqrt{x}+y\sqrt{y}+y\sqrt{x}+x\sqrt{y}}{2\left(\sqrt{x}+\sqrt{y}\right)^2}$$

$$=\frac{(x+y)\left(\sqrt{x}+\sqrt{y}\right)}{2\left(\sqrt{x}+\sqrt{y}\right)^2}=\frac{1}{2}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$$

$$\Rightarrow x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=\frac{1}{2}f \quad \text{...(4)}$$

$(1)=(4)\Rightarrow$ Theorem is verified. **Ans.**

Ex.8. If $u=\log\left(\frac{x^4+y^4}{x+y}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=3$.

Sol. Here $e^u = \dfrac{x^4+y^4}{x+y} = z$ (Let)

$$z = x^3\left(\frac{1+\left(\frac{y}{x}\right)^4}{1+\frac{y}{x}}\right) = x^3\phi\left(\frac{y}{x}\right)$$

$\therefore$ by Euler's theorem $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} = 3z$

or, $x\dfrac{\partial}{\partial x}\left(e^u\right) + y\dfrac{\partial}{\partial x}\left(e^u\right) = 3e^u$

or, $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 3$ **Proved**

Ex.9. If $u = \sec^{-1}\left(\dfrac{x^3+y^3}{x+y}\right)$ proved that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2\cot u$.

Sol. Let $\sec u = \dfrac{x^3+y^3}{x+y} = z$, say;

$$\Rightarrow \quad z = x^2\left[\frac{1+\left(\frac{y}{x}\right)^3}{\left(1+\frac{y}{x}\right)}\right] = x^3\phi\left(\frac{y}{x}\right)$$

$\Rightarrow$ z is a homogeneous function of x and y of degree 2.

$\therefore$ by Euler's theorem :

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z$$

$\Rightarrow$ $x\dfrac{\partial}{\partial x}(\sec u) + y\dfrac{\partial}{\partial y}(\sec u) = 2\sec u$

$\Rightarrow$ $x\sec u\tan u\dfrac{\partial u}{\partial x} + y\sec u\tan u\dfrac{\partial u}{\partial y} = 2\sec u$

$\Rightarrow$ $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = 2\cot u$. **Proved**

Ex.10. If $u = \sin^{-1}\left(\dfrac{x^{\frac{1}{4}}+y^{\frac{1}{4}}}{x^{\frac{1}{5}}+y^{\frac{1}{5}}}\right)$ prove that $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{20}\tan u$.

Sol. $\sin u = \dfrac{x^{1/4}+y^{1/4}}{x^{1/5}+y^{1/5}} = z$ (Let)

$$\therefore \quad z = x^{1/20}\left\{\frac{1+\left(\frac{y}{x}\right)^{1/4}}{1+\left(\frac{y}{x}\right)^{1/5}}\right\} = x^{1/20}f\left(\frac{y}{x}\right)$$

$\therefore$ by Euler's theorem

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{1}{20}z$$

or, $x\dfrac{\partial}{\partial x}(\sin u) + y\dfrac{\partial}{\partial y}(\sin u) = \dfrac{1}{20}\sin u$

$\Rightarrow$ $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = \dfrac{1}{20}\tan u$ **Proved**

Ex.11. If $u = \tan^{-1}\left\{\dfrac{(x^3+y^3)}{(x-y)}\right\}$, prove that $x\cdot\dfrac{\partial u}{\partial x} + y\cdot\dfrac{\partial u}{\partial y} = \sin 2u$.

Sol. Here $u = \tan^{-1}\left(\dfrac{x^3+y^3}{x-y}\right)$

$\Rightarrow$ $\tan u = \dfrac{x^3+y^3}{x-y} = z$ (let)

$$\therefore \quad z = x^2\frac{\left(1+\frac{y^3}{x^3}\right)}{\left(1-\frac{y}{x}\right)} = x^2\phi\left(\frac{y}{x}\right)$$

$\Rightarrow$ z is a homogenoues function of x and y of degree 2.

$\therefore$ by Euler's theorem :

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=2z$$

$$\Rightarrow \quad x\frac{\partial}{\partial x}(\tan u)+y\frac{\partial}{\partial y}(\tan u)=2\tan u$$

$$\Rightarrow \quad \sec^2 u\left(x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}\right)=2\tan u$$

or $\quad x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=2\sin u\cos u=\sin 2u$ **Proved.**

EXERCISE–3(B)

Ex.1. **If $z = x^2-y^2$, where $x=e^t\cos t, y=e^t\sin t$, find $\dfrac{dz}{dt}$.**

Sol. Since, $\dfrac{dz}{dt}=\dfrac{\partial z}{\partial x}\cdot\dfrac{dx}{dt}+\dfrac{\partial z}{\partial y}\cdot\dfrac{dy}{dt}$ ....(1)

Now, $\dfrac{\partial z}{\partial x}=2x;\ \dfrac{\partial z}{\partial y}=-2y;\ \dfrac{dx}{dt}=e^t(\cos t-\sin t);$

$$\frac{dy}{dt}=e^t(\sin t+\cos t)$$

$\therefore$ (1) gives

$$\frac{dz}{dt}=2xe^t(\cos t-\sin t)+(-2y)e^t(\sin t+\cos t)$$

$$=2(e^t\cos t)e^t(\cos t-\sin t)-2(e^t\sin t)e^t(\sin t+\cos t)$$

$$=2e^{2t}(\cos^2 t-\cos t\sin t-\sin^2 t-\sin t\cos t)$$

$$=2e^{2t}(\cos^2 t-\sin^2 t-2\sin t\cos t)$$

$\Rightarrow \quad \dfrac{dz}{dt}=2e^{2t}(\cos 2t-\sin 2t)$ **Ans.**

Ex.2. **If $u=x^2-y^2+\sin yz$, where $y=e^x$, $z=\log x$, find $\dfrac{du}{dx}$.**

Sol. Since, $\dfrac{du}{dx}=\dfrac{\partial u}{\partial x}+\dfrac{\partial u}{\partial y}\dfrac{dy}{dx}+\dfrac{\partial u}{\partial z}\dfrac{dz}{dx}$

$$=2x+(-2y+z\cos yz)e^x+(y\cos yz)\frac{1}{x}$$

$\Rightarrow \quad \dfrac{du}{dx}=2x+\left[-2e^x+\log x\cos(e^x\log x)\right]e^x+\dfrac{e^x}{x}\cos(e^x\log x)$

Ans.

Ex.3. **Find $\dfrac{dy}{dx}$, if $y^x+x^y=c$**

Sol. Let $f(x,y)=y^x+x^y-c=0$ then, $\dfrac{dy}{dx}=\dfrac{-\partial f/\partial x}{\partial f/\partial y}$

Now, $\dfrac{\partial f}{\partial x}=y^x\log y+yx^{y-1};\ \dfrac{\partial f}{\partial y}=xy^{x-1}+x^y\log x$

$\therefore \quad \dfrac{dy}{dx}=-\dfrac{y^x\log y+yx^{y-1}}{x^y\log x+xy^{x-1}}$ **Ans.**

Ex.4. **If $z=x^5y^4$, where $x=t^3, y=t^2$, find $\dfrac{dz}{dt}$.**

Sol. Since, $\dfrac{dz}{dt}=\dfrac{\partial z}{\partial x}\dfrac{dx}{dt}+\dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$

$$=5x^4y^4\cdot 3t^2+4x^5y^3\cdot 2t=15(t^3)^4(t^2)^4t^2+8(t^3)^5(t^2)^3t$$

$=15t^{22}+8t^{22}\Rightarrow \dfrac{dz}{dt}=23t^{22}$ **Ans.**

Ex.5. **Find dy/dx when $ax^2+2hxy+by^2=1$**

Sol. $f(x,y)=ax^2+2hxy+by^2=1$ [const.]

$\therefore \quad \dfrac{dy}{dx}=-\dfrac{\partial f/\partial x}{\partial f/\partial y}=-\dfrac{2ax+2hy}{2by+2hx}$

$\therefore \quad \dfrac{dy}{dx}=-\dfrac{ax+hy}{hx+by}$ **Ans.**

Ex.6. **Prove that $\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=\dfrac{\partial^2 u}{\partial \xi^2}+\dfrac{\partial^2 u}{\partial \eta^2}$ where**

$x=\xi\cos\alpha-\eta\sin\alpha,\ y=\xi\sin\alpha+\eta\cos\alpha$. [Raj. Uni., 2000]

Sol. $\frac{\partial u}{\partial \xi} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \xi} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \xi} = \cos\alpha\cdot\frac{\partial u}{\partial x} + \sin\alpha\frac{\partial u}{\partial y}$

$$\Rightarrow \frac{\partial^2 u}{\partial \xi^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial \xi}\cos\alpha + \sin\alpha\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial \xi}$$

$$\therefore \frac{\partial^2 u}{\partial \xi^2} = \frac{\partial^2 u}{\partial x^2}\cos^2\alpha + \frac{\partial^2 u}{\partial y^2}\sin^2\alpha \qquad ....(1)$$

Now, $\frac{\partial u}{\partial \eta} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \eta} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \eta} = -\sin\alpha\frac{\partial u}{\partial x} + \cos\alpha\cdot\frac{\partial u}{\partial y}$

$$\therefore \frac{\partial^2 u}{\partial \eta^2} = -\sin\alpha\cdot\frac{\partial^2 u}{\partial x^2}\cdot\frac{\partial x}{\partial \eta} + \cos\alpha\frac{\partial^2 u}{\partial y^2}\cdot\frac{\partial y}{\partial \eta}$$

$$\therefore \frac{\partial^2 u}{\partial \eta^2} = \sin^2\alpha\frac{\partial^2 u}{\partial x^2} + \cos^2\alpha\frac{\partial^2 u}{\partial y^2} \qquad ....(2)$$

$$\therefore (1)+(2) \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2}$$ **Proved.**

Ex.7. If $u = f(y-z, z-x, x-y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

Sol. Let $r_1 = y - z; r_2 = z - x; r_3 = x - y$

$\therefore \quad u = f(r_1, r_2, r_3)$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r_1}\cdot\frac{\partial r_1}{\partial x} + \frac{\partial u}{\partial r_2}\cdot\frac{\partial r_2}{\partial x} + \frac{\partial u}{\partial r_3}\cdot\frac{\partial r_3}{\partial x} = -\frac{\partial u}{\partial r_2} + \frac{\partial u}{\partial r_3} \qquad ....(1)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r_1}\cdot\frac{\partial r_1}{\partial y} + \frac{\partial u}{\partial r_2}\cdot\frac{\partial r_2}{\partial y} + \frac{\partial u}{\partial r_3}\cdot\frac{\partial r_3}{\partial y} = \frac{\partial u}{\partial r_1} - \frac{\partial u}{\partial r_3} \qquad ....(2)$$

and $$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r_1}\cdot\frac{\partial r_1}{\partial z} + \frac{\partial u}{\partial r_2}\cdot\frac{\partial r_2}{\partial z} + \frac{\partial u}{\partial r_3}\cdot\frac{\partial r_3}{\partial z} = -\frac{\partial u}{\partial r_1} + \frac{\partial u}{\partial r_2} \qquad ....(3)$$

$$\therefore (1)+(2)+(3) \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$ **Proved**

Ex.8. If $x = r\cos\theta, y = r\sin\theta$, find the values of

(i) $\left(\frac{\partial x}{\partial \theta}\right)_{r=\text{constant}}$

(ii) $\left(\frac{\partial \theta}{\partial x}\right)_{y=\text{constant}}$ and prove that

(iii) $\frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$

Sol. (i) $x = r\cos\theta$

$$\Rightarrow \left(\frac{\partial x}{\partial \theta}\right)_{r=\text{constant}} = -r\sin\theta = -y$$ **Ans.**

(ii) Since, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$$\therefore \left(\frac{\partial \theta}{\partial x}\right)_{y=\text{constant}} = \frac{\partial}{\partial x}\left[\tan^{-1}\left(\frac{y}{x}\right)\right] = \frac{1}{\left(1+\frac{y^2}{x^2}\right)}\left(-\frac{y}{x^2}\right)$$

$$= -\frac{y}{(x^2+y^2)}$$ **Ans.**

(iii) Since, $\frac{\partial \theta}{\partial x} = -\frac{y}{x^2+y^2}$

$$\Rightarrow \frac{\partial^2 \theta}{\partial x^2} = \frac{2xy}{(x^2+y^2)^2} \qquad ....(1)$$

Also, $\frac{\partial \theta}{\partial y} = \frac{1}{\left(1+\frac{y^2}{x^2}\right)}\cdot\frac{1}{x} = \frac{x}{x^2+y^2}$

$$\Rightarrow \frac{\partial^2 \theta}{\partial y^2} = -\frac{2xy}{(x^2+y^2)^2} \qquad ....(2)$$

$(1)+(2) \Rightarrow \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$ **Proved**

Ex.9. If $r^2 = x^2 + y^2$, prove that

$$\frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}=\frac{1}{r}\left\{\left(\frac{\partial r}{\partial x}\right)^2+\left(\frac{\partial r}{\partial y}\right)^2\right\}$$

Sol. $r^2 = x^2 + y^2$

$$\Rightarrow \quad \frac{\partial r}{\partial x}=\frac{x}{r} \qquad \text{....(1)}$$

$$\Rightarrow \quad \frac{\partial^2 r}{\partial x^2}=\frac{r-x\cdot\partial r/\partial x}{r^2}$$

$$\therefore \quad \frac{\partial^2 r}{\partial x^2}=\frac{r-x\cdot x/r}{r^2}=\frac{r^2-x^2}{r^3}=\frac{y^2}{r^3} \qquad \text{....(2)}$$

$$\text{and} \quad \frac{\partial r}{\partial y}=\frac{y}{r} \qquad \text{....(3)}$$

$$\Rightarrow \quad \frac{\partial^2 r}{\partial y^2}=\frac{r-y\cdot\partial r/\partial y}{r^2}=\frac{r^2-y^2}{r^3}=\frac{x^2}{r^3} \qquad \text{....(4)}$$

$\therefore$ (2) + (4)

$$\Rightarrow \quad \frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}=\frac{x^2+y^2}{r^3}=\frac{1}{r} \qquad \text{....(5)}$$

and $(1)^2 + (3)^2$

$$\Rightarrow \quad \left(\frac{\partial r}{\partial x}\right)^2+\left(\frac{\partial r}{\partial y}\right)^2=\frac{x^2+y^2}{r^2}=1 \qquad \text{....(6)}$$

$\therefore$ (5) and (6)

$$\Rightarrow \quad \frac{\partial^2 r}{\partial x^2}+\frac{\partial^2 r}{\partial y^2}=\frac{1}{r}\left\{\left(\frac{\partial r}{\partial x}\right)^2+\left(\frac{\partial r}{\partial y}\right)^2\right\} \qquad \textbf{Proved}$$

Ex.10. If $u = f(r)$, where $r^2 = x^2 + y^2$, prove that

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=f''(r)+\frac{1}{r}f'(r).$$

Sol. $u = f(r); \quad r^2 = x^2 + y^2$

$$\therefore \quad \frac{\partial u}{\partial x}=\frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x}=\frac{x}{r}f'(r)$$

$$\frac{\partial^2 u}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{x}{r}f'(r)\right)=\frac{f'(r)}{r}+x\frac{\partial}{\partial r}\left\{\frac{f'(r)}{r}\right\}\cdot\frac{\partial r}{\partial x}$$

$$=\frac{f'(r)}{r}+x\left\{\frac{rf''(r)-f'(r)}{r^2}\right\}\cdot\frac{x}{r}$$

$$=\frac{f'(r)}{r}+x^2\frac{f''(r)}{r^2}-\frac{x^2}{r^3}f'(r) \qquad \text{....(1)}$$

$$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial y}=\frac{y}{r}f'(r)$$

$$\therefore \quad \frac{\partial^2 u}{\partial y^2}=\frac{1}{r}f'(r)+y\cdot\frac{\partial}{\partial r}\left\{\frac{f'(r)}{r}\right\}\frac{\partial r}{\partial y}$$

$$=\frac{1}{r}f'(r)+\frac{y^2}{r}\left\{\frac{rf''(r)-f'(r)}{r^2}\right\}$$

$$=\frac{1}{r}f'(r)+\frac{y^2}{r^2}f''(r)-\frac{y^2}{r^3}f'(r) \qquad \text{....(2)}$$

$\therefore$ (1) + (2) $\Rightarrow$

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=\frac{2f'(r)}{r}+\frac{f''(r)}{r^2}(x^2+y^2)-\frac{f'(r)}{r^3}(x^2+y^2)$$

$$=\frac{2f'(r)}{r}+f''(r)-\frac{f'(r)}{r}$$

$$\Rightarrow \quad \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=f''(r)+\frac{1}{r}f'(r) \qquad \textbf{Proved}$$

EXERCISE–3(C)

Ex.1. Find the percentage error in the area of an ellipse when an error of 1% is made in measuring each of the semi-axes.

Sol. Area of an ellipse with semi-major axis and semi-minor axis x and y respectively, is given by :

$$A = \pi x y$$

$\Rightarrow$ $\log A = \log(\pi x y) = \log \pi + \log x + \log y$

on differentiating partially –

$$\frac{1}{A}\delta A = \frac{1}{x}\delta x + \frac{1}{y}\delta y$$

or, $$100 \times \frac{\delta A}{A} = 100 \times \frac{\delta x}{x} + 100\frac{\delta y}{y} \quad \text{...(1)}$$

It is given that $100 \times \dfrac{\delta x}{x} = 1 = 100 \times \dfrac{\delta y}{y}$

$\therefore$ $(1) \Rightarrow 100 \times \dfrac{\delta A}{A} = 1 + 1 = 2$

$\therefore$ The percentage error in Area = 2 **Ans.**

Ex.2. The angles of a triangle are calculated from the sides, a, b, c; if small changes δa, δb, δc are made in the sides, show that approximately,

$$\delta A = \frac{1}{2}\frac{a(\delta a - \delta b\cos C - \delta c\cos B)}{\Delta}$$

Where Δ is the area of the triangle. Also verify that

$$\delta A + \delta B + \delta C = 0$$

Sol. We know that $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

$\Rightarrow$ $a^2 = b^2 + c^2 - 2bc\cos A = u$, say

Then $u = a^2 \Rightarrow \delta u = \dfrac{du}{da}\delta a = 2a\delta a$

Also, $u = b^2 + c^2 - 2bc\cos A$

Now, $\delta u = \dfrac{\partial u}{\partial b}\delta b + \dfrac{\partial u}{\partial c}\delta c + \dfrac{\partial u}{\partial A}\delta A$

$\Rightarrow$ $\delta u = (2b - 2c\cos A)\delta b + (2c - 2b\cos A)\delta c + (2bc\sin A)\delta A$

$= (2a\cos C)\delta b + (2a\cos B)\delta c + 2bc\sin A\delta A$

$[\because\ b = c\cos A + a\cos C \ \text{ and } \ c = b\cos A + a\cos B]$

$\therefore$ $\delta u = 2a\delta a = 2a\cos C\delta b + 2a\cos B\delta c + 2bc\sin A\delta A$

or, $2a\delta a = 2a\cos C\delta b + 2a\cos B\delta c + 4\Delta\delta A \quad \left(\because D = \frac{1}{2}bc\sin A\right)$

or, $a\delta a = a\cos C\delta b + a\cos B\delta c + 2\Delta\delta A$

or, $\delta A = \dfrac{1}{2\Delta}a(\delta a - \delta b\cos C - \delta c\cos B)$ Proved

similarly,

$$\delta B = \frac{1}{2\Delta}b(\delta b - \delta c\cos A - \delta a\cos C)$$

and $$\delta C = \frac{1}{2\Delta}c(\delta c - \delta a\cos B - \delta b\cos A)$$

Now,

$$\delta A + \delta B + \delta c = \frac{1}{2\Delta}\big[(a - b\cos C - c\cos B)\delta a + (b - a\cos C - c\cos A)$$

$$\delta b + (c - a\cos B - b\cos A)\delta c\big] = 0 \quad \text{Proved}$$

Ex.3. Find the error in volume and total surface (including the base) of a right circular cone whose radius is 4 cm. and altitude 6 cm., if there be shortage of 0.01 cm. percent in the measure used.

Sol. Volume of a cone $(V) = \dfrac{1}{3}\pi r^2 h$;

given that $r = 4$ cm; $h = 6$ cm.

Also, $\dfrac{\delta r}{r} = 0.1 = \dfrac{\delta h}{h}$ (given) ...(1)

Now, $\delta V = \dfrac{\partial V}{\partial r}\delta r + \dfrac{\partial V}{\partial h}\delta h = \dfrac{\pi}{3}(2rh\delta r + r^2\delta h)$

$$= \frac{\pi}{3}r^2h\left(\frac{2}{r}\delta r + \frac{\delta h}{h}\right) = V\left(\frac{2}{r}\delta r + \frac{\delta h}{h}\right)$$

$$\Rightarrow \quad \frac{\delta V}{V} = \frac{2}{r}\delta r + \frac{\delta h}{h} \Rightarrow \frac{\delta V}{V} = 2(.01) + (.01) \qquad \text{(using 1)}$$

$$\Rightarrow \quad \frac{\delta V}{V} \times = 0.03$$

$\therefore$ error in $V = 0.03$ **Ans.**

Again,

Total surface area of a cone $S = \pi r(l + r)$; where $l^2 = r^2 + h^2$

$$\Rightarrow \quad S = \pi r\sqrt{r^2 + h^2} + \pi r^2$$

$$\Rightarrow \quad \delta S = \pi\left[\delta r\sqrt{r^2 + h^2} + r\left(\frac{r\delta r + h\delta h}{\sqrt{(r^2 + h^2)}}\right) + 2r\delta r\right]$$

$$= \pi\left[4 \times 0.01 \times \sqrt{36 + 16} + \frac{4(4 \times 4 \times 0.01 + 6 \times 6 \times 0.01)}{\sqrt{16 + 36}} + 2 \times 4 \times 4 \times 0.01\right]$$

$$= \pi\left[4\sqrt{52} + 4\sqrt{52} + 32\right] \times 0.01$$

$$= \pi(8\sqrt{52} + 32) \times 0.01 = \pi(16\sqrt{13} + 32) \times 0.01$$

$$= \pi[16 \times 3.6 + 32] \times 0.01 = \pi[57.6 + 32] \times 0.01 = \pi[89.6] \times 0.01$$

$$= 3.14 \times 0.896 = 3.14 \times 0.9 = 2.826$$

$\Rightarrow \quad \delta S = 2.82$ sq. cm **Ans.**

Ex.4. ***ABC* is an acute-angled triangle with fixed base *BC*. If δb, δc, δA and δB are small increments in *b*, *c*, *A* and *B* respectively when the vertex *A* is given a small dispalcement parallel to BC. Prove that**

$$\delta A = h\delta x\ (1/c^2 - 1/b^2)$$

where *h* is the unaltered height of the triangle.

Sol. Given $BC = a$ is fixed.

Let $BM = x$

$\therefore$ From Δ AMC

$$\cot C = \frac{MC}{AM} = \frac{a - x}{AM}$$

or, $a - x = AM \cot C$

or, $-\delta x = -AM \operatorname{cosec}^2 C\delta C \qquad (\because a \text{ and } AM \text{ is constant})$

or, $\delta x = AM\left(\frac{AC}{AM}\right)^2 \delta C = \frac{AC^2}{AM}\delta C = \frac{b^2}{h}\delta C$

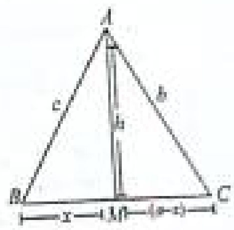
$$\Rightarrow \quad \delta C = \frac{h\delta x}{b^2} \qquad \text{...(1)}$$

Again, $\cot B = \frac{BM}{AM} = \frac{x}{h}$

$\Rightarrow \quad x = h\cot B$

$\Rightarrow \quad \delta x = -h\operatorname{cosec}^2 B\delta B \qquad (\delta h = 0, \because h \text{ is constant})$

$$= -h\left(\frac{AB}{AM}\right)^2 \delta B$$



$$= -h\frac{c^2}{h^2}\delta B = -\frac{c^2}{h}\delta B$$

$$\Rightarrow \quad \delta B = -\frac{h\delta x}{c^2} \qquad \text{...(2)}$$

Since, $A + B + C = \pi$

$\Rightarrow \quad \delta A + \delta B + \delta C = 0$

or, $\delta A = -(\delta B + \delta C)$

$$= -\left(-\frac{h\delta x}{c^2} + \frac{h\delta x}{b^2}\right) \qquad \text{(using (1) and (2))}$$

or, $\delta A = h\delta x\left(\frac{1}{c^2} - \frac{1}{b^2}\right)$ **Proved**

Ex.5. **The sides of an acute-angled triangle *ABC* are measured. Prove that the increment in *A* due to small increments in *a*, *b*, *c* is given by the equation**

$$bc \sin A\delta A = -a\ (\cos C\ \delta b + \cos B\ \delta c - \delta a)$$

Supposing that the limits of error in the length of any side are $\pm\mu$ percent, where μ is small, prove that the limits of error in A are approximately, $\pm$ 1.15 ($\mu a^2/bc \sin A$) degrees.

Sol. We know that in a $\Delta\, ABC$

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$\Rightarrow$ $a^2 = b^2+c^2-2bc\cos A$

or, $2a\delta a = 2b\delta b + 2c\delta c - 2b\cos A\delta c - 2c\cos A\delta b + 2bc\sin A\,\delta A$

or, $bc\sin A\delta A = a\delta a - b\delta b - c\delta c + b\cos A\delta c + c\cos A\delta b$

$= a\delta a + [c\cos A - b]\delta b + [b\cos A - c]\delta c$

Since, $a = b\cos C + c\cos B, b = a\cos C + c\cos A$

and $c = b\cos A + a\cos B$

$\therefore$ $bc\sin A\delta A = a\delta a + (-a\cos C)\delta b + (-a\cos B)\delta c$

or, $bc\sin A\delta A = -a(\cos C\delta b + \cos B\delta c - \delta a)$ **Proved**

Now, it is given that $\frac{\delta a}{a}\times 100 = \pm\mu = \frac{\delta b}{b}\times 100 = \frac{\delta c}{c}\times 100$

$\Rightarrow$ $\delta a = \pm\frac{\mu a}{100}$; $\delta b = \pm\frac{\mu b}{100}$; $\delta c = \pm\frac{\mu c}{100}$

$$\therefore\ bc\sin A\delta A = -a\left[\cos C\left(\pm\frac{\mu b}{100}\right) + \cos B\left(\pm\frac{\mu c}{100}\right) - \left(\pm\frac{\mu a}{100}\right)\right]$$

$$= -a\left(\frac{\mu}{100}\right)[\pm(b\cos C + c\cos B) - (\pm a)]$$

$$= -\frac{a\mu}{100}[(\pm a)\mp(a)]$$

For finding the limits of error in A, both 'a' in bracket must have take the same signs.

$$\therefore\ bc\sin A\delta A = \pm\frac{2\mu a^2}{100}\text{ radian} = \pm\frac{\mu a^2}{50}\frac{180}{\pi}\text{ degree}$$

$$= \pm 1{\cdot}15\,\mu a^2\text{ degree}$$

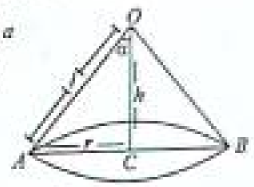
$\Rightarrow$ $\delta A = \frac{\pm 1{\cdot}15\mu a^2}{bc\sin A}$ degree **Proved**

Ex.6. **The height h and the semi-vertical angle α of a cone are measured and from them A, the total area of the cone, including the base is calculated. If h and α are in error by small quantities δh and $\delta\alpha$ respectively, find the correspoding error in the area.**

Also, show that if $\alpha = \frac{\pi}{6}$, an error of + 1% in h will be approximately comensated by an error of – 0.33° in α.

Sol. Total area of a cone $A = \pi r l + \pi r^2$

in $\Delta\, OAC \rightarrow AC = r = h\tan\alpha$ and $OA = l = h\sec\alpha$



$\therefore$ $A = \pi h^2\tan\alpha\sec\alpha + \pi h^2\tan^2\alpha$

$\therefore$ $A = \pi h^2(\tan\alpha\sec\alpha + \tan^2\alpha)$

Now,

$$\delta A = \frac{\partial A}{\partial h}\delta h + \frac{\partial A}{\partial\alpha}\delta\alpha;$$

Given $\rightarrow$ $\alpha = \frac{\pi}{6}$ and $\frac{\delta h}{h}\times 100 = 1 \Rightarrow \delta h = \frac{h}{100}$

$\therefore$ $\delta A = 2\pi h(\tan\alpha\sec\alpha + \tan^2\alpha)\delta h + \pi h^2$

$(\sec^3\alpha + \tan^2\alpha\sec\alpha + 2\tan\alpha\sec^2\alpha)\delta\alpha$ Ans.

$$= 2\pi h^2\times(.01) + \pi h^2\left(\frac{8}{3\sqrt{3}} + \frac{1}{3}\cdot\frac{2}{\sqrt{3}} + 2\cdot\frac{1}{\sqrt{3}}\cdot\frac{4}{3}\right)\delta\alpha$$

$$2\pi h^2\times(0.01) + \pi h^2\frac{18}{3\sqrt{3}}\delta\alpha$$

$$= 2\pi h^2\times .01 + \frac{18\sqrt{3}}{9}\pi h^2\times\delta\alpha = 2\pi h^2\times 0.01 + 2\times 1.732\pi h^2\times\delta\alpha$$

$\Rightarrow$ $\delta A = \pi h^2(.02 + 3.464\delta\alpha)$

If $\delta A = 0$

$\Rightarrow$ $0 = \pi h^2(.02 + 3.464\,\delta\alpha)$

$\Rightarrow$ $\delta\alpha = -\frac{0.02}{3.464}$ radain

$$= -\frac{0.02}{3.464}\times\frac{180}{\pi}\text{ degree}\quad\left(\because 1\text{ radian} = \frac{180}{\pi}\text{ degree}\right)$$

$$= -\frac{0.02}{3.464} \times 57.3 \text{ degree}$$

$\Rightarrow \quad \delta a = -0.33^\circ$. **Proved**

Ex.7. In a plane triangle, if the angles and sides receive small variations, prove that:

(i) $(\delta a)\cos C + (\delta c)\cos A = 0$; b, B being constants

(ii) $c\delta A + (\delta C)\, a\cos B = 0$; a, b being constants

Sol. (i) We know that

$$b = c\cos A + a\cos C$$

$\Rightarrow$ b is a function of c, A, a, C

$$\therefore \quad \delta b = \frac{\partial b}{\partial c}\delta c + \frac{\partial b}{\delta A}\delta A + \frac{\partial b}{\delta a}\delta a + \frac{\partial b}{\delta C}\delta C$$

$\delta b = 0$, $\because$ b is constant

$$\therefore \quad 0 = (\cos A)\delta c + (-c\sin A)\delta A + (\delta a)\cos C - a\sin C\delta C$$

$$\Rightarrow \quad (\delta c)\cos A + (\delta a)\cos C = c\sin A\delta A + a\sin C\delta C$$

$$\because \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow a\sin C = c\sin A$$

$$\therefore \quad (\delta c)\cos A + (\delta a)\cos C = c\sin A(\delta A + \delta C)$$

$$= -c\sin A(\delta B)$$

$$= 0, \quad \because \delta B = 0$$

$[\because A + B + C = 180^\circ \Rightarrow \delta A + \delta B + \delta C = 0$

$\Rightarrow \quad \delta A + \delta C = -\delta B]$ **Proved**

(B is given to be a constant)

(ii) Here a and b are constant prove we know that

$$a = b\cos C + c\cos B.$$

$$\therefore \quad \delta a = \frac{\partial a}{\partial b}\delta b + \frac{\partial a}{\partial C}\delta C + \frac{\partial a}{\partial c}\delta c + \frac{\partial a}{\partial B}\delta B$$

$\because$ a being a constant $\Rightarrow \delta a = 0$

$$\Rightarrow o = (\cos C)\delta b + (-b\sin C)\delta C + (\cos B)\delta c + (-c\sin B)\delta B$$

$$\Rightarrow \quad \delta c\cos B = \delta cb\sin C + \delta Bc\sin B \qquad (\because b\sin C = c\sin B)$$

$$= b\sin C(-\delta A) \qquad (\because \delta A + \delta B + \delta C = 0)$$

$$\Rightarrow \quad \delta c\cos B = -b\sin C\delta A \qquad \text{...(1)}$$

Now $\quad c = a\cos B + b\cos A \qquad \text{...(2)}$

$$\Rightarrow \quad \delta c = (\delta a)\cos B - a\sin B\delta B + (\delta b)\cos A - (b\sin A)\cdot\delta A$$

$$= -a\sin B\delta B - b\sin A\delta A; \ \delta a = 0 = \delta b$$

$$= -a\sin B(\delta B + \delta A) \qquad (\because a\&b \text{ are constants})$$

$$\Rightarrow \quad \delta c = a\sin B(\delta c) \qquad (\because \delta A + \delta B + \delta C = 0$$

$$\Rightarrow \quad \delta B + \delta A = -\delta C)$$

Putting this value of δc in (I) we get :

$$(a\sin B\delta C)\cos B = -b\sin C\delta A$$

$$\because \quad \frac{b}{\sin B} = \frac{c}{\sin C} \text{ i.e. } b\sin C = c\sin B$$

It gives

$$a\sin B\cos B\delta C = -c\sin B\delta A$$

$\Rightarrow \quad c\delta A + a\cos B\delta C = 0$ **Proved**

Scanned with

12. Linear Differential Equations with Constant Coefficients

Linear differential equation with constant coefficients

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots\ldots + a_n y = Q(x)$$

where a_0, a_1, a_2 .... a_n are constants

The solution of this equation is :

$y = C.F. + P.I.$

Methods for finding C.F. (Complementary Function) :

Let differential equation be

$$a_0 \frac{d^4 y}{dx^4} + a_1 \frac{d^3 y}{dx^3} + a_2 \frac{d^2 y}{dx^2} + a_3 \frac{dy}{dx} + a_4 = Q(x)$$

Then to find the C.F. we substitude 'm' in place of $\frac{dy}{dx} = Dy$ in the

equation $a_0 \frac{d^4 y}{dx^4} + a_1 \frac{d^3 y}{dx^3} + a_2 \frac{d^2 y}{dx^2} + a_3 \frac{dy}{dx} + a_4 = 0$ [taking $Q(x) = 0$]

And thus find the Auxiliary equation (AE)

$\therefore$ AE here is

$a_0 m^4 + a_1 m^3 + a_2 m^2 + a_3 m + a_4 = 0$

This equation will have four roots :

(a) If the roots of this AE are real and different say m_1, m_2, m_3, m_4

then C.F. = $C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$

(b) If the roots of AE are such that two roots are consident i.e. m_1, m_1, m_2, m_3 are the roots

Then C.F. = $(C_0 + C_1x)e^{m_1x} + C_2e^{m_2x} + C_3e^{m_3x}$

If three roots are coincident i.e. roots are = m_1, m_1, m_1, m_2

Then C.F. = $(C_0 + C_1x + C_2x^2)e^{m_1x} + C_3e^{m_2x}$

(c) If the roots of AE are real and in surds

i.e. let the roots are $m_1 \pm \sqrt{m_2}, m_3, m_4$

Then C.F. = $e^{m_1x}\left[C_1e^{\sqrt{m_2}x} + C_2e^{-\sqrt{m_2}x}\right] + C_3e^{m_3x} + C_4e^{m_4x}$

or C.F. = $e^{m_1x}\left[C_1\cosh\sqrt{m_2}x + C_2\sinh\sqrt{m_2}x\right] + C_3e^{m_3x} + c_4e^{m_4x}$

or C.F. = $Ae^{m_1x}\cosh\left(\sqrt{m_2}x + B\right) + C_3e^{m_3x} + C_4e^{m_4x}$

or C.F. = $Ce^{m_1x}\sinh\left(\sqrt{m_2}x + D\right) + C_3e^{m_3x} + C_4e^{m_4x}$

(d) If any of roots of the AE are complex

i.e. Let the roots be $m_1 \pm i\,m_2, m_3, m_4$

Then

C.F. = $e^{m_1x}(C_1\cos m_2x + C_2\sin m_2x) + C_3e^{m_3x} + C_4e^{m_4x}$

or C.F. = $Ae^{m_1x}\cos(m_2x + B) + C_3e^{m_3x} + C_4e^{m_4x}$

or C.F. = $Ce^{m_1x}\sin(m_2x + D) + C_3e^{m_3x} + C_4e^{m_4x}$

(e) If the complex roots of AE are repeated i.e. roots are $m_1 \pm i\,m_2$, Then

C.F. = $e^{m_1x}\left[(C_1 + C_2x)\cos m_2x + (C_3 + C_4x)\sin m_2x\right]$

Methods for finding Particular Integrals (P.I.)

For the differential equation $(D - \alpha)\,y = Q(x)$,

$$\text{P.I.} = \frac{1}{D-\alpha}Q(x) = e^{\alpha x}\int e^{-\alpha x}Q(x)\,dx$$

If the differential equation is $f(D)\,y = Q(x)$, then we have

$$\text{P.I.} = \frac{1}{f(D)}Q(x)$$

then resolve $f(D)$ into partial fractions and find P.I. by above given formula.

Special cases for finding P.I.

In the differential equation $f(D)\,y = Q(x)$ we have some special formula for particular forms of $Q(x)$

(i) When $Q(x) = e^{ax}$.

(a) $\text{P.I.} = \frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}$, where $f(a) \neq 0$

(b) $\text{P.I.} = \frac{1}{\phi(D)(D-a)^r}e^{ax} = \frac{1}{\phi(a)}\frac{x^r}{\lfloor r}e^{ax}$, where $\phi(a) \neq 0$

(ii) When $Q(x) = \cos ax$ or $\sin ax$.

(a) $\text{P.I.} = \frac{1}{f(D^2)}\cos ax = \frac{1}{f(-a^2)}\cos ax$, where $f(-a^2) \neq 0$

& $\text{P.I.} = \frac{1}{f(D^2)}\sin ax = \frac{1}{f(-a^2)}\sin ax$, where $f(-a^2) \neq 0$

(b) $\text{P.I.} = \frac{1}{D^2+a^2}\cos ax = \frac{x}{2a}\sin ax$

& $\text{P.I.} = \frac{1}{D^2+a^2}\sin ax = -\frac{x}{2a}\cos ax$

(iii) When $Q(x) = x^m$

$$\text{P.I.} = \frac{1}{f(D)}x^m$$

We break $f(D)$ into partial fractions and solve as :

$\text{P.I.} = \frac{1}{f(D)}x^m = \frac{1}{D-a}x^m$ [considering $f(D) = (D - a)$]

$\Rightarrow$ P.I. $= -\frac{1}{a}\left[1-\frac{D}{a}\right]^{-1} x^m$

We expand $\left(1-\frac{D}{a}\right)^{-1}$ in ascending powers of D till D^m and operate x^m by the various terms :

(iv) When $Q(x) = e^{ax} \cdot U(x)$

Then $\frac{1}{f(D)} e^{ax} U(x) = e^{ax} \frac{1}{f(D+a)} U(x)$

(v) $Q(x) = x\, U(x)$

Then $\frac{1}{f(D)}(xU) = x\frac{1}{f(D)}U + \frac{d}{dD}\left[\frac{1}{f(D)}\right]U$

Note :

(i) $\cosh ax = \frac{e^{ax}+e^{-ax}}{2}$

(ii) $\sinh ax = \frac{e^{ax}-e^{-ax}}{2}$

(iii) $\cos ax = \frac{e^{iax}+e^{-iax}}{2}$

(iv) $\sin ax = \frac{e^{iax}-e^{-iax}}{2i}$

(v) $\cos ax$ = Real part of (e^{iax}) and $\sin ax$ = Imaginary part of (e^{iax})

EX-33

Exercise 12(A)

Solve :

Ex.1 $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = 0$.

Sol. It can be written as :

$(D^2 - 4D + 1)\,y = 0$

$\therefore$ A. E. is :

$m^2 - 4m + 1 = 0$

$\Rightarrow$ $m = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

$\therefore$ Solution is :

$y = e^{m_1 x}[c_1 \cosh m_2 x + c_2 \sinh m_2 x]$

Here, $m_1 = 2$, $m_2 = \sqrt{3}$

$\therefore$ $y = e^{2x}(c_1 \cosh\sqrt{3}x + c_2 \sinh\sqrt{3}x)$ **Ans.**

Ex.2 $\frac{d^2y}{dx^2} + y = 0$, given $y = 2$ for $x = 0$ and $y = -2$ for $x = \frac{\pi}{2}$.

Sol. Given equation is :

$(D^2 + 1)y = 0$; $D = \frac{d}{dx}$

$\therefore$ A. E. $\to m^2 + 1 = 0 \Rightarrow m = \pm i$

$\therefore$ Solution is :

$y = c_1 \cos x + c_2 \sin x$ ...(1)

given, $y = 2$ for $x = 0$

$\therefore$ (1) gives $\to 2 = c_1$ ...(2)

and $y = -2$ for $x = \frac{\pi}{2}$

$\therefore$ (1) gives $\to -2 = c_2$ ...(3)

using (2) and (3) in (1), we get :

$$y = 2\cos x - 2\sin x = 2\sqrt{2}\cos\left(x+\frac{\pi}{4}\right)$$

$$\left[\because \cos\left(x+\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}(\cos x - \sin x)\right] \quad \textbf{Ans.}$$

Ex.3 $\dfrac{d^2y}{dx^2} + 2p\dfrac{dy}{dx} + (p^2+q^2)y = 0.$

Sol. Given equation is :

$$\left[D^2 + 2pD + (p^2+q^2)\right]y = 0;\ D = \frac{d}{dx}$$

$\therefore$ A. E. is : $m^2 + 2mp + (p^2+q^2) = 0$

$$\Rightarrow \quad m = \frac{-2p \pm \sqrt{4p^2 - 4(p^2+q^2)}}{2} = -p \pm iq$$

$\therefore$ Solution is :

$$y = e^{m_1 x}(c_1\cos m_2 x + c_2 \sin m_2 x)$$

Here $m_1 = -p$, $m_2 = q$

$\therefore$ $y = e^{-px}(c_1\cos qx + c_2\sin qx)$ **Ans.**

Ex.4 $\dfrac{d^2y}{dx^2} - 7\dfrac{dy}{dx} + 12y = 0.$

Sol. Given equation can be written as

$$(D^2 - 7D + 12)\,y = 0;\quad D = \frac{d}{dx}$$

$\therefore$ Auxiliary equation (A. E.) is –

$m^2 - 7m + 12 = 0 \Rightarrow (m-4)(m-3) = 0 \Rightarrow m = 4, 3$

$\therefore$ solution is :

$y = c_1 e^{3x} + c_2 e^{4x}$ **Ans.**

Ex.5 $\dfrac{d^2x}{dt^2} - 3\dfrac{dx}{dt} + 2x = 0$; given $t = 0$, $x = 0$ and $\dfrac{dx}{dt} = 0$.

Sol. It can be written as :

$$(D^2 - 3D + 2)x = 0;\ D = \frac{d}{dt}$$

$\therefore$ A. E. is :

$$m^2 - 3m + 2 = 0 \Rightarrow (m-1)(m-2) = 0 \Rightarrow m = 1, 2$$

$\therefore$ solution is :

$x = c_1 e^t + c_2 e^{2t}$ ...(1)

$\Rightarrow$ $\dfrac{dx}{dt} = c_1 e^t + 2c_2 e^{2t}$ ...(2)

Given, $t = 0$, $x = 0$

$\therefore$ (1) gives, $0 = c_1 + c_2$ ...(3)

and $t = 0$, $\dfrac{dx}{dt} = 0$, (2) gives, $0 = c_1 + 2c_2$ ...(4)

$\therefore$ (3) and (4) $\Rightarrow$ $c_1 = 0 = c_2$

$\therefore$ (1) gives :

$x = 0$ **Ans.**

Ex.6 $(D^3 - D^2 - D + 1)\,y = 0.$

Sol. A. E. is :

$m^3 - m^2 - m + 1 = 0 \Rightarrow (m^2-1)(m-1) = 0 \Rightarrow m = 1, 1, -1$

$\therefore$ Solution is :

$y = (C_1 + C_2 x)\,e^x + C_3 e^{-x}$ **Ans.**

Ex.7 $\dfrac{d^3y}{dx^3} - 6\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} - 6y = 0.$

Sol. In can be written as

$$(D^3 - 6D^2 + 11D - 6)\,y = 0;\ D = \frac{d}{dx}$$

A. E. is

$$m^3 - 6m^2 + 11m - 6 = 0$$

$\Rightarrow$ $(m-1)(m-2)(m-3) = 0$

$\Rightarrow$ $m = 1, 2, 3$

$\therefore$ Solution is–

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x} \quad \textbf{Ans.}$$

Ex.8 $(D^4 + 8D^2 + 16)\,y = 0.$

Sol. A. E. is :

$$m^4 + 8m^2 + 16 = 0 \Rightarrow (m^2+4)^2 = 0 \Rightarrow m = \pm 2i, \pm 2i$$

$\therefore$ Solution is :

$$y = e^{m_1 x}[(c_1 + c_2 x)\cos m_2 x + (c_3 + c_4 x)\sin m_2 x]$$

Here, $m = \pm 2i, \pm 2i \Rightarrow m_1 = 0$ and $m_2 = 2$

$\therefore$ $y = (c_1 + c_2 x)\cos 2x + (c_3 + c_4 x)\sin 2x$ **Ans.**

Ex.9 $(D^4 - 2D^2 + 1)\, y = 0.$

Sol. A. E. is :

$$m^4 - 2m^2 + 1 = 0 \Rightarrow (m^2-1)^2 = 0 \Rightarrow m = 1, 1, -1, -1$$

$\therefore$ Solution is :

$$y = (c_1 + c_2 x)e^x + (c_3 + c_4 x)e^{-x}$$ **Ans.**

Ex-34

Exercise 12(b)

Solve :

Ex.1 $\dfrac{d^2y}{dx^2} - 4y = e^x + \sin 2x$

Sol. Given equation is :

$$(D^2 - 4)y = e^x + \sin 2x$$

A. E. $\rightarrow$ $m^2 - 4 = 0 \Rightarrow m = \pm 2$

$\therefore$ $C.F. = c_1 e^{2x} + c_2 e^{-2x}$ ....(1)

$$\text{P. I.} = \frac{1}{(D^2-4)}e^x + \frac{1}{(D^2-4)}\sin 2x$$

$$= \frac{1}{1-4}e^x + \frac{1}{-4-4}\sin 2x = -\frac{1}{3}e^x - \frac{1}{8}\sin 2x \quad ....(2)$$

$\therefore$ Complete solution is :

$y = C.F. + P.I.$

$\Rightarrow$ $y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{3}e^x - \frac{1}{8}\sin 2x$ **Ans.**

Ex.2 $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + y = a\sin 2x.$

Sol. Given equation is :

$(D^2 - 4D + 1)\, y = a \sin 2x$

$A.E. \rightarrow m^2 - 4m + 1 = 0 \Rightarrow m = 2 \pm \sqrt{3}$

$\therefore$ C. F. $= c_1 e^{2x}\cosh\left(\sqrt{3}x + c_2\right)$ ....(1)

$$\text{P. I.} = \frac{1}{D^2 - 4D + 1} a\sin 2x$$

$$= a\frac{1}{-4-4D+1}\sin 2x = \frac{-a}{(4D+3)}\left(\frac{4D-3}{4D-3}\right)\sin 2x$$

$$= \frac{-a(4D-3)}{(16D^2-9)}\sin 2x = \frac{-a(4D-3)}{-64-9}\sin 2x$$

$$= \frac{a}{73}(8\cos 2x - 3\sin 2x) \quad ....(2)$$

$\therefore$ Complete solution is :

$y = C.F. + P.I.$

$\Rightarrow$ $y = c_1 e^{2x}\cosh\left(\sqrt{3}x + c_2\right) + \frac{a}{73}(8\cos 2x - 3\sin 2x)$ **Ans.**

Ex.3 $(D^2 - 3D + 2)y = \sin 3x.$

Sol. $A.E.$ is :

$$m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0 \Rightarrow m = 1, 2$$

$C.F. = c_1 e^x + c_2 e^{2x}$ ....(1)

$$P.I. = \frac{1}{D^2 - 3D + 2}\sin 3x$$

$$= \frac{1}{-9-3D+2}\sin 3x \qquad \left[\because \frac{1}{f(D^2)}\sin ax = \frac{1}{f(-a^2)}\sin ax\right]$$

$$= -\frac{1}{(3D+7)}\times\left(\frac{3D-7}{3D-7}\right)\sin 3x = -\frac{(3D-7)}{9D^2-49}\sin 3x$$

$$= -\frac{(3D-7)}{9(-9)-49}\sin 3x = \frac{1}{130}(3D-7)\sin 3x$$

$\therefore \quad P.I. = \frac{1}{130}(9\cos 3x - 7\sin 3x)$ ....(2)

$\therefore$ Complete solution is :

$y = C.F. + P.I$

$\Rightarrow \quad y = c_1 e^x + c_2 e^{2x} + \frac{1}{130}(9\cos 3x - 7\sin 3x)$ **Ans.**

Ex.4 $(D^3 + 1)y = \cos 2x$.

Sol. A. E. is :

$$m^3 + 1 = 0 \Rightarrow (m+1)(m^2 - m + 1) = 0$$

$\Rightarrow \quad m = -1,$ and $m^2 - m + 1 = 0$

$\Rightarrow \quad m = \frac{1 \pm i\sqrt{3}}{2}$

$\therefore \quad m = -1, \frac{1 \pm i\sqrt{3}}{2}$

$\therefore \quad C.F. = c_1 e^{-x} + e^{\frac{x}{2}}\left(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x\right)$ ....(1)

$$P.I. = \frac{1}{D^3+1}\cos 2x = \frac{1}{D^2 D + 1}\cos 2x = \frac{1}{-4D+1}\cos 2x$$

or,

$$P.I. = \frac{1}{(1-4D)}\left(\frac{1+4D}{1+4D}\right)\cos 2x = \frac{1+4D}{1-16D^2}\cos 2x = \frac{(1+4D)}{1+64}\cos 2x$$

$$= \frac{1}{65}(\cos 2x - 8\sin 2x)$$

$\therefore$ Complete solution is :

$y = C.F. + P.I.$

or, $\quad y = c_1 e^{-x} + e^{\frac{x}{2}}\left(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x\right)$

$+ \frac{1}{65}(\cos 2x - 8\sin 2x)$ **Ans.**

Ex.5 $(D^2 - 4D + 1)y = \cos 2x$.

Sol. A. E. is :

$$m^2 - 4m + 1 = 0 \Rightarrow m = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$$

$\therefore \quad C.F. = e^{2x}(c_1 \cosh\sqrt{3}x + c_2 \sinh\sqrt{3}x)$ ....(1)

$$P.I. = \frac{1}{D^2 - 4D + 1}\cos 2x$$

$$= \frac{1}{-4 - 4D + 1}\cos 2x = \frac{-1}{3+4D}\cos 2x$$

$$= -\frac{1}{(3+4D)}\left(\frac{3-4D}{3-4D}\right)\cos 2x = \frac{-(3-4D)}{9-16D^2}\cos 2x$$

$$= \frac{-(3-4D)}{9+64}\cos 2x = -\frac{1}{73}(3\cos 2x + 8\sin 2x)$$

$\therefore$ Complete solution is :

$y = C.F. + P.I.$

$\Rightarrow \quad y = e^{2x}\left(c_1 \cosh\sqrt{3}x + c_2 \sinh\sqrt{3}x\right) - \frac{1}{73}(3\cos 2x + 8\sin 2x)$ **Ans.**

Ex.6 $(D^2 - 4D + 4)\, y = e^{-4x} + 5\cos 3x$.

Sol. A. E. is :

$$m^2 - 4m + 4 = 0 \Rightarrow (m-2)^2 = 0 \Rightarrow m = 2, 2$$

$\therefore \quad C.F. = (c_1 + c_2 x)e^{2x}$

$$P.I. = \frac{1}{D^2 - 4D + 4}e^{-4x} + 5\frac{1}{D^2 - 4D + 4}\cos 3x$$

$$= \frac{1}{16+16+4}e^{-4x} + 5\frac{1}{-9-4D+4}\cos 3x$$

$$= \frac{1}{36}e^{-4x} - 5\frac{1}{(4D+5)}\frac{(4D-5)}{(4D-5)}\cos 3x = \frac{1}{36}e^{-4x} - \frac{5(4D-5)}{16D^2-25}\cos 3x$$

$$= \frac{1}{36}e^{-4x} - \frac{5(4D-5)}{-144-25}\cos 3x = \frac{1}{36}e^{-4x} - \frac{5}{169}(12\sin 3x + 5\cos 3x)$$

$\therefore$ Complete solution is :

$y = C.F. + P.I \Rightarrow$

$$y = (c_1 + c_2 x)e^{2x} + \frac{1}{36}e^{-4x} - \frac{5}{169}(12\sin 3x + 5\cos 3x) \quad \textbf{Ans.}$$

Ex.7 $\dfrac{d^2y}{dx^2} + y = \operatorname{cosec} x$, **[MNIT., 2003]**

Sol. It can be written as

$(D^2+1)y = \operatorname{cosec} x$; $D = \dfrac{d}{dx}$ A. E. is –

A.E. $\rightarrow$ $m^2 + 1 = 0$ $m = \pm i$

$\therefore$ $C.F. = c_1 \cos x + c_2 \sin x$

$$\text{P. I.} = \frac{1}{(D^2+1)}\operatorname{cosec} x = \frac{1}{(D+i)(D-i)}\operatorname{cosec} x$$

$$= \frac{1}{2i}\left[\frac{1}{(D-i)} - \frac{1}{(D+i)}\right]\operatorname{cosec} x$$

$$= \frac{1}{2i}\left[e^{ix}\int e^{-ix}\operatorname{cosec} x\,dx - e^{-ix}\int e^{ix}\operatorname{cosec} x\,dx\right]$$

$$= \frac{1}{2i}\left[e^{ix}\int (\cos x - i\sin x)\operatorname{cosec} x\,dx - e^{-ix}\right.$$

$$\left.\int (\cos x + i\sin x)\operatorname{cosec} x\,dx\right]$$

$$= \frac{1}{2i}\left[e^{ix}\int (\cot x - i)dx - e^{-ix}\int (\cot x + i)dx\right]$$

$$= \frac{1}{2i}\left[(e^{ix} - e^{-ix})\int \cot x\,dx - i(e^{ix} + e^{-ix})\int dx\right]$$

$$= \frac{1}{2i}\left[(2i\sin x)(\log\sin x) - i(2\cos x)(x)\right]$$

$= \sin x \log \sin x - x\cos x$

$C.S.$ is

$y = C.F. + P.I.$

or, $y = C_1 \cos x + C_2 \sin x + \sin x \log \sin x - x\cos x$ **Ans.**

Ex.8 $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{5x}$. [Raj., 2002]

Sol. It can written as

$(D^2 - 3D + 2)y = e^{5x}$

$\therefore$ A. E. $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$

$\therefore$ C. F. $= c_1 e^x + c_2 e^{2x}$

$$\text{P. I.} = \frac{1}{(D^2 - 3D + 2)}e^{5x}$$

$$= \frac{1}{(25 - 15 + 2)}e^{5x}$$

$$= \frac{1}{12}e^{5x}$$

$\therefore$ C. S is –

$y = \text{C. F.} + \text{P. I.}$

or, $y = c_1 e^x + c_2 e^{2x} + \dfrac{1}{12}e^{5x}$ **Ans.**

Ex.9 $\dfrac{d^2y}{dx^2} - 7\dfrac{dy}{dx} + 6y = e^{2x}$; **$y = 0$ when $x = 0$.**

Sol. Given equation is :

$$(D^2 - 7D + 6)y = e^{2x}$$

A. E. is :

$$m^2 - 7m + 6 = 0 \Rightarrow (m-6)(m-1) = 0 \Rightarrow m = 1, 6$$

$C.F. = c_1 e^x + c_2 e^{6x}$ ...(1)

$$P.I. = \frac{1}{(D^2 - 7D + 6)}e^{2x} = \frac{1}{4 - 14 + 6}e^{2x} = -\frac{1}{4}e^{2x} \quad \text{...(2)}$$

$\therefore$ Complete solution is :

$y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow \quad y = c_1e^{x} + c_2e^{6x} - \frac{1}{4}e^{2x} \qquad ...(3)$$

given $x = 0,\ y = 0 \Rightarrow 0 = c_1 + c_2 - \frac{1}{4} \Rightarrow c_1 = \frac{1}{4} - c_2$

$\therefore$ (3) gives

$$y = \left(\frac{1}{4} - c_2\right)e^{x} + c_2e^{6x} - \frac{1}{4}e^{2x}$$

$$= c(e^{x} - e^{6x}) + \frac{1}{4}e^{x}(1 - e^{x})\ ;\ c = -c_2 \qquad \textbf{Ans.}$$

Ex.10 $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 5y = \sin 3x$.

Sol. Given equation is :

$$(D^2 - 2D + 5)y = \sin 3x$$

A. E. $\rightarrow$ $m^2 - 2m + 5 = 0$

$$\Rightarrow \quad m = \frac{2 \pm \sqrt{4 - 20}}{2} = 1 \pm 2i$$

$\therefore$ C. F. $= c_1e^{x}\cos(2x + c_2)$ ...(1)

$$P.I. = \frac{1}{D^2 - 2D + 5}\sin 3x$$

$$= \frac{1}{-9 - 2D + 5}\sin 3x = -\frac{1}{2D + 4}\sin 3x$$

$$= -\frac{1}{2}\frac{1}{(D + 2)}\left(\frac{D - 2}{(D - 2)}\right)\sin 3x$$

$$= \frac{-(D - 2)}{2(D^2 - 4)}\sin 3x = -\frac{(D - 2)}{2(-9 - 4)}\sin 3x$$

$$\text{P.I.} = \frac{1}{26}(3\cos 3x - 2\sin 3x)$$

$\therefore$ **Complete solution is**

$$y = C.F. + P.I.$$

$$\Rightarrow \quad y = c_1e^{x}\cos(2x + c_2) + \frac{3}{26}\cos 3x - \frac{1}{13}\sin 3x \qquad \textbf{Ans.}$$

Ex.11 $\dfrac{d^2y}{dx^2} - 8\dfrac{dy}{dx} + 9y = 40\sin 5x$.

Sol. Given equation is :

$$(D^2 - 8D + 9)y = 40\sin 5x$$

A. E. is :

$$m^2 - 8m + 9 = 0$$

$$\therefore \quad m = \frac{8 \pm \sqrt{64 - 36}}{2} = 4 \pm \sqrt{7}$$

$$\therefore \quad C.F. = c_1e^{4x}\cosh\left(\sqrt{7}x + c_2\right) \qquad ...(1)$$

$$P.I. = \frac{1}{(D^2 - 8D + 9)}40\sin 5x$$

$$= 40\frac{1}{-25 - 8D + 9}\sin 5x = -40\frac{1}{8D + 16}\sin 5x$$

$$= -5\frac{1}{(D + 2)}\left(\frac{D - 2}{D - 2}\right)\sin 5x$$

$$P.I. = -5\frac{(D - 2)}{D^2 - 4}\sin 5x = \frac{-5(D - 2)}{-25 - 4}\sin 5x$$

$$= \frac{5}{29}(5\cos 5x - 2\sin 5x)$$

$\therefore$ Complete solution is :

$$y = C.F. + P.I.$$

$$\Rightarrow \quad y = c_1e^{4x}\cosh\left(\sqrt{7}x + c_2\right) + \frac{25}{29}\cos 5x - \frac{10}{29}\sin 5x \qquad \textbf{Ans.}$$

Ex.12 $\dfrac{d^2y}{dx^2} + 9y = \cos 2x + \sin 2x$.

Sol. Given equation is :

$$(D^2 + 9)y = \cos 2x + \sin 2x$$

A. E. is $\rightarrow$ $m^2 + 9 = 0 \Rightarrow m = \pm 3i$

$\therefore \quad C.F. = c_1 \cos 3x + c_2 \sin 3x \qquad$ ....(1)

$$P.I. = \frac{1}{(D^2+9)}\cos 2x + \frac{1}{(D^2+9)}\sin 2x$$

$$= \frac{1}{-4+9}\cos 2x + \frac{1}{-4+9}\sin 2x = \frac{1}{5}(\cos 2x + \sin 2x) \qquad \text{....(2)}$$

$\therefore$ Complete solution is –

$$y = C.F. + P.I.$$

$$\Rightarrow \quad y = c_1\cos 3x + c_2 \sin 3x + \frac{1}{5}(\cos 2x + \sin 2x) \qquad \textbf{Ans.}$$

Ex.13 $\dfrac{d^2y}{dx^2} + 4y = \tan 2x$. **[Raj., 2001]**

Sol. Given equation is

$$(D^2+4)y = \tan 2x$$

$\therefore$ A. E. is –

$$m^2 + 4 = 0 \Rightarrow m = \pm 2i$$

$\therefore \quad$ C. F. $= c_1 \cos 2x + c_2 \sin 2x \qquad$ ....(1)

$$\text{P. I.} = \frac{1}{(D^2+4)}\tan 2x$$

$$= \frac{1}{4i}\left[\frac{1}{D-2i} - \frac{1}{D+2i}\right]\tan 2x$$

$$= \frac{1}{4i}\left[e^{2ix}\int e^{-2ix}\tan 2x\,dx - e^{-2ix}\int (e^{2ix})\tan 2x\,dx\right]$$

$$= \frac{1}{4i}\left[e^{2ix}\int (\cos 2x - i\sin 2x)\tan 2x\,dx - e^{-2ix}\int(\cos 2x + i\sin 2x)\tan 2x\,dx\right]$$

$$= \frac{1}{4i}\left[\left(e^{2ix} - e^{-2ix}\right)\int \sin 2x\,dx - i(e^{2ix}+e^{-2ix})\int \frac{\sin^2 2x}{\cos 2x}dx\right]$$

$$= \frac{1}{4i}\left[(2i\sin 2x)\left(-\frac{1}{2}\cos 2x\right) - i(2\cos 2x)\int \frac{1-\cos^2 2x}{\cos 2x}dx\right]$$

$$= \frac{-1}{4}\sin 2x\cos 2x - \frac{1}{2}\cos 2x\left\{\int \sec 2x\,dx - \int \cos 2x\,dx\right\}$$

$$= \frac{-1}{4}\sin 2x\cos 2x - \frac{1}{2}\cos 2x\left\{\frac{1}{2}\log(\sec 2x + \tan 2x) - \frac{1}{2}\sin 2x\right\}$$

$$= \frac{-1}{4}\cancel{\sin 2x\cos 2x} - \frac{1}{4}\cos 2x\log(\sec 2x + \tan 2x) + \frac{1}{4}\cancel{\cos 2x\sin 2x}$$

$\therefore \quad$ P. I. $= -\dfrac{1}{4}\cos 2x \log(\sec 2x + \tan 2x) \qquad$ ....(2)

$\therefore \quad$ complete solution is–

$$y = C.F. + P.I$$

or, $\quad y = c_1 \cos 2x + c_2 \sin 2x - \dfrac{1}{4}\cos 2x \log(\sec 2x + \tan 2x) \quad$ **Ans.**

Ex.14 $\dfrac{d^2x}{dt^2} + 2n\cos\alpha\dfrac{dx}{dt} + n^2x = a\cos nt$, such that $t = 0, x = 0$,

$\dfrac{dx}{dt} = 0$.

Sol. Given equation is :

$$(D^2 + 2n\cos\alpha\, D + n^2)x = a\cos nt, \quad D = \frac{d}{dt}$$

$$A.E. \rightarrow \quad m^2 + (2n\cos\alpha)m + n^2 = 0$$

$$\Rightarrow \quad m = \frac{-2n\cos\alpha \pm \sqrt{4n^2\cos^2\alpha - 4n^2}}{2} = -n\cos\alpha \pm in\sin\alpha$$

$\therefore \quad$ C. F. $= e^{(-n\cos\alpha)t}\left[c_1\cos(n\sin\alpha)t + c_2\sin(n\sin\alpha)t\right] \qquad$ ....(1)

$$P.I. = \frac{1}{D^2 + (2n\cos\alpha)D + n^2}a\cos nt$$

$$= \frac{a}{-n^2 + (2n\cos\alpha)D + n^2}\cos nt$$

$$= \frac{a}{2n\cos\alpha}\frac{1}{D}\cos nt = \frac{a}{2n^2\cos\alpha}\sin nt \qquad \text{....(2)}$$

$\therefore$ Complete solution is–

$x = C.F. + P.I.$

$$\Rightarrow \quad x = e^{-nt\cos\alpha}\left[c_1\cos(n\sin\alpha)t + c_2\sin(n\sin\alpha)t\right]$$

$$+\frac{a}{2n^2\cos\alpha}\sin nt \qquad \text{...(3)}$$

Also, $\frac{dx}{dt} = e^{-nt\cos\alpha}$

$$(-n\cos\alpha)\left[c_1\cos(n\sin\alpha)t + c_2\sin(n\sin\alpha)t\right]$$

$$+e^{-nt\cos\alpha}\left[-c_1\sin(n\sin\alpha)t + c_2\cos(n\sin\alpha)t\right](n\sin\alpha)$$

$$+\frac{an}{2n^2\cos\alpha}\cos nt \qquad \text{...(4)}$$

Given, $t = 0, x = 0$

$\therefore$ (3) gives :

$0 = c_1 \qquad \text{...(5)}$

Also, $t = 0,\ \frac{dx}{dt} = 0$ and from (5), $c_1 = 0$

$\therefore$ (4) gives :

$$0 = (-n\cos\alpha)\ (0+0) + n\sin\alpha[c_2] + \frac{an}{2n^2\cos\alpha}$$

$$\Rightarrow \quad c_2 = -\frac{a}{2n^2\cos\alpha\sin\alpha} = -\frac{a}{n^2\sin 2\alpha} \qquad \text{...(6)}$$

using (5) and (6) in (3), we get :

$$x = -ae^{-nt\cos\alpha}\left[\frac{\sin(n\sin\alpha)t}{n^2\sin 2\alpha}\right] + \frac{a\sin nt}{2n^2\cos\alpha} \qquad \textbf{Ans.}$$

Ex.15 $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^{-x}$.

Sol. Given equation is :

$(D^2 + D + 1)y = e^{-x}$

$\therefore$ A. E. is

$m^2 + m + 1 = 0$

$$\Rightarrow \quad m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\therefore \quad C.F. = c_1 e^{-\frac{x}{2}}\cos\left(\frac{\sqrt{3}}{2}x + c_2\right) \qquad \text{...(1)}$$

$$P.I. = \frac{1}{D^2 + D + 1}e^{-x} = \frac{1}{1-1+1}e^{-x} = e^{-x} \qquad \text{...(2)}$$

$\therefore$ solution is :

$$y = C.F. + P.I. = c_1 e^{-\frac{x}{2}}\cos\left(\frac{\sqrt{3}}{2}x + c_2\right) + e^{-x} \qquad \textbf{Ans.}$$

Ex.16 $\frac{d^2y}{dx^2} + 2p\frac{dy}{dx} + (p^2 + q^2)y = e^{ax}$.

Sol. Give equation is

$$\left[D^2 + 2pD + (p^2 + q^2)\right]y = e^{ax}$$

A. E. is $\rightarrow m^2 + 2pm + (p^2 + q^2) = 0$

$\therefore \quad m = -p \pm iq$

$$\therefore \quad C.F. = c_1 e^{-px}\cos(qx + c_2) \qquad \text{...(1)}$$

$$P.I. = \frac{1}{D^2 + 2pD + (p^2 + q^2)}e^{ax}$$

$$= \frac{1}{a^2 + 2pa + p^2 + q^2}e^{ax} = \frac{1}{(p+a)^2 + q^2}e^{ax}$$

$\therefore$ solution is

$y = C.F. + P.I.$

$$\Rightarrow \quad y = c_1 e^{-px}\cos(qx + c_2) + \frac{1}{(p+a)^2 + q^2}e^{ax} \qquad \textbf{Ans.}$$

Ex.17 $(D^3 + 1)\ y = (e^x + 1)^2$.

Sol. A. E. is :

$$m^2+1=0 \Rightarrow m=-1, \frac{1 \pm i\sqrt{3}}{2}$$

$\therefore$ $$C.F. = c_1e^{-x} + e^{x/2}\left(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x\right) \qquad \text{....(1)}$$

$$P.I. = \frac{1}{D^3+1}\left(e^x+1\right)^2 = \frac{1}{D^3+1}\left(e^{2x}+2e^x+1\right)$$

$$= \frac{1}{D^3+1}e^{2x} + 2\frac{1}{D^3+1}e^x + \frac{1}{D^3+1}e^{0x}$$

or, $$P.I = \frac{1}{8+1}e^{2x} + \frac{2}{1+1}e^x + \frac{1}{0+1}e^{0x}$$

$$= \frac{1}{9}e^{2x} + e^x + 1$$

$\therefore$ Complete solution is :

$y = C.F. + P.I.$

$\Rightarrow$ $$y = c_1e^{-x} + e^{x/2}\left(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x\right) + \frac{1}{9}e^{2x} + e^x + 1$$

Ans.

Exercise - 12(C)

Ex-35

Ex.1 $\dfrac{d^2y}{dx^2} - 2\dfrac{dy}{dx} + 4y = e^x$

Sol. Given equation is :

$(D^2 - 2D + 4)y = e^x$

A. E. $\rightarrow$ $m^2 - 2m + 4 = 0$

$\Rightarrow$ $$m = \frac{2 \pm \sqrt{4-16}}{2} = 1 \pm i\sqrt{3}$$

$\therefore$ C.F. $= e^x(C_1 \cos\sqrt{3}x + C_2 \sin\sqrt{3}x)$

P.I. $$= \frac{1}{(D^2-2D+4)}e^x = \frac{1}{1-2+4}e^x = \frac{1}{3}e^x$$

$\therefore$ C.S. is :

$y = $ C.F. + P.I.

$\Rightarrow$ $$y = e^x(C_1 \cos\sqrt{3}x + C_2 \sin\sqrt{3}x) + \frac{1}{3}e^x \quad \text{Ans.}$$

Ex.2 $\dfrac{d^3y}{dx^3} + 2\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = e^{2x} + x^2 + x$.

Sol. Given equation is :

$(D^3 + 2D^2 + D)y = e^{2x} + x^2 + x$

A. E. $\rightarrow$ $m^3 + 2m^2 + m = 0 \Rightarrow m(m+1)^2 = 0 \Rightarrow m = 0, -1, -1$

$\therefore$ C.F. $= C_1 + (C_2 + C_3x)e^{-x}$

P.I. $$= \frac{1}{(D^3+2D^2+D)}e^{2x} + \frac{1}{(D^3+2D^2+D)}(x^2+x)$$

$$= \frac{1}{(8+8+2)}e^{2x} + \frac{1}{D(1+2D+D^2)}(x^2+x)$$

$$= \frac{1}{18}e^{2x} + \frac{1}{D}(1+2D+D^2)^{-1}(x^2+x)$$

$$= \frac{1}{18}e^{2x} + \frac{1}{D}\left[1-(2D+D^2)+(2D+D^2)^2-...\right](x^2+x)$$

$$= \frac{1}{18}e^{2x} + \frac{1}{D}\left[1-2D-D^2+4D^2+.....\right](x^2+x)$$

$$= \frac{1}{18}e^{2x} + \frac{1}{D}\left[x^2+x-2(2x+1)-(2)+4(2)\right]$$

$$= \frac{1}{18}e^{2x} + \frac{1}{D}\left[x^2-3x+4\right]$$

$$= \frac{1}{18}e^{2x} + \frac{x^3}{3} - \frac{3x^2}{2} + 4x$$

$$= \frac{1}{18}e^{2x} + \frac{1}{6}x(2x^2-9x+24)$$

$\therefore$ C.S. is :

$y = $ C.F. + P.I.

$$\Rightarrow \quad y = C_1 + (C_2 + C_3 x)e^{-x} + \frac{1}{18}e^{2x} + \frac{1}{6}x\,(2x^2 - 9x + 24) \quad \textbf{Ans.}$$

Ex.3 $\dfrac{d^3y}{dx^3} - 4\dfrac{d^2y}{dx^2} + 5\dfrac{dy}{dx} - 2 = 0.$

Sol. Given equation is :

$$(D^3 - 4D^2 + 5D)\,y = 2$$

A.E. $\rightarrow m^3 - 4m^2 + 5m = 0 \quad \Rightarrow m(m^2 - 4m + 5) = 0$

$\Rightarrow m = 0$ and $m^2 - 4m + 5 = 0 \quad \Rightarrow m = 2 \pm i$

$\therefore$ C.F. $= C_1 + C_2 e^{2x} \cos(x + C_3)$ ....(1)

$$\text{P.I.} = \frac{2}{D^3 - 4D^2 + 5D}$$

$$= \frac{2}{D(D^2 - 4D + 5)}e^{0x} = \frac{2}{D(0 - 0 + 5)}e^{0x} = \frac{2}{5D}e^{0x} = \frac{2}{5D}(1)$$

$$= \frac{2}{5}x \qquad \left(\because \frac{1}{D}(1) = x\right)$$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$$\Rightarrow \quad y = C_1 + C_2 e^{2x}\cos(x + C_3) + \frac{2x}{5} \quad \textbf{Ans.}$$

Ex.4 $\dfrac{d^3y}{dx^3} - \dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} = 1 + x^2.$

Sol. Given equation is :

$$(D^3 - D^2 - 6D)y = 1 + x^2$$

A.E. $\rightarrow m^3 - m^2 - 6m = 0 \quad \Rightarrow m(m^2 - m - 6) = 0$

$\Rightarrow m(m - 3)(m + 2) = 0 \quad \Rightarrow m = 0, -2, 3$

$\therefore$ C.F. $= C_1 + C_2 e^{-2x} + C_3 e^{3x}$

$$\text{P.I.} = \frac{1}{D^3 - D^2 - 6D}(1 + x^2)$$

$$= -\frac{1}{6D\left(1 - \frac{D^2 - D}{6}\right)}(1 + x^2)$$

$$= -\frac{1}{6D}\left(1 - \frac{D^2 - D}{6}\right)^{-1}(1 + x^2)$$

$$= -\frac{1}{6D}\left[1 + \frac{D^2 - D}{6} + \left(\frac{D^2 - D}{6}\right)^2 + \ldots\right](1 + x^2)$$

$$= -\frac{1}{6D}\left[1 + \frac{D^2 - D}{6} + \frac{D^2}{36} + \ldots\right](1 + x^2)$$

$$= -\frac{1}{6D}\left[(1 + x^2) + \frac{1}{6}(2 - 2x) + \frac{2}{36}\right]$$

$$= -\frac{1}{6D}\left[1 + x^2 + \frac{14}{36} - \frac{x}{3}\right]$$

$$= -\frac{1}{6}\left[x + \frac{x^3}{3} + \frac{14x}{36} - \frac{x^2}{6}\right] = -\frac{1}{6}\left(\frac{x^3}{3} - \frac{x^2}{6} + \frac{50}{36}x\right)$$

$$= -\frac{1}{18}x^3 + \frac{x^2}{36} - \frac{25}{108}x$$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$$y = C_1 + C_2 e^{-2x} + C_3 e^{3x} - \frac{1}{18}x^3 + \frac{1}{36}x^2 - \frac{25}{108}x \quad \textbf{Ans.}$$

Ex.5 $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = x^2.$

Sol. It can be written as

$$(D^2 - 4D + 4)y = x^2$$

A. E. is

$$m^2 - 4m + 4 = 0 \Rightarrow (m - 2)^2 = 0 \Rightarrow m = 2, 2$$

$\therefore$ C. F. $= (C_1 + C_2 x)e^{2x}$

$$\text{P. I.} = \frac{1}{(D - 2)^2}x^2$$

$$=\frac{1}{4}\left(1-\frac{D}{2}\right)^{-2}x^2$$

$$=\frac{1}{4}\left(1+D+\frac{3D^2}{4}+\ldots\ldots\right)x^2$$

$$=\frac{1}{4}\left(x^2+2x+\frac{3}{2}\right)$$

$\therefore$ C. S. is–

$y = C.F. + P.I.$

or, $y=(c_1+c_2x)e^{2x}+\frac{1}{4}\left(x^2+2x+\frac{3}{2}\right)$ **Ans.**

Ex.6 $(D^2+D-2)\,y=e^x$.

Sol. A. E. is –

$m^2+m-2=0$

$\Rightarrow$ $m=1,-2$

$\therefore$ C. F. $=c_1e^x+c_2e^{-2x}$

$$\text{P. I.}=\frac{1}{(D^2+D-2)}e^x$$

$$=\frac{1}{(D-1)(D+2)}e^x$$

$$=\frac{1}{(D-1)(1+2)}e^x$$

$$=\frac{1}{3}\frac{1}{(D-1)}e^x$$

$$=\frac{1}{3}xe^x \qquad \left(\because \frac{1}{(D-a)^r}e^{ax}=\frac{x^re^{ax}}{r!}\right)$$

$\therefore$ C. S. is –

$y=C.F.+P.I.=c_1e^x+c_2e^{-2x}+\frac{1}{3}xe^x$ **Ans.**

Ex.7 $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=x\sin x$. **[Raj., 2003; JNVU, 1998]**

Sol. Given equation is :

$$(D^2-2D+1)y=x\sin x$$

A. E. $\rightarrow$ $m^2-2m+1=0 \Rightarrow (m-1)^2=0 \Rightarrow m=1,1$

$\therefore$ C.F. $=(C_1+C_2x)e^x$

$$\text{P.I.}\quad=\frac{1}{D^2-2D+1}x\sin x=\frac{1}{(D-1)^2}x\sin x$$

$$=x\frac{1}{(D-1)^2}\sin x-\frac{2(D-1)}{(D-1)^4}\sin x$$

$$\left[\because \frac{1}{f(D)}xU=x\frac{1}{f(D)}U-\frac{f'(D)}{[f(D)]^2}U\right]$$

$$=x\frac{1}{D^2-2D+1}\sin x-\frac{2}{(D-1)^3}\sin x$$

$$=x\frac{1}{-1-2D+1}\sin x-\frac{2}{D^3-1-3D^2+3D}\sin x$$

$$=-\frac{x}{2D}\sin x-\frac{2}{-D-1+3+3D}\sin x$$

$$=\frac{x}{2}\cos x-\frac{2}{2D+2}\sin x=\frac{x}{2}\cos x-\frac{1}{(D+1)}\sin x$$

or, $\text{P.I.}=\frac{x}{2}\cos x-\frac{D-1}{D^2-1}\sin x=\frac{x}{2}\cos x-\frac{(D-1)}{-1-1}\sin x$

$$=\frac{x}{2}\cos x+\frac{1}{2}(\cos x-\sin x)$$

$\therefore$ C.S. is :

$y=$ C.F. + P.I.

$\Rightarrow$ $y=(C_1+C_2x)e^x+\frac{1}{2}(x\cos x+\cos x-\sin x)$ **Ans.**

Ex.8 $\frac{d^2y}{dx^2}+y=x^2\sin 2x$.

Sol. Given equation is :

$$(D^2+1)\,y = x^2 \sin 2x$$

A. E. $\rightarrow$ $m^2+1=0 \Rightarrow m=\pm i$

$\therefore$ C.F. $= C_1 \cos(x + C_2)$

P.I. $= \dfrac{1}{(D^2+1)} x^2 \sin 2x$

$= \text{I.P.} \dfrac{1}{(D^2+1)} x^2 e^{2ix}$

$[\because\ e^{2ix} = \cos 2x + i \sin 2x \Rightarrow \sin 2x = \text{Imaginary part of } e^{2ix}]$

$= \text{I.P.} e^{2ix} \dfrac{1}{(D+2i)^2+1} x^2$

$= \text{I.P.} e^{2ix} \dfrac{1}{(D^2+4Di-3)} x^2 = \text{I.P.} e^{2ix} \dfrac{1}{-3\left(1-\dfrac{D^2+4Di}{3}\right)} x^2$

$= \text{I.P.} \dfrac{e^{2ix}}{-3} \left(1-\dfrac{D^2+4Di}{3}\right)^{-1} x^2$

$= -\dfrac{1}{3} \text{I.P.} e^{2ix} \left[1+\dfrac{D^2+4Di}{3}+\left(\dfrac{D^2+4Di}{3}\right)^2+\ldots\right] x^2$

$= -\dfrac{1}{3} \text{I.P.} e^{2ix} \left[1+\dfrac{D^2+4Di}{3}-\dfrac{16D^2}{9}+\ldots\ldots\right] x^2$

$= -\dfrac{1}{3} \text{I.P.} e^{2ix} \left[x^2+\dfrac{2}{3}+\dfrac{8xi}{3}-\dfrac{32}{9}\right]$

$= -\dfrac{1}{3} \text{I.P.} (\cos 2x + i \sin 2x)\left(x^2-\dfrac{26}{9}+i\dfrac{8x}{3}\right)$

$= -\dfrac{1}{3}\left[\left(x^2-\dfrac{26}{9}\right)\sin 2x+\dfrac{8x}{3}\cos 2x\right]$

or, P.I. $= \dfrac{8}{3} x \cos 2x + \dfrac{1}{27}\left(26-9x^2\right)\sin 2x$

$\therefore$ C.S. is :
$y =$ C.F. + P.I.

$\Rightarrow\quad y = C_1 \cos(x + C_2) + \dfrac{8}{3} x \cos 2x + \dfrac{1}{27}(26-9x^2)\sin 2x$ **Ans.**

Ex.9 $\dfrac{d^3y}{dx^3} + y = 3e^{-x} + 5e^{2x}$.

Sol. Given equation is :

$$(D^3+1)\,y = 3e^{-x} + 5e^{2x}$$

A. E. $\rightarrow$ $m^3+1=0 \Rightarrow (m+1)(m^2-m+1)=0$

$\Rightarrow m = -1, \dfrac{1 \pm i\sqrt{3}}{2}$

$\therefore$ C.F. $= C_1 e^{-x} + e^{x/2}\left(C_2 \cos \dfrac{\sqrt{3}}{2} x + C_3 \sin \dfrac{\sqrt{3}}{2} x\right)$

P.I. $= \dfrac{1}{(D^3+1)}(3e^{-x}+5e^{2x})$

$= 3\dfrac{1}{(D^3+1)} e^{-x} + 5\dfrac{1}{(D^3+1)} e^{2x}$

$= 3\dfrac{1}{(D+1)(D^2-D+1)} e^{-x} + 5\dfrac{1}{(8+1)} e^{2x}$

$= \dfrac{3}{3(D+1)} e^{-x} + \dfrac{5}{9} e^{2x} = x e^{-x} + \dfrac{5}{9} e^{2x}$

$\therefore$ C.S. is :
$y =$ C.F. + P.I.

$\Rightarrow\quad y = C_1 e^{-x} + e^{x/2}\left(C_2 \cos \dfrac{\sqrt{3}}{2} x + C_3 \sin \dfrac{\sqrt{3}}{2} x\right) + x e^{-x} + \dfrac{5}{9} e^{2x}$

Ans.

Ex.10 $(D^2-2D+1)\,y = xe^x \sin x$. **[Raj., 1996]**

Sol. A. E. $\rightarrow$ $m^2-2m+1=0 \Rightarrow (m-1)^2=0 \Rightarrow m=1,1$

$\therefore$ C.F. $= (C_1 + C_2 x)e^x$

P.I. $= \dfrac{1}{(D^2 - 2D + 1)} xe^x \sin x$

$= \dfrac{1}{(D-1)^2} xe^x \sin x$

$= e^x \dfrac{1}{(D+1-1)^2} x \sin x = e^x \dfrac{1}{D^2} x \sin x$

$= e^x \left[x \dfrac{1}{D^2} \sin x - \dfrac{2}{D^3} \sin x \right]$

$$\left[\because \frac{1}{f(D)} xU = x \frac{1}{f(D)} U - \frac{f'(D)}{[f(D)]^2} U \right]$$

$= e^x [-x \sin x - 2\cos x]$

$\therefore$ C.S. is :
$y =$ C.F. + P.I.

$\Rightarrow \quad y = (C_1 + C_2 x)e^x - e^x (x \sin x + 2\cos x)$ **Ans.**

Ex.11 $(D^2 - 1)y = \cosh x \cos x + a^x$.

Sol. A. E. $\to m^2 - 1 = 0 \Rightarrow m = 1, -1$

$\therefore$ C.F. $= C_1 e^x + C_2 e^{-x}$

P.I. $= \dfrac{1}{(D^2 - 1)} \cosh x \cos x + \dfrac{1}{(D^2 - 1)} a^x$

$= \dfrac{1}{(D^2 - 1)} \left(\dfrac{e^x + e^{-x}}{2} \right) \cos x + \dfrac{1}{(D^2 - 1)} e^{x \log a}$

$$\left(\because e^{x \log a} = e^{\log a^x} = a^x \right)$$

$= \dfrac{1}{2} \left[\dfrac{1}{(D^2 - 1)} e^x \cos x + \dfrac{1}{(D^2 - 1)} e^{-x} \cos x \right] + \dfrac{1}{(\log a)^2 - 1} e^{x \log a}$

$= \dfrac{1}{2} \left[e^x \dfrac{1}{(D+1)^2 - 1} \cos x + e^{-x} \dfrac{1}{(D-1)^2 - 1} \cos x \right] + \dfrac{1}{(\log a)^2 - 1} a^x$

$= \dfrac{1}{2} \left[e^x \dfrac{1}{(D^2 + 2D)} \cos x + e^{-x} \dfrac{1}{(D^2 - 2D)} \cos x \right] + \dfrac{1}{[(\log a)^2 - 1]} a^x$

$= \dfrac{1}{2} \left[e^x \dfrac{1}{(2D - 1)} \cos x + e^{-x} \dfrac{1}{-(1 + 2D)} \cos x \right] + \dfrac{1}{[(\log a)^2 - 1]} a^x$

$= \dfrac{1}{2} \left[e^x \dfrac{(2D+1)}{(4D^2 - 1)} \cos x - e^{-x} \dfrac{(1 - 2D)}{(1 - 4D^2)} \cos x \right] + \dfrac{1}{[(\log a)^2 - 1]} a^x$

$= \dfrac{1}{2} \left[e^x \dfrac{(2D+1)}{(-4-1)} \cos x - e^{-x} \dfrac{(1-2D)}{(1+4)} \cos x \right] + \dfrac{1}{[(\log a)^2 - 1]} a^x$

$= \dfrac{1}{2} \left[-\dfrac{1}{5} e^x (-2\sin x + \cos x) - \dfrac{e^{-x}}{5} (\cos x + 2\sin x) \right] + \dfrac{a^x}{[(\log a)^2 - 1]}$

$= \dfrac{1}{2} \left[\dfrac{2}{5} (e^x - e^{-x}) \sin x - \dfrac{1}{5} (e^x + e^{-x}) \cos x \right] + \dfrac{a^x}{[(\log a)^2 - 1]}$

$= \dfrac{1}{2} \left[\dfrac{4}{5} \sinh x \sin x - \dfrac{2}{5} \cosh x \cos x \right] + \dfrac{a^x}{[(\log a)^2 - 1]}$

$\Rightarrow$ P.I. $= \dfrac{2}{5} \sinh x \sin x - \dfrac{1}{5} \cosh x \cos x + \dfrac{a^x}{[(\log a)^2 - 1]}$

$\therefore$ C.S. is :
$y =$ C.F. + P.I.

$\Rightarrow y = C_1 e^x + C_2 e^{-x} + \dfrac{2}{5} \sinh x \sin x - \dfrac{1}{5} \cosh x \cos x + \dfrac{a^x}{[(\log a)^2 - 1]}$

Ans.

Ex.12 $\dfrac{d^2 y}{dx^2} - 2\dfrac{dy}{dx} + y = x^2 e^{3x}$.

Sol. Given equation is :

$$(D^2-2D+1)y = x^2e^{3x}$$

$A.\ E. \to m^2-2m+1=0 \Rightarrow (m-1)^2=0 \Rightarrow m=1,\ 1$

$\therefore$ C. F. $= (c_1+c_2x)\,e^x$

$$P.I. = \frac{1}{(D^2-2D+1)}x^2e^{3x}$$

$$= \frac{1}{(D-1)^2}x^2e^{3x} = e^{3x}\frac{1}{(D+3-1)^2}x^2$$

$$= e^{3x}\frac{1}{(D+2)^2}x^2 = \frac{e^{3x}}{4}\left(1+\frac{D}{2}\right)^{-2}x^2$$

$$= \frac{1}{4}e^{3x}\left[1-D+\frac{3D^2}{4}.....\right]x^2 = \frac{1}{4}e^{3x}\left(x^2-2x+\frac{3}{2}\right)$$

$\Rightarrow \quad P.I. = \frac{1}{8}e^{3x}(2x^2-4x+3)$

$\therefore \quad C.\ S.$ is :

$y = C.\ F. + P.\ I.$

$\Rightarrow \quad y = (c_1+c_2x)e^x + \frac{1}{8}e^{3x}(2x^2-4x+3)$ **Ans.**

Ex.13 $\dfrac{d^2y}{dx^2}+a^2y=\sin ax$. **[MREC (Auto), 2001]**

Sol. Given equation is–

$$(D^2+a^2)y = \sin ax$$

$A.\ E. \to \quad m^2+a^2=0 \Rightarrow m=\pm ia$

$\therefore \quad C.F. = c_1\cos ax + c'_1\sin ax$

$$P.I. = \frac{1}{(D^2+a^2)}\sin ax$$

or, $\quad P.I. = \frac{1}{2ia}\left[\left(\frac{1}{D-ia}\right)-\left(\frac{1}{D+ia}\right)\right]\sin ax$

$$= \frac{1}{2ia}\left[e^{iax}\int e^{-iax}\sin ax\,dx - e^{-iax}\int e^{iax}\sin ax\,dx\right]$$

$$= \frac{1}{2ia}\left[e^{iax}\int(\cos ax - i\sin ax)\sin ax\,dx - e^{-iax}\int(\cos ax + i\sin ax)\sin ax\,dx\right]$$

$$= \frac{1}{2ia}\left[(e^{iax}-e^{-iax})\int\cos ax\sin ax\,dx - i(e^{iax}+e^{-iax})\int\sin^2 ax\,dx\right]$$

$$= \frac{1}{2ia}\left[(2i\sin ax)\int\frac{1}{2}\sin 2ax\,dx - 2i\cos ax\int\left(\frac{1-\cos 2ax}{2}\right)dx\right]$$

$$= \frac{1}{a}\left[\frac{1}{2}\sin ax\left(\frac{-\cos 2ax}{2a}\right)-\frac{1}{2}\cos ax\left(x-\frac{\sin 2ax}{2a}\right)\right]$$

$$= \frac{1}{a}\left[-\frac{1}{4a}\sin ax\cos 2ax - \frac{x}{2}\cos ax + \frac{1}{4a}\cos ax\sin 2ax\right]$$

$$= \frac{1}{a}\left[\frac{1}{4a}(\sin 2ax\cos ax - \sin ax\cos 2ax) - \frac{x}{2}\cos ax\right]$$

$$= \frac{1}{4a^2}\sin(2ax-ax) - \frac{x}{2a}\cos ax$$

$\Rightarrow \quad$ P. I. $= \frac{1}{4a^2}\sin ax - \frac{x}{2a}\cos ax$

$\therefore \quad C.\ S.$ is –

$y = C.\ F. + P.\ I.$

$\Rightarrow \quad y = c_1\cos ax + c_1'\sin ax + \frac{1}{4a^2}\sin ax - \frac{x}{2a}\cos ax$

$$= c_1\cos ax + \left(c_1' + \frac{1}{4a^2}\right)\sin ax - \frac{x}{2a}\cos ax$$

or, $\quad y = c_1\cos ax + c_2\sin ax - \frac{x}{2a}\cos ax\ ; \quad c_2 = c'_1 + \frac{1}{4a^2}$ **Ans.**

Ex.14 $\dfrac{d^2y}{dx^2}-4y=x+\sin 2x+\cos^2 x$.

Sol. Given equation is :

$$(D^2-4)y=x+\sin 2x+\cos^2 x$$

A. E. $\rightarrow$ $m^2-4=0 \Rightarrow m=\pm 2$

$\therefore$ C. F. $=c_1e^{2x}+c_2e^{-2x}$

$$P.I.=\frac{1}{(D^2-4)}x+\frac{1}{(D^2-4)}\sin 2x+\frac{1}{(D^2-4)}\cos^2 x$$

$$=-\frac{1}{4}\left(1-\frac{D^2}{4}\right)^{-1}+\frac{1}{(-4-4)}\sin 2x+\frac{1}{(D^2-4)}\left(\frac{1+\cos 2x}{2}\right)$$

$$=-\frac{1}{4}\left(1+\frac{D^2}{4}+.....\right)x-\frac{1}{8}\sin 2x+\frac{1}{2}$$

$$\left[\frac{1}{(D^2-4)}e^{0x}+\frac{1}{(D^2-4)}\cos 2x\right]$$

$$=-\frac{1}{4}(x)-\frac{1}{8}\sin 2x+\frac{1}{2}\left[-\frac{1}{4}+\frac{1}{(-4-4)}\cos 2x\right]$$

$$=-\frac{x}{4}-\frac{1}{8}\sin 2x-\frac{1}{8}-\frac{1}{16}\cos 2x$$

$\therefore$ C. S. is :

$y=C.F.+P.I.$

or, $y=c_1e^{2x}+c_2e^{-2x}-\frac{1}{4}x-\frac{1}{8}\sin 2x-\frac{1}{16}\cos 2x-\frac{1}{8}$ **Ans.**

Ex.15 $(D^2-1)\,y=x\sin x+(1+x^2)\,e^x$.

Sol. $A.E. \rightarrow m^2-1=0 \Rightarrow m=1,-1$

$\therefore$ C. F. $=c_1\,e^x+c_2e^{-x}$

$$P.I.=\frac{1}{(D^2-1)}x\sin x+\frac{1}{(D^2-1)}(1+x^2)e^x$$

$$=x\frac{1}{(D^2-1)}\sin x-\frac{2D}{(D^2-1)^2}\sin x+e^x\frac{1}{\left[(D+1)^2-1\right]}(1+x^2)$$

$$=x\frac{1}{(-1-1)}\sin x-\frac{2D}{(-1-1)^2}\sin x+e^x\frac{1}{(D^2+2D)}(1+x^2)$$

$$=-\frac{x}{2}\sin x-\frac{1}{2}\cos x+e^x\frac{1}{2D}\left(1+\frac{D}{2}\right)^{-1}(1+x^2)$$

$$=-\frac{x}{2}\sin x-\frac{1}{2}\cos x+e^x\frac{1}{2D}\left(1-\frac{D}{2}+\frac{D^2}{4}.......\right)(1+x^2)$$

$$=-\frac{x}{2}\sin x-\frac{1}{2}\cos x+e^x\frac{1}{2D}\left(1+x^2-x+\frac{1}{2}\right)$$

$$=-\frac{x}{2}\sin x-\frac{1}{2}\cos x+\frac{e^x}{2}\left(\frac{x^3}{3}-\frac{x^2}{2}+\frac{3x}{2}\right)$$

$\Rightarrow$ $P.I.=-\frac{1}{2}(x\sin x+\cos x)+\frac{1}{12}(2x^2-3x+9)xe^x$

$\therefore$ C. S. is :

$y=C.F.+P.I$

$\Rightarrow$ $y=c_1e^x+c_2e^{-x}-\frac{1}{2}(x\sin x+\cos x)+\frac{1}{12}(2x^2-3x+9)xe^x$

Ans.

Ex.16 $(D^2-2D+5)y=e^{2x}\sin x$.

Sol. Given equation is :

A. E. $\rightarrow$ $m^2-2m+5=0 \Rightarrow m=1\pm 2i$

$\therefore$ C.F. $=e^x(C_1\cos 2x+C_2\sin 2x)$

$$P.I.=\frac{1}{(D^2-2D+5)}e^{2x}\sin x$$

$$=e^{2x}\frac{1}{(D+2)^2-2(D+2)+5}\sin x=e^{2x}\frac{1}{(D^2+2D+5)}\sin x$$

$$=e^{2x}\frac{1}{(-1+2D+5)}\sin x=\frac{e^{2x}}{2}\frac{1}{(D+2)}\sin x$$

$$= \frac{1}{2}e^{2x}\frac{(D-2)}{(D^2-4)}\sin x = \frac{1}{2}e^{2x}\frac{(D-2)}{(-1-4)}\sin x$$

$$= -\frac{1}{10}e^{2x}(\cos x - 2\sin x)$$

∴ C.S. is :

$y =$ C.F. + P.I.

$$\Rightarrow \quad y = e^{x}(C_1\cos 2x + C_2\sin 2x) - \frac{1}{10}e^{2x}(\cos x - 2\sin x) \quad \textbf{Ans.}$$

Ex.17 $(D^2+4D+4)y = 2\sinh 2x$.

Sol. A. E. → $m^2+4m+4=0 \Rightarrow (m+2)^2 = 0 \Rightarrow m = -2, -2$

∴ C.F. $= (C_1 + C_2 x)e^{-2x}$

$$\text{P.I.} = \frac{1}{(D^2+4D+4)}2\sinh 2x$$

$$= \frac{2}{(D+2)^2}\left(\frac{e^{2x}-e^{-2x}}{2}\right) = \frac{1}{(D+2)^2}e^{2x} - \frac{1}{(D+2)^2}e^{-2x}$$

$$= \frac{1}{16}e^{2x} - \frac{x^2e^{-2x}}{2!} \qquad \left[\because \frac{1}{(D-a)^r}e^{ax} = \frac{x^r e^{ax}}{r!}\right]$$

$$\therefore \text{P.I.} = \frac{1}{16}e^{2x} - \frac{1}{2}x^2e^{-2x}$$

∴ C.S. is :

$y =$ C.F. + P.I.

$$\Rightarrow \quad y = (C_1 + C_2 x)e^{-2x} + \frac{1}{16}e^{2x} - \frac{1}{2}x^2e^{-2x} \quad \textbf{Ans.}$$

Ex.18 $\frac{d^3y}{dx^3} - y = (e^x+1)^2$.

Sol. Given equation is :

$$(D^3-1)y = (e^x+1)^2$$

A. E. → $m^3 - 1 = 0 \Rightarrow m = 1, \frac{-1 \pm i\sqrt{3}}{2}$

$$\therefore \quad \text{C.F.} = C_1e^{x} + C_2e^{-x/2}\cos\left(\frac{\sqrt{3}}{2}x + C_3\right)$$

$$\text{P.I.} = \frac{1}{(D^3-1)}(e^x+1)^2 = \frac{1}{(D^3-1)}\left(e^{2x}+2e^x+1\right)$$

$$= \frac{1}{(D^3-1)}e^{2x} + \frac{1}{(D^3-1)}2e^x + \frac{1}{(D^3-1)}e^{0x}$$

$$= \frac{1}{(2^3-1)}e^{2x} + \frac{2}{(D-1)(D^2+D+1)}e^x + \frac{1}{(0-1)}e^{0x}$$

$$= \frac{1}{7}e^{2x} + \frac{2}{3}\cdot\frac{1}{(D-1)}e^x - 1$$

$$\therefore \text{P.I.} = \frac{1}{7}e^{2x} + \frac{2}{3}xe^x - 1 \qquad \left[\because \frac{1}{(D-a)^r}e^{ax} = \frac{x^r e^{ax}}{r!}\right]$$

∴ C.S. is :

$y =$ C.F. + P.I.

$$\Rightarrow \quad y = C_1e^{x} + C_2e^{-x/2}\cos\left(\frac{\sqrt{3}}{2}x + C_3\right) + \frac{1}{7}e^{2x} + \frac{2}{3}xe^x - 1 \quad \textbf{Ans.}$$

Ex.19 $(D^2-4D+3)y = e^x\cos 2x + \cos 3x$.

Sol. A. E. → $m^2 - 4m + 3 = 0 \Rightarrow (m-3)(m-1) = 0 \Rightarrow m = 1, 3$

∴ C. F. $= c_1e^x + c_2e^{3x}$

$$P.I. = \frac{1}{(D^2-4D+3)}e^x\cos 2x + \frac{1}{(D^2-4D+3)}\cos 3x$$

$$= e^x\frac{1}{\left[(D+1)^2 - 4(D+1) + 3\right]}\cos 2x + \frac{1}{(-9-4D+3)}\cos 3x$$

$$= e^x\frac{1}{(D^2-2D)}\cos 2x - \frac{1}{2(2D+3)}\cos 3x$$

$$\text{or,} \quad P.I. = e^x\frac{1}{(-4-2D)}\cos 2x - \frac{1}{2}\frac{(2D-3)}{(4D^2-9)}\cos 3x$$

$$= -\frac{e^x}{2}\frac{(2-D)}{(4-D^2)}\cos 2x - \frac{1}{2}\frac{(2D-3)}{(-36-9)}\cos 3x$$

$$= -\frac{e^x}{2}\frac{(2-D)}{(4+4)}\cos 2x + \frac{1}{90}(-6\sin 3x - 3\cos 3x)$$

$$= -\frac{e^x}{16}(2\cos 2x + 2\sin 2x) - \frac{1}{15}\sin 3x - \frac{1}{30}\cos 3x$$

or, $P.I. = -\frac{1}{8}e^x(\cos 2x + \sin 2x) - \frac{1}{30}(2\sin 3x + \cos 3x)$

$\therefore$ C. S. is :

$y = C.F. + P.I.$

$$\Rightarrow \quad y = c_1e^x + c_2e^{3x} - \frac{1}{8}e^x(\cos 2x + \sin 2x) - \frac{1}{30}(2\sin 3x + \cos 3x)$$

Ans.

Ex.20 $(D^2 + 2D + 1)y = e^x + x^2 - \sin x$.

Sol. A. E. $\rightarrow m^2 + 2m + 1 = 0 \Rightarrow (m+1)^2 = 0 \Rightarrow m = -1, -1$

$\therefore$ C. F. $= (c_1 + c_2x)e^{-x}$

$$P.I. = \frac{1}{(D^2+2D+1)}e^x + \frac{1}{(D^2+2D+1)}x^2 - \frac{1}{(D^2+2D+1)}\sin x$$

$$= \frac{1}{(1+2+1)}e^x + \left[1 + (D^2+2D)\right]^{-1}x^2 - \frac{1}{(-1+2D+1)}\sin x$$

$$= \frac{1}{4}e^x + [1 - (D^2+2D) + (D^2+2D)^2 \ldots]x^2 - \frac{1}{2D}\sin x$$

$$= \frac{1}{4}e^x + (1 - D^2 - 2D + 4D^2 \ldots)x^2 + \frac{1}{2}\cos x$$

$$\Rightarrow \quad P.I. = \frac{1}{4}e^x + (x^2 + 6 - 4x) + \frac{1}{2}\cos x$$

$\therefore$ C. S. is –

$y = C.F. + P.I.$

$$\Rightarrow \quad y = (c_1 + c_2x)e^{-x} + \frac{1}{4}e^x + x^2 - 4x + 6 + \frac{1}{2}\cos x \quad \text{Ans.}$$

Ex.21 $(D^2 - 3D + 2)\,y = \sin 3x + x^2 + x + e^{4x}$.

Sol. $A.\,E. \rightarrow m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0 \Rightarrow m = 1, 2$

$\therefore \quad C.F. = c_1e^x + c_2e^{2x}$

$$P.I. = \frac{1}{(D^2-3D+2)}\sin 3x + \frac{1}{(D^2-3D+2)}(x^2+x) + \frac{1}{(D^2-3D+2)}e^{4x}$$

$$= \frac{1}{(-9-3D+2)}\sin 3x + \frac{1}{2}\left[1 + \frac{D^2-3D}{2}\right]^{-1}(x^2+x) + \frac{1}{6}e^{4x}$$

$$= \frac{-1}{(3D+7)}\sin 3x + \frac{1}{2}\left[1 - \frac{D^2-3D}{2} + \left(\frac{D^2-3D}{2}\right)^2 \ldots\right](x^2+x) + \frac{1}{6}e^{4x}$$

$$= -\frac{(3D-7)}{(9D^2-49)}\sin 3x + \frac{1}{2}\left[(x^2+x) - \frac{1}{2}(2 - 6x - 3) + \frac{9}{4}(2)\right] + \frac{1}{6}e^{4x}$$

$$= -\frac{(3D-7)}{(-81-49)}\sin 3x + \frac{1}{2}\left(x^2 + x - 1 + 3x + \frac{3}{2} + \frac{9}{2}\right) + \frac{1}{6}e^{4x}$$

or, $P.I. = \frac{1}{130}(9\cos 3x - 7\sin 3x) + \frac{1}{2}\left(x^2 + 4x + 5\right) + \frac{1}{6}e^{4x}$

$\therefore$ C. S. is :

$y = C.\,F. + P.\,I.$

$$\Rightarrow \quad y = c_1e^x + c_2e^{2x} + \frac{1}{130}(9\cos 3x - 7\sin 3x) + \frac{1}{2}\left(x^2 + 4x + 5\right) + \frac{1}{6}e^{4x} \quad \text{Ans}$$

Ex.22 $\frac{d^2y}{dx^2} = a + bx + cx^2$, Given that $y = d$, $\frac{dy}{dx} = 0$ when $x = 0$.

Sol. Given, $\frac{d^2y}{dx^2} = a + bx + cx^2$ ....(1)

on integration, w. r. t. 'x', we get

$$\frac{dy}{dx} = ax + \frac{bx^2}{2} + \frac{cx^3}{3} + c_1 \qquad \text{...(2)}$$

Now, $\frac{dy}{dx} = 0$ when $x = 0$ (given)

$\therefore$ (3) gives $\rightarrow 0 = c_1 \Rightarrow c_1 = 0$

$$\therefore \quad \frac{dy}{dx} = ax + \frac{bx^2}{2} + \frac{cx^3}{3}$$

Again integrate w. r. t. x, we get–

$$y = \frac{ax^2}{2} + \frac{bx^3}{6} + \frac{cx^4}{12} + c_2 \qquad \text{...(3)}$$

given that, $y = d$, when $x = 0$

$\therefore$ (3) gives $\rightarrow d = c_2$

$\therefore$ (3) reduces to

$$y = \frac{ax^2}{2} + \frac{bx^3}{6} + \frac{cx^4}{12} + d$$

or, $y = \frac{1}{12}(6ax^2 + 2bx^3 + cx^4 + 12d)$ **Ans.**

Ex.23 $(D^3 + a^2 D)\, y = \sin ax.$

Sol. A. E. $\rightarrow m^3 + a^2 m = 0 \Rightarrow m(m^2 + a^2) = 0 \Rightarrow m = 0, \pm ia$

$\therefore$ $C.F. = c_1 + c'_2 \cos ax + c_3 \sin ax$

$$P.I. = \frac{1}{(D^3 + a^2 D)} \sin ax = \frac{1}{D(D^2 + a^2)} \sin ax$$

$$= \frac{1}{D}\left(-\frac{x}{2a}\cos ax\right) = -\frac{1}{2a}\int x \cos ax\, dx$$

$$= \frac{-1}{2a}\left[\frac{x \sin ax}{a} + \frac{\cos ax}{a^2}\right] = -\frac{x \sin ax}{2a^2} - \frac{\cos ax}{2a^3}$$

$\therefore$ C. S. is –

$y = C.F. + P.I.$

$$\Rightarrow \quad y = c_1 + c_2' \cos ax + c_3 \sin ax - \frac{x}{2a^2}\sin ax - \frac{1}{2a^3}\cos ax$$

$$\text{or,} \quad y = c_1 + \left(c_2' - \frac{1}{2a^3}\right)\cos ax + c_3 \sin ax - \frac{x \sin ax}{2a^2}$$

$$\text{or,} \quad y = c_1 + c_2 \cos ax + c_3 \sin ax - \frac{1}{2a^2} x \sin ax\ ; \quad c_2 = c'_2 - \frac{1}{2a^3}$$

Ans.

Ex.24 $(D-2)^2 y = 8x^2 e^{2x} \sin 2x.$ [MNIT, 2003]

Sol. A. E. $\rightarrow (m-2)^2 = 0 \Rightarrow m = 2, 2$

$\therefore$ C.F. $= (C_1 + C_2 x)e^{2x}$

$$\text{P.I.} = \frac{1}{(D-2)^2} 8x^2 e^{2x} \sin 2x$$

$$= 8e^{2x}\frac{1}{(D+2-2)^2} x^2 \sin 2x$$

$$= 8e^{2x}\frac{1}{D^2} x^2 \sin 2x$$

$$= 8e^{2x}\frac{1}{D}\left[\int x^2 \sin 2x\, dx\right]$$

$$= 8e^{2x}\frac{1}{D}\left[-\frac{x^2}{2}\cos 2x + \int x \cos 2x\, dx\right]$$

$$= 8e^{2x}\frac{1}{D}\left[-\frac{x^2 \cos 2x}{2} + \frac{x \sin 2x}{2} + \frac{\cos 2x}{4}\right]$$

$$= 8e^{2x}\left[-\frac{1}{2}\int x^2 \cos 2x\, dx + \frac{1}{2}\int x \sin 2x\, dx + \frac{1}{4}\int \cos 2x\, dx\right]$$

$$= 8e^{2x}\left[-\frac{1}{2}\left\{\frac{x^2 \sin 2x}{2} - \int x \sin 2x\, dx\right\} + \frac{1}{2}\int x \sin 2x\, dx + \frac{\sin 2x}{8}\right]$$

$$= 8e^{2x}\left[-\frac{1}{4}x^2 \sin 2x + \int x \sin 2x\, dx + \frac{\sin 2x}{8}\right]$$

$$= 8e^{2x}\left[-\frac{1}{4}x^2 \sin 2x - \frac{x \cos 2x}{2} + \frac{\sin 2x}{4} + \frac{\sin 2x}{8}\right]$$

$\therefore$ P.I. $= e^{2x}(3\sin 2x - 4x\cos 2x - 2x^2 \sin 2x)$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$\Rightarrow \quad y = (C_1 + C_2 x)e^{2x} + e^{2x}(3\sin 2x - 4x\cos 2x - 2x^2\sin 2x)$

Ans.

Ex.25 $(D^3 + D^2 - D - 1)\, y = \cos 2x.$ **[Raj., 2009]**

Sol. A. E. $\rightarrow m^3 + m^2 - m - 1 = 0$

$\Rightarrow \quad (m^2 - 1)(m+1) = 0 \Rightarrow m = 1, -1, -1$

$\therefore$ C. F. $= c_1 e^x + (c_2 + c_3 x)e^{-x}$

$\therefore$ P. I. $= \dfrac{1}{(D^3 + D^2 - D - 1)}\cos 2x$

$$= \frac{1}{-4D - 4 - D - 1}\cos 2x = \frac{-1}{5(D+1)}\cos 2x$$

$$= -\frac{(D-1)}{5(D^2-1)}\cos 2x = -\frac{(D-1)}{5(-4-1)}\cos 2x$$

$\Rightarrow$ P. I. $= \dfrac{1}{25}(-2\sin 2x - \cos 2x)$

$\therefore$ C. S. is :

$y = C.\ F. + P.\ I.$

$\Rightarrow \quad y = c_1 e^x + (c_2 + c_3 x)e^{-x} - \dfrac{2}{25}\sin 2x - \dfrac{1}{25}\cos 2x$ **Ans.**

Ex.26 $(D^2 + 4)y = x\sin x.$

Sol. A. E. $\rightarrow m^2 + 4 = 0 \Rightarrow m = \pm 2i$

$\therefore$ C.F. $= C_1\cos 2x + C_2 \sin 2x$

P.I. $= \dfrac{1}{(D^2+4)}x\sin x$

$$= x\frac{1}{(D^2+4)}\sin x - \frac{2D}{(D^2+4)^2}\sin x$$

$$\left[\because \frac{1}{f(D)}xU = x\frac{1}{f(D)}U - \frac{f'(D)}{[f(D)]^2}U\right]$$

$$= x\frac{1}{(-1+4)}\sin x - \frac{2D}{(-1+4)^2}\sin x$$

$$= \frac{x}{3}\sin x - \frac{2}{9}D(\sin x)$$

$\Rightarrow$ P.I. $= \dfrac{x}{3}\sin x - \dfrac{2}{9}\cos x$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$\Rightarrow \quad y = C_1\cos 2x + C_2\sin 2x + \dfrac{1}{3}x\sin x - \dfrac{2}{9}\cos x$ **Ans.**

Ex.27 $(D^2 - 1)y = x^2\cos x.$ **[MREC (Auto), 2001]**

Sol. A. E. $\rightarrow m^2 - 1 = 0 \Rightarrow m = 1, -1$

$\therefore$ C.F. $= C_1 e^x + C_2 e^{-x}$

P.I. $= \dfrac{1}{(D^2-1)}x^2\cos x$

$$= \text{R.P.}\frac{1}{(D^2-1)}x^2 e^{ix} = \text{R.P.}\, e^{ix}\frac{1}{\{(D+i)^2 - 1\}}x^2$$

$$= \text{R.P.}\, e^{ix}\frac{1}{(D^2 + 2Di - 2)}x^2$$

$$= \text{R.P.}\frac{e^{ix}}{-2}\left(1 - \frac{D^2 + 2Di}{2}\right)^{-1}x^2$$

$$= \text{R.P.}\frac{e^{ix}}{-2}\left[1 + \frac{D^2+2Di}{2} + \left(\frac{D^2+2Di}{2}\right)^2 + \ldots\right]x^2$$

$$= \text{R.P.}\frac{e^{ix}}{-2}\left(1 + \frac{D^2+2Di}{2} - D^2 + \ldots\right)x^2$$

$$= \text{R.P.}\frac{e^{ix}}{-2}[x^2+1+2xi-2]$$

$$= \text{R.P.}\frac{(\cos x + i\sin x)}{-2}(x^2-1+i2x)$$

$$= -\frac{1}{2}(x^2-1)\cos x + x\sin x$$

$\therefore$ C.S. is :
y = C.F. + P.I.

$$\Rightarrow \quad y = C_1e^{x} + C_2e^{-x} + x\sin x + \frac{1}{2}(1-x^2)\cos x \quad \textbf{Ans.}$$

Ex.28 $\dfrac{d^3y}{dx^3} - 7\dfrac{dy}{dx} - 6y = e^{2x}(1+x)$.

Sol. Given equation is :

$$(D^3-7D-6)y = e^{2x}(1+x)$$

A. E. $\to$ $m^3-7m-6=0 \Rightarrow (m+1)(m+2)(m-3)=0$

$$\Rightarrow m = -1, -2, 3$$

$\therefore$ C.F. $= C_1e^{-x} + C_2e^{-2x} + C_3e^{3x}$

P.I. $= \dfrac{1}{D^3-7D-6}e^{2x}(1+x)$

$$= e^{2x}\frac{1}{(D+2)^3-7(D+2)-6}(1+x)$$

$$= e^{2x}\frac{1}{D^3+6D^2+12D+8-7D-14-6}(1+x)$$

$$= e^{2x}\frac{1}{(D^3+6D^2+5D-12)}(1+x)$$

$$= e^{2x}\frac{1}{-12}\left(1-\frac{D^3+6D^2+5D}{12}\right)^{-1}(1+x)$$

$$= -\frac{1}{12}e^{2x}\left(1+\frac{D^3+6D^2+5D}{12}+\ldots\right)(1+x)$$

$$= -\frac{1}{12}e^{2x}\left(1+x+\frac{5}{12}\right) = -\frac{1}{12}e^{2x}\left(x+\frac{17}{12}\right)$$

$\therefore$ C.S. is :
y = C.F. + P.I.

$$\Rightarrow \quad y = C_1e^{-x} + C_2e^{-2x} + C_3e^{3x} - \frac{1}{12}e^{2x}\left(x+\frac{17}{12}\right) \quad \textbf{Ans.}$$

Ex.29 $(D^4+2D^2+1)y = x^2\cos x$.

Sol. A. E. $\to$ $m^4+2m^2+1=0 \Rightarrow (m^2+1)^2=0 \Rightarrow m = \pm i, \pm i$

$\therefore$ C.F. $= (C_1+C_2x)\cos x + (C_3+C_4x)\sin x$

P.I. $= \dfrac{1}{D^4+2D^2+1}x^2\cos x$

$$= \text{Real part}\frac{1}{D^4+2D^2+1}x^2e^{ix} = \text{R.P.}\frac{1}{(D^2+1)^2}x^2e^{ix}$$

$$= \text{R.P.}e^{ix}\frac{1}{[(D+i)^2+1]^2}x^2 = \text{R.P.}\ e^{ix}\frac{1}{(D^2+2Di)^2}x^2$$

or, P.I. $= \text{R.P.}e^{ix}\dfrac{1}{-4D^2}\left(1+\dfrac{D}{2i}\right)^{-2}x^2$

$$= -\frac{1}{4}\text{R.P.}\ e^{ix}\frac{1}{D^2}\left(1-\frac{D}{i}-\frac{3D^2}{4}\ldots\right)x^2$$

$$= -\frac{1}{4}\text{R.P.}\ e^{ix}\frac{1}{D^2}\left(x^2-\frac{2x}{i}-\frac{3}{2}\right)$$

$$= -\frac{1}{4}\text{R.P.}\ e^{ix}\left(\frac{x^4}{12}-\frac{x^3}{3i}-\frac{3}{4}x^2\right)$$

$$= -\frac{1}{4}\text{R.P.}\ (\cos x + i\sin x)\left(\frac{x^4}{12}-\frac{x^3}{3i}-\frac{3}{4}x^2\right)$$

$$= -\frac{1}{4}\left[\left(\frac{x^4}{12}-\frac{3}{4}x^2\right)\cos x - \frac{x^3}{3}\sin x\right]$$

$\therefore$ C.S. is :

$y = \text{C.F.} + \text{P.I.}$

$\Rightarrow \quad y = (C_1 + C_2 x)\cos x + (C_3 + C_4 x)\sin x$

$$+\frac{1}{12}x^3 \sin x + \frac{1}{48}(9x^2 - x^4)\cos x \quad \text{Ans}$$

Ex.30 $(D-1)^2(D^2+1)^2 y = \sin^2 \frac{x}{2} + e^x$.

Sol. $A.E. \rightarrow \quad (m-1)^2(m^2+1)^2 = 0$

$\Rightarrow \quad m = 1, 1, \pm i, \pm i$

$\therefore \quad C.F. = (c_1 + c_2 x)e^x + (c_3 + c_4 x)\cos x + (c_5 + c_6 x)\sin x$

$$P.I. = \frac{1}{(D-1)^2(D^2+1)^2}\sin^2\frac{x}{2} + \frac{1}{(D-1)^2(D^2+1)^2}e^x$$

$$= \frac{1}{(D^2-2D+1)(D^2+1)^2}\left(\frac{1-\cos x}{2}\right) + \frac{1}{4(D-1)^2}e^x$$

$$= \frac{1}{2}\left[\frac{1}{(D^2-2D+1)(D^2+1)^2}e^{0x} - \frac{1}{(D^2-2D+1)(D^2+1)^2}\cos x\right] + \frac{1}{4}\left(\frac{x^2 e^x}{2}\right)$$

$$= \frac{1}{2}\left[\left(\frac{1}{1}\right)e^{0x} - \frac{1}{(-1-2D+1)(D^2+1)^2}\cos x\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2}\left[1 + \frac{1}{2D(D^2+1)^2}\cos x\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2}\left[1 + \frac{1}{2(D^2+1)^2}\sin x\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2}\left[1 + \frac{1}{2(D^2+1)}\frac{1}{(D^2+1)}\sin x\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2}\left[1 + \frac{1}{2(D^2+1)}\left(-\frac{x}{2}\cos x\right)\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2}\left[1 - \frac{1}{4}\frac{1}{(D^2+1)}x\cos x\right] + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - \frac{1}{8(D^2+1)}x\cos x + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{1}{8(D^2+1)}xe^{ix} + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot Pe^{ix}\frac{1}{8\{(D+i)^2+1\}}x + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{e^{ix}}{8}\left(\frac{1}{D^2+2Di}\right)x + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{e^{ix}}{8}\cdot\frac{1}{2Di}\left(1+\frac{D}{2i}\right)^{-1}x + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{e^{ix}}{16Di}\left(1-\frac{D}{2i}+\ldots\right)x + \frac{1}{8}x^2 e^x$$

or, $\quad P.I. = \frac{1}{2} - R\cdot P\frac{e^{ix}}{16iD}\left(x - \frac{1}{2i}\right) + \frac{1}{8}x^2 e^x$

$$= \frac{1}{2} - R\cdot P\frac{e^{ix}}{16i}\left(\frac{x^2}{2} - \frac{x}{2i}\right) + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{e^{ix}}{32}\left(\frac{x^2}{i} + x\right) + \frac{1}{8}x^2 e^x$$

$$= \frac{1}{2} - R\cdot P\frac{1}{32}(\cos x + i\sin x)\left(\frac{x^2}{i} + x\right) + \frac{1}{8}x^2 e^x$$

or, $\quad P.I. = \frac{1}{2} - \frac{1}{32}\left[x\cos x + x^2\sin x\right] + \frac{1}{8}x^2 e^x$

$\therefore$ C. S. is :

$y = C.F. + P.I.$

$$\Rightarrow \quad y = (c_1 + c_2 x)e^x + (c_3 + c'_4 x)\cos x + (c_5 + c_6 x)\sin x$$

$$+\frac{1}{2} - \frac{1}{32}x\cos x - \frac{1}{32}x^2 \sin x + \frac{1}{8}x^2 e^x$$

or $\quad y = (c_1 + c_2 x)e^x + (c_3 + c_4 x)\cos x + (c_5 + c_6 x)\sin x + \frac{1}{2}$

$+\frac{1}{32}\left[-x^2 \sin x + 4x^2 e^x\right]$; (where $c_4 = c'_4 - \frac{1}{32}$) **Ans.**

Ex.31 $(D^4 + D^2 + 1)\,y = ax^2 + be^{-x} \sin 2x.$

Sol. A. E. is –

$$m^4 + m^2 + 1 = 0$$

$$\Rightarrow \quad (m^2 + m + 1)(m^2 - m + 1) = 0$$

$$\Rightarrow \quad m^2 + m + 1 = 0 \text{ or } m^2 - m + 1 = 0$$

$$\therefore \quad m = \frac{-1 \pm i\sqrt{3}}{2} \text{ or } m = \frac{1 \pm i\sqrt{3}}{2}$$

$$\therefore \quad \text{C. F.} = e^{-\frac{x}{2}}\left(c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x\right)$$

$$+e^{\frac{x}{2}}\left(c_3 \cos\frac{\sqrt{3}}{2}x + c_4 \sin\frac{\sqrt{3}}{2}x\right)$$

$$\text{P. I.} = \frac{1}{(D^4 + D^2 + 1)}ax^2 + \frac{1}{(D^4 + D^2 + 1)}be^{-x}\sin 2x$$

$$= a\left[1 + (D^4 + D^2)\right]^{-1} x^2$$

$$+be^{-x}\frac{1}{\left[(D-1)^4 + (D-1)^2 + 1\right]}\sin 2x$$

$$= a\left[1 - (D^4 + D^2) + \ldots\ldots\right]x^2 + be^{-x}$$

$$\frac{1}{\left[(D^4 - 4D^3 + 6D^2 - 4D + 1) + (D^2 - 2D + 1) + 1\right]}\sin 2x$$

$$= a(x^2 - 2) + be^{-x}\frac{1}{\left[(-4)^2 - 4D(-4) + 7(-4) - 6D + 3\right]}\sin 2x$$

$$= a(x^2 - 2) + be^{-x}\frac{1}{(10D - 9)}\sin 2x$$

$$= a(x^2 - 2) + be^{-x}\frac{(10D + 9)}{(100D^2 - 81)}\sin 2x$$

$$= a(x^2 - 2) + be^{-x}\frac{(10D + 9)}{(100(-4) - 81)}\sin 2x$$

$$= a(x^2 - 2) - \frac{be^{-x}}{481}(20\cos 2x + 9\sin 2x)$$

$\therefore$ C. S. is –

$y = C.F. + P.I.$

$$\text{or,} \quad y = e^{-\frac{x}{2}}\left(c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x\right)$$

$$+e^{-\frac{x}{2}}\left(c_3 \cos\frac{\sqrt{3}}{2}x + c_4 \sin\frac{\sqrt{3}}{2}x\right) + a(x^2 - 2)$$

$$-\frac{1}{481}be^{-x}(20\cos 2x + 9\sin 2x) \qquad \textbf{Ans.}$$

Ex.32 $(D^3 + 8)y = x^4 + 2x + 1.$

Sol. A. E. $\rightarrow$ $m^3 + 8 = 0 \Rightarrow (m + 2)(m^2 - 2m + 4) = 0$

$$\Rightarrow m = -2, m^2 - 2m + 4 = 0 \Rightarrow m = \frac{2 \pm \sqrt{4 - 16}}{2} = 1 \pm i\sqrt{3}$$

$$\therefore \quad \text{C.F.} = C_1 e^{-2x} + e^x(C_2 \cos\sqrt{3}x + C_3 \sin\sqrt{3}x)$$

Scanned with

$$\text{P.I.} = \frac{1}{(D^3+8)}(x^4+2x+1)$$

$$= \frac{1}{8}\left(1+\frac{D^3}{8}\right)^{-1}(x^4+2x+1)$$

$$= \frac{1}{8}\left(1-\frac{D^3}{8}+.....\right)(x^4+2x+1)$$

$$= \frac{1}{8}(x^4+2x+1-3x) = \frac{1}{8}(x^4-x+1)$$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$\Rightarrow \quad y = C_1e^{-2x} + e^x(C_2\cos\sqrt{3}x + C_3\sin\sqrt{3}x) + \frac{1}{8}(x^4-x+1)$ Ans.

Ex.33 $(D^2-3D+2)y = xe^x$.

Sol. A. E. $\rightarrow m^2-3m+2=0 \Rightarrow (m-2)(m-1)=0 \Rightarrow m=1,2$

$\therefore$ C.F. $= C_1e^x + C_2e^{2x}$

$$\text{P.I.} = \frac{1}{D^2-3D+2}xe^x$$

$$= e^x\frac{1}{(D+1)^2-3(D+1)+2}x = e^x\frac{1}{(D^2-D)}x$$

$$= e^x\frac{1}{-D(1-D)}x = -e^x\frac{1}{D}(1-D)^{-1}x$$

$$= -e^x\frac{1}{D}(1+D+D^2+.....)x = -e^x\frac{1}{D}(x+1)$$

$$= -e^x\left(\frac{x^2}{2}+x\right) = -\frac{1}{2}xe^x(x+2)$$

$\therefore$ C.S. is :

$y =$ C.F. + P.I.

$\Rightarrow \quad y = C_1e^x + C_2e^{2x} - \frac{1}{2}xe^x(x+2)$ **Ans.**